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W1 Lesson 1

Introduction to Probability



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Introduction to probability

What is Probability?

Introduction to Probability



Find the probability that a child picked at random plays soccer.

Introduction to Probability



Find the probability that a child picked at random plays soccer.

The probability that a child picked at random plays soccer.

$P(\text{soccer})$

Introduction to Probability

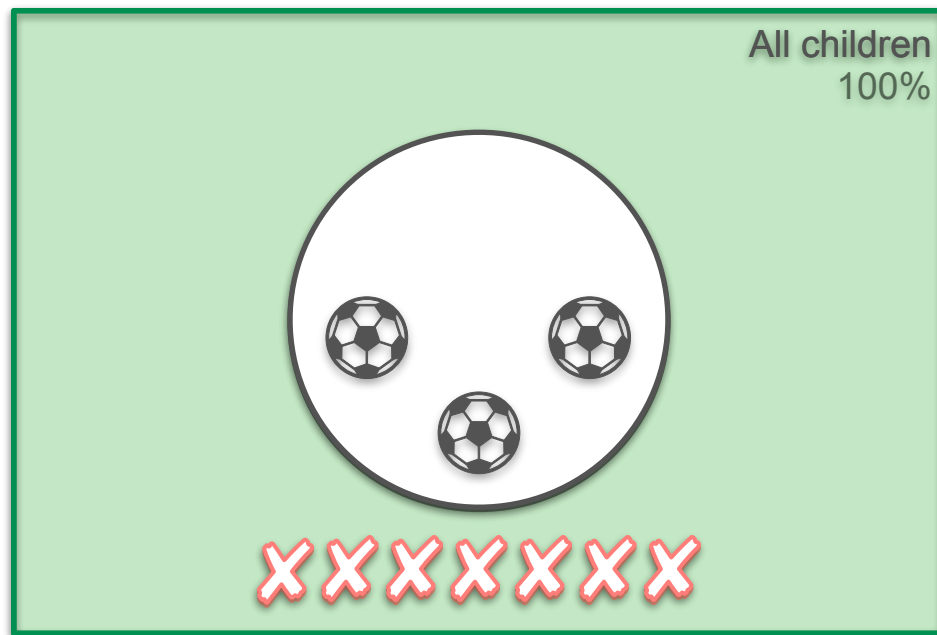


Find the probability that a child picked at random plays soccer.

$$P(\text{soccer}) = \frac{\text{soccer}}{\text{total}} = \frac{\text{Event}}{\text{Sample space}} = \frac{3}{10} = 0.3$$

The diagram illustrates the probability calculation. A large blue rounded rectangle contains 10 icons: 3 soccer balls and 7 red X marks. A smaller blue rounded rectangle above it contains only the 3 soccer balls. An arrow points from the label "Event" to the smaller rectangle, and another arrow points from the label "Sample space" to the larger rectangle.

Introduction to Probability: Venn Diagram



Sample Space

Event

Introduction to Probability: Coin Example 1



Experiment

Probability of landing on heads

$P(\text{heads})$

Introduction to Probability: Coin Example 1



50% 50%

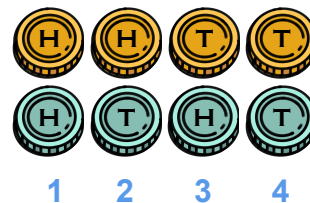
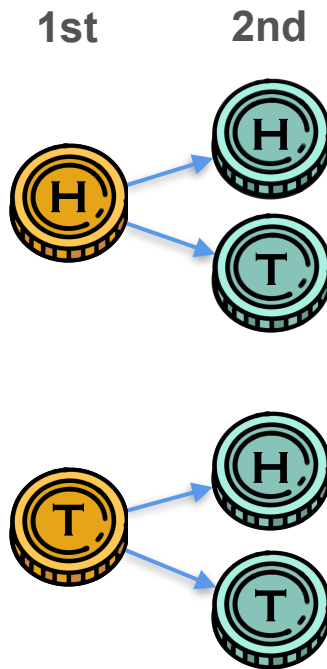
$$P(\text{heads}) = \frac{\text{H}}{\text{H} \quad \text{T}} = \frac{1}{2} = 0.5$$

Introduction to Probability: Coin Example 2



50% 50%

What is the probability of landing on heads twice?



Introduction to Probability: Coin Example 2



50% 50%



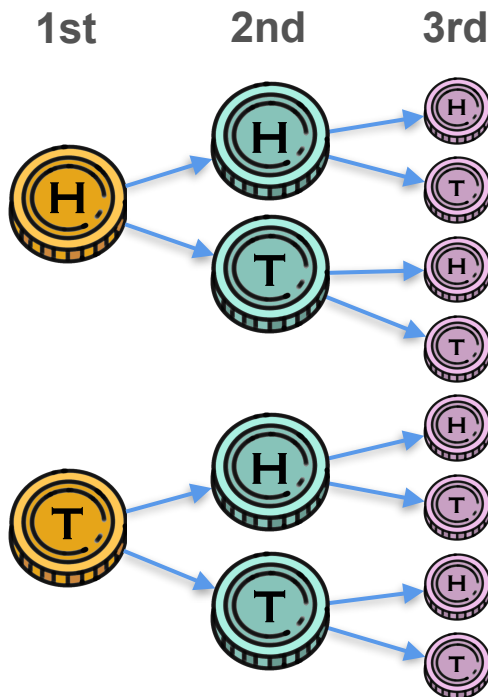
$$P(HH) = \text{—————} = \frac{1}{4} = 0.25$$

Introduction to Probability: Coin Example 3



50% 50%

What is the probability of landing on heads 3 times?

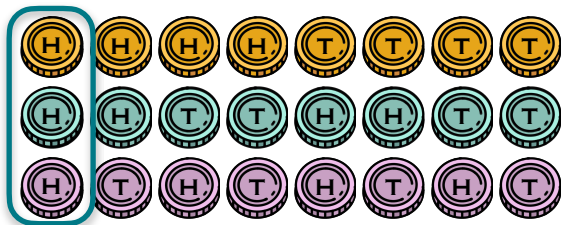


1 2 3 4 5 6 7 8

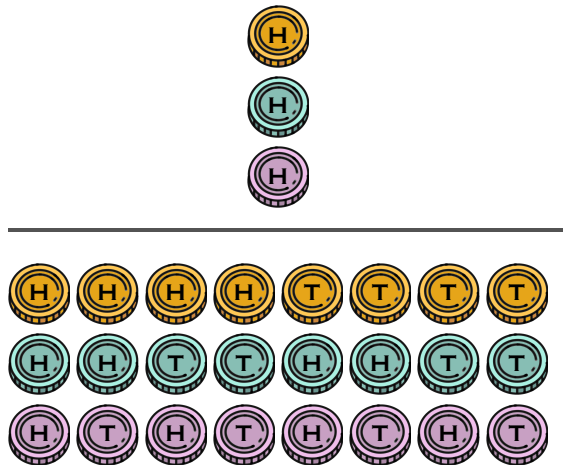
Introduction to Probability: Coin Example 3



50% 50%



$P(HHH) =$



$$= \frac{1}{8} = 0.125$$

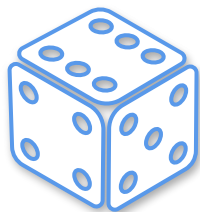


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Introduction to Probability

What is Probability? - Dice Example

Introduction to Probability: Dice Example 1

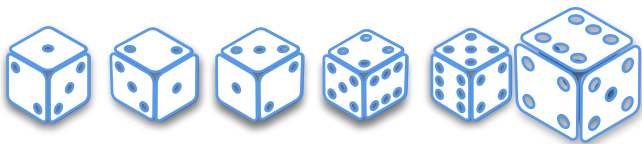


What is the probability of obtaining 6?

Introduction to Probability: Dice Example 1

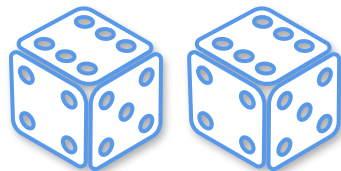


What is the probability of obtaining 6?



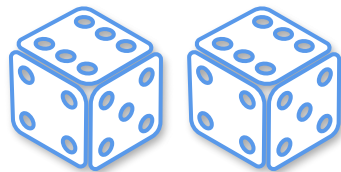
$$P(6) = \text{_____} = \frac{1}{6}$$

Introduction to Probability: Dice Example 2



What is the probability of obtaining 6,6?

Introduction to Probability: Dice Example 2



What is the probability of obtaining 6,6?

A vertical column of six blue-outlined dice icons is positioned to the left of the table. Each die shows a different face, representing the rows of the table.

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$$P(6,6) = \frac{\text{Number of outcomes where both dice are 6}}{\text{Total number of outcomes}} = \frac{1}{36}$$

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6



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Introduction to probability

Complement of Probability

Complement of Probability



30%

What is the probability of a child NOT playing soccer?

Complement of Probability

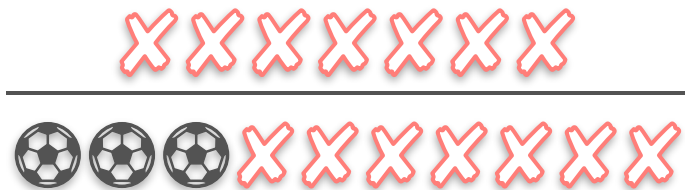


30%

What is the probability of a child NOT playing soccer?

$$P(\text{not soccer}) = \frac{\text{not soccer}}{\text{total}} = \frac{\text{X X X X X X X X}}{\text{⚽ ⚽ ⚽ X X X X X X X X}} = \frac{7}{10} = 0.7$$

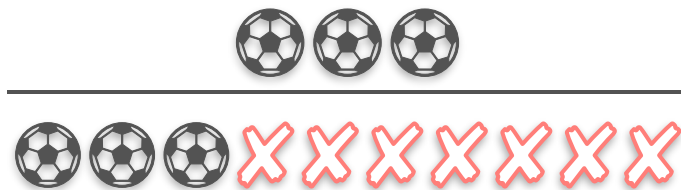
Complement of Probability



$P(\text{not soccer})$

0.7

+



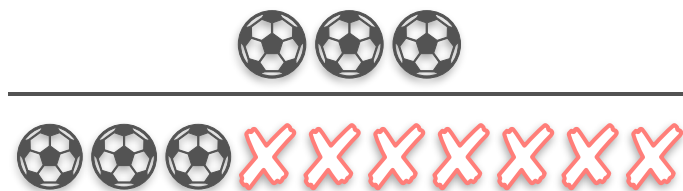
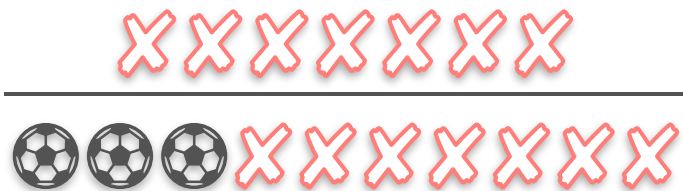
$P(\text{soccer})$

0.3

=

1

Complement of Probability



$P(\text{not soccer})$

$P(\text{soccer})$

0.7

=

1

-

0.3

Complement Rule

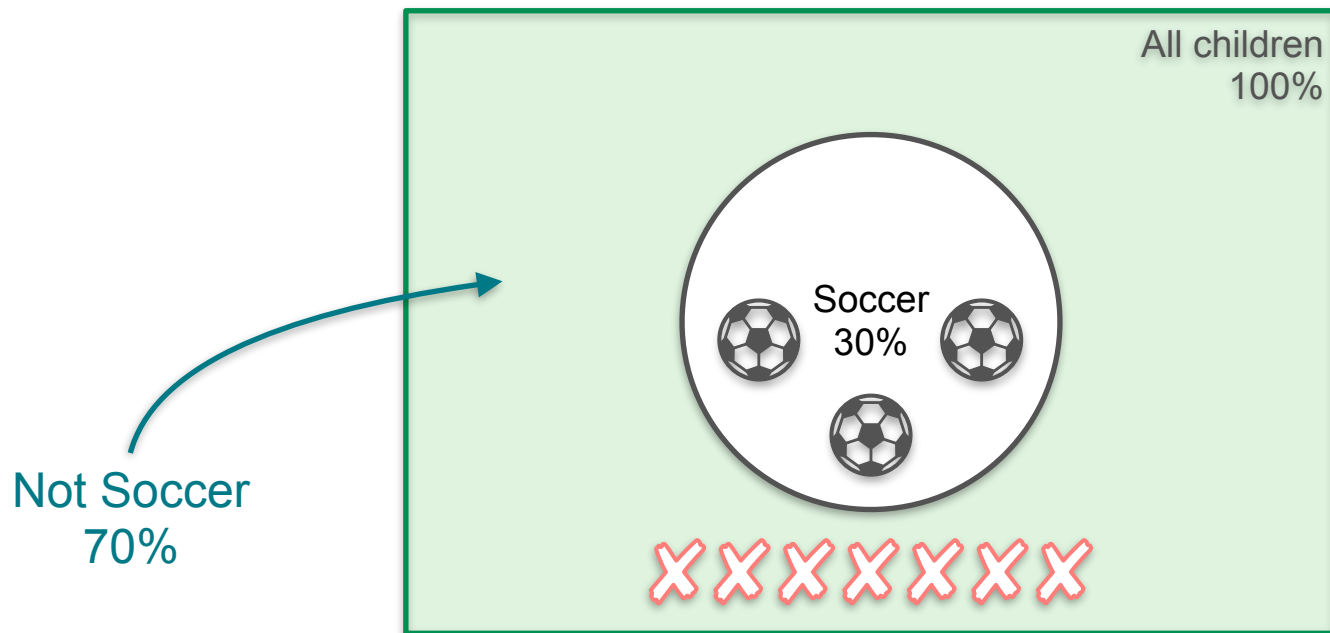
Complement of Probability

$$P(\text{not soccer}) = 1 - P(\text{soccer})$$

$$P(A') = 1 - P(A)$$

Complement Rule

Complement of Probability: Venn Diagram



Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

$$P(\text{not } HHH) = 1 - P(HHH)$$

$$P(HHH) = \frac{\begin{array}{c} \text{H} \\ \text{H} \\ \text{H} \end{array}}{\begin{array}{ccccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{T} & \text{T} & \text{T} & \text{T} \\ \text{H} & \text{H} & \text{T} & \text{T} & \text{H} & \text{H} & \text{T} & \text{T} \\ \text{H} & \text{T} & \text{H} & \text{T} & \text{H} & \text{T} & \text{H} & \text{T} \end{array}}$$

Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

$$P(\text{not } HHH) = 1 - P(HHH)$$

$$P(\text{not } HHH) = 1 - \frac{1}{8}$$
A diagram illustrating the 8 possible outcomes for three coins. The outcomes are arranged in three rows. The top row has three coins stacked vertically, all showing heads (H). The middle row has eight coins in a single row, showing the sequence H, H, H, T, T, T, T, T. The bottom row has eight coins in a single row, showing the sequence H, T, H, T, H, T, H, T. The coins are colored yellow, teal, and purple from left to right in each row.

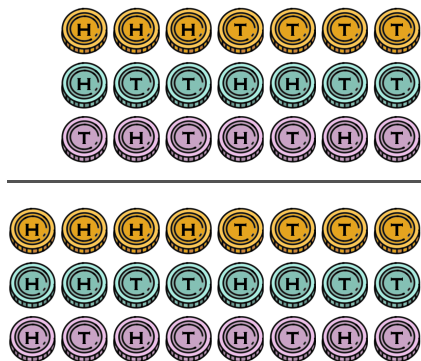
Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

$$P(\text{not } HHH) = 1 - P(HHH)$$

$$\begin{aligned} P(\text{not } HHH) &= 1 - \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$



Complement of Probability: Dice Example 1



What is the probability of obtaining anything other than 6?

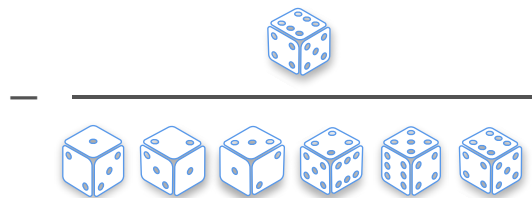
Complement of Probability: Dice Example 1



What is the probability of obtaining anything other than 6?



$P(\text{not } 6) =$



$=$

$$= \frac{5}{6}$$

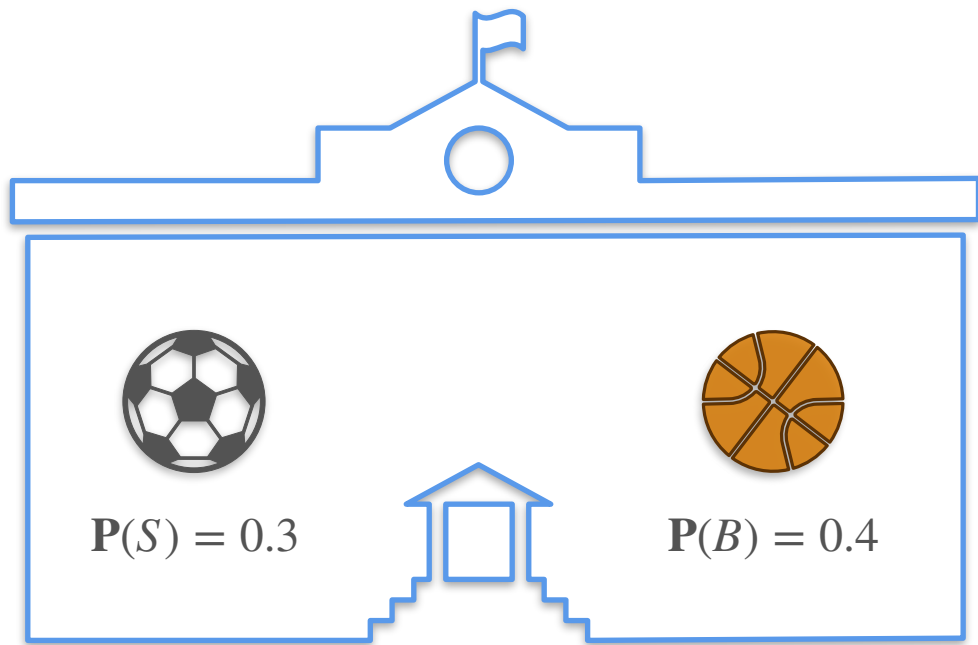


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Introduction to probability

Sum of Probabilities (Disjoint Events)

Sum of Probabilities: Quiz 1

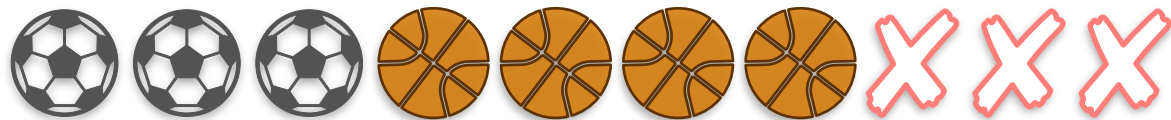


At a school, kids can only play one sport.

What is the probability that a kid plays soccer or basketball?

Hint: What if there were only 10 kids?

Sum of Probabilities: Quiz 1 Solution



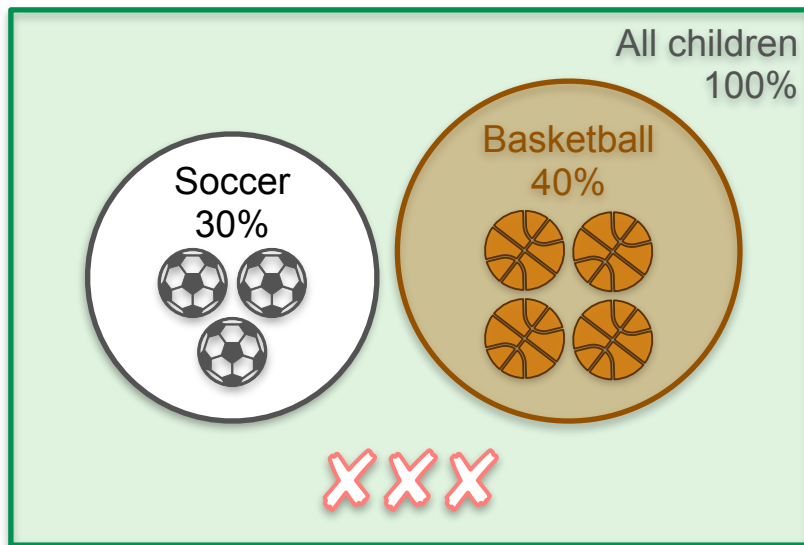
└── 30% ──┐ ┌── 40% ──┐

└────────── 70% ─────────┘

$$P(\text{soccer or basketball}) = \frac{\text{soccer or basketball}}{\text{total}} = \frac{3 + 4}{10} = 0.7$$

$$P(\text{soccer or basketball}) = P(\text{soccer}) + P(\text{basketball})$$

Sum of Probabilities: Quiz 1 Solution



$$P(\text{soccer} \cup \text{basketball}) = P(\text{soccer}) + P(\text{basketball})$$

$$P(A \cup B) = P(A) + P(B)$$

Sum of Probabilities: Dice Example 1



What is the probability of obtaining an even number or a 5?

Sum of Probabilities: Dice Example 1

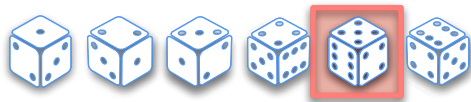


What is the probability of obtaining an even number or a 5?

A



B

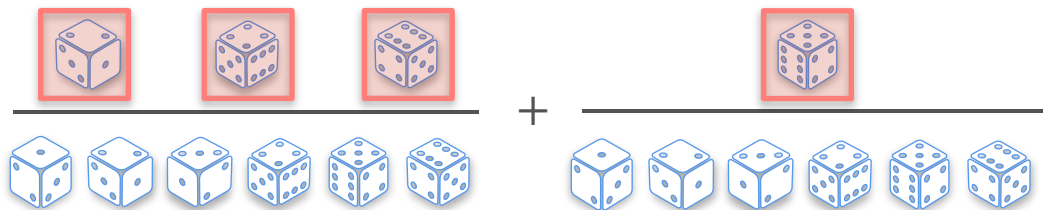


_____ + _____

Sum of Probabilities: Dice Example 1

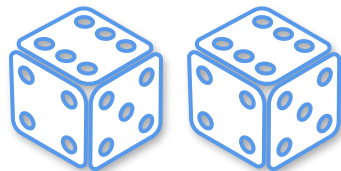


What is the probability of obtaining an even number or a 5?



$$P(\text{even number or } 5) = P(\text{even number}) + P(5) = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

Sum of Probabilities: Dice Example 2



What is the probability of obtaining a sum of 7 or a sum of 10?

Sum of Probabilities: Dice Example 2

A

sum of 7



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

B

sum of 10

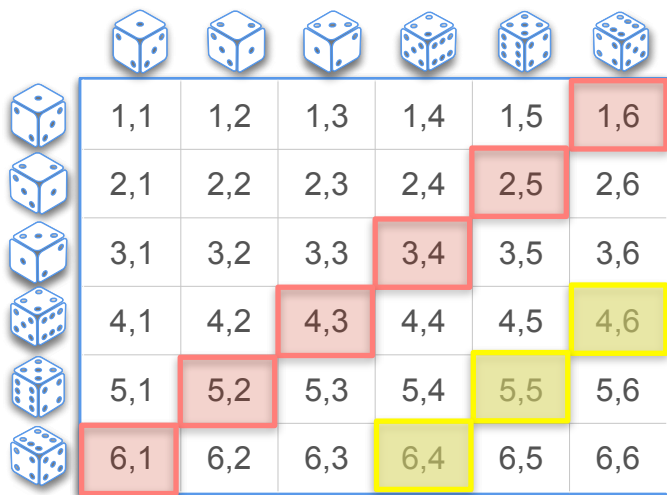


1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Sum of Probabilities: Dice Example 2

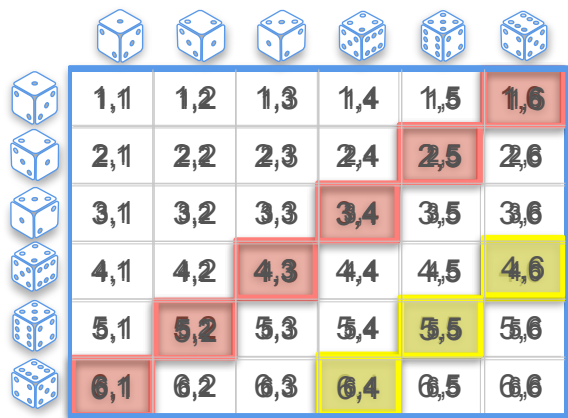
A or B

sum of 7 or sum of 10



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Sum of Probabilities: Dice Example 2



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

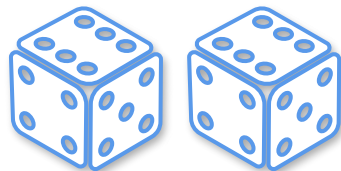
P(sum of 7 or sum of 10)

$$= \boxed{P(\text{sum of 7})} + \boxed{P(\text{sum of 10})}$$

$$= \frac{6}{36} + \frac{3}{36}$$

$$= \frac{9}{36} = \frac{1}{4}$$

Sum of Probabilities: Dice Example 3



What is the probability of obtaining a difference of 2 or a difference of 1?

Sum of Probabilities: Dice Example 3

A

diff = 2



	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

B

diff = 1

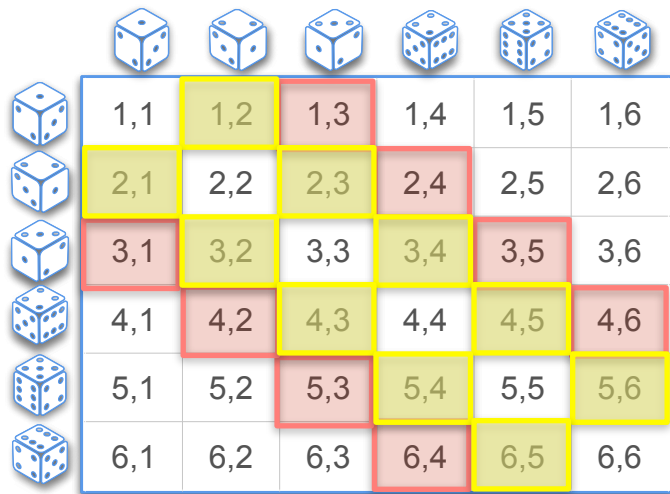


	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

Sum of Probabilities: Dice Example 3

A or B

diff = 2 or diff = 1



A 6x6 grid representing all possible outcomes of two dice rolls. The rows and columns are labeled with dice icons and numbers 1 through 6. The cells are color-coded based on the absolute difference between the two dice values:

- Yellow cells (diff = 1): (1,2), (2,1), (2,3), (3,2), (3,4), (4,3), (4,5), (5,4), (5,6), (6,5)
- Red cells (diff = 2): (1,3), (2,4), (3,5), (4,6), (5,3), (6,4)
- White cells (diff = 0 or 3 or 4 or 5): (1,1), (1,4), (1,5), (1,6), (2,2), (2,5), (2,6), (3,1), (3,3), (3,6), (4,1), (4,2), (4,4), (4,6), (5,1), (5,2), (5,5), (5,6), (6,1), (6,2), (6,3), (6,6)

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

Sum of Probabilities: Dice Example 3

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$P(\text{diff} = 2 \text{ or } \text{diff} = 1)$

$$= \boxed{P(\text{diff} = 2)} + \boxed{P(\text{diff} = 1)}$$

$$= \frac{8}{36} + \frac{10}{36}$$

$$= \frac{18}{36}$$

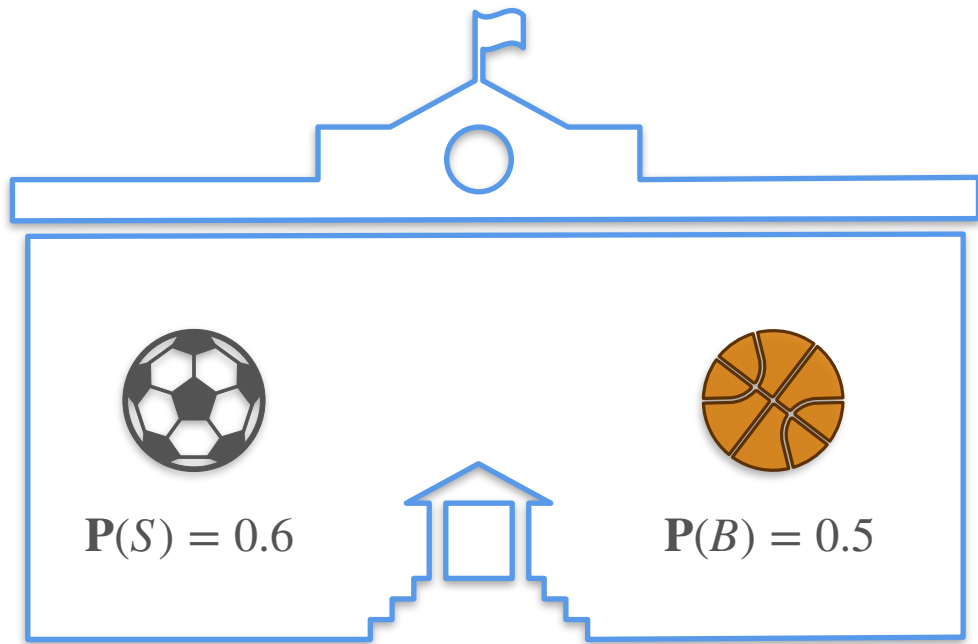


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Introduction to probability

Sum of Probabilities (Joint Events)

Sum of Probabilities (Joint Events): Quiz 1

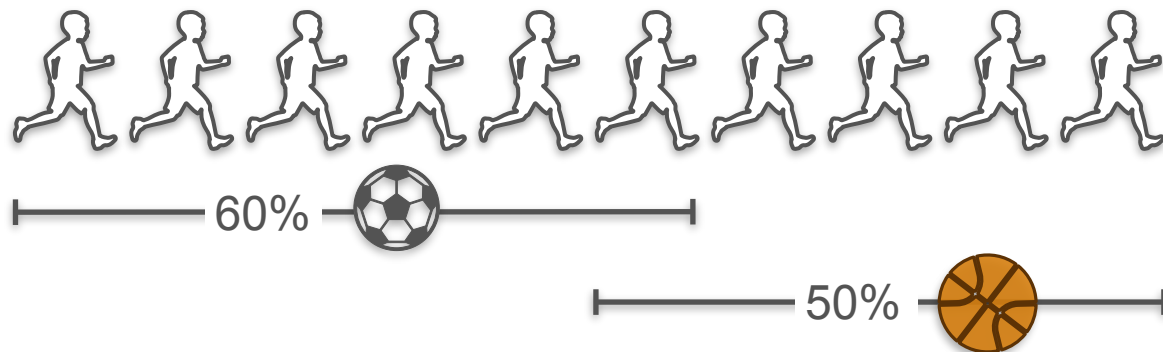


At a school, kids can play as many sports as they want.

What is the probability that a kid plays soccer or basketball?

Hint: What if there were only 10 kids?

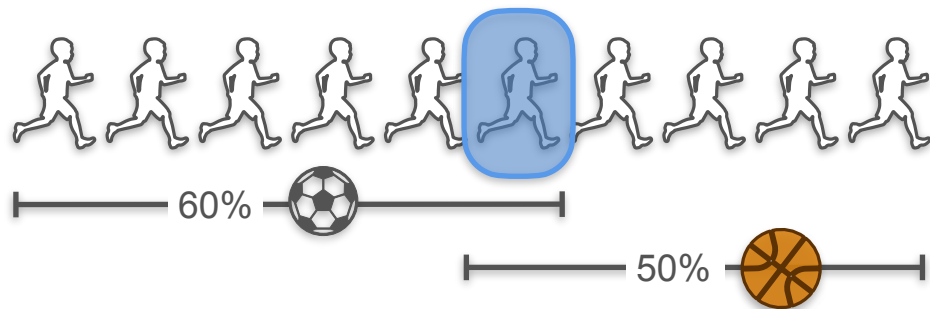
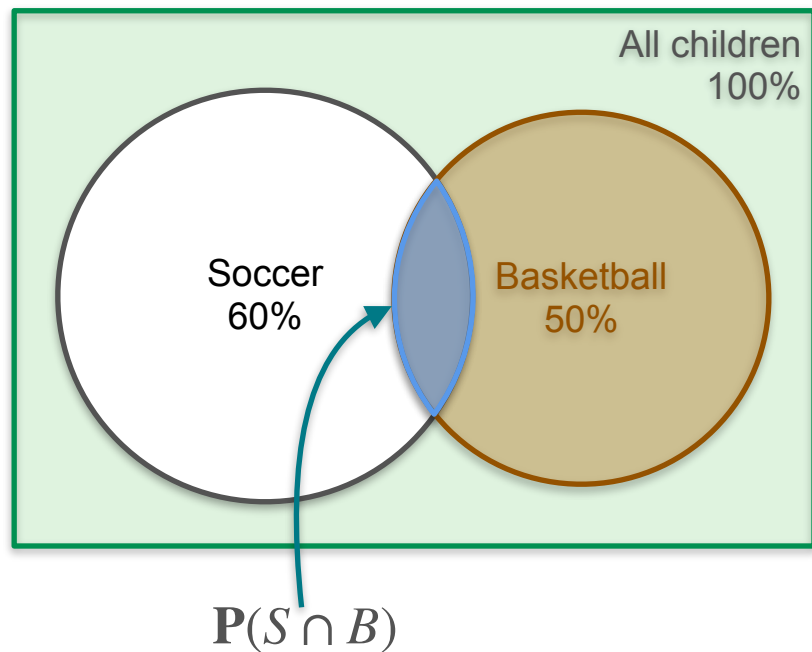
Sum of Probabilities (Joint Events): Quiz 1 Solution



$$P(\text{soccer} \cup \text{basketball}) = ?$$

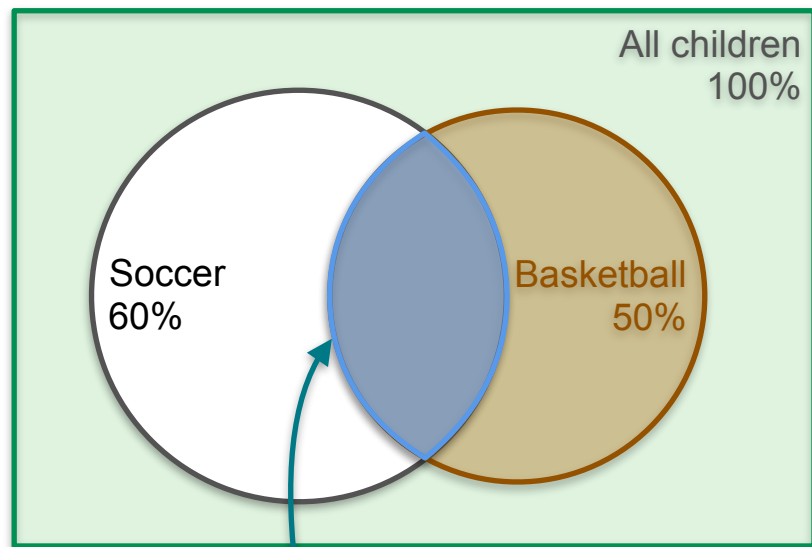
We don't know how many children play multiple sports

Sum of Probabilities (Joint Events): Quiz 1 Solution

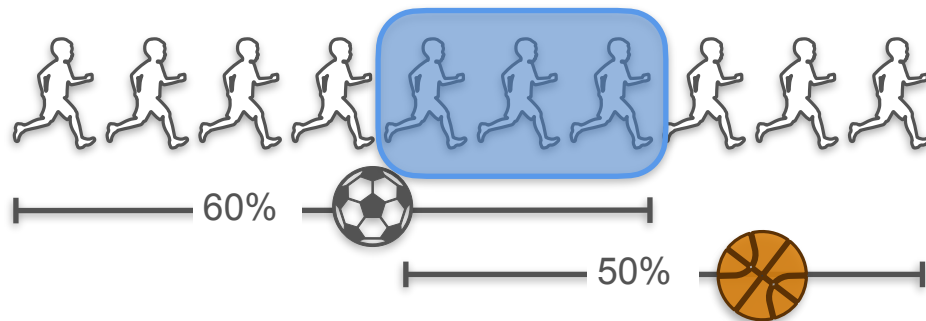


$$P(S \cup B) = P(S) + P(B)$$

Sum of Probabilities (Joint Events): Quiz 1 Solution

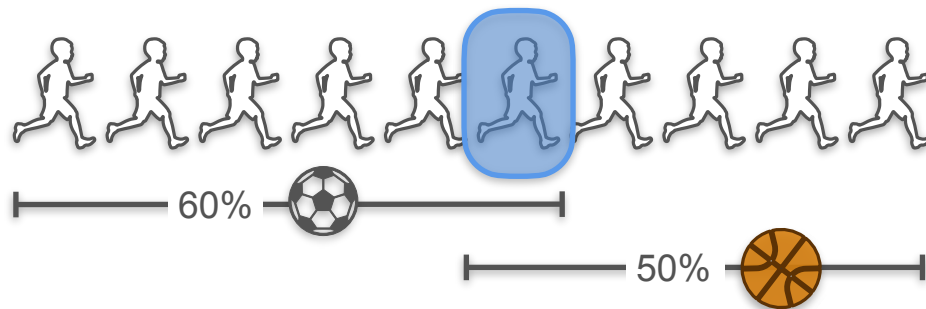
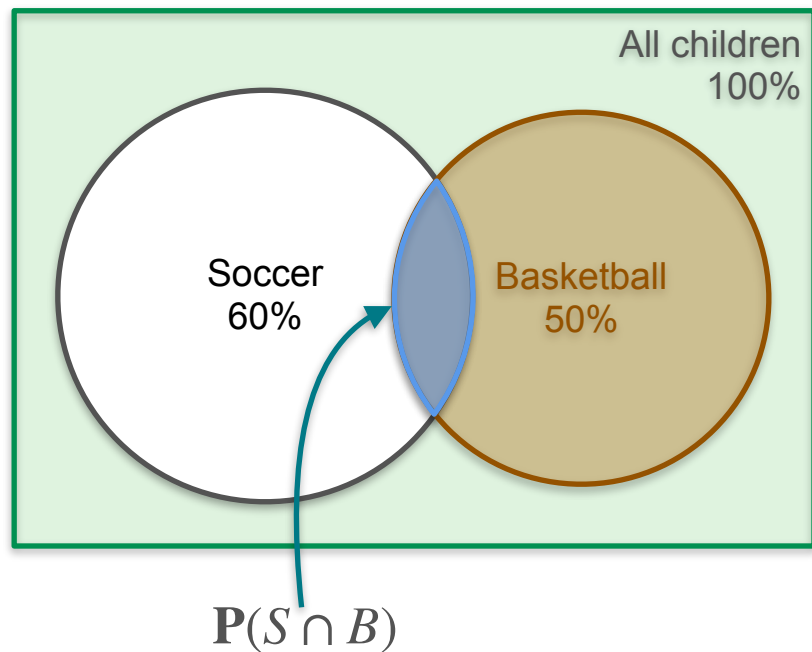


$P(S \cap B)$



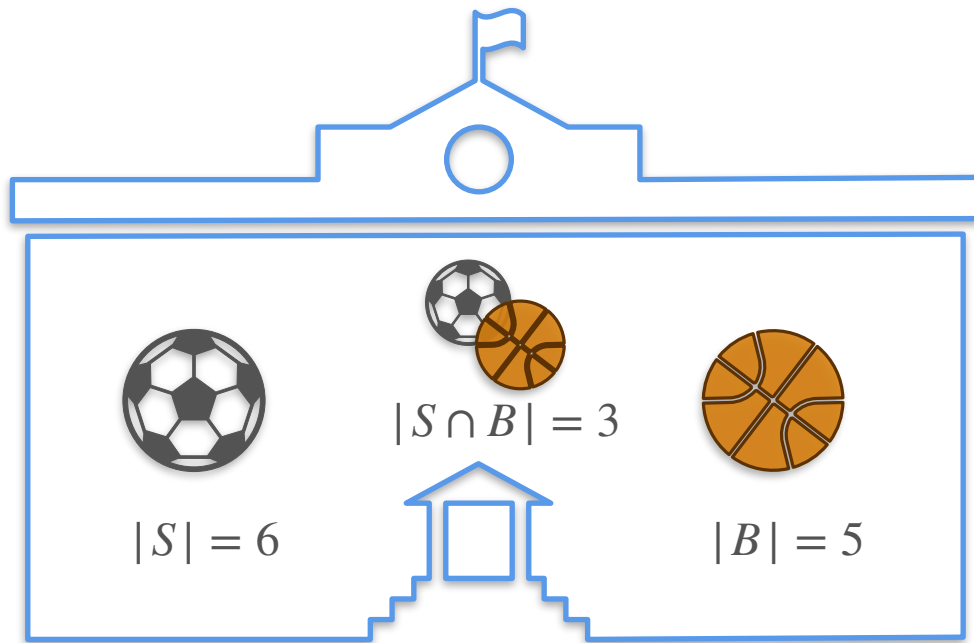
$$P(S \cup B) = P(S) + P(B)$$

Sum of Probabilities (Joint Events): Quiz 1 Solution



$$P(S \cup B) = P(S) + P(B)$$

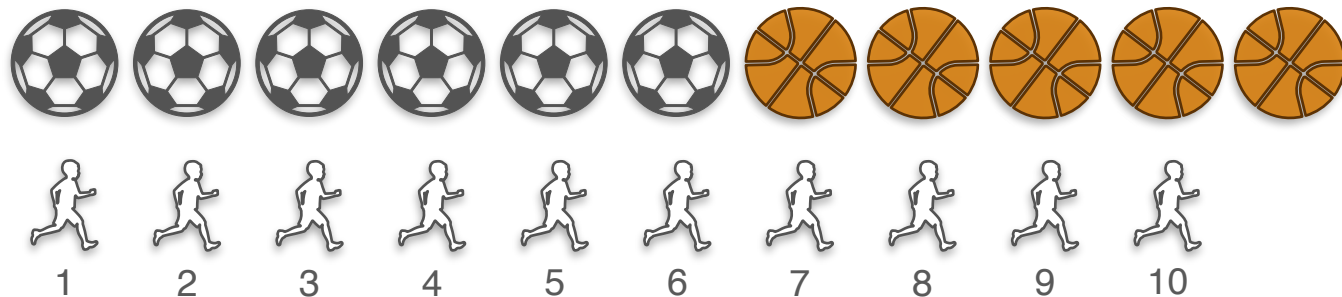
Sum of Probabilities (Joint Events): Quiz 2



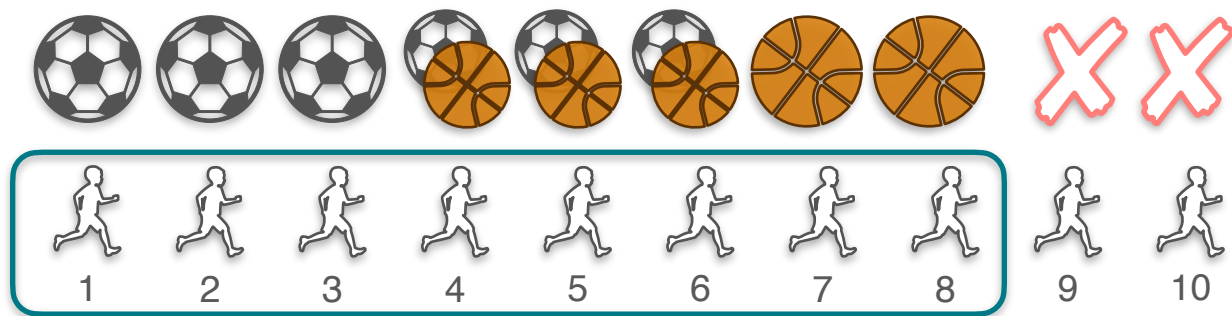
How many kids play soccer or basketball?

Hint: What if there were only 10 kids?

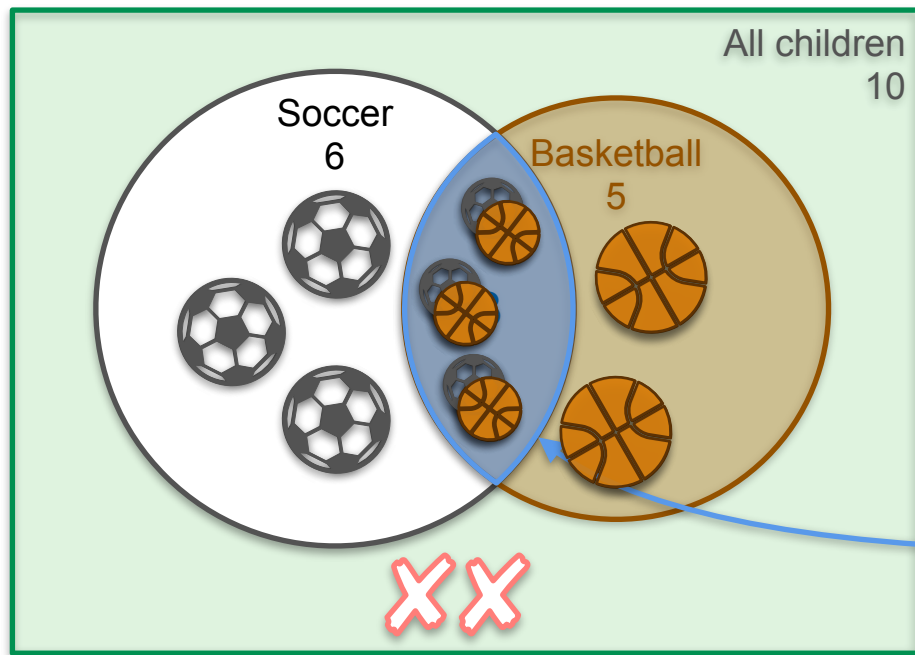
Sum of Probabilities (Joint Events): Quiz 2 Solution



Sum of Probabilities (Joint Events): Quiz 2 Solution



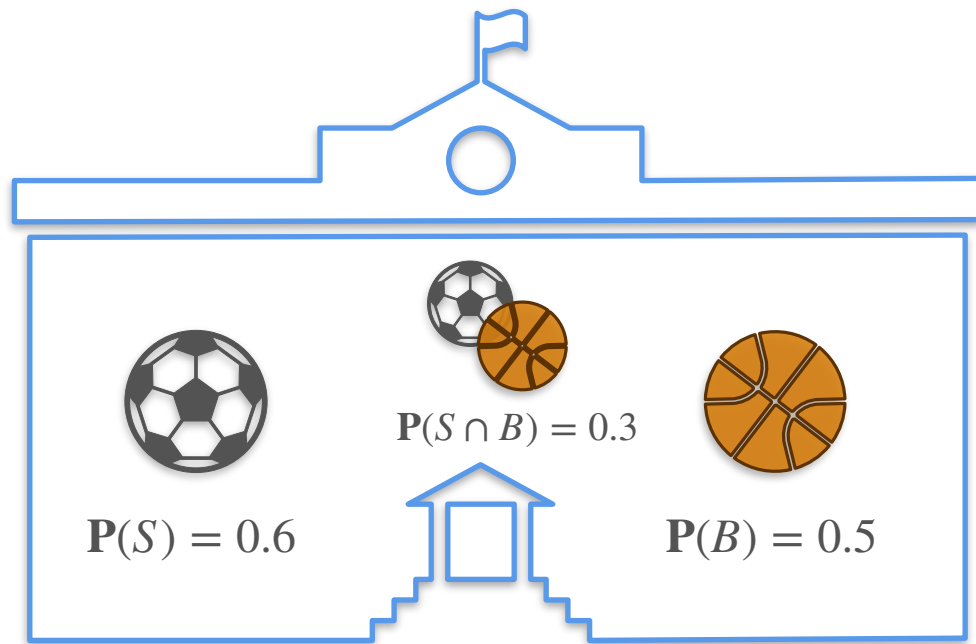
Sum of Probabilities (Joint Events): Venn Diagram



$$\begin{aligned} |S \cup B| &= |S| + |B| - |S \cap B| \\ &= 6 + 5 - 3 \\ &= 8 \end{aligned}$$

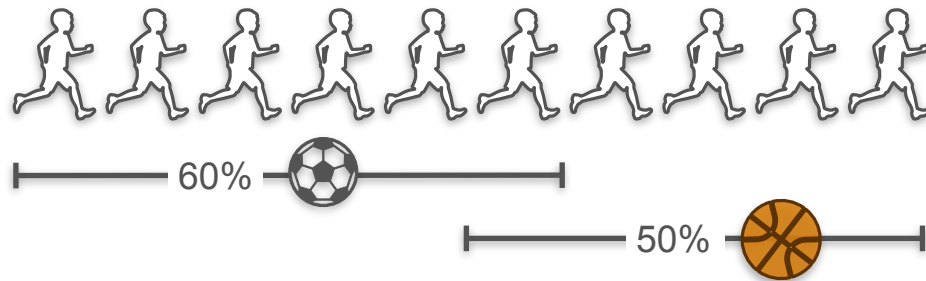
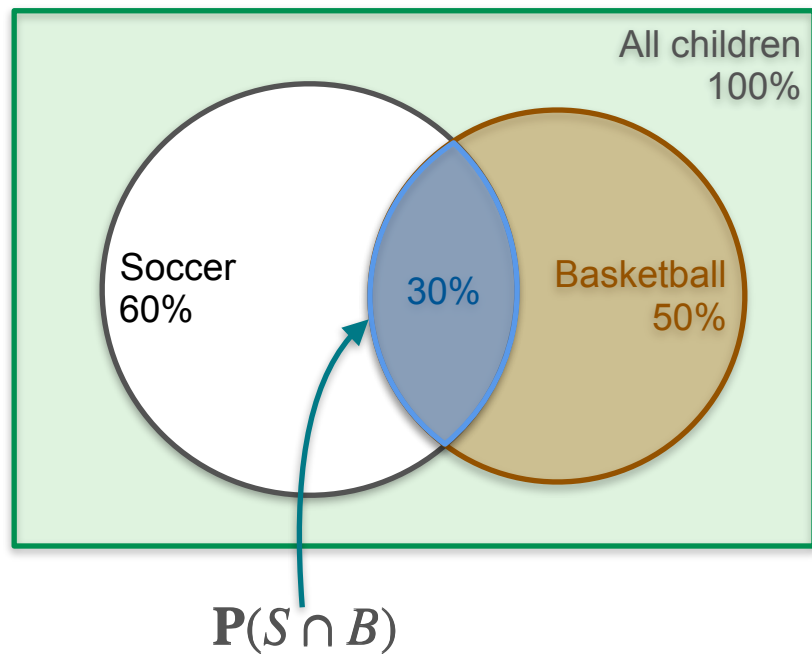
Soccer and Basketball

Sum of Probabilities (Joint Events): Quiz 3



What is the probability that a child plays soccer or basketball?

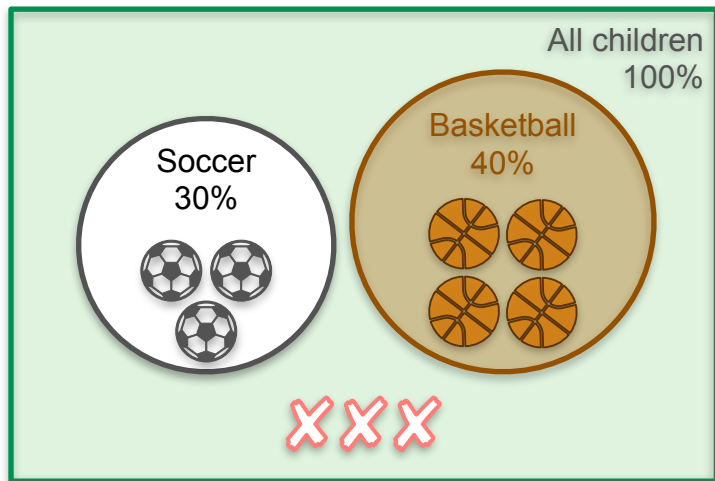
Sum of Probabilities (Joint Events): Quiz 3 Solution



$$\begin{aligned} P(S \cup B) &= P(S) + P(B) - P(S \cap B) \\ &= 0.6 + 0.5 - 0.3 \\ &= 0.8 \end{aligned}$$

Disjoint Events vs Joint Events

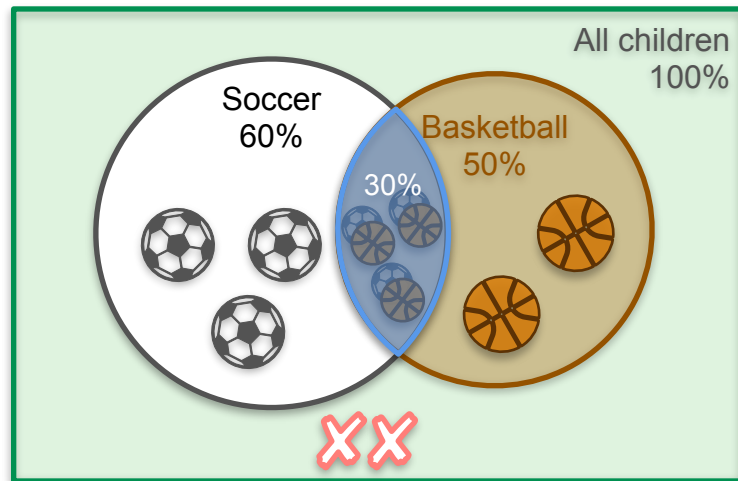
Disjoint



Mutually exclusive

$$\mathbf{P}(S \cup B) = \mathbf{P}(S) + \mathbf{P}(B)$$

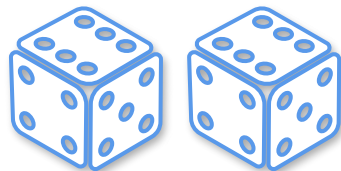
Joint



Non-mutually exclusive

$$\mathbf{P}(S \cup B) = \mathbf{P}(S) + \mathbf{P}(B) - \mathbf{P}(S \cap B)$$

Sum of Probabilities (Joint Events): Dice Example 1



What is the probability of obtaining a sum of 7 or a difference of 1?

Sum of Probabilities (Joint Events): Dice Example 1

A

sum = 7



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

B

diff = 1



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Sum of Probabilities (Joint Events): Dice Example 1

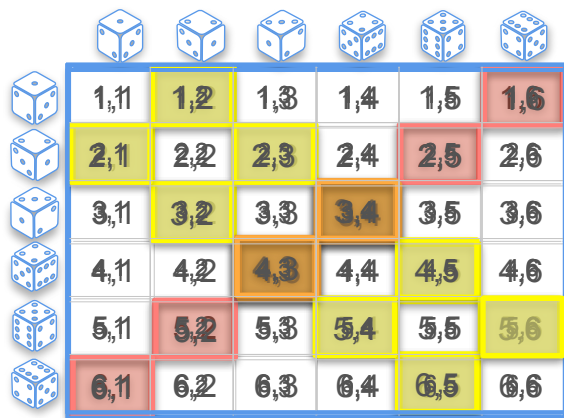
A or B

sum = 7 or diff = 1

	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

sum = 7 and diff = 1

Sum of Probabilities (Joint Events): Dice Example 1



A 6x6 grid representing the outcomes of two dice. The first die is on the left, and the second die is on top. The grid is color-coded: yellow for sum=7, red for diff=1, and orange for both. The cells are as follows:

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

$P(\text{sum} = 7 \text{ or } \text{diff} = 1)$

$$= \boxed{P(\text{sum} = 7)} + \boxed{P(\text{diff} = 1)} - \boxed{P(\text{sum} = 7 \cap \text{diff} = 1)}$$

$$= \frac{6}{36} + \frac{10}{36} - \frac{2}{36}$$
$$= \frac{14}{36}$$

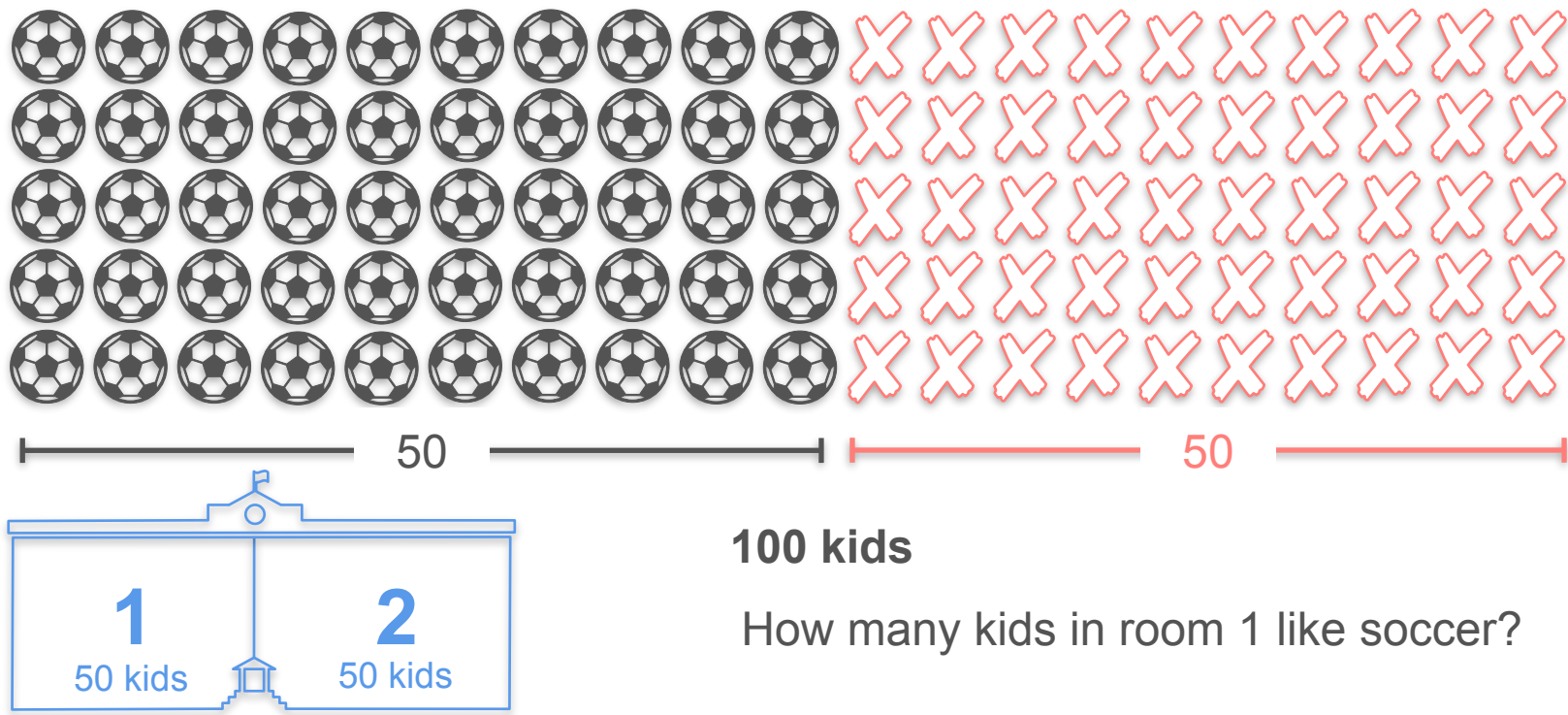


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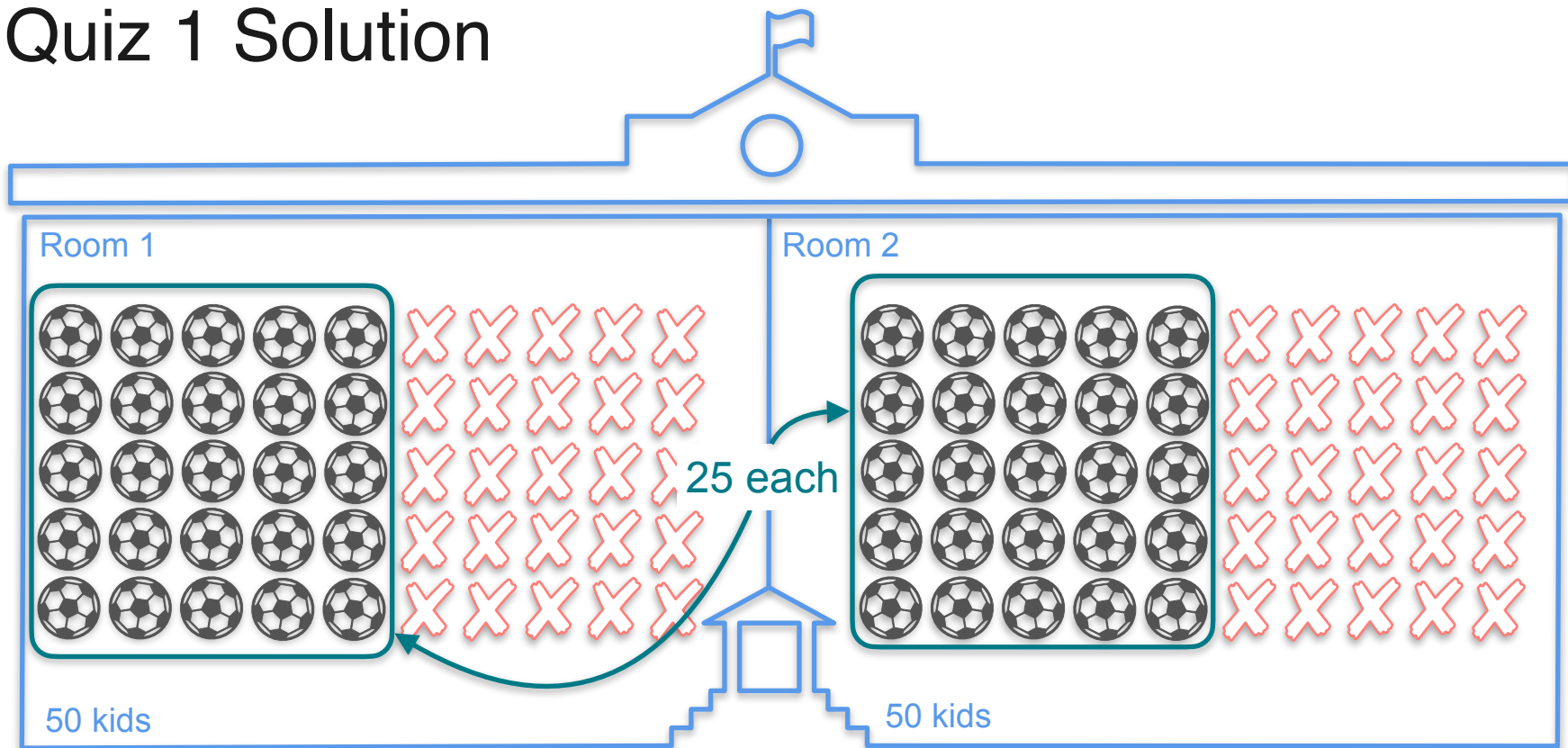
Introduction to probability

Independence

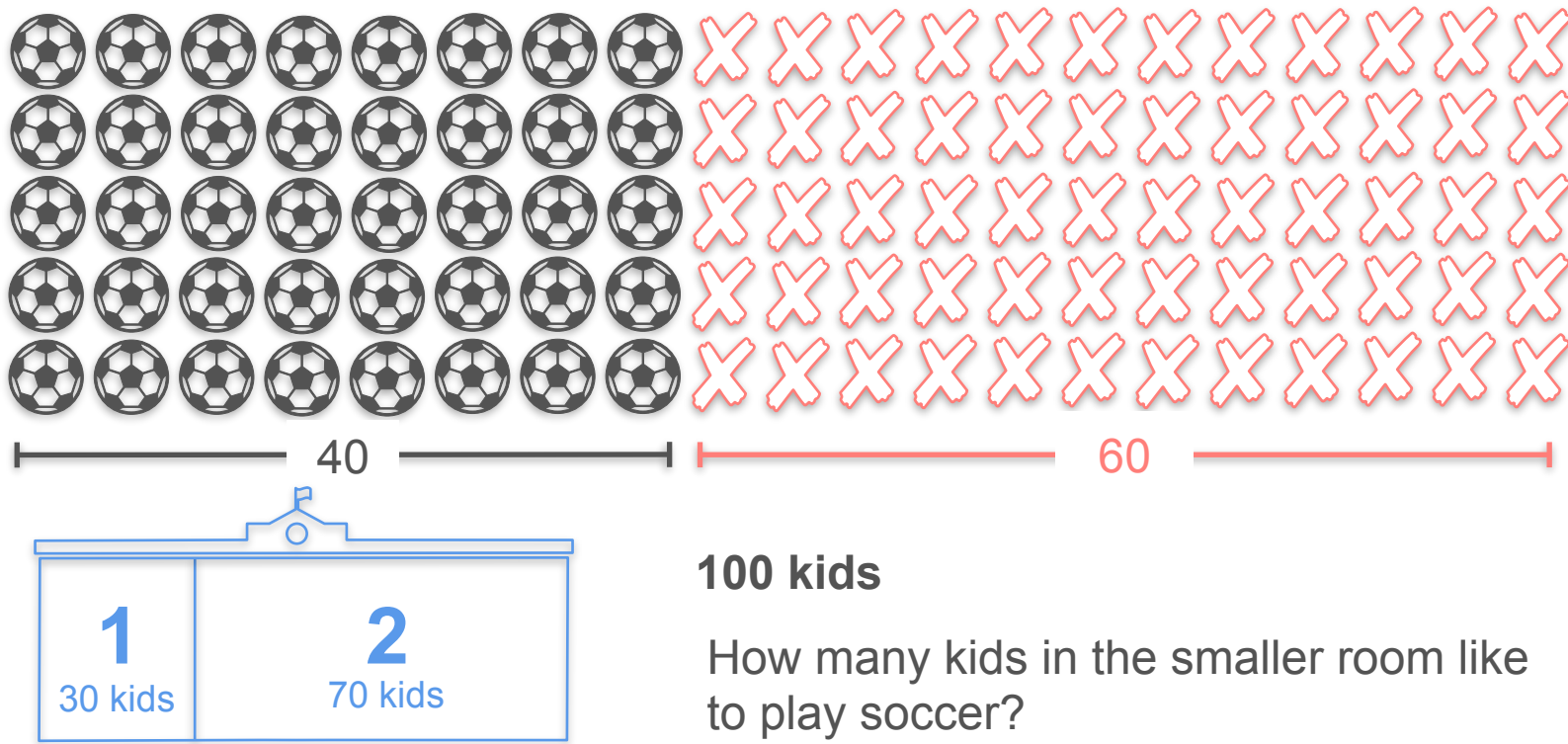
Independence: Quiz 1



Quiz 1 Solution



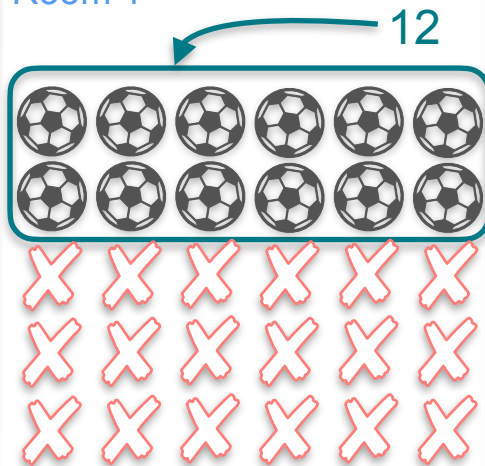
Independence: Quiz 2



Quiz 2 Solution

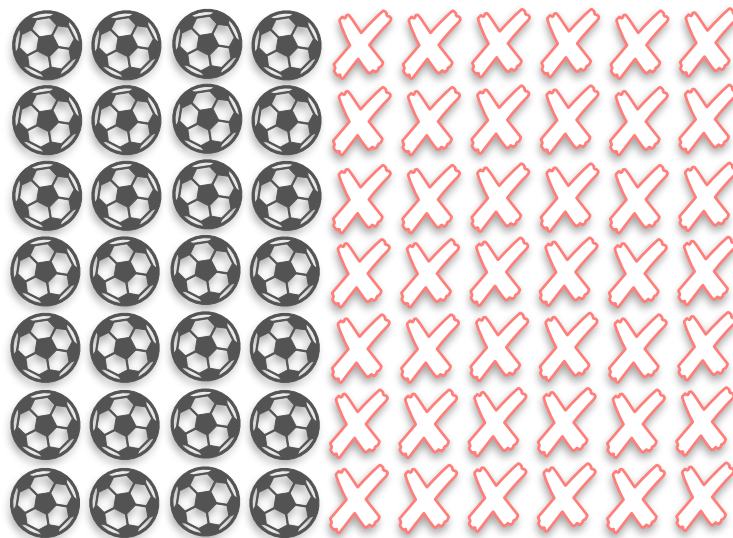


Room 1



30 kids

Room 2



70 kids

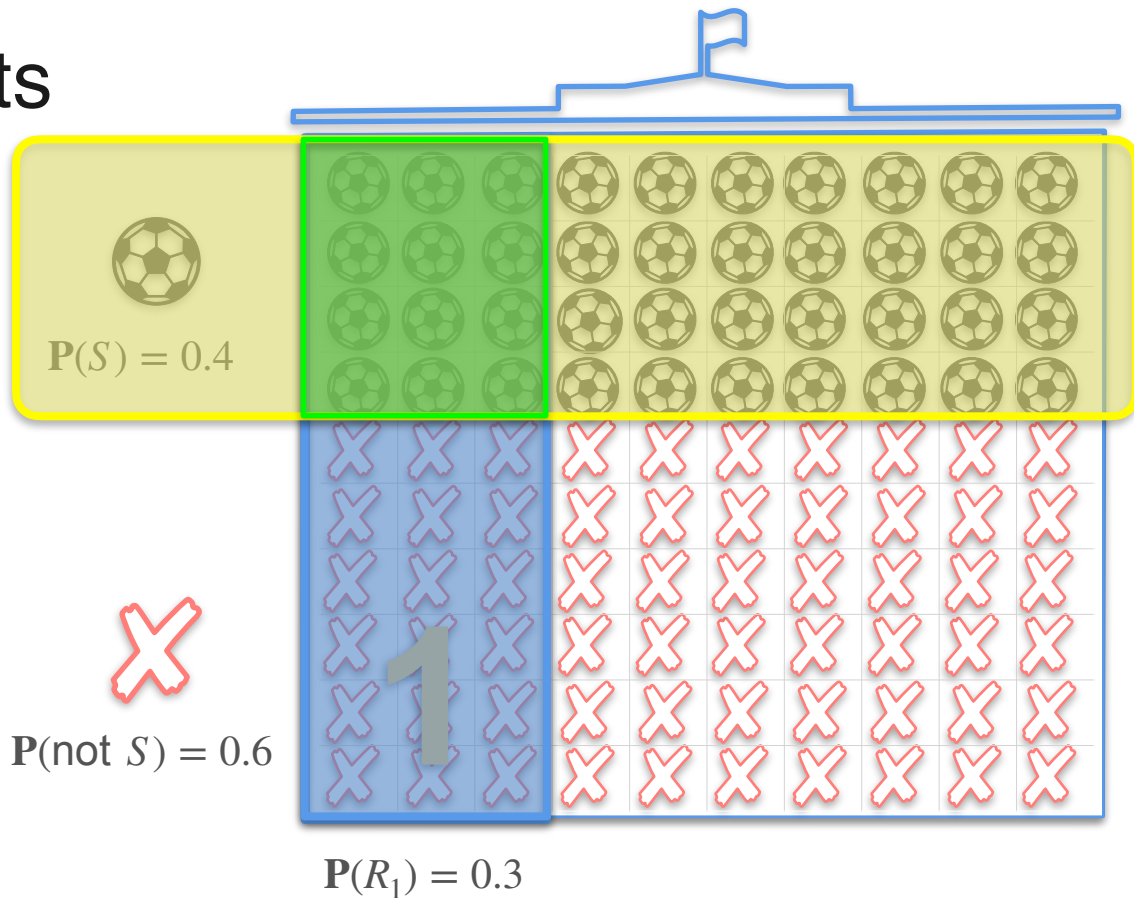
Independent Events

$P(\text{Soccer and Room 1})$

$$P(S \cap R_1) = P(S) \bullet P(R_1)$$

$$= 0.4 \bullet 0.3$$

$$= 0.12$$



Product Rule (for Independent Events)

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

Independent Events: Coin Example 1

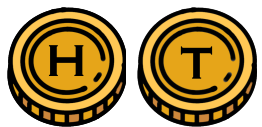


50% 50%

What is the probability of landing on heads five times?



Independent Events: Coin Example 1



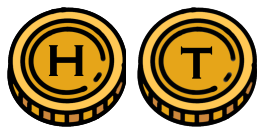
50% 50%

What is the probability of landing on heads five times?

$$P(5 \text{ heads}) = \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}}$$

$\frac{1}{2} \quad \cdot \quad \frac{1}{2} \quad \cdot \quad \frac{1}{2} \quad \cdot \quad \frac{1}{2} \quad \cdot \quad \frac{1}{2}$

Independent Events: Coin Example 2



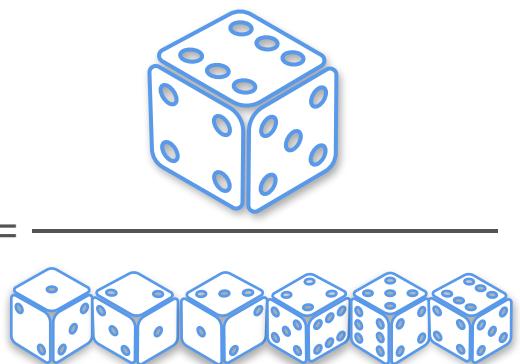
50% 50%

What is the probability of landing 5n heads five times?

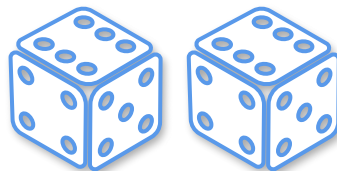
$$P(5 \text{ heads}) = \left(\frac{\text{H}}{\text{H} \quad \text{T}} \right)$$

$$\left(\frac{1}{2} \right)^5 = \frac{1}{32}$$

Independent Events: Dice Example 1


$$P(6) = \frac{1}{6}$$

Independent Events: Dice Example 1

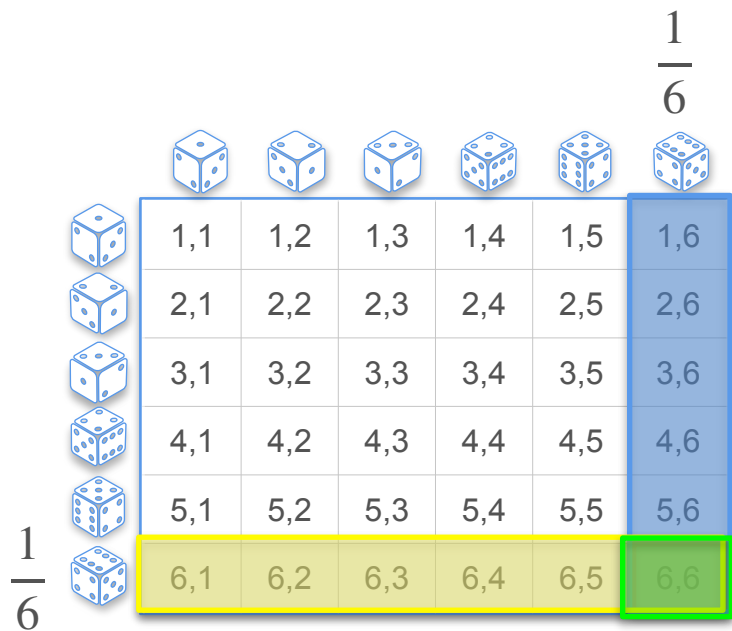


2 dice

						
	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

$$P(6,6) = \frac{1}{36}$$

Independent Events: Dice Example 1



A 6x6 grid of dice outcomes. The top row is highlighted in blue, and the bottom row is highlighted in yellow. The cell (6,6) is highlighted in green.

						
	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

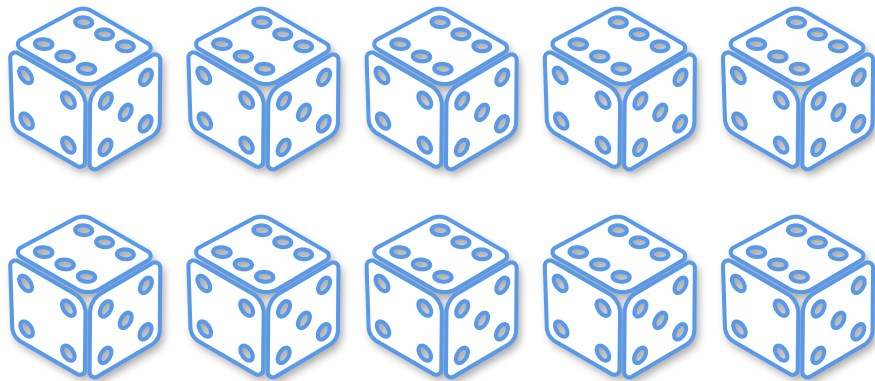
$\frac{1}{6}$

2 dice

$$P(6,6) = \frac{\text{1 die showing 6}}{\text{6 possible outcomes}} \cdot \frac{\text{1 die showing 6}}{\text{6 possible outcomes}}$$
$$= \left(\frac{1}{6}\right) \cdot \left(\frac{1}{6}\right) = \frac{1}{36}$$

Independent Events: Dice Example 2

What is the probability of getting 10 sixes?



Independent Events: Dice Example 2

What is the probability of getting 10 sixes?

$$\mathbf{P(10\ sixes)} = \left(\frac{\img alt="A single blue die showing a six" data-bbox="475 455 525 545}}{\img alt="Ten blue dice arranged in a 2x5 grid, all showing a six" data-bbox="460 575 540 665}} \right)^{10}$$
$$= \left(\frac{1}{6} \right)^{10}$$



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Introduction to probability

Birthday problem

Quiz

- You have 30 friends at a party. What do you think is more likely:
 - That there exist two people with the same birthday
 - That no two of them have the same birthday
- (Assume the year has 365 days, nobody has a birthday on Feb 29).

Quiz

- Answer: It's more likely that 2 people have the same birthday.
- In fact, the probability of no two people having the same birthday is around 0.3.

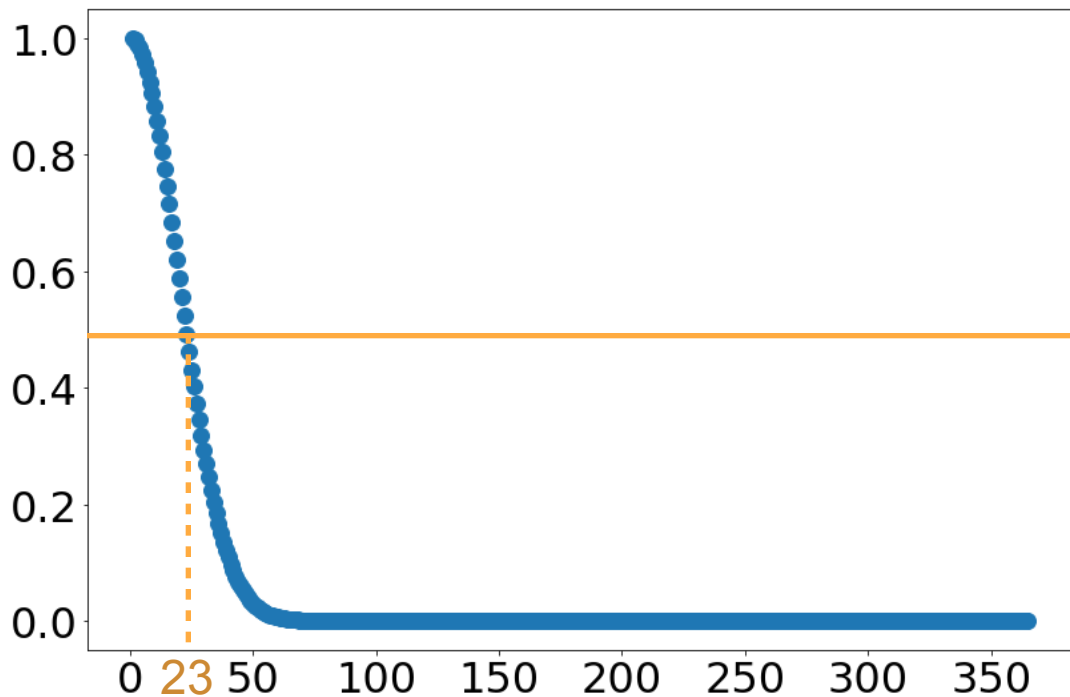
Probability That Everyone Has a Different Birthday



$$\frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365} \cdot \frac{361}{365} \cdot \frac{360}{365} \cdot \frac{359}{365} \cdot \frac{358}{365} \cdot \frac{357}{365} = 0.905$$

Probability That no Two People Have the Same Birthday

1 person: 1
2 people: 0.997
3 people: 0.992
4 people: 0.984
5 people: 0.973
10 people: 0.883
20 people: 0.589
23 people: 0.493
30 people: 0.294
50 people: 0.030
100 people: 0.0000003
365 people: 0





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Introduction to Probability

Conditional probability

Conditional Probability: Coin Example 1



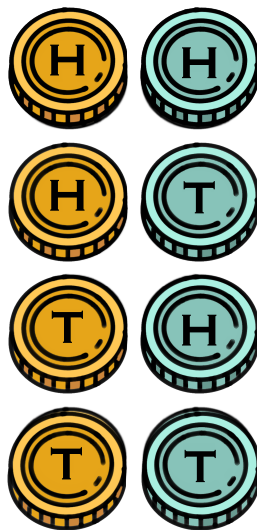
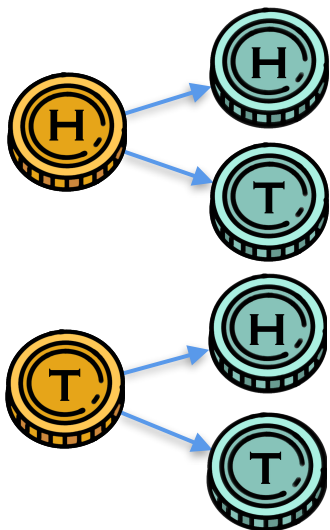
50% 50%

What is the probability of landing on heads twice?

1st

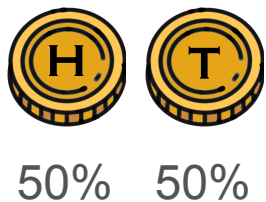
2nd

GIVEN that the first one is heads



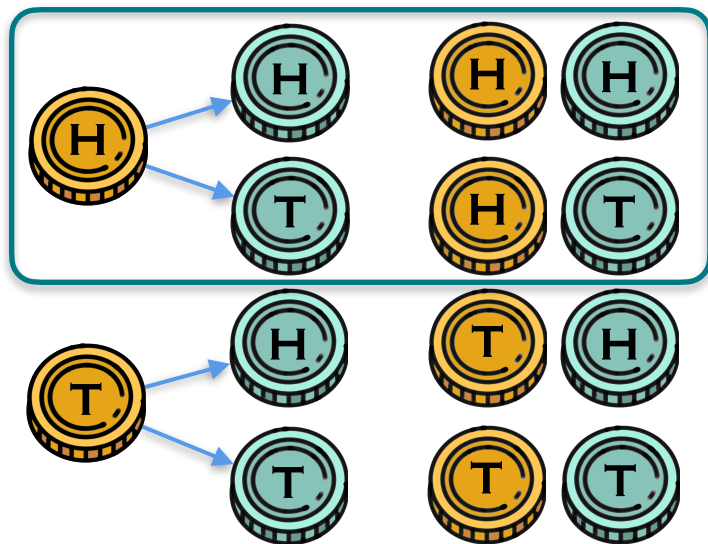
$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Total number of outcomes}} = \frac{1}{4}$$

Conditional Probability: Coin Example 1



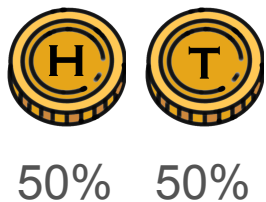
What is the probability of landing on heads twice?

1st 2nd **GIVEN that the first one is heads**



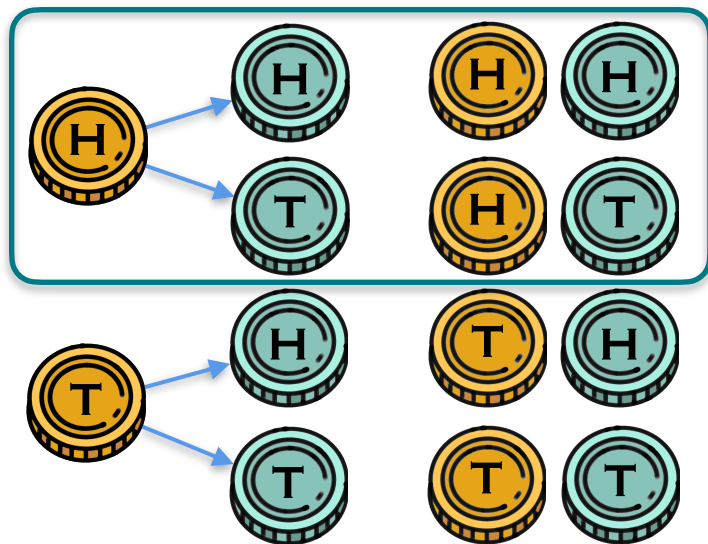
$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Total number of outcomes}} = \frac{1}{4}$$

Conditional Probability: Coin Example 1



What is the probability of landing on heads twice?

1st 2nd **GIVEN that the first one is heads**



$$P(HH \mid 1st \text{ is } H) = \frac{\text{Number of outcomes where both are heads}}{\text{Number of outcomes where the first is heads}} = \frac{1}{2}$$

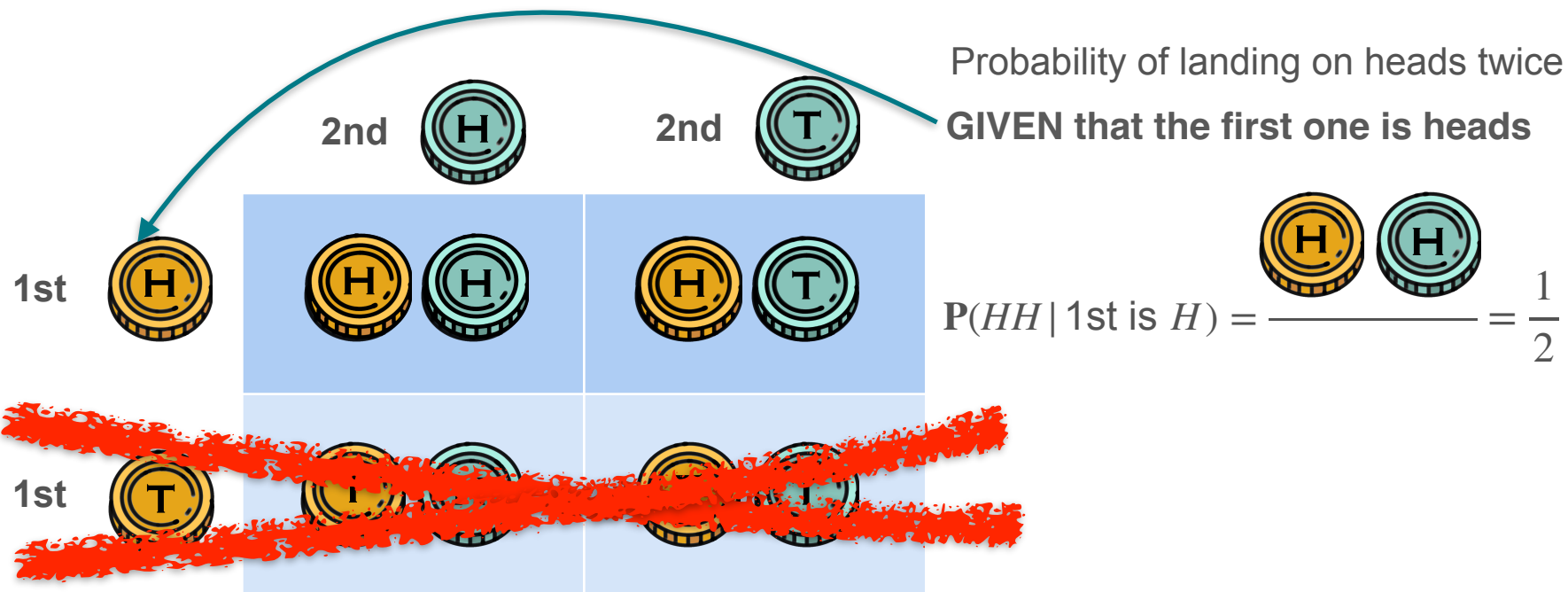
Conditional Probability: Coin Example 1

What is the probability of landing on heads twice?

GIVEN that the first one is heads


$$\mathbf{P}(HH \mid \text{1st is } H)$$

Conditional Probability: Coin Example 1



Conditional Probability: Coin Example 2



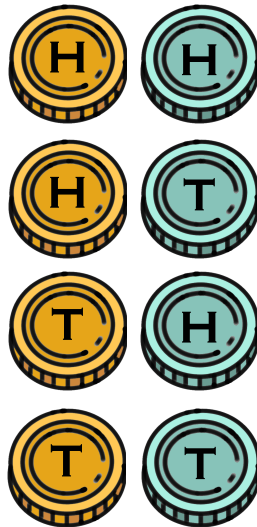
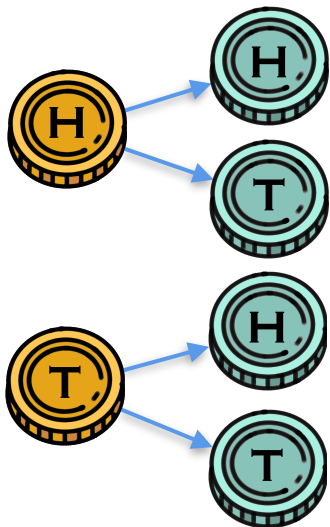
50% 50%

What is the probability of landing on heads twice?

1st

2nd

GIVEN that the first one is tails



$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Number of outcomes where the first is tails}} = \frac{1}{4}$$

Conditional Probability: Coin Example 2



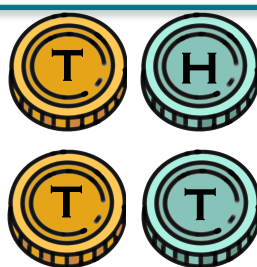
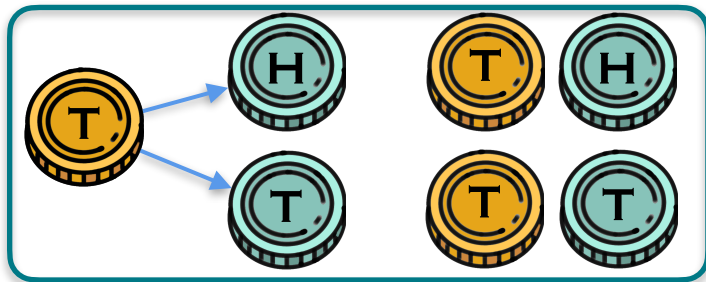
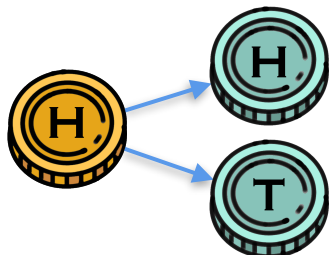
50% 50%

What is the probability of landing on heads twice?

1st

2nd

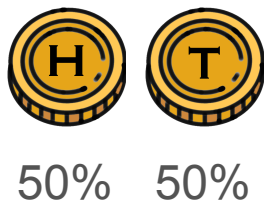
GIVEN that the first one is tails



$$P(HH) = \frac{\text{Number of outcomes with 2 heads}}{\text{Total number of outcomes}} = \frac{1}{4}$$



Conditional Probability: Coin Example 2

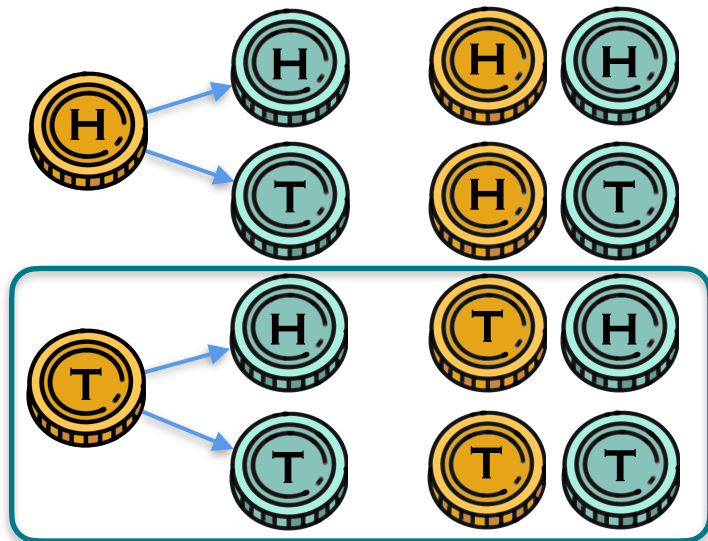


What is the probability of landing on heads twice?

1st

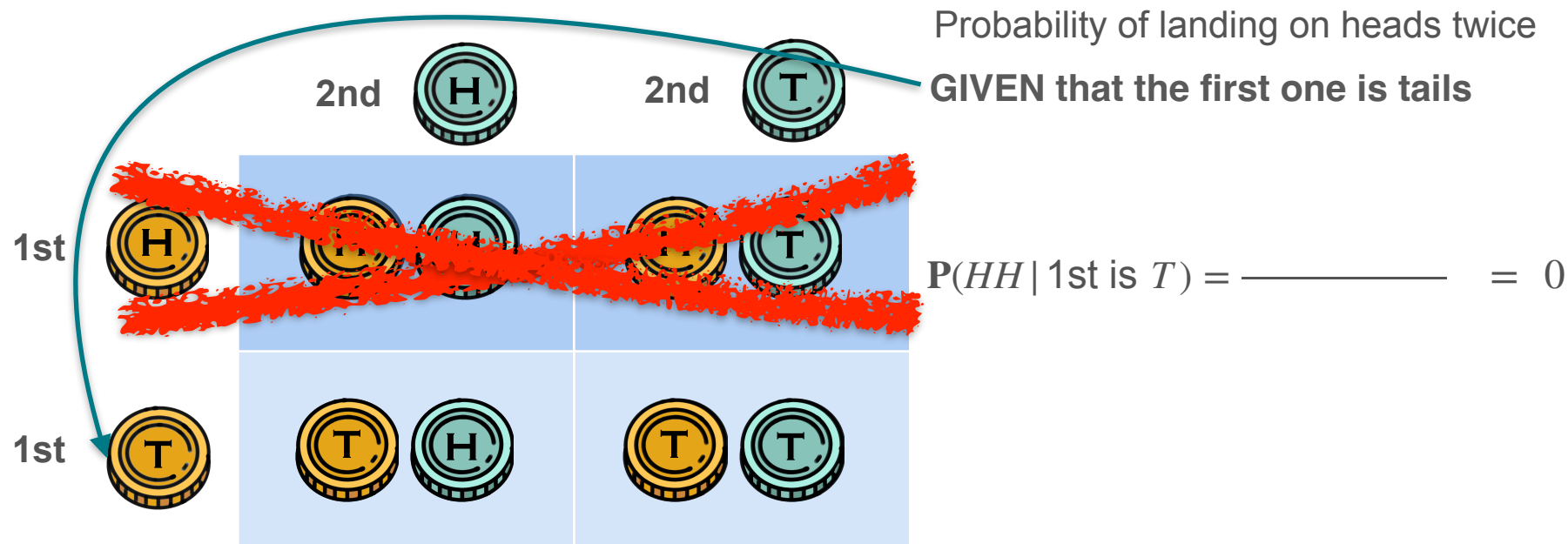
2nd

GIVEN that the first one is tails



$$P(HH | 1st \text{ is } T) = \frac{0}{\text{4 outcomes}} = 0$$

Conditional Probability: Coin Example 2

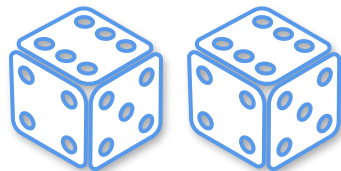


Product Rule (for Independent Events)

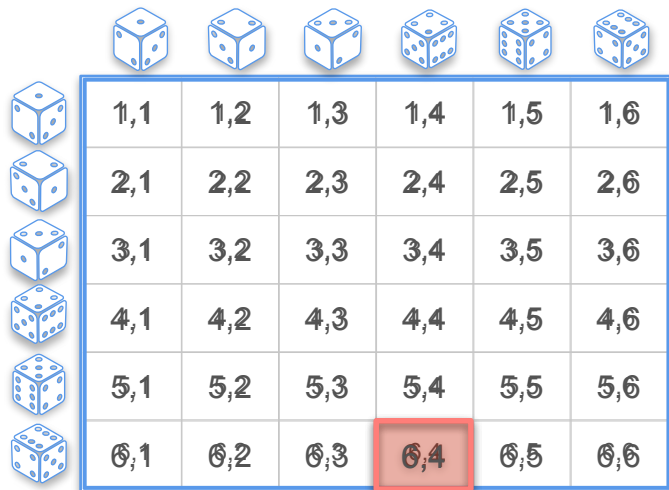
When A and B independent

$$\mathbf{P(A \cap B) = P(A) \cdot P(B)}$$

Conditional Probability: Dice Example 3



What is the probability that
the first is 6 **AND** the sum = 10?

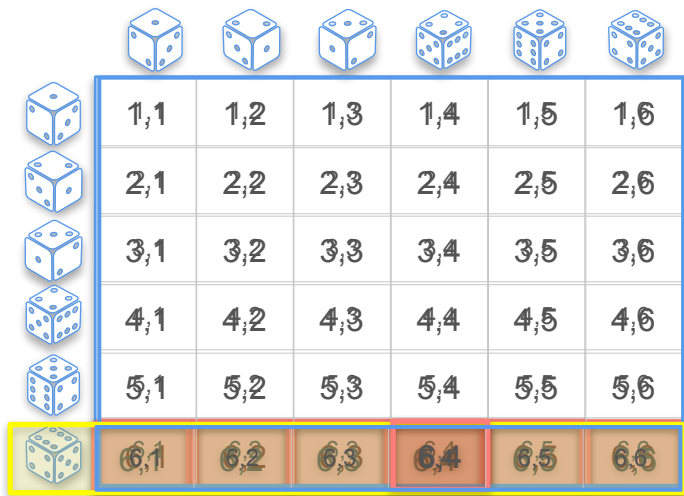
Six blue-outlined dice are arranged in a vertical column to the left of the table. Each die shows a different face: 1, 2, 3, 4, 5, and 6 from top to bottom.

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$$P(\text{1st is 6} \cap \text{sum} = 10) = \underline{\hspace{2cm}}$$

$$= \frac{1}{36}$$

Conditional Probability: Dice Example 3



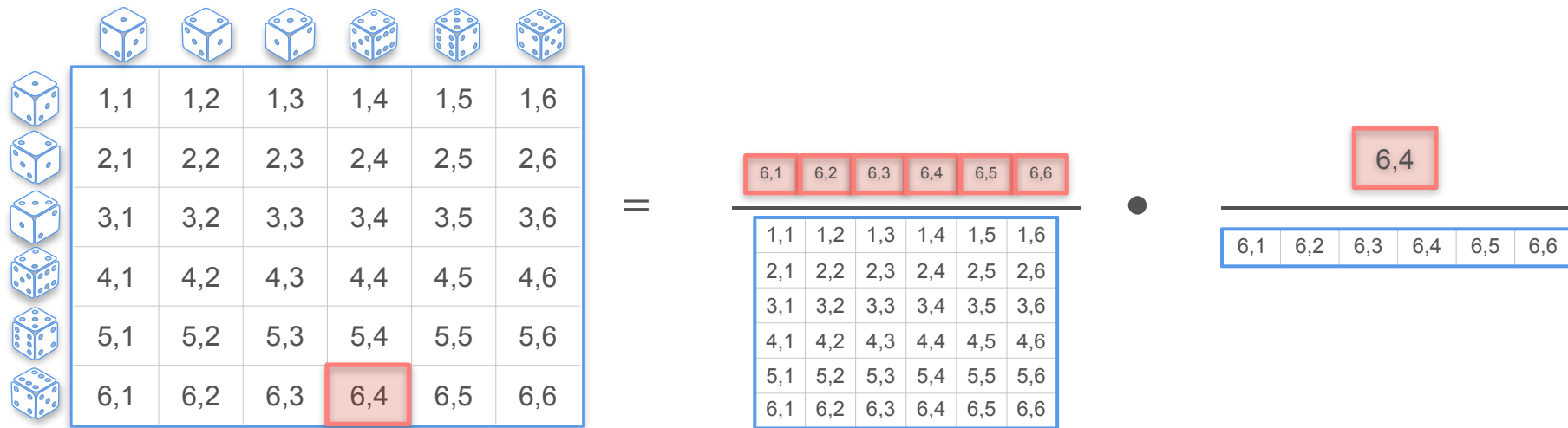
1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$P(\text{1st is 6} \cap \text{sum} = 10) =$

$P(\text{1st is 6}) \quad \bullet \quad P(\text{sum} = 10 \mid \text{1st 6})$

$$\frac{6}{36} \quad \bullet \quad \frac{1}{6} = \frac{1}{36}$$

Conditional Probability: Dice Example 3



The diagram illustrates the calculation of conditional probability for two dice. It shows a 6x6 grid of outcomes, a row of outcomes where the first die is 6, and a single outcome (6,4) where the sum is 10.

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

=

6,1	6,2	6,3	6,4	6,5	6,6
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•

6,1	6,2	6,3	6,4	6,5	6,6
-----	-----	-----	-----	-----	-----

$$\mathbf{P(1st\ is\ 6 \cap\ sum = 10) = P(1st\ is\ 6) \cdot P(sum = 10 \mid 1st\ 6)}$$

$$\mathbf{P(A \cap B) = P(A) \cdot P(B \mid A)}$$

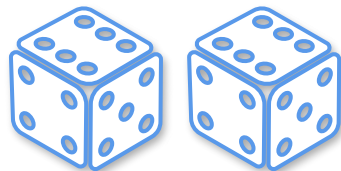
The General Product Rule

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

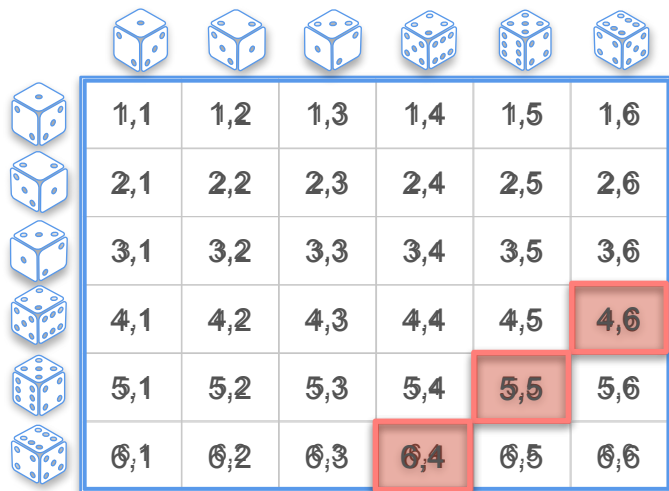
When independent

$$\mathbf{P}(B | A) = \mathbf{P}(B)$$

Conditional Probability: Dice Example 1



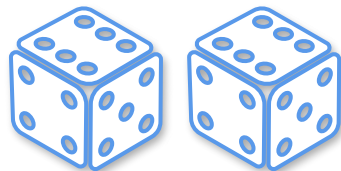
What is the probability that the sum is 10?

A vertical column of six blue-outlined dice icons is positioned to the left of the table, each corresponding to a row.

						
	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

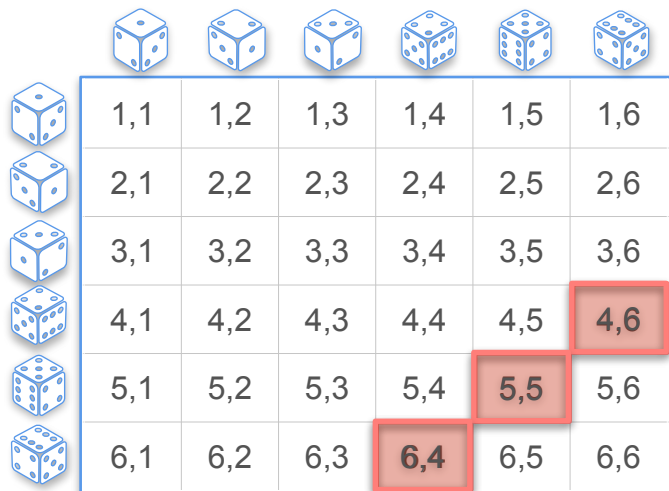
$$P(\text{sum} = 10) = \frac{3}{36}$$

Conditional Probability: Dice Example 1



What is the probability that the sum is 10?

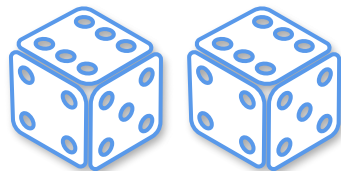
GIVEN that the first one is 6

Six small blue-outlined dice icons are arranged vertically to the left of the table, each corresponding to a row of the table.

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

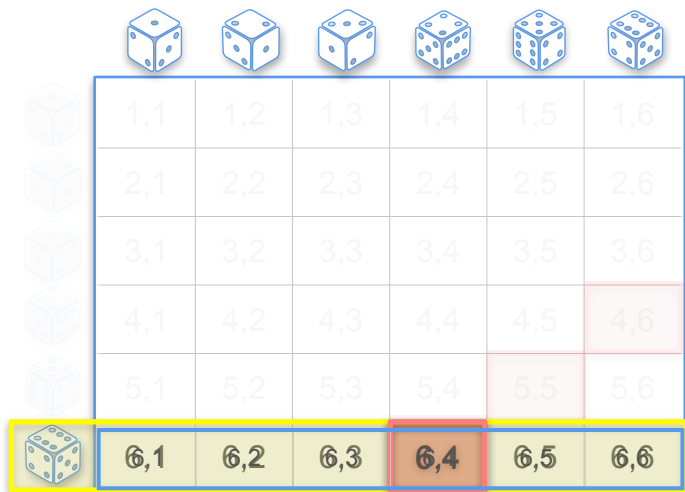
$P(\text{sum} = 10 \mid \text{1st is 6}) = \underline{\hspace{2cm}}$

Conditional Probability: Dice Example 1



What is the probability that the sum is 10?

GIVEN that the first one is 6

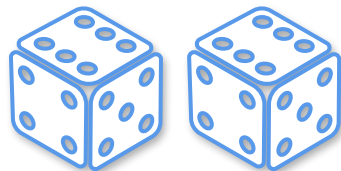


1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

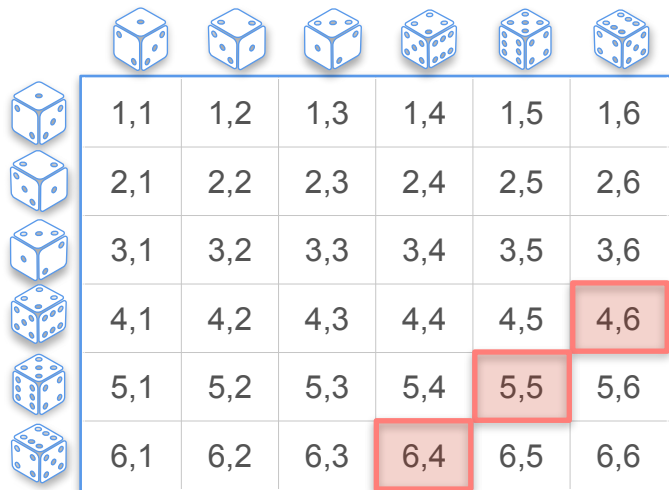
$P(\text{sum} = 10 \mid \text{1st is 6}) = \underline{\hspace{2cm}}$

$$= \frac{1}{6}$$

Conditional Probability: Dice Example 2



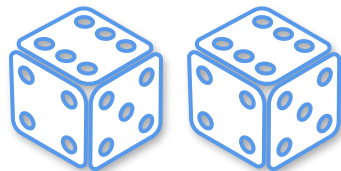
What is the probability that the sum is 10?



						
	1,1	1,2	1,3	1,4	1,5	1,6
	2,1	2,2	2,3	2,4	2,5	2,6
	3,1	3,2	3,3	3,4	3,5	3,6
	4,1	4,2	4,3	4,4	4,5	4,6
	5,1	5,2	5,3	5,4	5,5	5,6
	6,1	6,2	6,3	6,4	6,5	6,6

$$P(\text{sum} = 10) = \frac{\begin{array}{c} \text{4,6} \\ \text{5,5} \\ \text{6,4} \end{array}}{\begin{array}{|c|c|c|c|c|c|} \hline 1,1 & 1,2 & 1,3 & 1,4 & 1,5 & 1,6 \\ \hline 2,1 & 2,2 & 2,3 & 2,4 & 2,5 & 2,6 \\ \hline 3,1 & 3,2 & 3,3 & 3,4 & 3,5 & 3,6 \\ \hline 4,1 & 4,2 & 4,3 & 4,4 & 4,5 & 4,6 \\ \hline 5,1 & 5,2 & 5,3 & 5,4 & 5,5 & 5,6 \\ \hline 6,1 & 6,2 & 6,3 & 6,4 & 6,5 & 6,6 \\ \hline \end{array}} = \frac{3}{36}$$

Conditional Probability: Dice Example 2



What is the probability that the sum is 10?

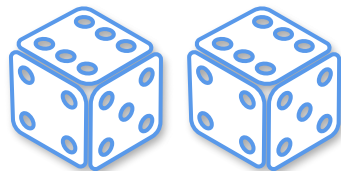
GIVEN that the first one is 1



					
1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$P(\text{sum} = 10 \mid \text{1st is } 1) = \underline{\hspace{2cm}}$

Conditional Probability: Dice Example 2



What is the probability that the sum is 10?

GIVEN that the first one is 1

					
 1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$$P(\text{sum} = 10 \mid \text{1st is } 1) = \frac{0}{\quad}$$

$$= 0$$

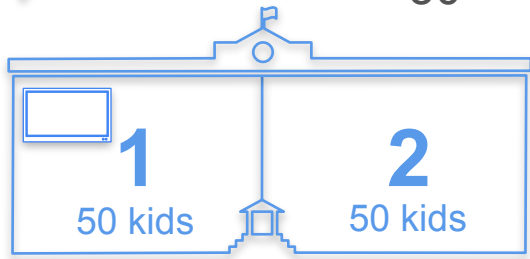
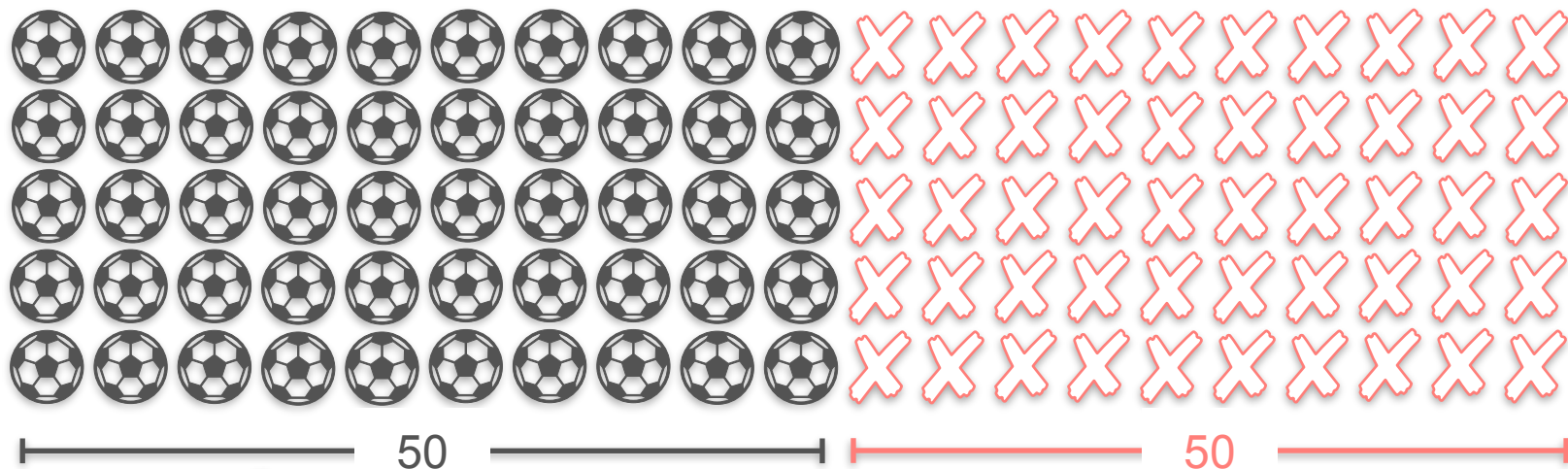


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Introduction to Probability

Conditional probability - Part 2

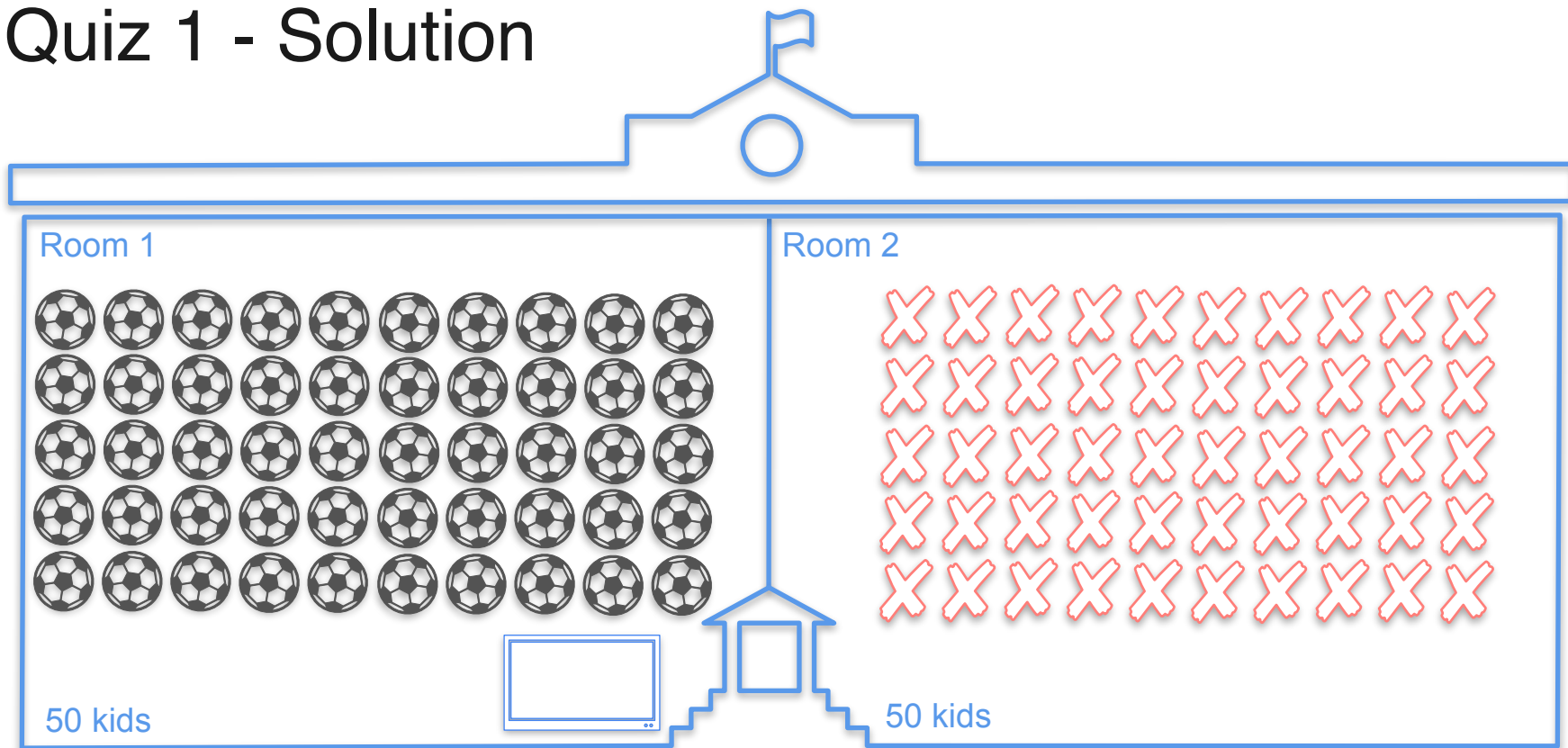
Quiz 1



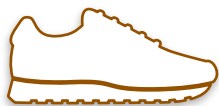
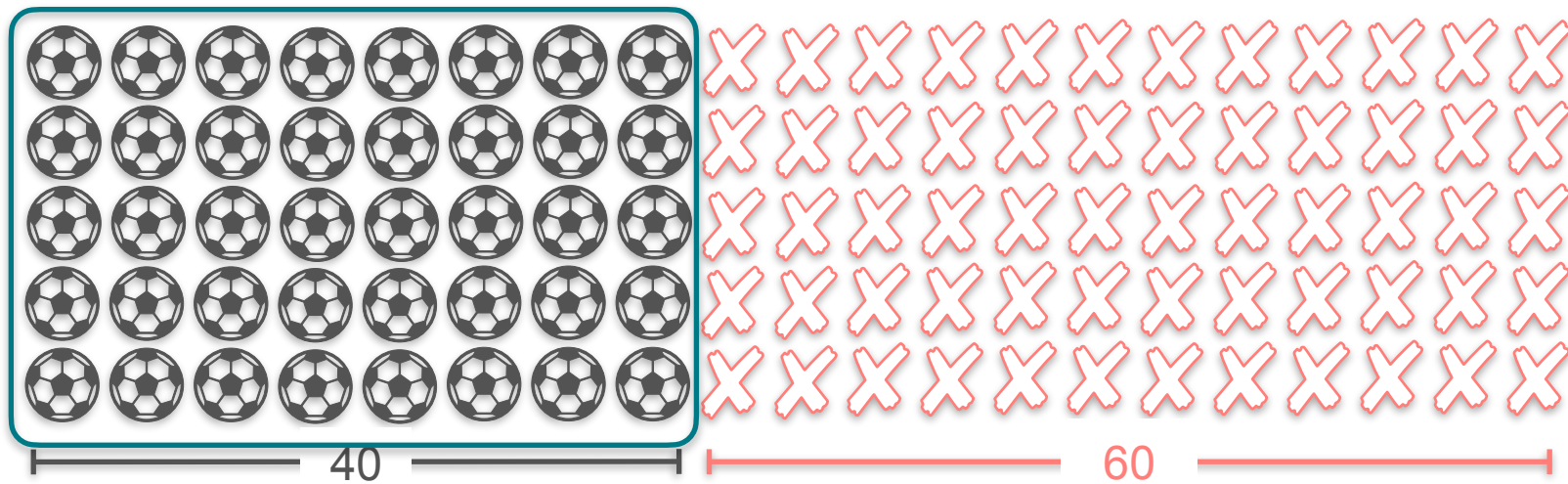
100 kids

How many kids in room 1 like soccer?

Quiz 1 - Solution



Quiz 2

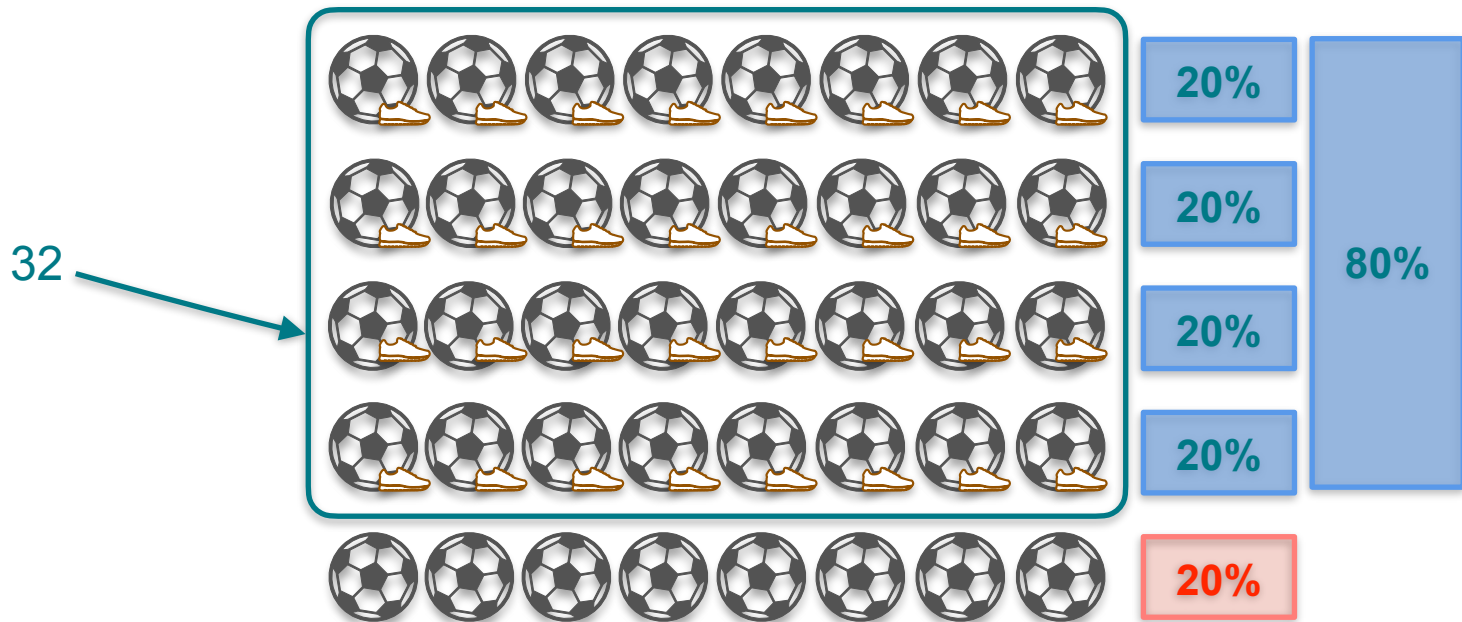


80%

100 kids

How many kids play soccer and wear running shoes

Quiz 2 - Solution



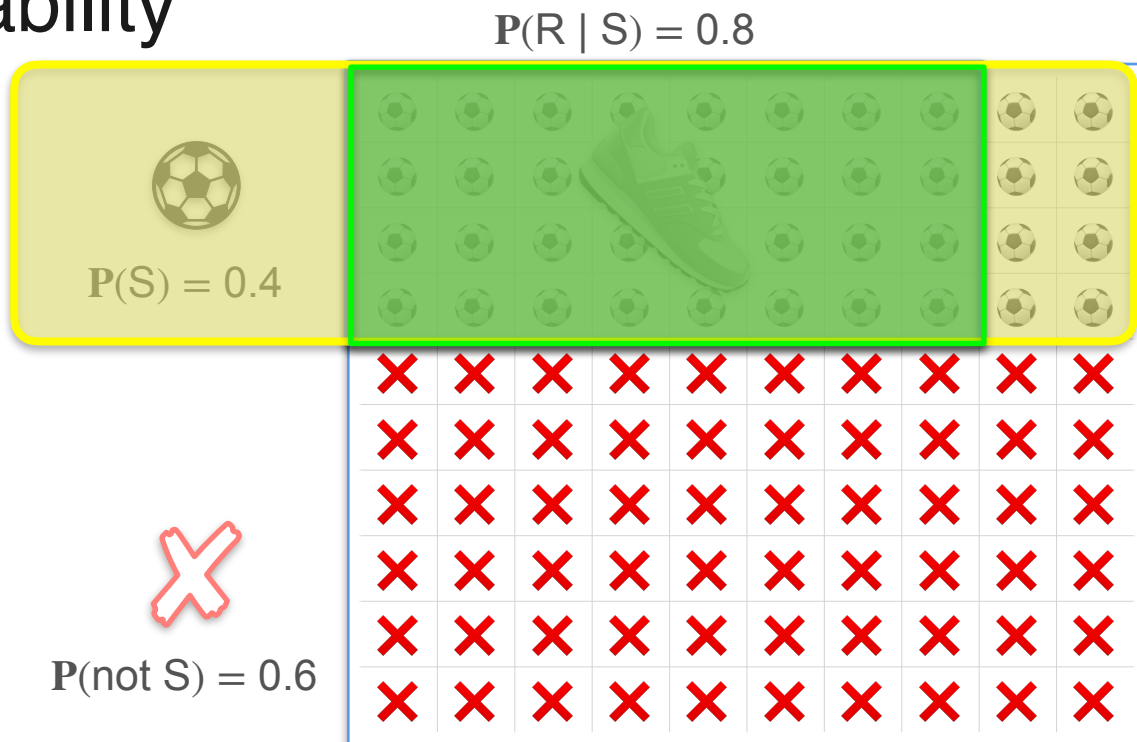
Conditional Probability

$P(\text{Soccer and Running shoes})$

$$P(S \cap R) = P(S) \bullet P(R|S)$$

$$= 0.4 \bullet 0.8$$

$$= 0.32$$



Conditional Probability

$P(\text{Soccer and Running shoes})$

$$P(S \cap R) = 0.32$$

$P(\text{not Soccer and Running shoes})$

$$P(\text{not } S \cap R) = P(\text{not } S) \bullet P(R | \text{not } S)$$

$$= 0.6 \bullet 0.5$$

$$= 0.3$$



$$P(S) = 0.4$$



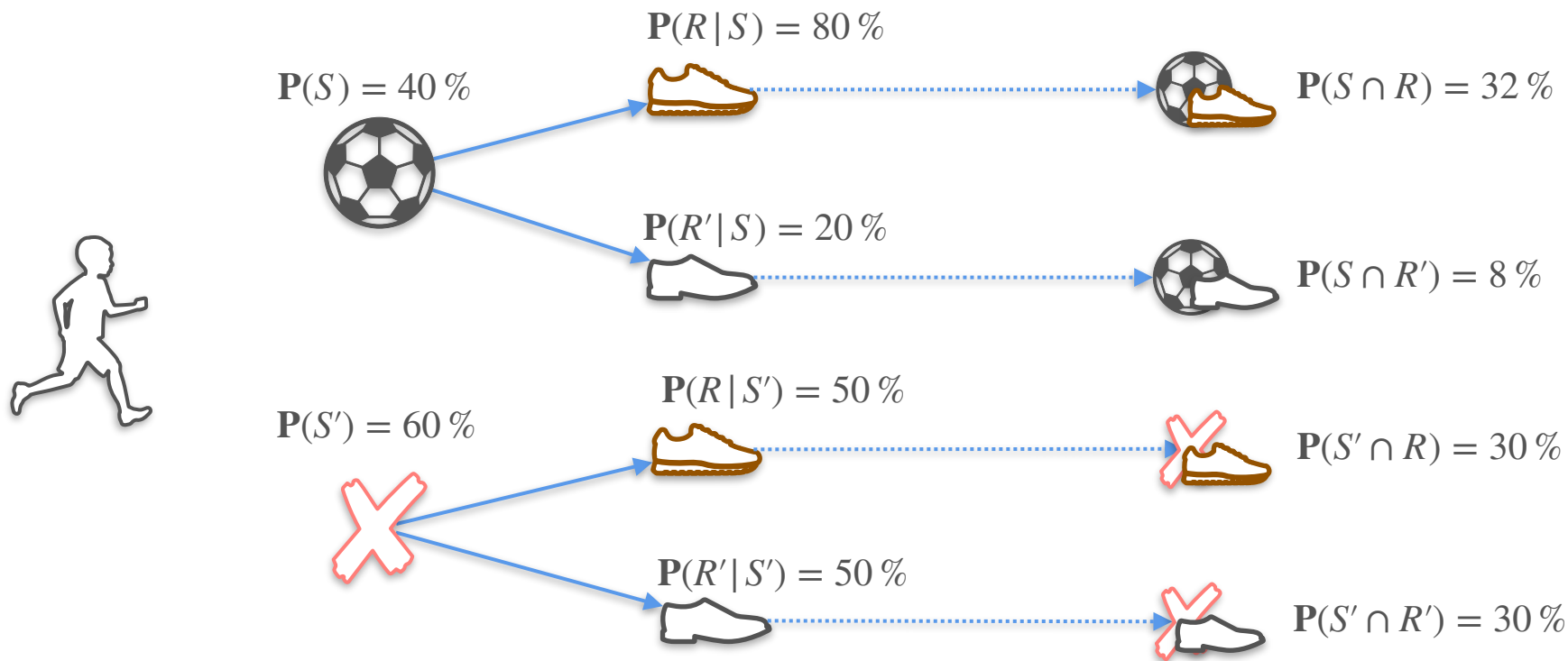
$$P(\text{not } S) = 0.6$$

$$P(R | S) = 0.8$$



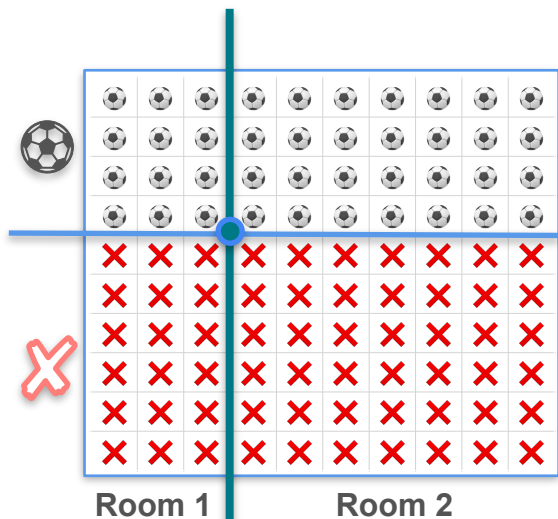
$$P(R | \text{not } S) = 0.5$$

Conditional Probability

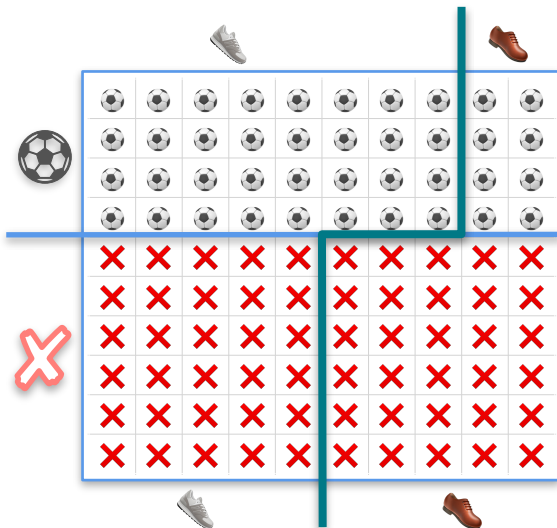


Independent vs Dependent Events

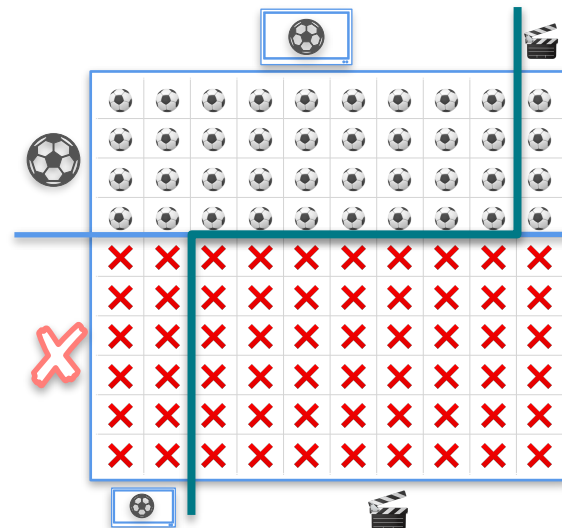
Independent



Dependent



Dependent



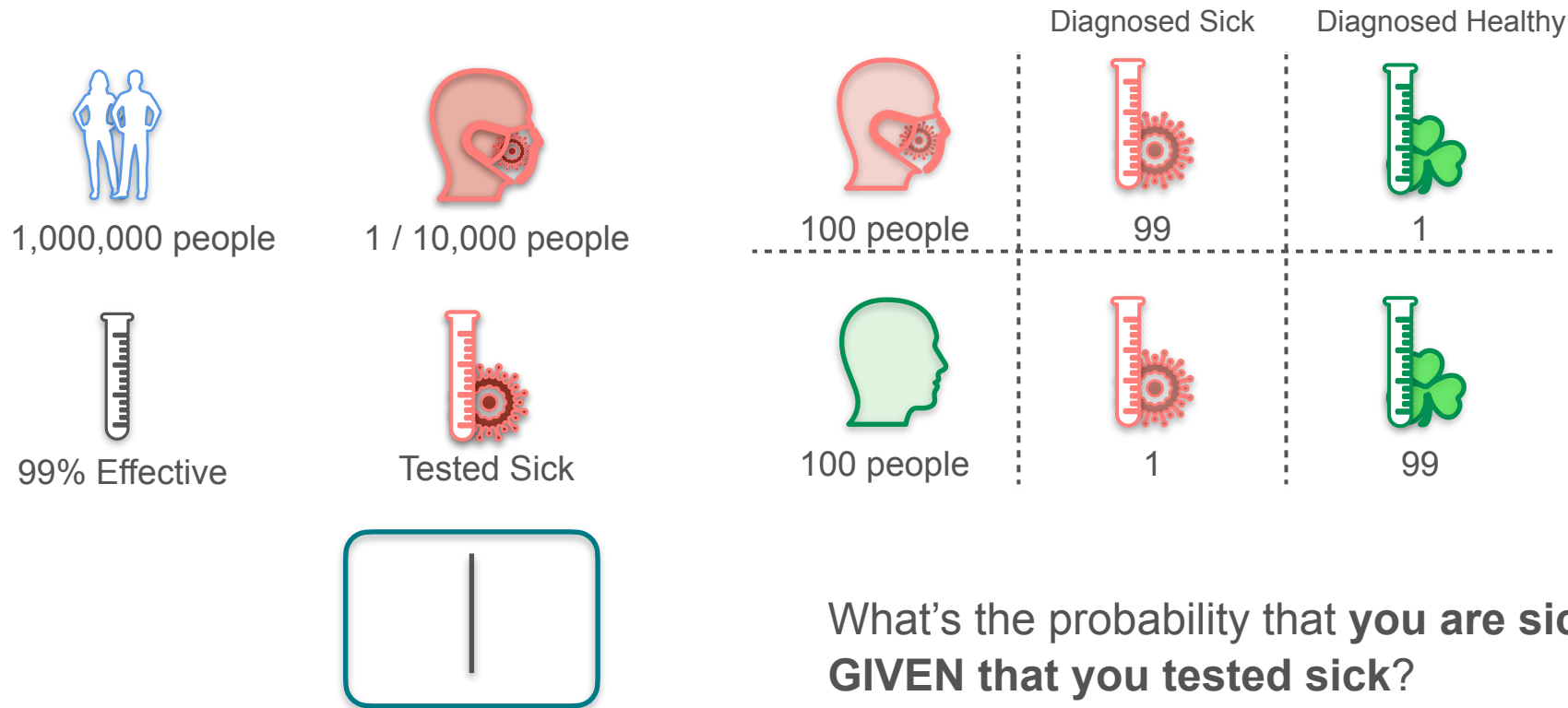


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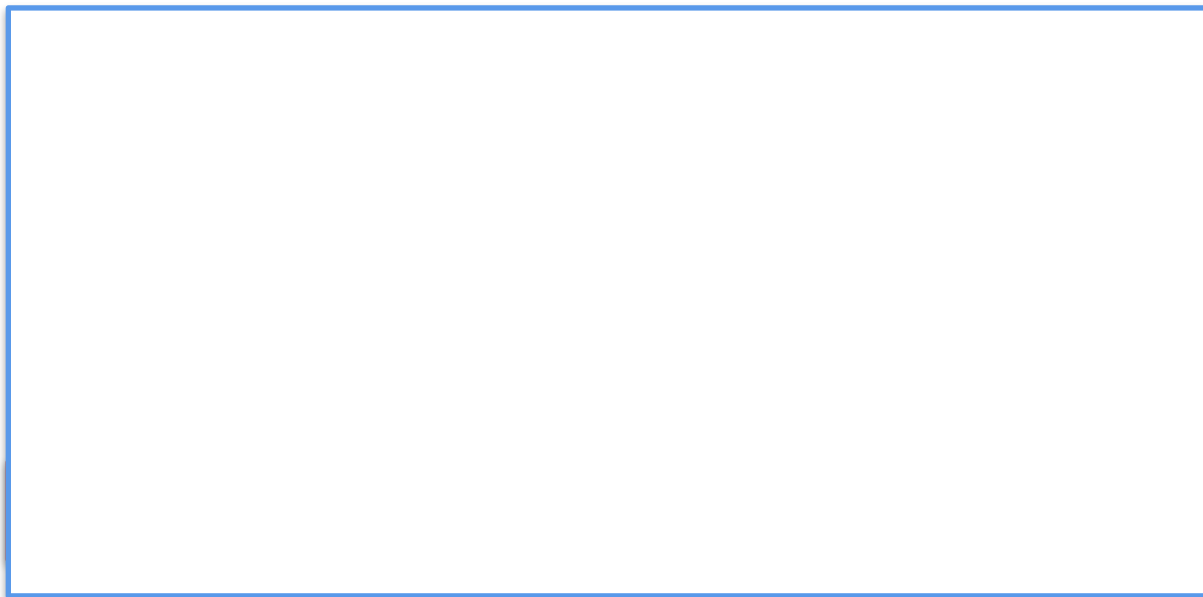
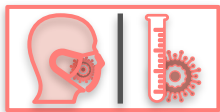
Introduction to probability

Bayes Theorem - Intuition

Bayes Theorem: Intuition

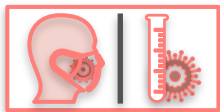


Bayes Theorem: Intuition



1,000,000 people

Bayes Theorem: Intuition



Healthy



999,900

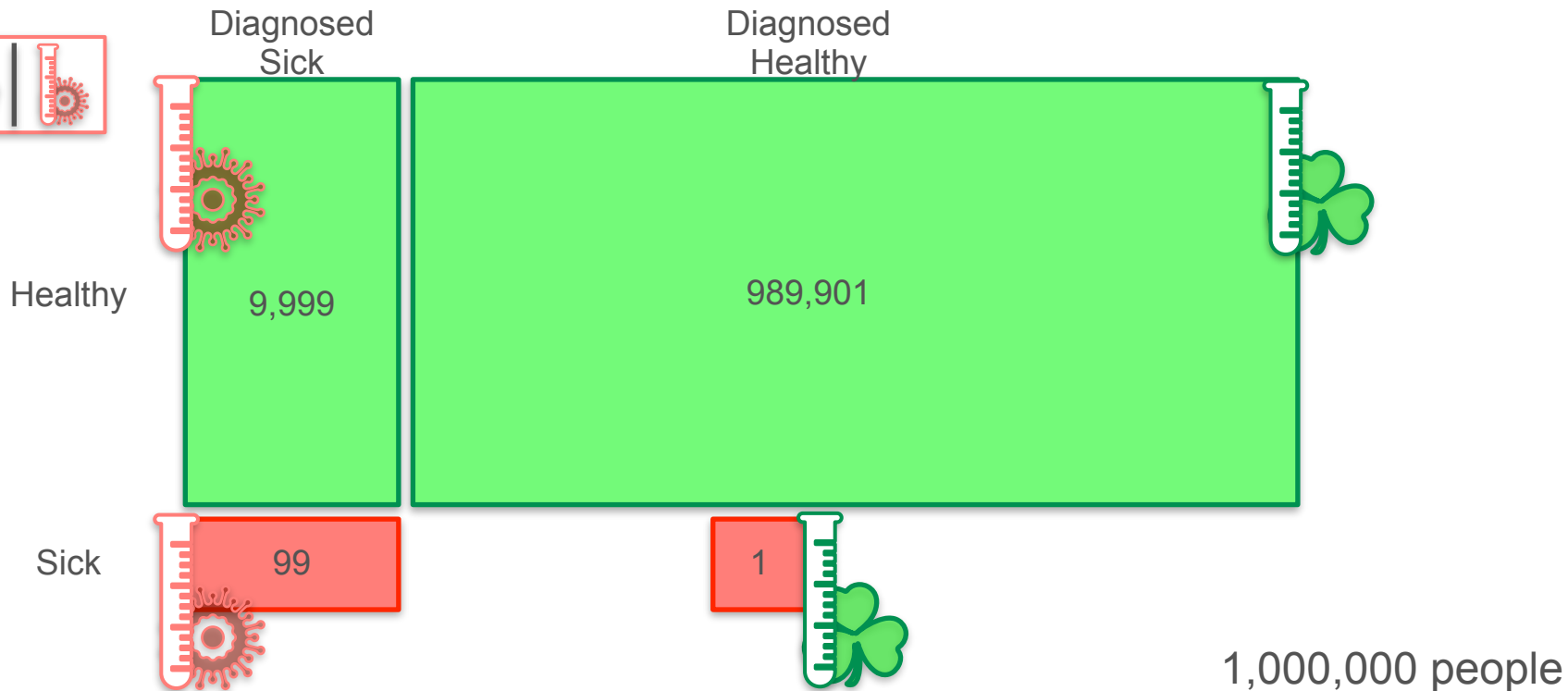
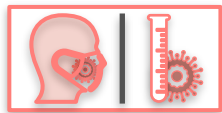
Sick



100

1,000,000 people

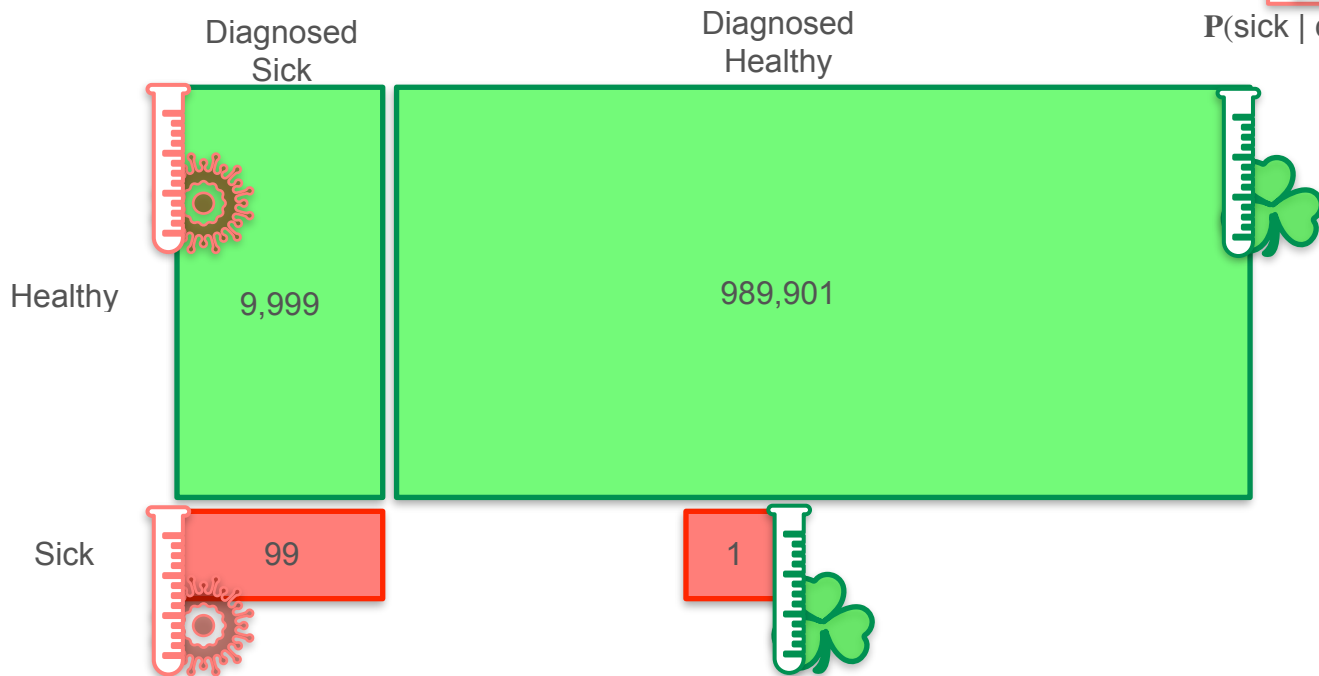
Bayes Theorem: Intuition



Bayes Theorem: Intuition



$P(\text{sick} \mid \text{diagnosed sick}) = \text{_____}$

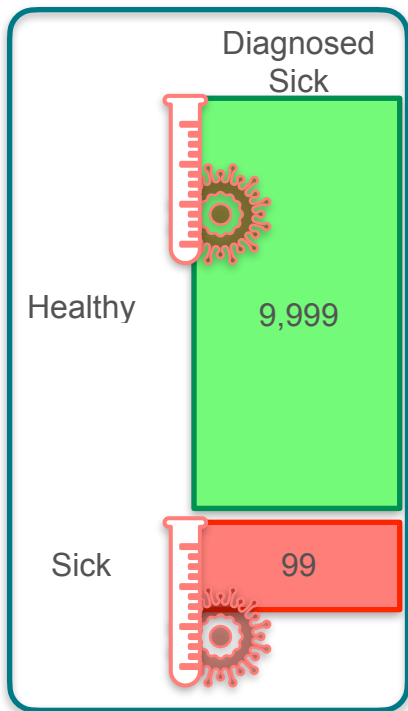


1,000,000 people

Bayes Theorem: Intuition

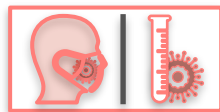
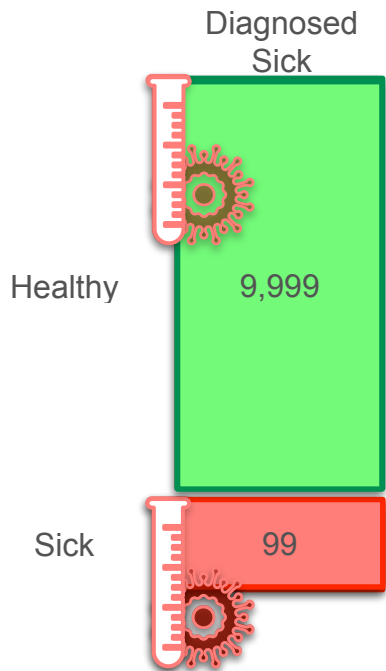


$P(\text{sick} \mid \text{diagnosed sick}) = \text{_____}$

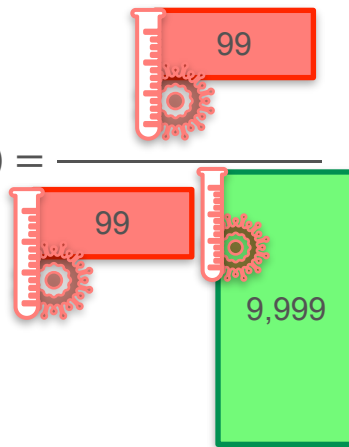


1,000,000 people

Bayes Theorem: Intuition



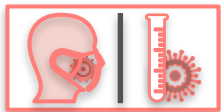
$P(\text{sick} \mid \text{diagnosed sick}) =$



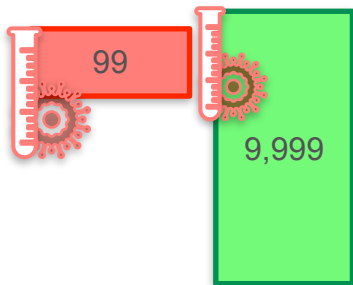
$$= \frac{99}{99 + 9999}$$
$$= \frac{99}{10098} = 0.0098$$

$$= \frac{\text{sick and diagnosed sick}}{\text{sick and diagnosed sick} + \text{healthy and diagnosed sick}}$$

Bayes Theorem: Intuition



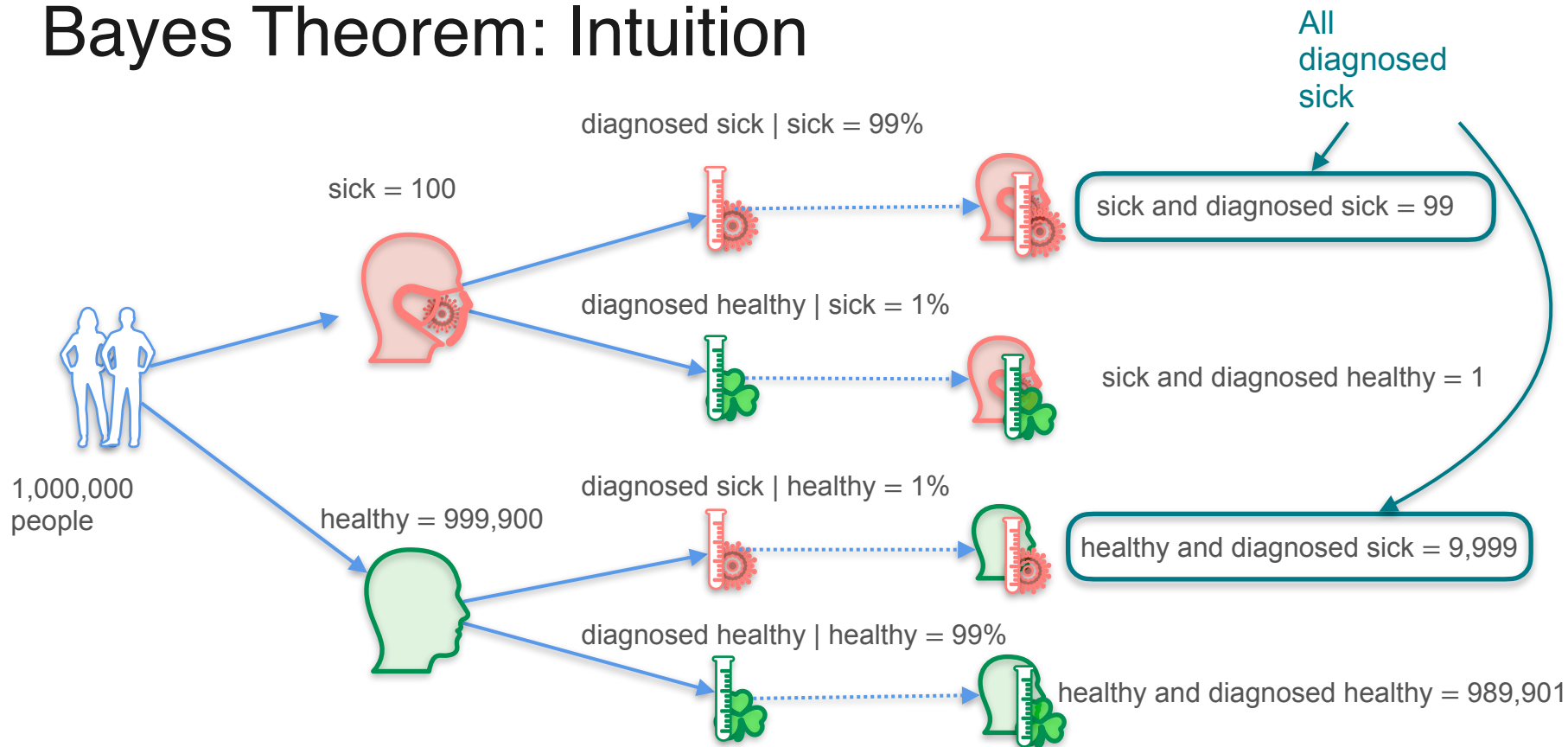
$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{\text{99}}{\text{99} + \text{9,999}}$$



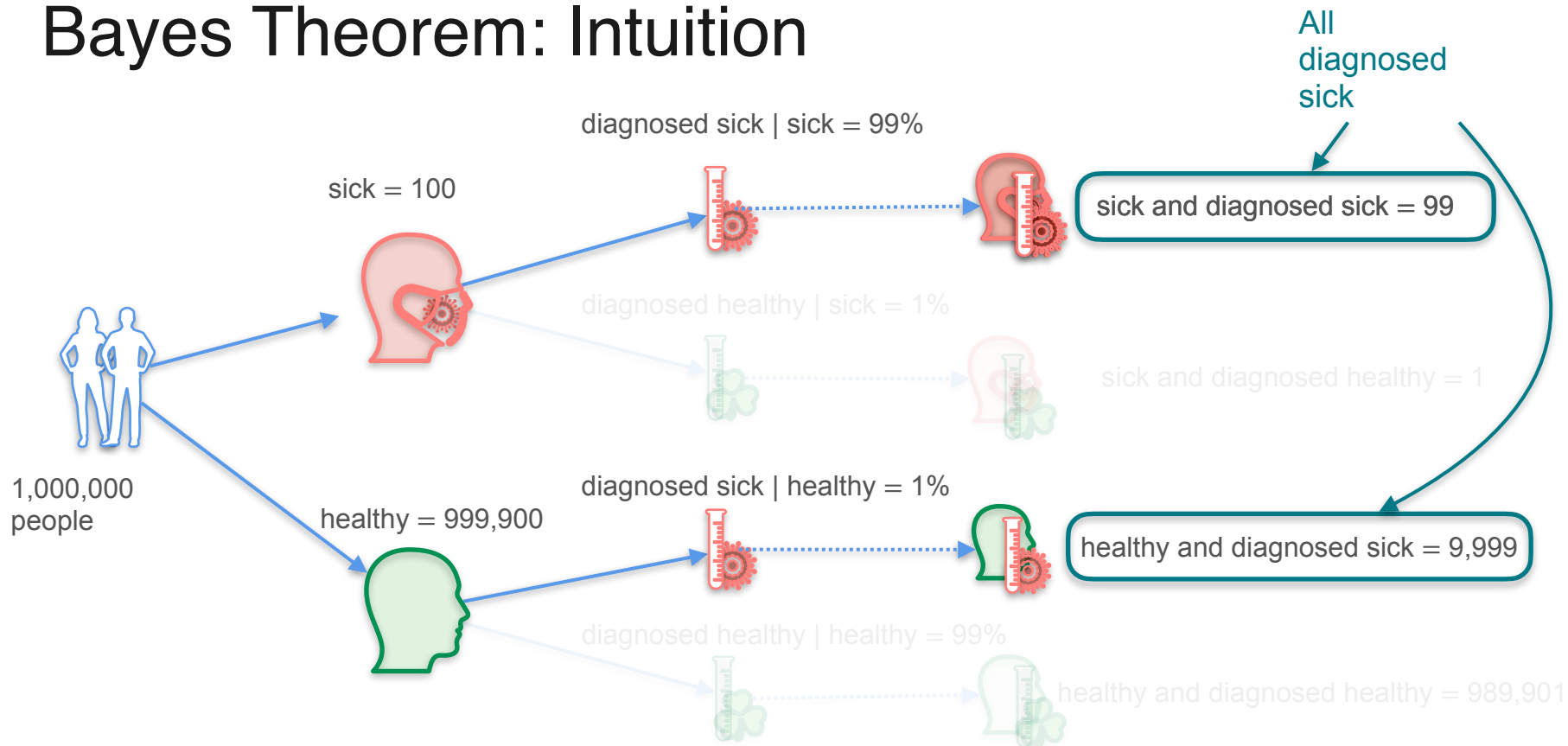
$$= \frac{\text{sick and diagnosed sick}}{\text{sick and diagnosed sick} + \text{healthy and diagnosed sick}}$$

$$= \frac{\text{sick and diagnosed sick}}{\text{everyone diagnosed sick}}$$

Bayes Theorem: Intuition



Bayes Theorem: Intuition



Bayes Theorem: Intuition

$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{\text{red head icon}}{\text{red head icon} + \text{green head icon}}$$

$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{\text{sick and diagnosed sick} = 99}{\text{healthy and diagnosed sick} = 9,999 + \text{sick and diagnosed sick} = 99}$$

$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{99}{10098} = 0.0098$$

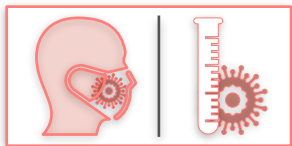
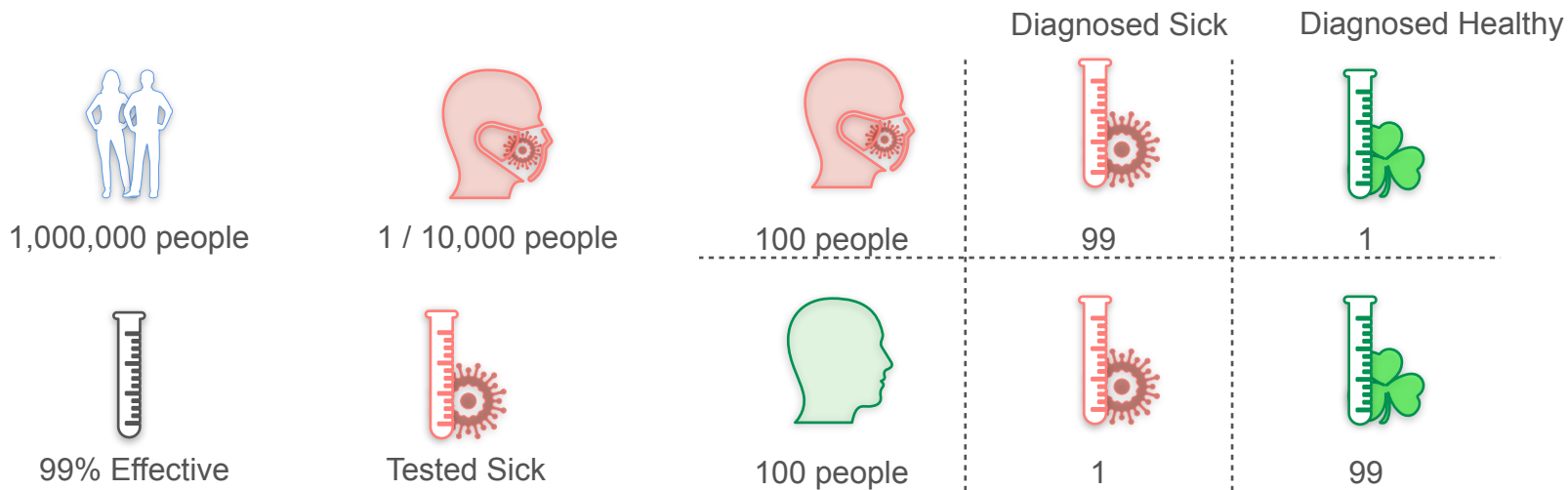


DeepLearning.AI

Introduction to probability

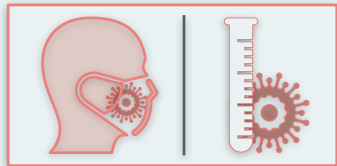
Bayes Theorem - Mathematical Formula

Bayes Theorem: Formula



The probability that **you are sick**
GIVEN that you tested sick

Bayes Theorem: Formula



$$P(\text{sick} \mid \text{diagnosed sick}) = ?$$

$$P(\text{sick}) = 0.01\%$$

$$P(\text{not sick}) = 99.99\%$$

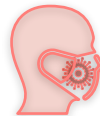
$$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$$

$$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$$

The probability that **you are sick**
GIVEN that you tested sick



1,000,000

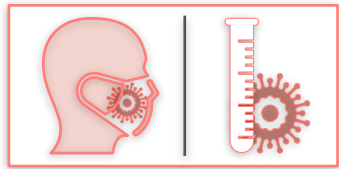


1 / 10,000



99% Effective

Bayes Theorem: Formula



A : sick

B : diagnosed sick

$$P(\text{sick} \mid \text{diagnosed sick}) = ?$$

$$P(A \mid B) = ?$$

$$P(\text{sick}) = 0.01\%$$

$$P(\text{not sick}) = 99.99\%$$

$$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$$

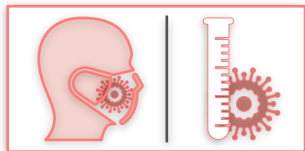
$$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$$

From Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{P(\text{sick and diagnosed sick})}{P(\text{diagnosed sick})}$$

Bayes Theorem: Formula



A : sick

B : diagnosed sick

$P(\text{sick} \mid \text{diagnosed sick}) = ?$

$P(\text{sick}) = 0.01\%$

$P(\text{not sick}) = 99.99\%$

$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$

$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

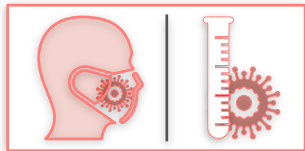
$$P(A \mid B) = \frac{P(\text{sick and diagnosed sick})}{P(\text{diagnosed sick})}$$

$P(\text{sick and diagnosed sick}) = ?$

$P(\text{diagnosed sick}) = ?$

BAYES THEOREM FORMULA CAN HELP

Bayes Theorem: Formula



A : sick

B : diagnosed sick

$P(\text{sick} \mid \text{diagnosed sick}) = ?$

$P(\text{sick}) = 0.01\%$

$P(\text{not sick}) = 99.99\%$

$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$

$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

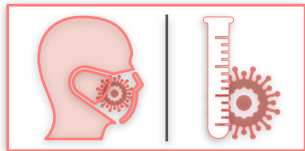
$$P(A \mid B) = \frac{P(\text{sick and diagnosed sick}) = ?}{P(\text{diagnosed sick}) = ?}$$

From Conditional Probability

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

$$P(\text{sick and diagnosed sick}) = P(\text{sick}) \cdot P(\text{diagnosed sick} \mid \text{sick})$$

Bayes Theorem: Formula



A : sick

B : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{diagnosed sick}) = ?}$$

$$P(\text{sick} | \text{diagnosed sick}) = ?$$

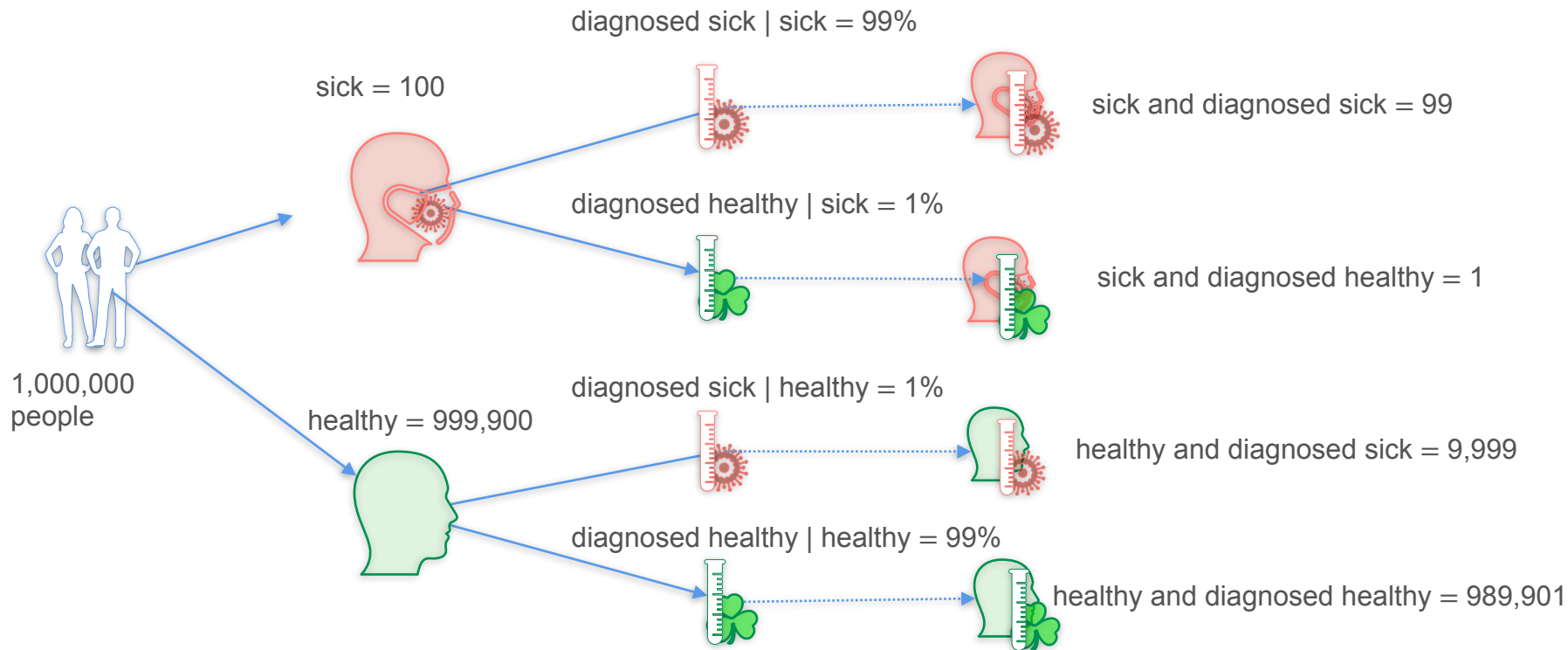
$$P(\text{sick}) = 0.01\%$$

$$P(\text{not sick}) = 99.99\%$$

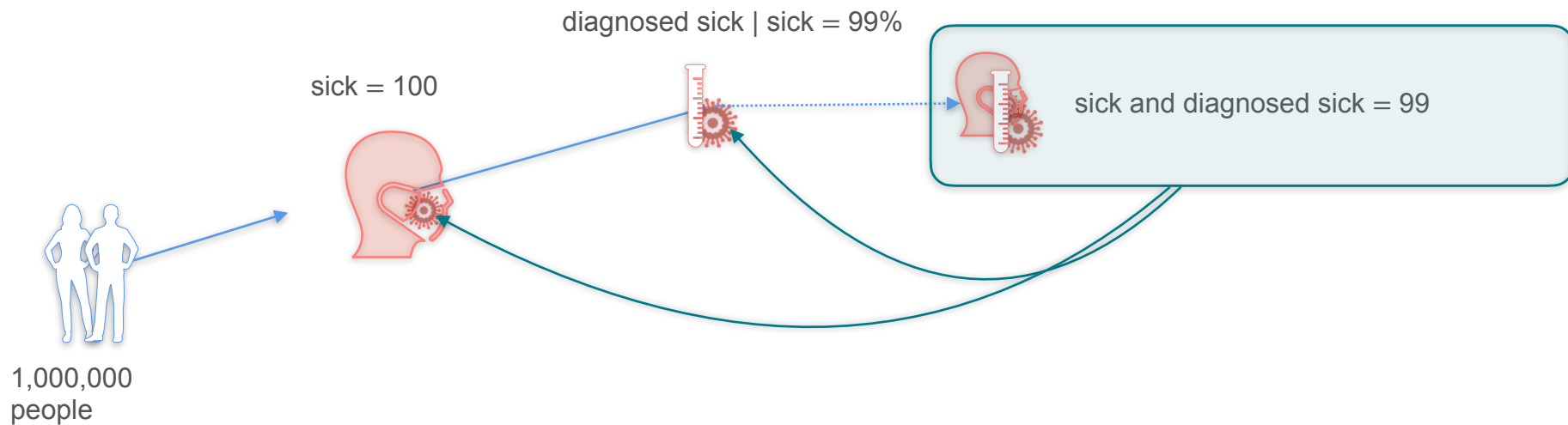
$$P(\text{diagnosed sick} | \text{sick}) = 99\%$$

$$P(\text{diagnosed sick} | \text{not sick}) = 1\%$$

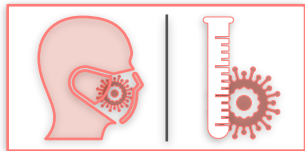
Bayes Theorem: Formula



Bayes Theorem: Formula



Bayes Theorem: Formula



A : sick

B : diagnosed sick

$P(\text{sick} \mid \text{diagnosed sick}) = ?$

$P(\text{sick}) = 0.01\%$

$P(\text{not sick}) = 99.99\%$

$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$

$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$

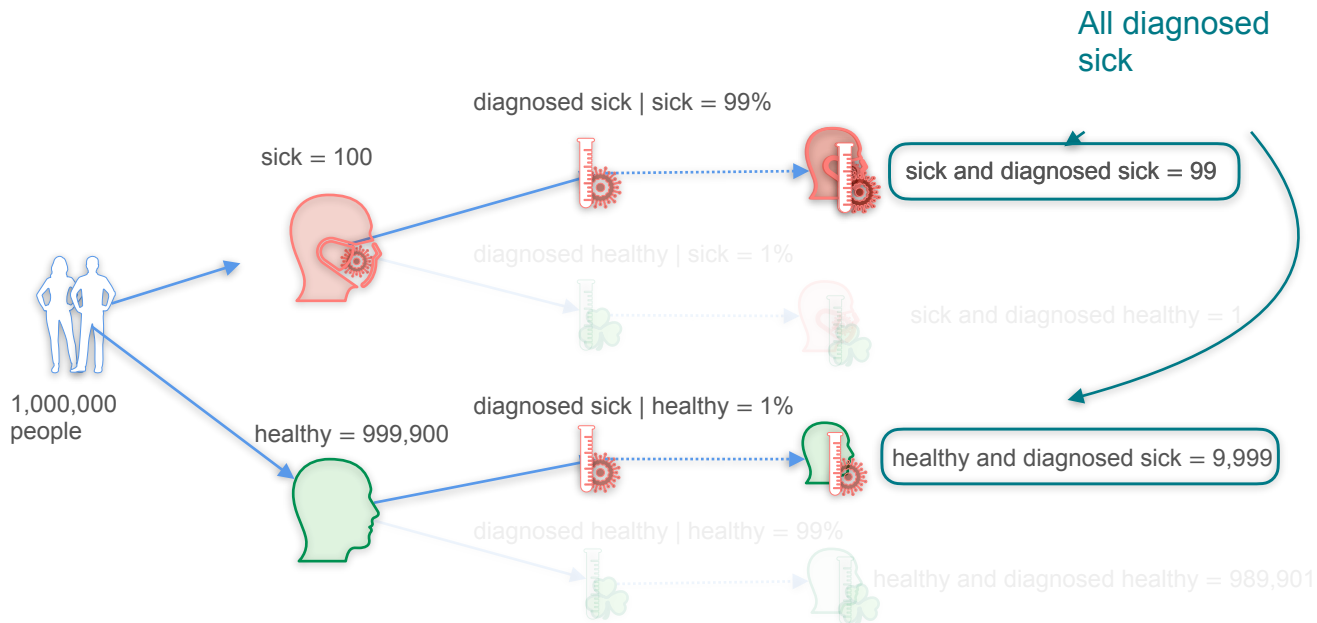
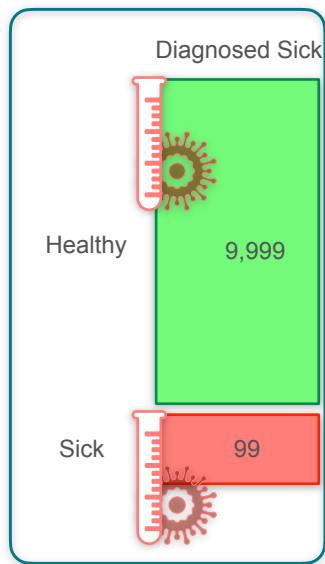
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} \mid \text{sick})}{P(\text{diagnosed sick}) = ?}$$

$P(\text{diagnosed sick}) =$

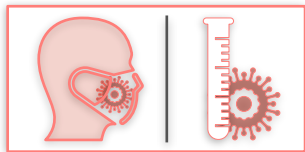
Bayes Theorem: Formula

$P(\text{diagnosed sick}) =$



$$P(\text{sick} \cap \text{diagnosed sick}) + P(\text{not sick} \cap \text{diagnosed sick})$$

Bayes Theorem: Formula



A : sick

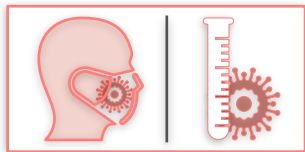
B : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{diagnosed sick}) = ?}$$

$$P(\text{diagnosed sick}) = P(\text{sick} \cap \text{diagnosed sick}) + P(\text{not sick} \cap \text{diagnosed sick})$$

Bayes Theorem: Formula



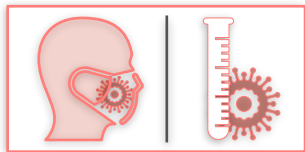
A : sick

B : diagnosed sick

$$\mathbf{P}(A | B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$$\mathbf{P}(A | B) = \frac{\mathbf{P}(\text{sick}) \bullet \mathbf{P}(\text{diagnosed sick} | \text{sick})}{\mathbf{P}(\text{sick} \cap \text{diagnosed sick}) + \mathbf{P}(\text{not sick} \cap \text{diagnosed sick})}$$

Bayes Theorem: Formula



A : sick

B : diagnosed sick

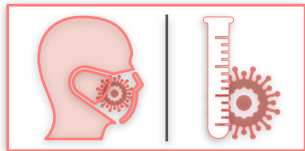
$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{sick} \cap \text{diagnosed sick}) + P(\text{not sick} \cap \text{diagnosed sick})}$$

$$\begin{aligned} P(\text{sick} \cap \text{diagnosed sick}) &= P(A \cap B) \\ &= P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick}) \end{aligned}$$

$$\begin{aligned} P(\text{not sick} \cap \text{diagnosed sick}) &= P(A' \cap B) \\ &= P(\text{not sick}) \cdot P(\text{diagnosed sick} | \text{not sick}) \end{aligned}$$

Bayes Theorem: Formula



A: sick

B: diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad ?$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick}) + P(\text{not sick}) \cdot P(\text{diagnosed sick} | \text{not sick})}$$

$$P(\text{sick}) = 0.01\%$$

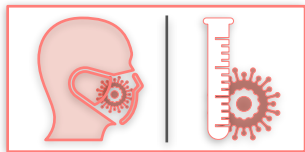
$$P(\text{not sick}) = 99.99\%$$

$$P(\text{diagnosed sick} | \text{sick}) = 99\%$$

$$P(\text{diagnosed sick} | \text{not sick}) = 1\%$$

**BAYES THEOREM
FORMULA**

Bayes Theorem: Formula



A : sick

B : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(A) \cdot P(B | A)}{P(A) \cdot P(B | A) + P(A') \cdot P(B | A')}$$

$$P(A) = 0.01\%$$

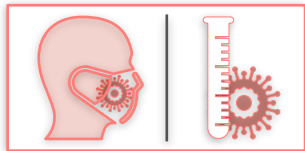
$$P(A') = 99.99\%$$

$$P(B | A) = 99\%$$

$$P(B | A') = 1\%$$

**BAYES THEOREM
FORMULA**

Bayes Theorem: Formula



A : sick

B : diagnosed sick

$$\mathbf{P(A | B) = \frac{P(A \cap B)}{P(B)}}$$

$$\mathbf{P(A | B) = \frac{P(A) \cdot P(B | A)}{P(A) \cdot P(B | A) + P(A') \cdot P(B | A')}}}$$

$$\mathbf{P(A) = 0.01\%}$$

$$\mathbf{P(A') = 99.99\%}$$

$$\mathbf{P(B | A) = 99\%}$$

$$\mathbf{P(B | A') = 1\%}$$

$$\mathbf{P(A | B) = \frac{0.0001 \times 0.99}{(0.0001 \times 0.99) + (0.9999 \times 0.01)}}$$

$$\mathbf{P(A | B) = 0.0098}$$

**BAYES THEOREM
FORMULA**



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Introduction to probability

Bayes Theorem - Spam Example

Bayes Theorem: Spam Example



Bayes Theorem: Spam Example



20 spam

80 not spam (ham)



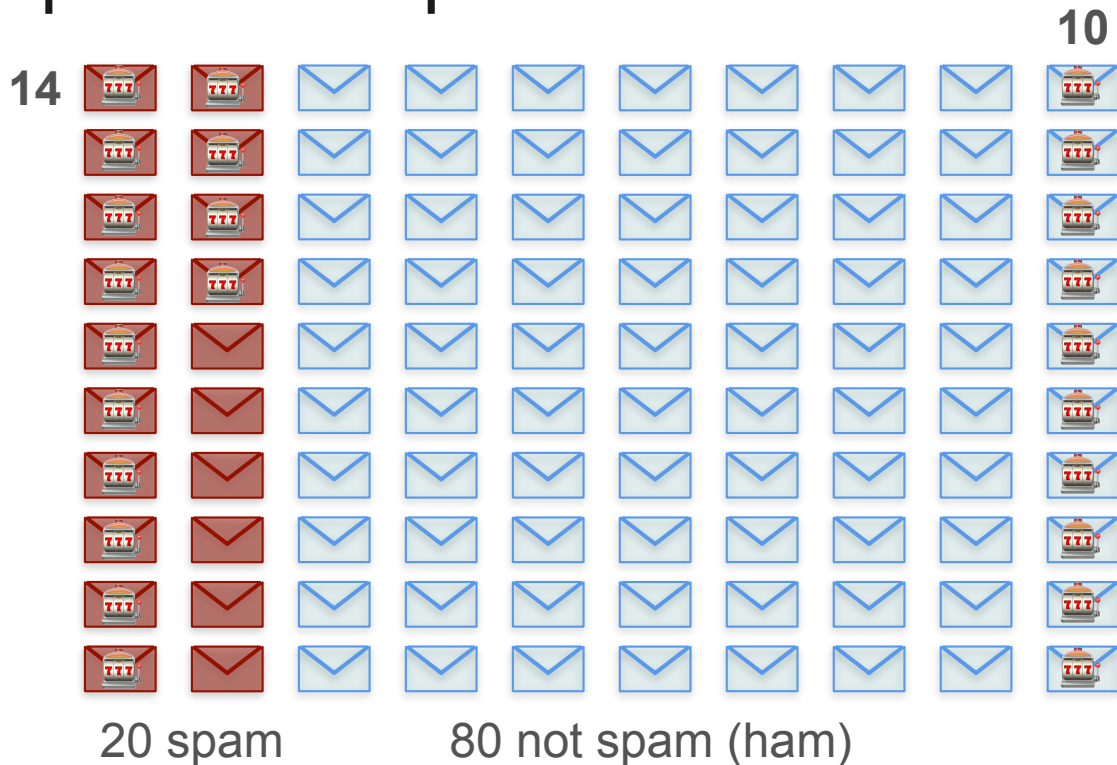
“lottery”

Bayes Theorem: Spam Example



What is the probability that an email containing lottery is a spam?

$$P(\text{spam} \mid \text{lottery})$$



Bayes Theorem: Spam Example (Intuition Solution)



$P(\text{spam} \mid \text{lottery})$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$



14

24 emails
containing lottery

$$P(\text{spam} \mid \text{lottery}) = \frac{\text{spam and lottery}}{\text{all lottery}}$$

$$= \frac{14}{24} \\ = \frac{7}{12} = 0.583$$



10

Bayes Theorem: Spam Example (Formula Solution)

$P(\text{spam} \mid \text{lottery})$

$$P(A \mid B) = \frac{P(A) \cdot P(B \mid A)}{P(A) \cdot P(B \mid A) + P(A') \cdot P(B \mid A')}$$

A : Email is spam B : Email contains lottery

$$P(\text{spam} \mid \text{lottery}) = \frac{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) + P(\text{not spam}) \cdot P(\text{lottery} \mid \text{not spam})}$$

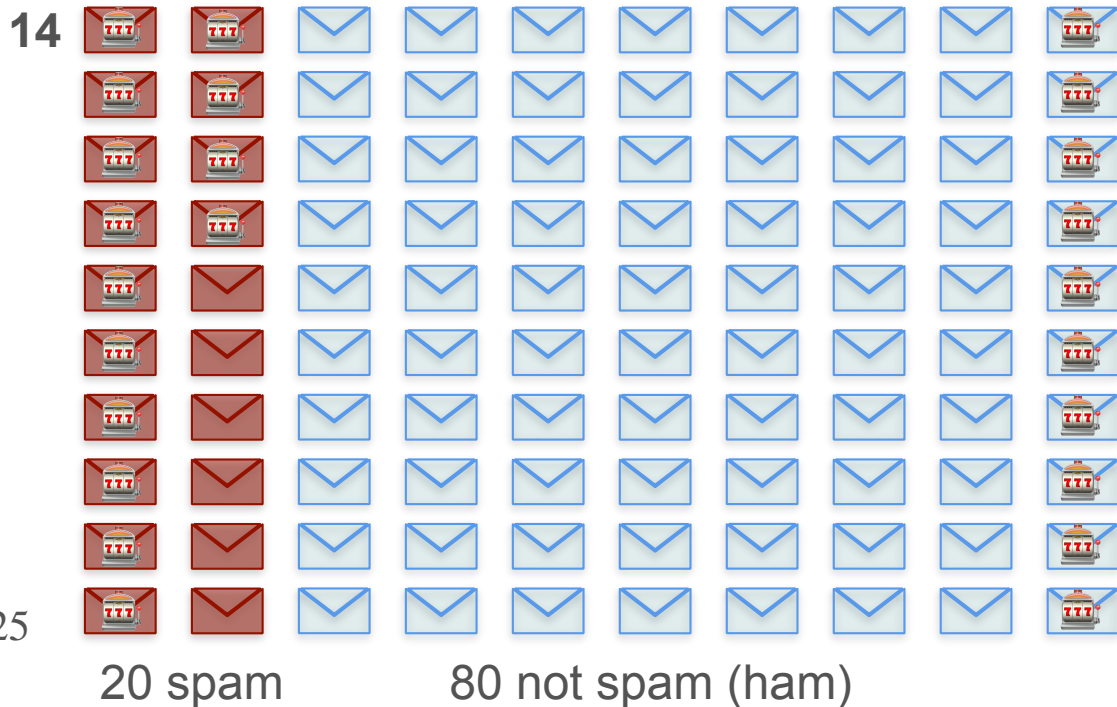
Bayes Theorem: Spam Example (Formula Solution)

$$P(\text{spam}) = \frac{20}{100} = 0.2$$

$$P(\text{not spam}) = \frac{80}{100} = 0.8$$

$$P(\text{lottery} \mid \text{spam}) = \frac{14}{20} = 0.7$$

$$P(\text{lottery} \mid \text{not spam}) = \frac{10}{80} = 0.125$$



Bayes Theorem: Spam Example (Formula Solution)

$$P(\text{spam}) = 0.2$$

$$P(\text{lottery} \mid \text{spam}) = 0.7$$

$$P(\text{not spam}) = 0.8$$

$$P(\text{lottery} \mid \text{not spam}) = 0.125$$

$$P(\text{spam} \mid \text{lottery}) = \frac{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) + P(\text{not spam}) \cdot P(\text{lottery} \mid \text{not spam})}$$

$$P(\text{spam} \mid \text{lottery}) = \frac{0.2 \times 0.7}{(0.2 \times 0.7) + (0.8 \times 0.125)} = 0.583$$



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Introduction to probability

Bayes Theorem - Prior and Posterior

Bayes Theorem

PRIOR

$$\mathbf{P}(A)$$

EVENT

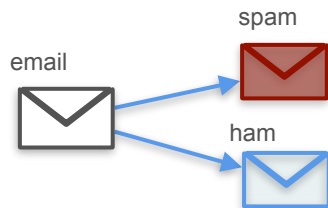
$$E$$

POSTERIOR

$$\mathbf{P}(A | E)$$

Prior and Posterior

PRIOR

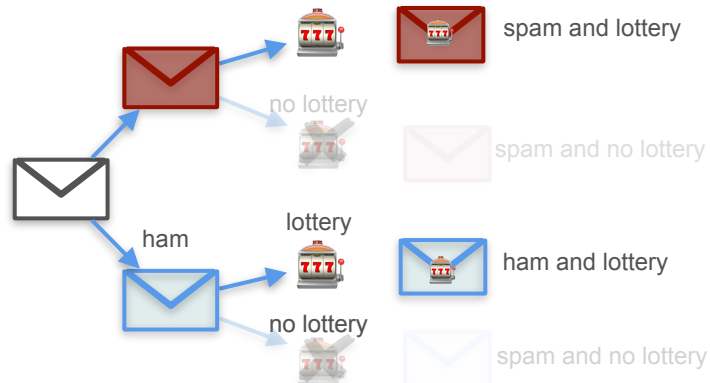


$$P(\text{spam}) = \frac{\text{red envelope icon}}{\text{red envelope icon} + \text{blue envelope icon}}$$

EVENT


Email contains lottery

POSTERIOR



$$P(\text{spam} \mid \text{lottery}) = \frac{\text{red envelope with lottery icon}}{\text{red envelope with lottery icon} + \text{blue envelope with lottery icon}}$$

Prior and Posterior

PRIOR



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$$P(\text{sum} = 10) = \frac{3}{36}$$

EVENT



1st dice is 6

POSTERIOR

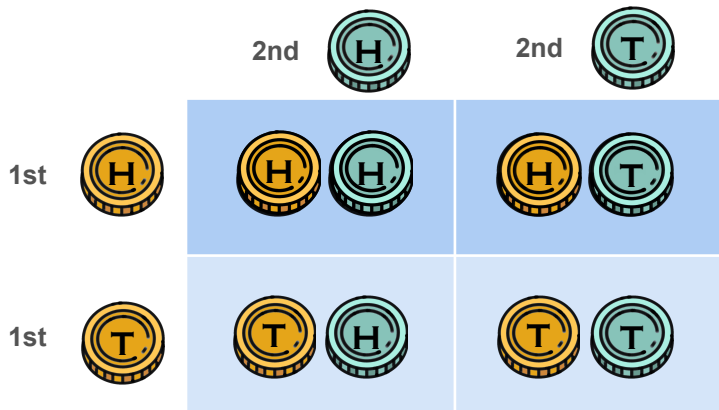


1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$$P(\text{sum} = 10 \mid \text{1st is 6}) = \frac{1}{6}$$

Prior and Posterior

PRIOR



$$P(HH) = \frac{1}{4}$$

EVENT



POSTERIOR



$$P(HH | 1st \text{ is } H) = \frac{1}{2}$$



DeepLearning.AI

Introduction to probability

Bayes Theorem - The Naive Bayes Model

What About 2 Events?

PRIOR

$$P(\text{spam}) = \frac{\text{Red Envelope}}{\text{Red Envelope} + \text{Blue Envelope}}$$

EVENT



Email contains 'lottery'



Email contains 'winning'



Email contains 'lottery' and 'winning'

POSTERIOR

$$P(\text{spam} \mid \text{lottery}) = \frac{\text{Red Envelope with Lottery Ticket}}{\text{Red Envelope with Lottery Ticket} + \text{Blue Envelope with Lottery Ticket}}$$

$$P(\text{spam} \mid \text{winning}) = \frac{\text{Red Envelope with Winning Ticket}}{\text{Red Envelope with Winning Ticket} + \text{Blue Envelope with Winning Ticket}}$$

?

What About 2 Events?

EVENT



Email contains 'lottery' and 'winning'

POSTERIOR

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{\text{[Red envelope with lottery ticket and winning ticket]}}{\text{[Red envelope with lottery ticket and winning ticket]} + \text{[Blue envelope with lottery ticket and winning ticket]}}$$

$$\frac{\text{\# Spam emails with 'lottery' and 'winning'}}{\text{\# Spam emails}}$$

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{P(\text{spam}) \cdot \text{P(lottery \& winning} \mid \text{spam)}}{P(\text{spam}) \cdot P(\text{lottery \& winning} \mid \text{spam}) + P(\text{ham}) \cdot P(\text{lottery \& winning} \mid \text{ham})}$$

A red arrow points from the fraction $\frac{\text{\# Spam emails with 'lottery' and 'winning'}}{\text{\# Spam emails}}$ to the term $P(\text{lottery \& winning} \mid \text{spam})$ in the denominator of the equation above. A red question mark is placed above the term $P(\text{lottery \& winning} \mid \text{spam})$.

What About More Than 2 Events?

EVENT

POSTERIOR

Email contains $w_1, w_2, \dots, w_{100}$

~~$\frac{\text{\# Spam emails with } w_1, \dots, w_{100}}{\text{\# Spam emails}}$~~ ⁰

$$\mathbf{P}(\text{spam} \mid w_1, \dots, w_{100}) = \frac{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1, \dots, w_{100} \mid \text{spam})}{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1, \dots, w_{100} \mid \text{spam}) + \mathbf{P}(\text{ham}) \cdot \mathbf{P}(w_1, \dots, w_{100} \mid \text{ham})}$$

[?]

Is There a Quicker Way To Estimate the Probability?

Naive assumption



The appearances of 'lottery' and 'winning' are independent

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{P(\text{spam}) \cdot P(\text{lottery \& winning} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery \& winning} \mid \text{spam}) + P(\text{ham}) \cdot P(\text{lottery \& winning} \mid \text{ham})}$$

Is There a Quicker Way To Estimate the Probability?

Naive assumption



The appearances of 'lottery' and 'winning' are independent

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

$$\mathbf{P}(\text{spam} \mid \text{lottery \& winning}) = \frac{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(\text{lottery} \mid \text{spam}) \cdot \mathbf{P}(\text{winning} \mid \text{spam})}{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(\text{lottery} \mid \text{spam}) \cdot \mathbf{P}(\text{winning} \mid \text{spam}) + \mathbf{P}(\text{ham}) \cdot \mathbf{P}(\text{lottery} \mid \text{ham}) \cdot \mathbf{P}(\text{winning} \mid \text{ham})}$$

Is There a Quicker Way To Estimate the Probability?

Naive assumption

The appearances of the words $w_1, w_2, \dots, w_n$ are independent

$$\mathbf{P}(\text{spam} \mid w_1, \dots, w_n) = \frac{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1 \mid \text{spam}) \cdots \mathbf{P}(w_n \mid \text{spam})}{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1 \mid \text{spam}) \cdots \mathbf{P}(w_n \mid \text{spam}) + \mathbf{P}(\text{ham}) \cdot \mathbf{P}(w_1 \mid \text{ham}) \cdots \mathbf{P}(w_n \mid \text{ham})}$$

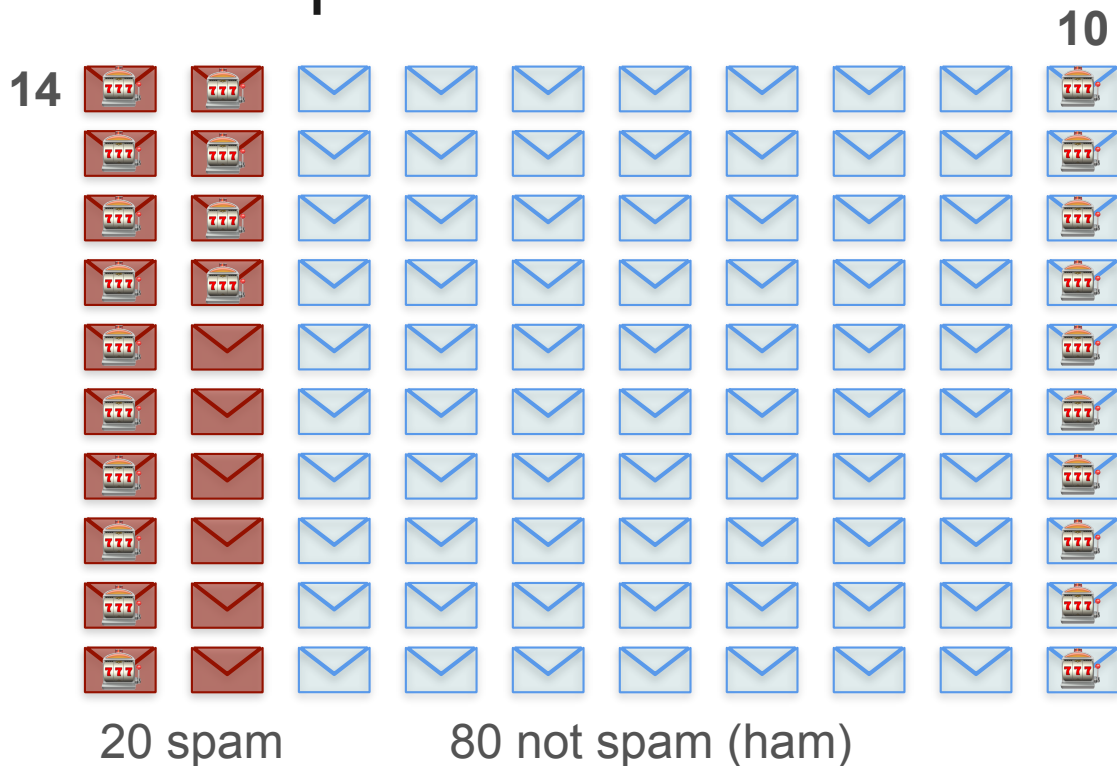
Naive Bayes: Spam Example

$$P(\text{spam}) = \frac{20}{100} = 0.2$$

$$P(\text{ham}) = \frac{80}{100} = 0.8$$

$$P(\text{lottery} \mid \text{spam}) = \frac{14}{20} = 0.7$$

$$P(\text{lottery} \mid \text{ham}) = \frac{10}{80} = 0.125$$



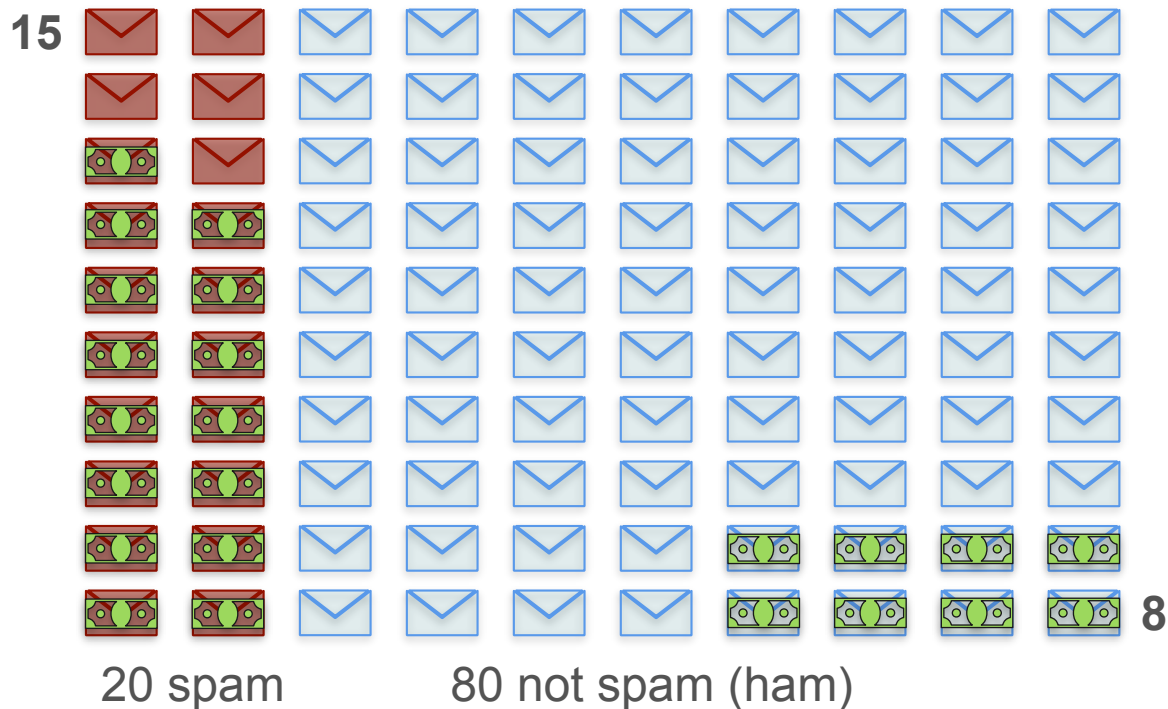
Naive Bayes: Spam Example

$$P(\text{spam}) = \frac{20}{100} = 0.2$$

$$P(\text{ham}) = \frac{80}{100} = 0.8$$

$$P(\text{winning} \mid \text{spam}) = \frac{15}{20} = 0.75$$

$$P(\text{winning} \mid \text{ham}) = \frac{8}{80} = 0.1$$



Naive Bayes: Spam Example

$$P(\text{spam}) = 0.2$$

$$P(\text{lottery} \mid \text{spam}) = 0.7$$

$$P(\text{winning} \mid \text{spam}) = 0.75$$

$$P(\text{not spam}) = 0.8$$

$$P(\text{lottery} \mid \text{ham}) = 0.125$$

$$P(\text{winning} \mid \text{ham}) = 0.1$$

$$P(\text{spam} \mid \text{lottery}) = \frac{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) \cdot P(\text{winning} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) \cdot P(\text{winning} \mid \text{spam}) + P(\text{ham}) \cdot P(\text{lottery} \mid \text{ham}) \cdot P(\text{winning} \mid \text{ham})}$$

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{0.2 \times 0.7 \times 0.75}{(0.2 \times 0.7 \times 0.75) + (0.8 \times 0.125 \times 0.1)} = 0.913$$



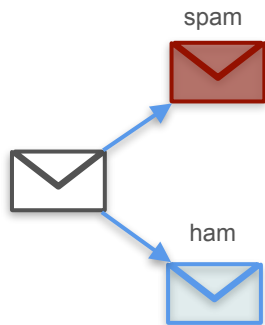
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Introduction to probability

Probability in Machine Learning

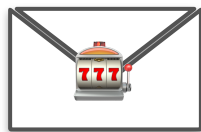
Bayes Theorem

PRIOR

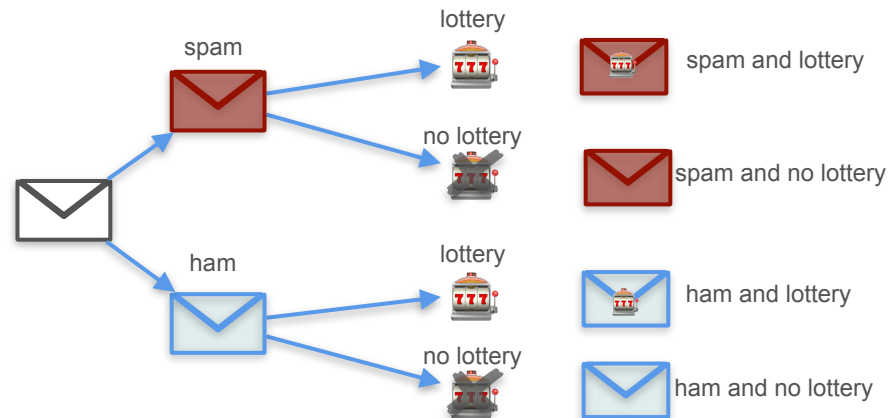


$$P(\text{spam}) = \frac{\text{red envelope icon}}{\text{red envelope icon} + \text{blue envelope icon}}$$

EVENT



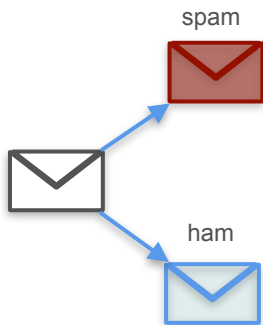
POSTERIOR



$$P(\text{spam} \mid \text{lottery}) =$$

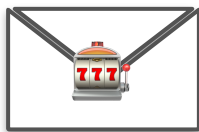
Bayes Theorem

PRIOR

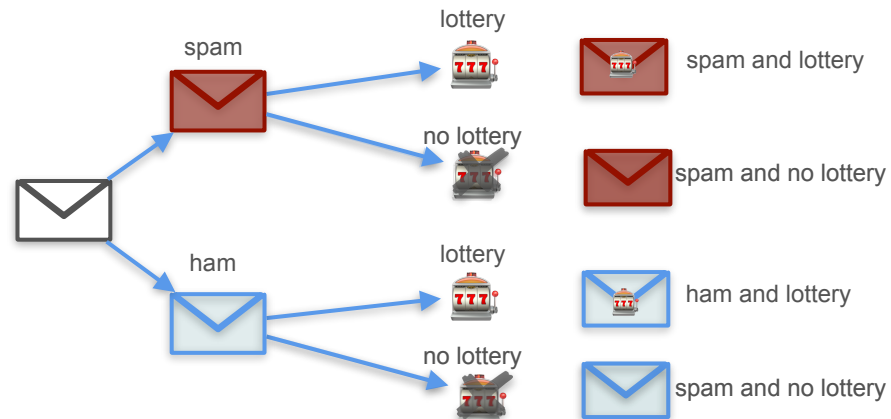


$$P(\text{spam}) = \frac{\text{spam icon}}{\text{spam icon} + \text{ham icon}}$$

EVENT



POSTERIOR



$$P(\text{spam} \mid \text{lottery}) = \frac{\text{spam and lottery icon}}{\text{spam and lottery icon} + \text{ham and lottery icon}}$$

Example Problem

Image recognition

- What is the probability that there is a cat in the image
- $\mathbf{P}(\text{cat} \mid \text{image}) = \mathbf{P}(\text{cat} \mid \text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_n)$



Example Problem

Classification

Patient 3	
Patient 2	
Patient 1	
A	Age
G	Gender
H	Height
W	Weight
S	Smoker
...	...
H	Heart rate
B	Blood pressure

- Is this patient healthy?
- Calculate $\mathbf{P}(\text{healthy} \mid \text{symptoms and history})$

Example Problem

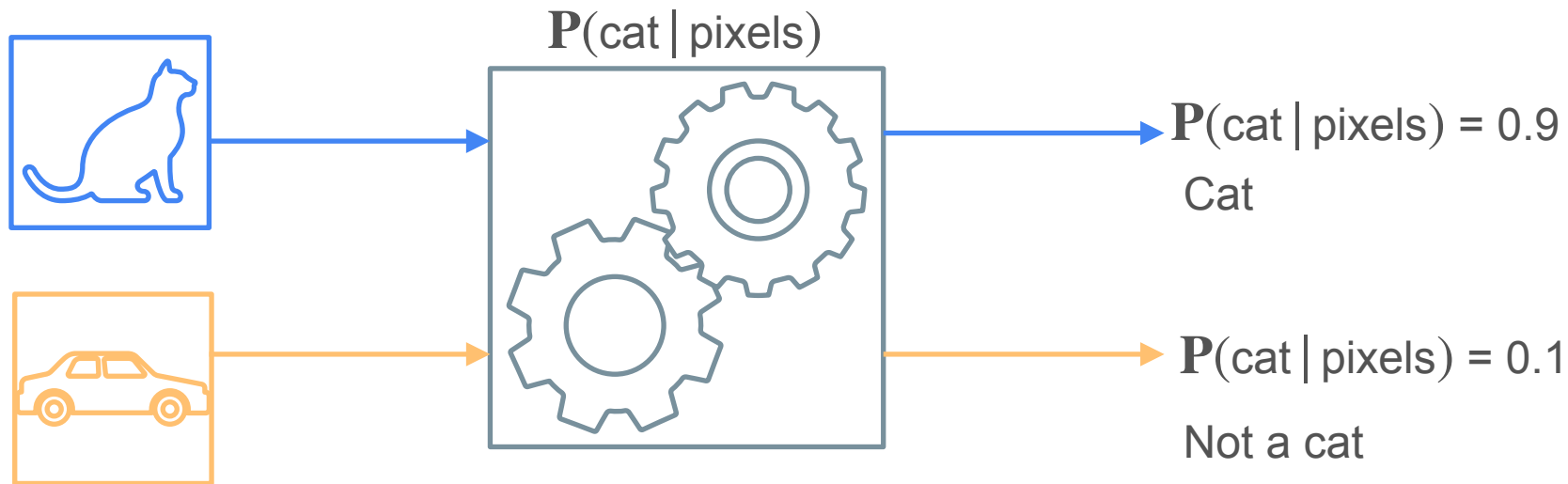
Sentiment analysis

the first cold shower
even the monkey seems to want
a little coat of straw

Matsuo Bashō

- Is this a happy sentence?
- Calculate $P(\text{happy} | \text{words in the sentence})$

Example Solution



Example Problem: Generative Models

Face generation

- Generate a group of pixels such that the resulting image looks like a human face.
- Goal: generate images such that $\mathbf{P}(\text{face} \mid \text{pixels})$ is high.



Image generated by a StyleGAN

W1 Lesson 2

Probability Distributions

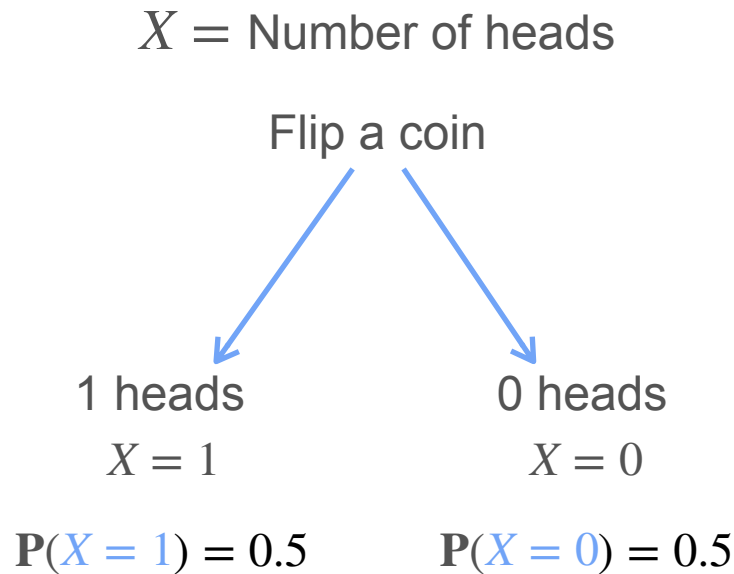
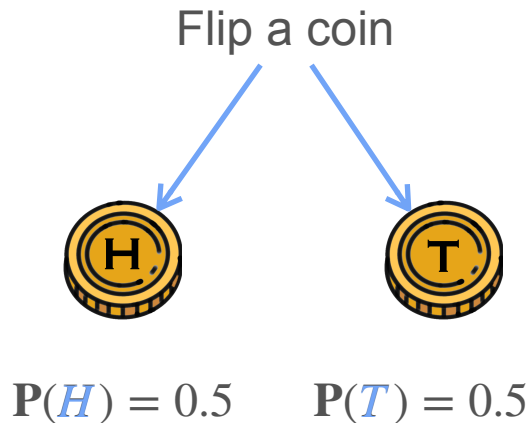


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Probability Distributions

Random Variables

From Events to Random Variables



X is a random variable

From Events to Random Variables

X = Number of heads in 10 coin tosses

$$P(H) = 0.5$$



$$X = 10$$

$$0.5^{10}$$

$$P(X = 0)?$$



$$X = 9$$

$$0.5^9 0.5$$

$$P(X = 1)?$$

...



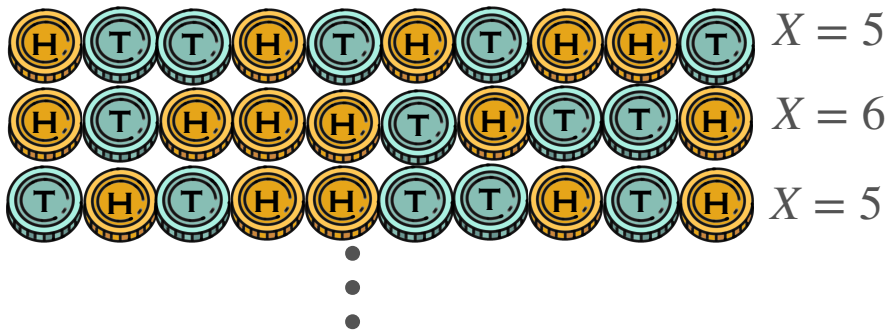
$$X = 9$$

$$0.5^9 0.5$$

$$P(X = 10)?$$

From Events to Random Variables

X = Number of heads in 10 coin tosses



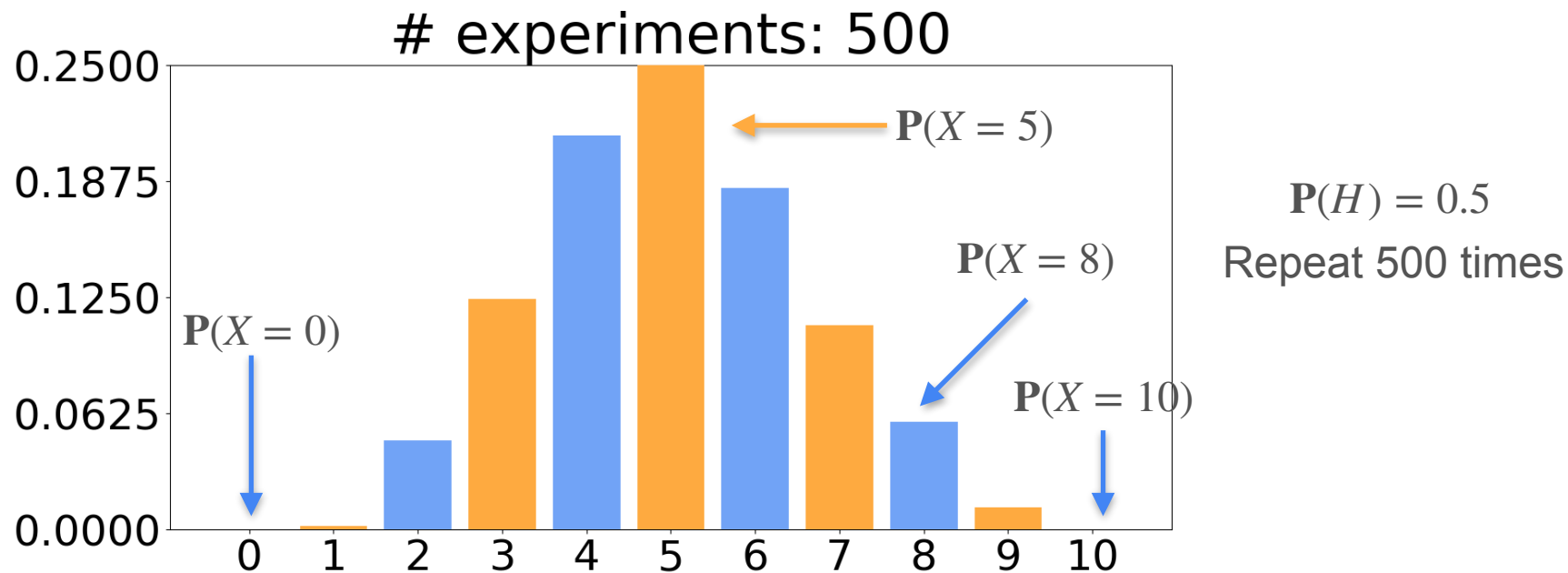
Possible outcomes:

0	4	8
1	5	9
2	6	10
3	7	

$$P(H) = 0.5$$

Repeat 500 times

Flipping a Fair Coin 500 Times



Why Random Variables?

- Random variables allow you to model the whole experiment at once



X = Number of heads



X = Number of 1's



X = Number of sick patients

?

$$P(X = 1) = 0.5$$

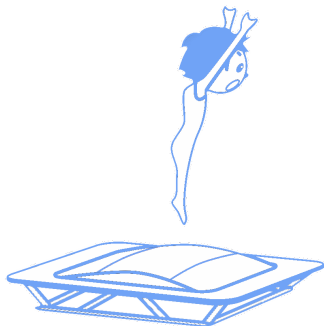
$$P(X = -7) = 0.2$$

$$P(X = 3.14159) = 0.3$$

Other Random Variables



Wait time until the
next bus arrives



Height of an
gymnast's jump



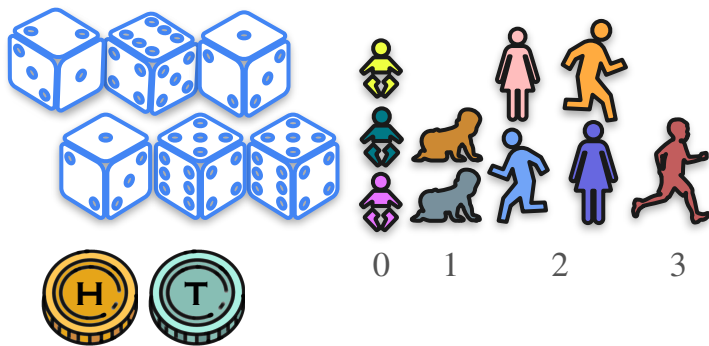
Number of
defective products
in a shipment



mm. of rain in
November

Discrete and Continuous Random Variables

Discrete random variables



~~Finite number of values~~

(Could be infinite too)

Can take only a **countable**
number of values

Continuous random variables



Infinite number of values

Takes values on an interval

Random Variable Vs. Deterministic Variable

Deterministic

$$x = 2, f(x) = x^2$$

Fixed outcome

Random

X = number of defective items in
a shipment

Uncertain outcome



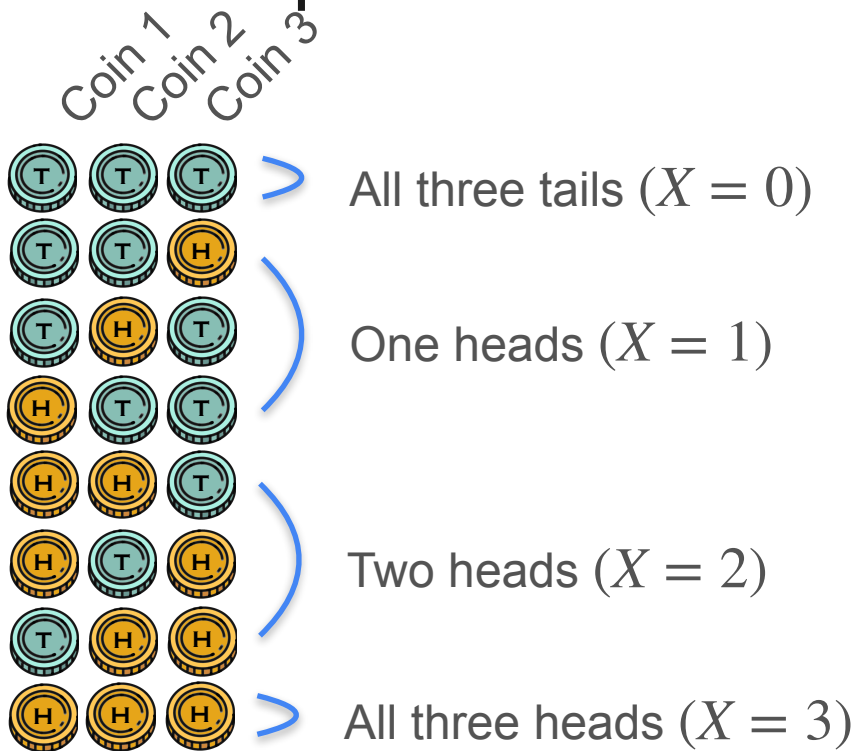
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Probability Distributions

Probability Distributions (Discrete)

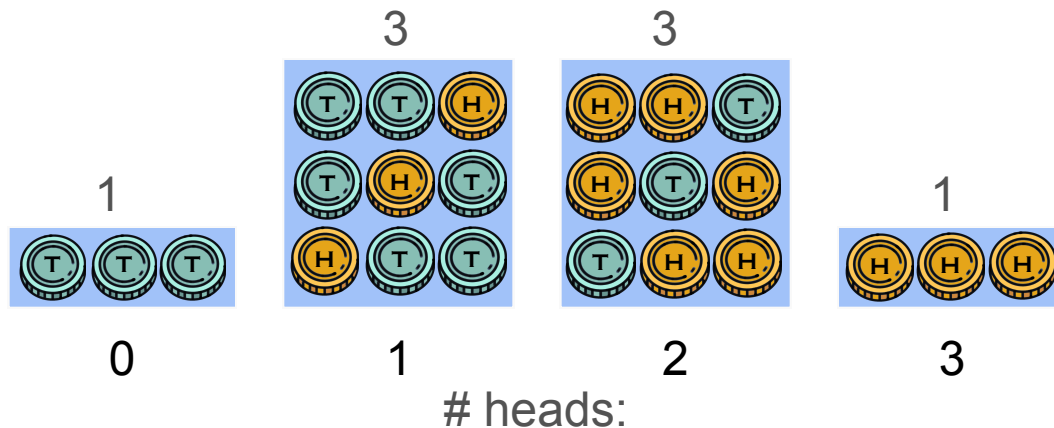
Discrete Distributions: Flip Three Coins

X_1 : number of heads in 3 coin tosses



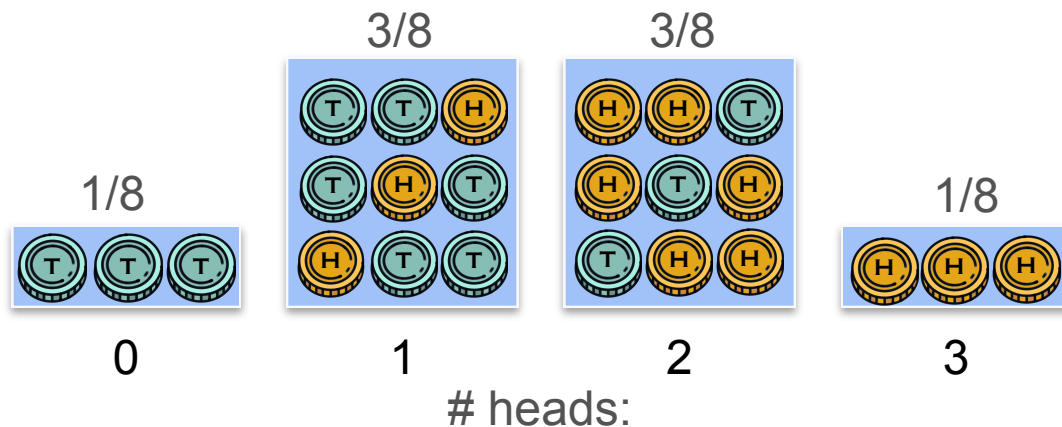
Discrete Distributions: Flip Three Coins

X_1 : number of heads in 3
coin tosses



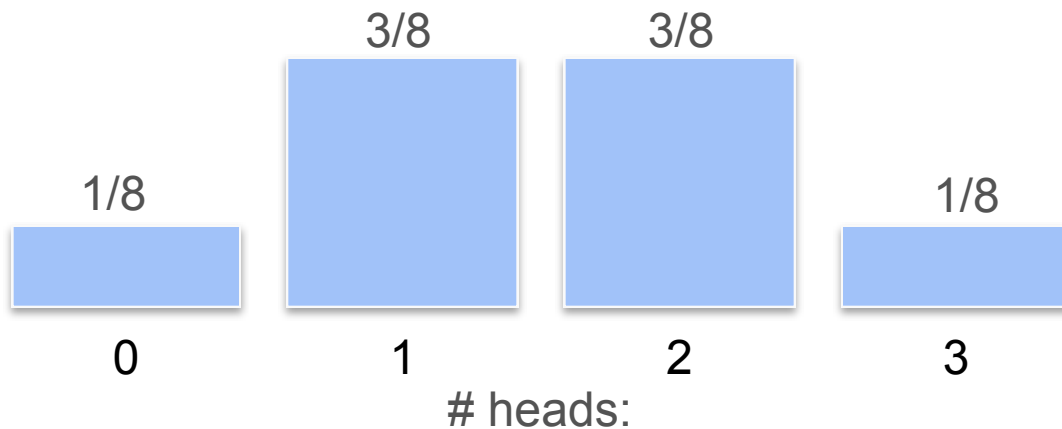
Discrete Distributions: Flip Three Coins

X_1 : number of heads in 3 coin tosses



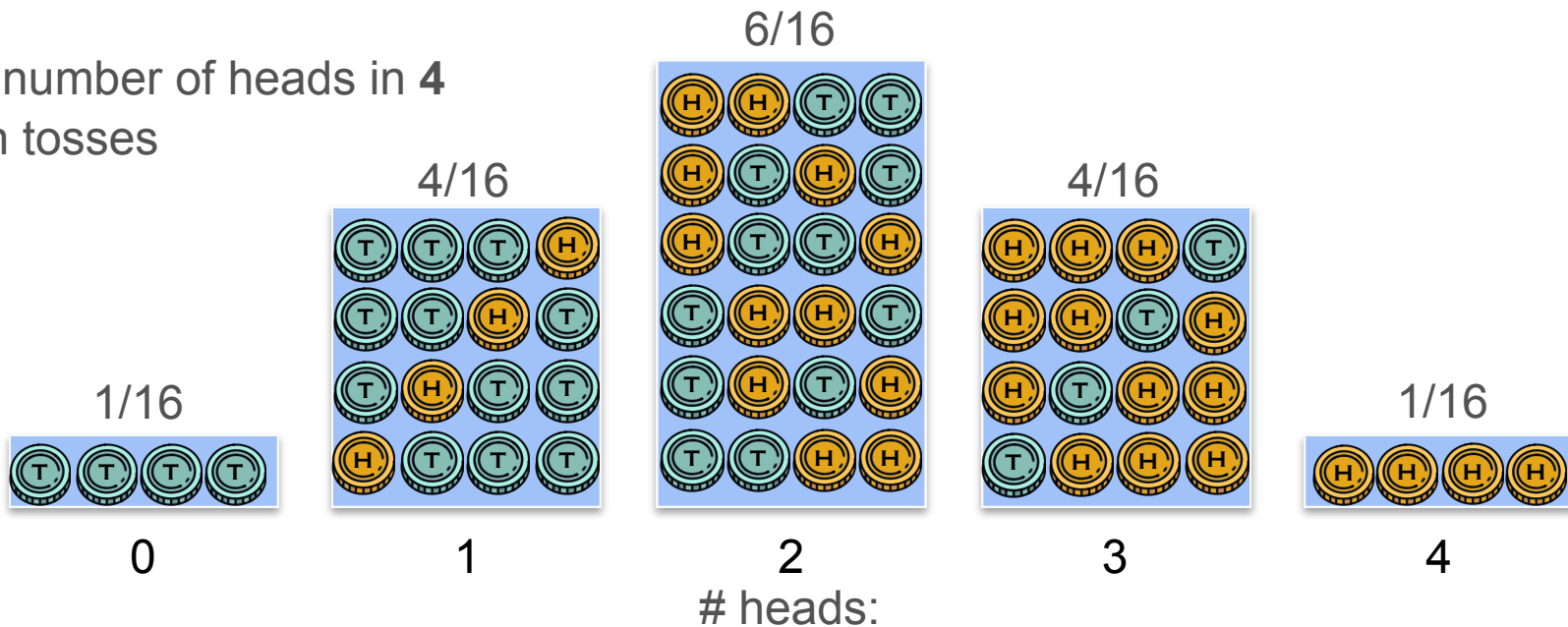
Discrete Distributions: Flip Three Coins

X_1 : number of heads in 3
coin tosses



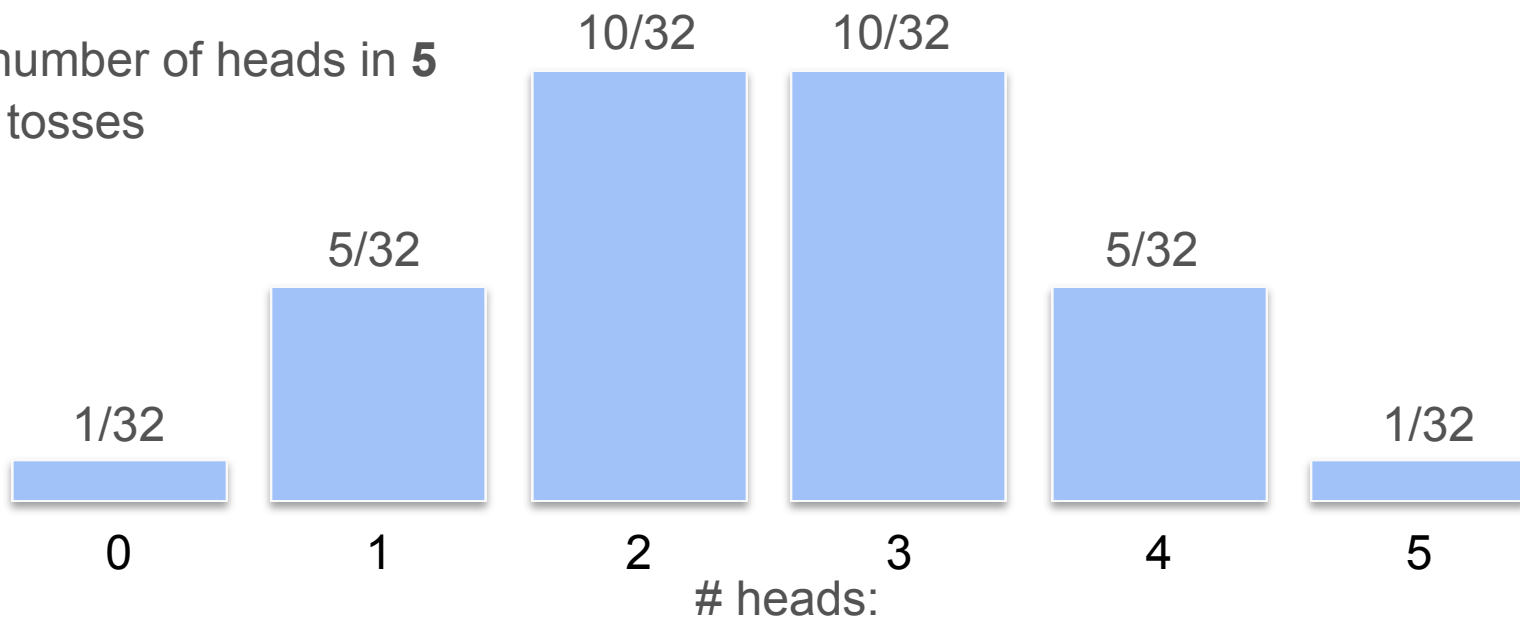
Discrete Distributions: Flip Four Coins

X_2 : number of heads in 4 coin tosses



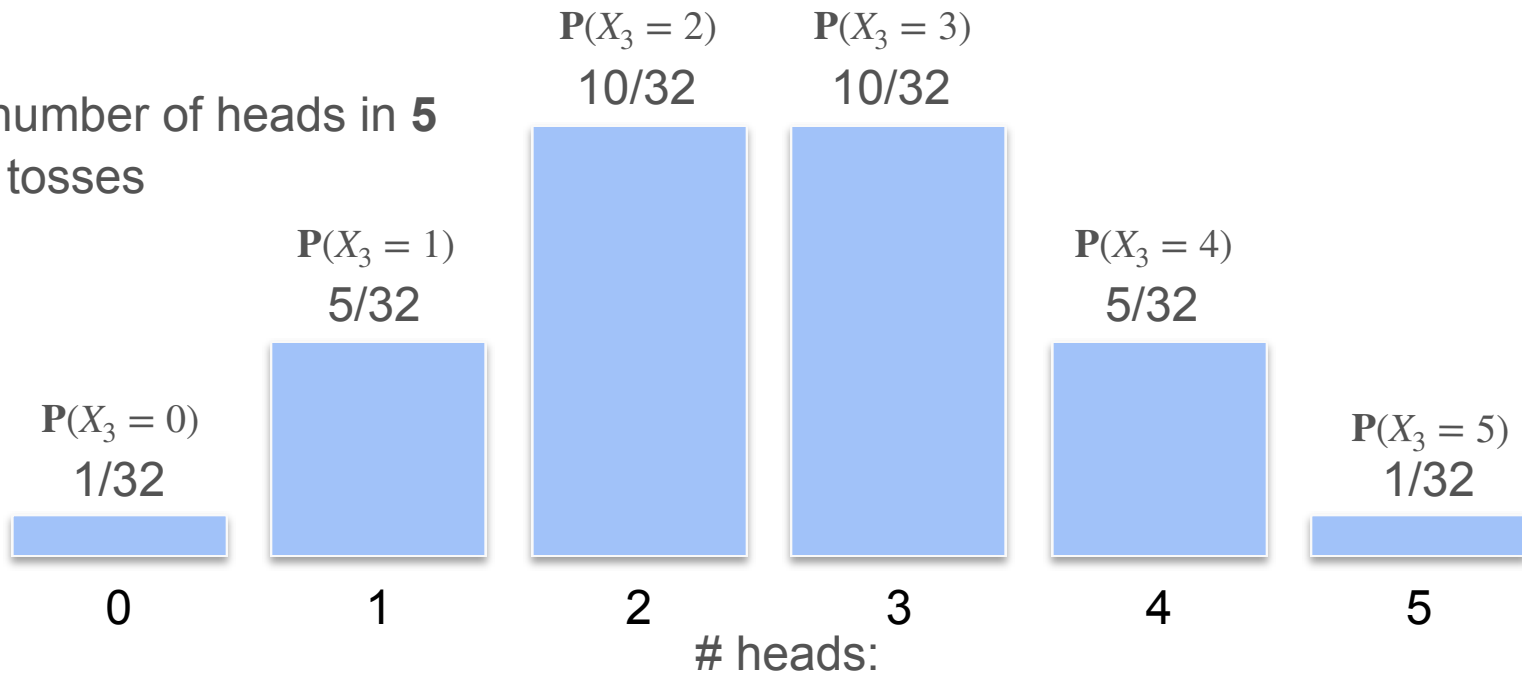
Discrete Distributions: Flip Five Coins

X_3 : number of heads in 5 coin tosses



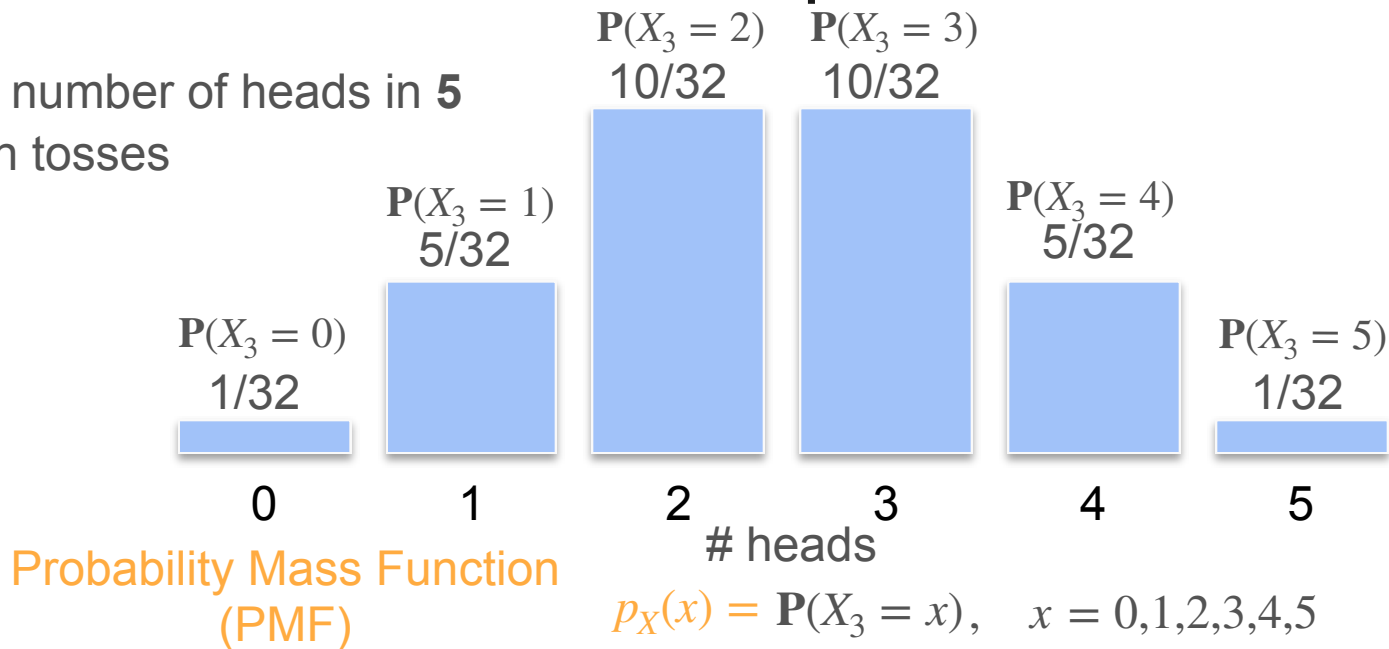
Discrete Distributions: Flip Five Coins

X_3 : number of heads in 5 coin tosses



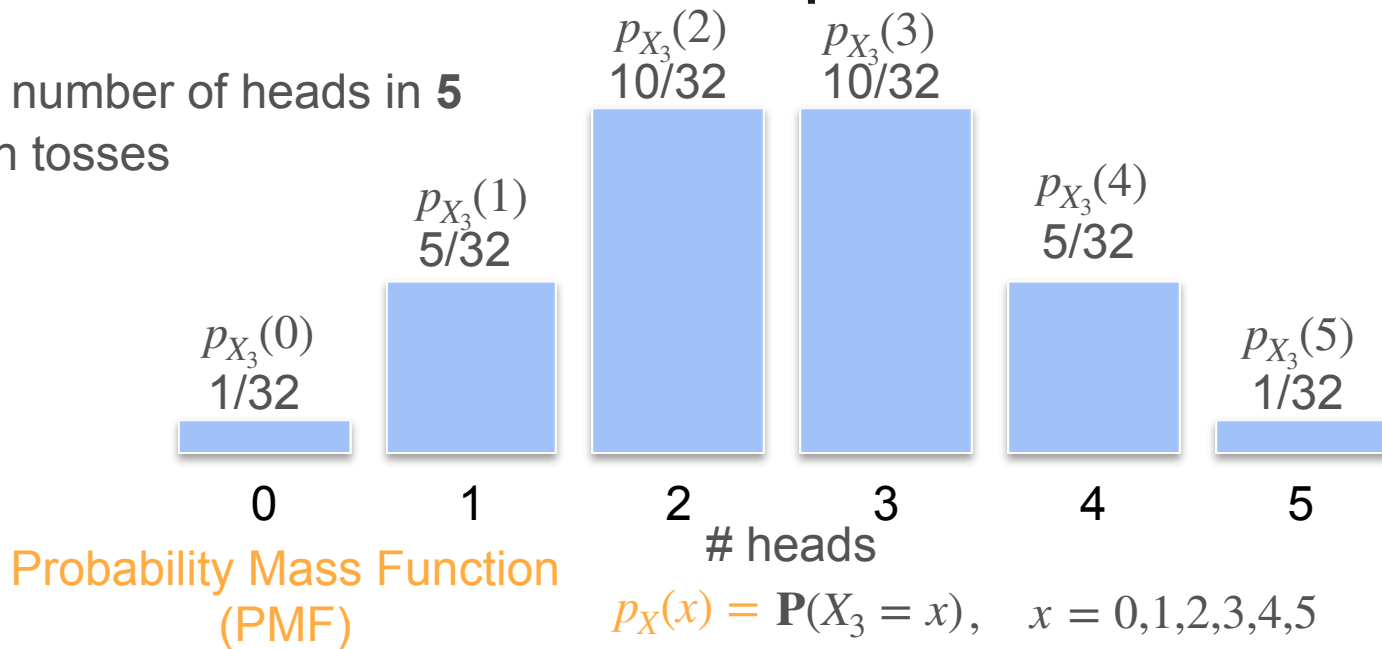
Discrete Distributions: Flip Five Coins

X_3 : number of heads in 5 coin tosses



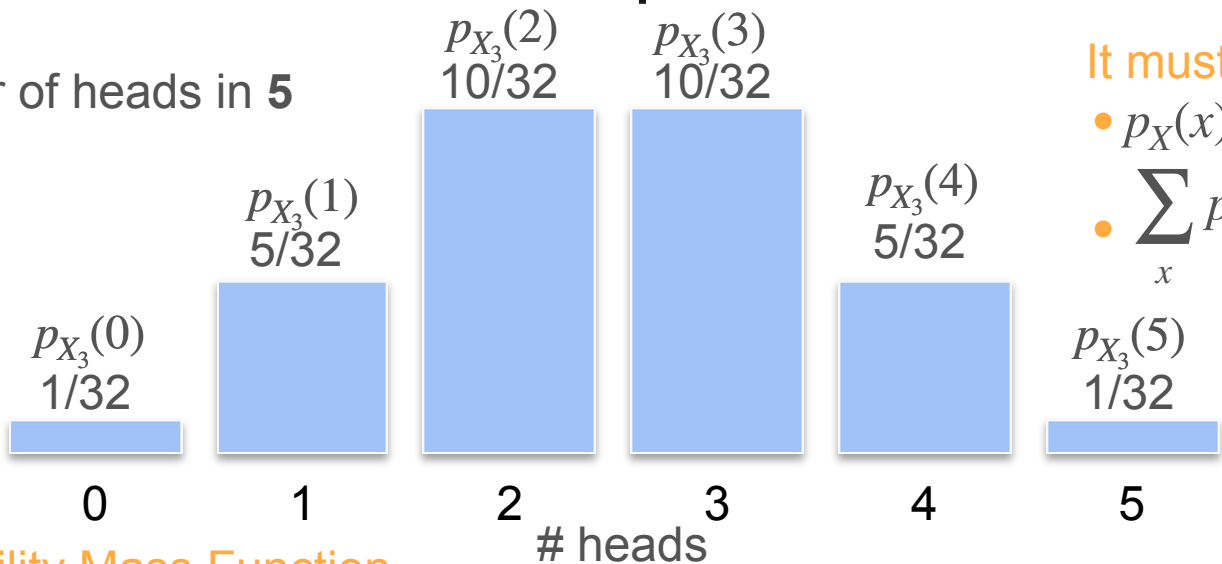
Discrete Distributions: Flip Five Coins

X_3 : number of heads in 5 coin tosses



Discrete Distributions: Flip Five Coins

X_3 : number of heads in 5 coin tosses



It must satisfy:

- $p_X(x) \geq 0$
- $\sum_x p_X(x) = 1$

Probability Mass Function (PMF)

$$p_X(x) = \mathbf{P}(X_4 = x), \quad x = 0, 1, 2, 3, 4, 5$$

$$\begin{aligned} &+ \quad + \quad + \quad + \quad + \quad = \frac{1}{32} + \frac{5}{32} + \frac{10}{32} + \frac{10}{32} + \frac{5}{32} + \frac{1}{32} = 1 \end{aligned}$$

Can You See a Pattern?

X_1, X_2, X_3, X_4 are very similar

They all represent **number of heads in n experiments**

The way the probability distributes along the possible outcomes seems to have a similar pattern

Could there be a **single model** to represent all this variables?



Binomial distribution



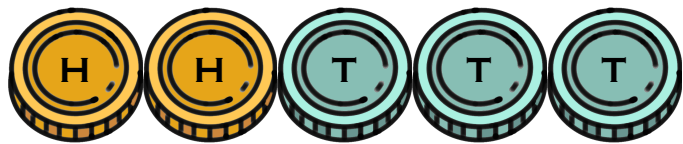
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Probability Distributions

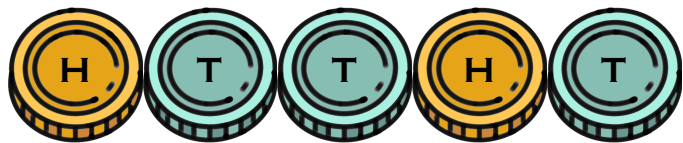
Binomial Distribution

Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



$$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 = \frac{1}{32}$$



$$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 = \frac{1}{32}$$

Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



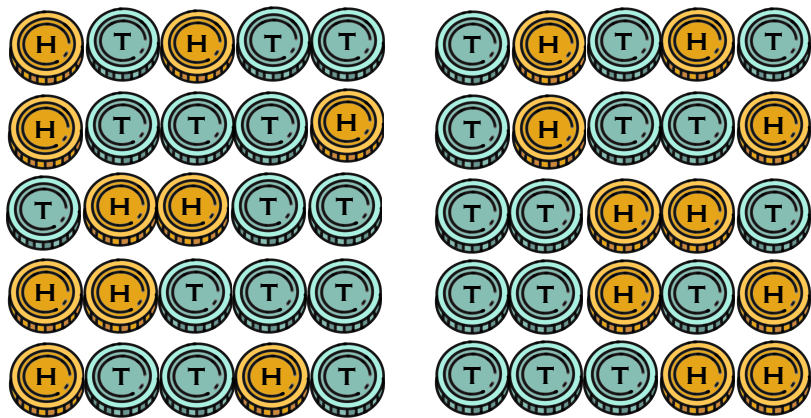
Number of ways
you can order 5
coins

$$10 = \frac{5!}{2!}$$

Number
of H

Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



$$10 = \frac{5!}{2!(5-2)!}$$

Number of ways you can order 5 coins

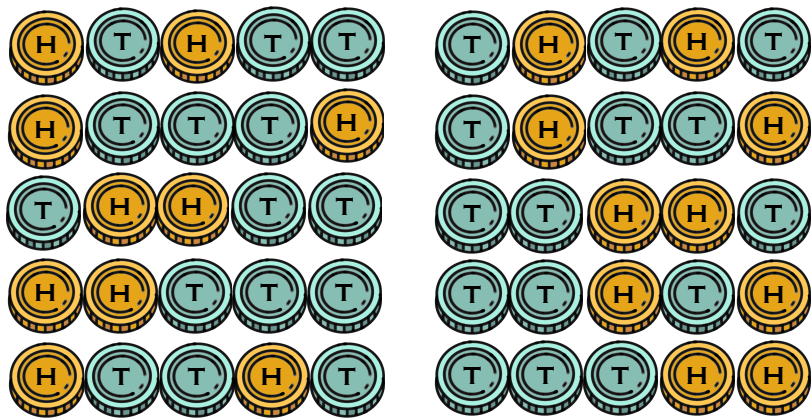
Number of H

Number of T



Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



$$10 = \frac{5!}{2!(5-2)!} = \binom{5}{2}$$

Binomial coefficient

Number of ways you can get 2
heads in 5 coin tosses

Binomial Distribution: Binomial Coefficient

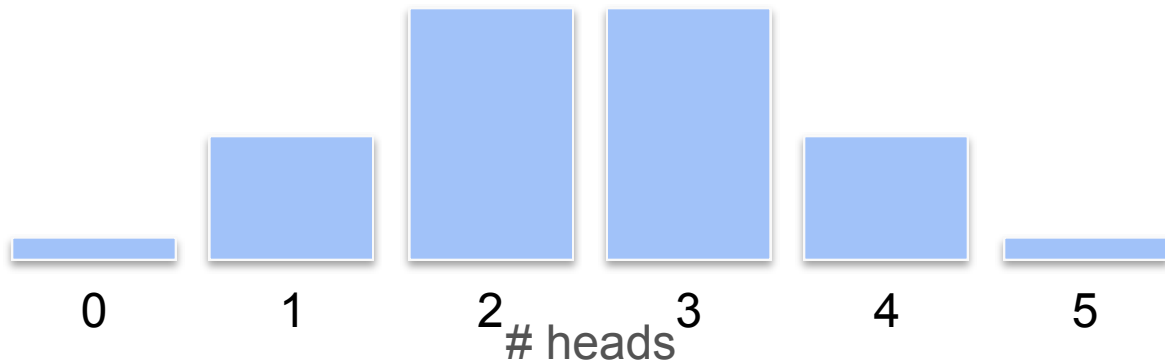
In general:

$\binom{n}{k}$ counts all the combinations for landing k heads in n coin tosses

Property:

$$\binom{n}{k} = \binom{n}{n-k}$$

The PMF with a fair coin is symmetrical

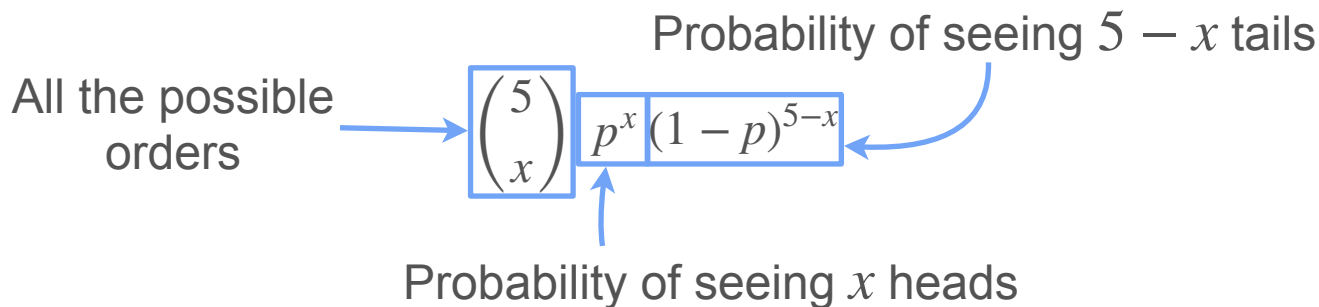


Binomial Distribution: Binomial Coefficient

General PMF for X : number of heads in 5 coin tosses?

Your coin has $\mathbf{P}(H) = p$

Event: $X = x$: x heads in 5 tosses



Binomial Distribution

General PMF for X : number of heads in 5 coin tosses?

Your coin has $\mathbf{P}(H) = p$

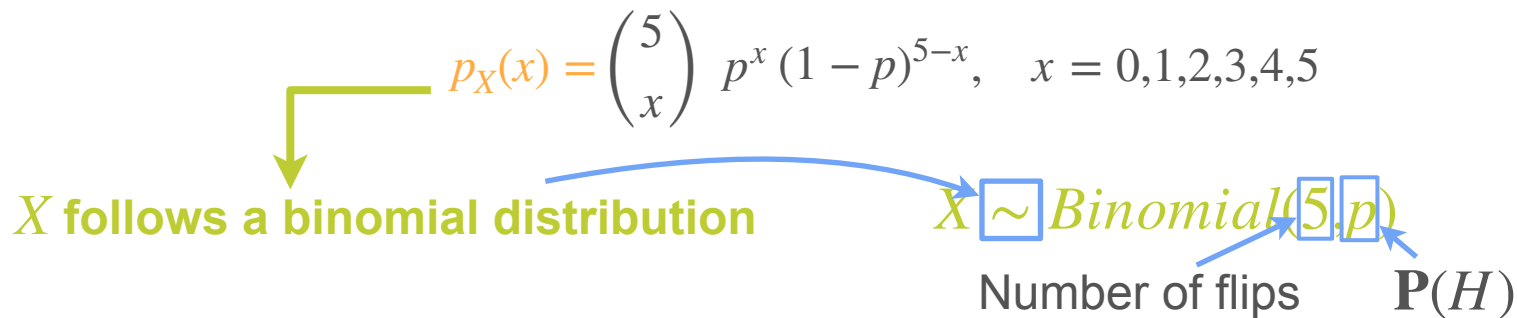
Event: $X = x$: x heads in 5 tosses

$$p_X(x) = \binom{5}{x} p^x (1-p)^{5-x}, \quad x = 0, 1, 2, 3, 4, 5$$

X follows a binomial distribution

$X \sim \text{Binomial}(5, p)$

Number of flips $\mathbf{P}(H)$

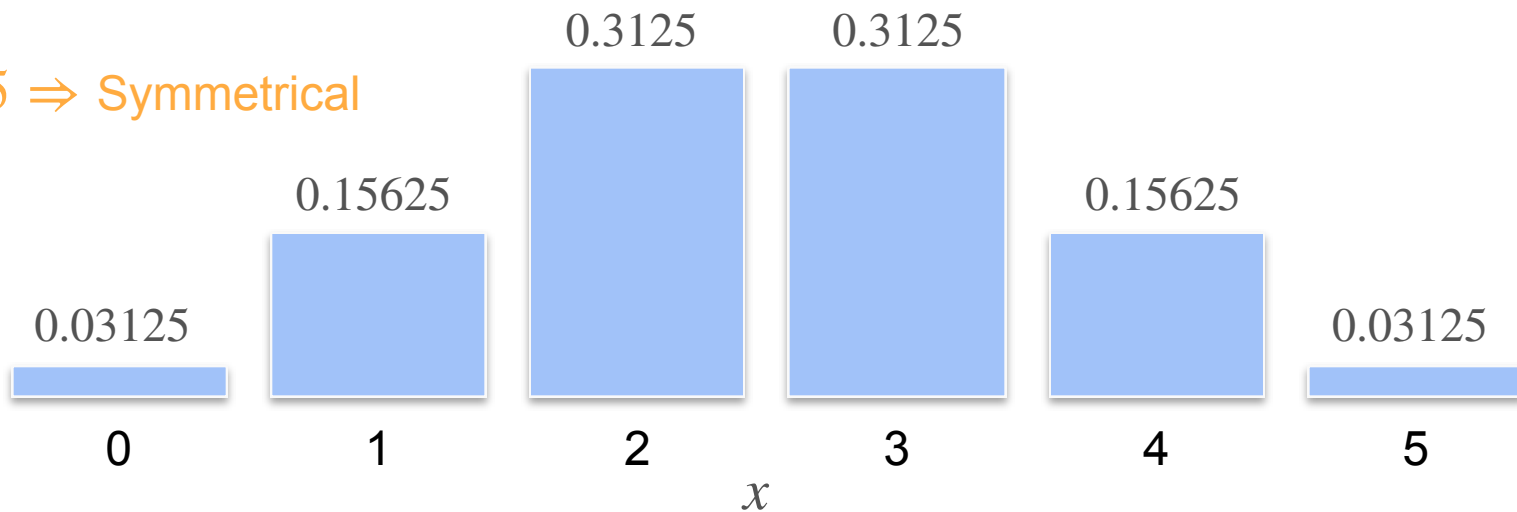


Binomial Distribution

$$\begin{matrix} n = 5 \\ p = 0.5 \end{matrix}$$

$$p_X(x) = \mathbf{P}(X = x) = \binom{5}{k} 0.5^k 0.5^{5-k}$$

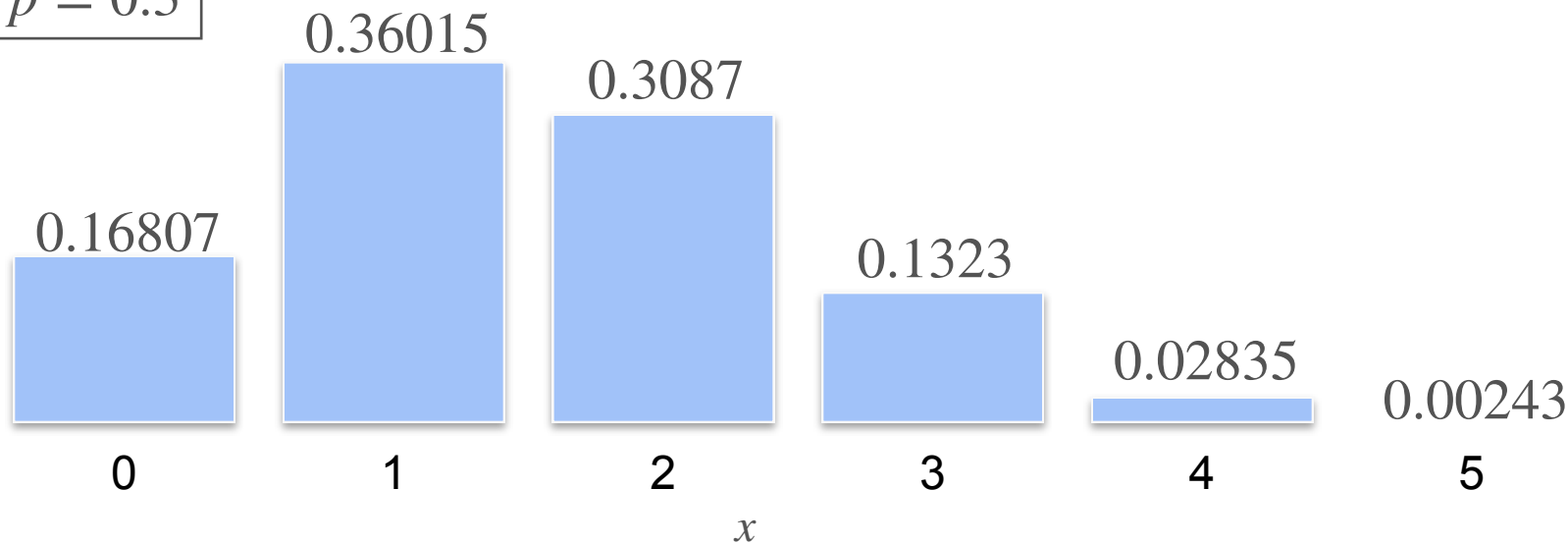
$p = 0.5 \Rightarrow \text{Symmetrical}$



Binomial Distribution

$$\begin{aligned} n &= 5 \\ p &= 0.3 \end{aligned}$$

$$p_X(x) = \mathbf{P}(X = x) = \binom{5}{k} 0.3^k 0.7^{5-k}$$



Binomial Distribution

General PMF for X : number of heads in 5 coin tosses?

Your coin has $\mathbf{P}(H) = p$

Event: $X = x$: x heads in 5 tosses

$$p_X(x) = \binom{5}{x} p^x (1-p)^{5-x}, \quad x = 0, 1, 2, 3, 4, 5$$

$$X \sim \textit{Binomial}(5, p)$$

Binomial Distribution

General PMF for X : number of heads in n coin tosses?

Your coin has $\mathbf{P}(H) = p$

Event: $X = x$: x heads in n tosses

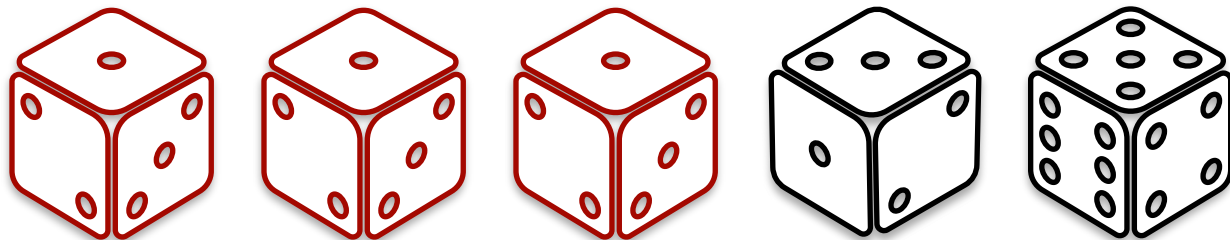
$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$X \sim \text{Binomial}(n, p)$

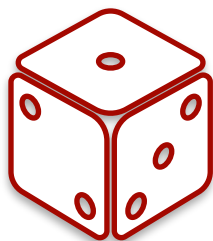
n and p are called the **parameters** of the binomial distribution

Binomial Distribution: Quiz

What is the probability of getting three ones when rolling a dice five times (no matter on which dice)?



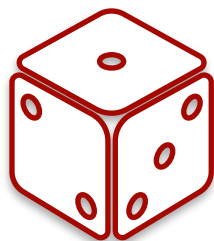
Binomial Distribution: Dice Is a Biased Coin!



one
 $p = \frac{1}{6}$



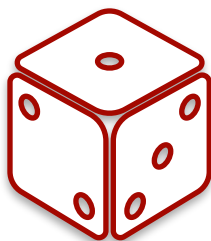
heads
 $p = \frac{1}{6}$



one
 $p = \frac{1}{6}$



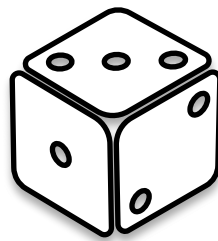
heads
 $p = \frac{1}{6}$



one
 $p = \frac{1}{6}$



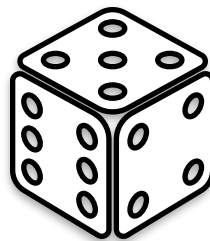
heads
 $p = \frac{1}{6}$



not one
 $p = \frac{5}{6}$



not heads
 $p = \frac{5}{6}$



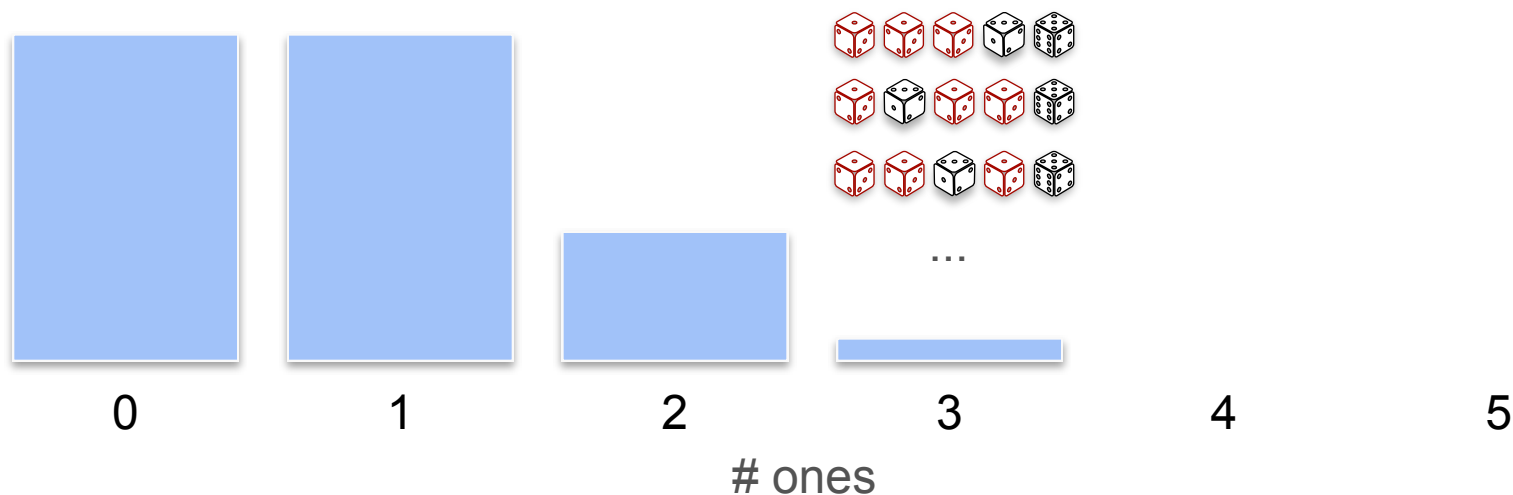
not one
 $p = \frac{5}{6}$



not heads
 $p = \frac{5}{6}$

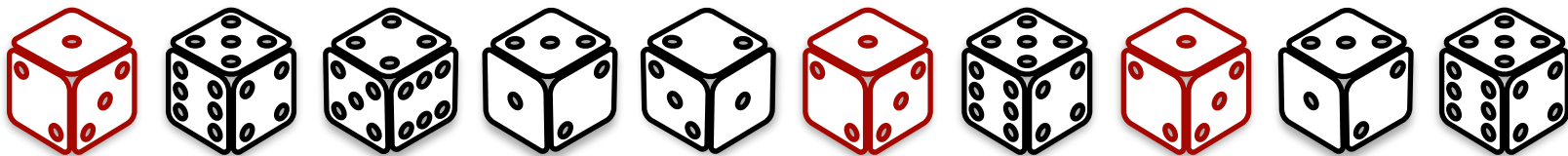
Binomial Distribution: Dice Is a Biased Coin!

$$n = 5$$
$$p = 0.1666$$



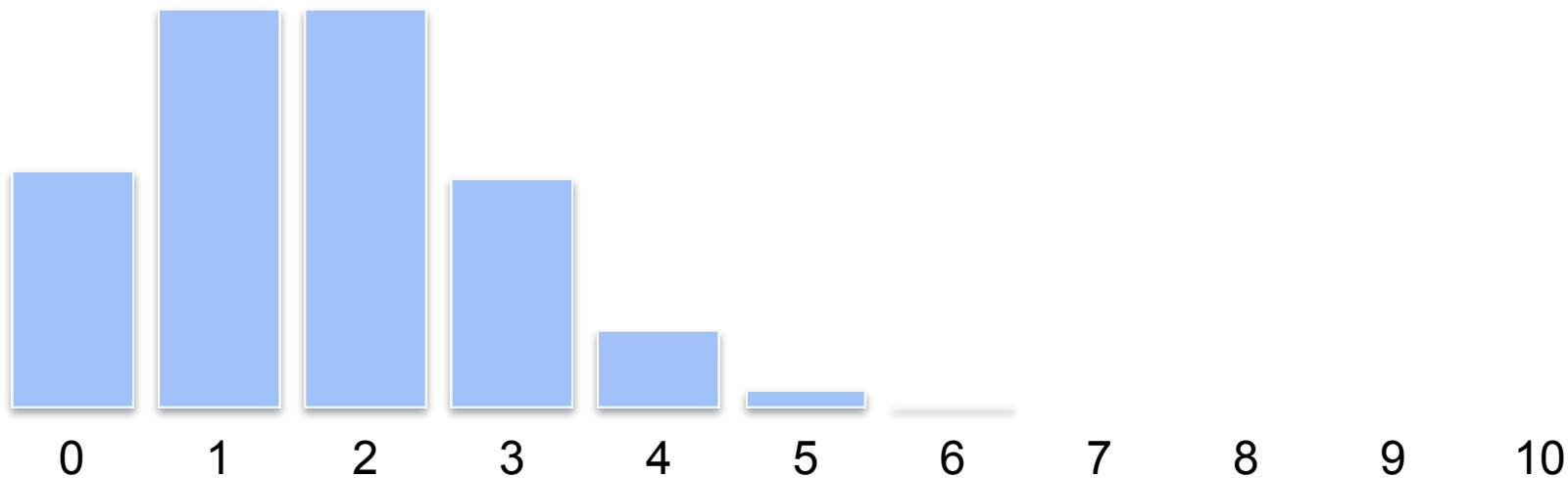
Binomial Distribution: Quiz

- Quiz: What are the parameters for the following binomial distribution:
 - I roll 10 dice
 - I want to record the number of times I obtain the number 1



Binomial Distribution: Quiz

$$n = 10$$
$$p = 0.1666$$



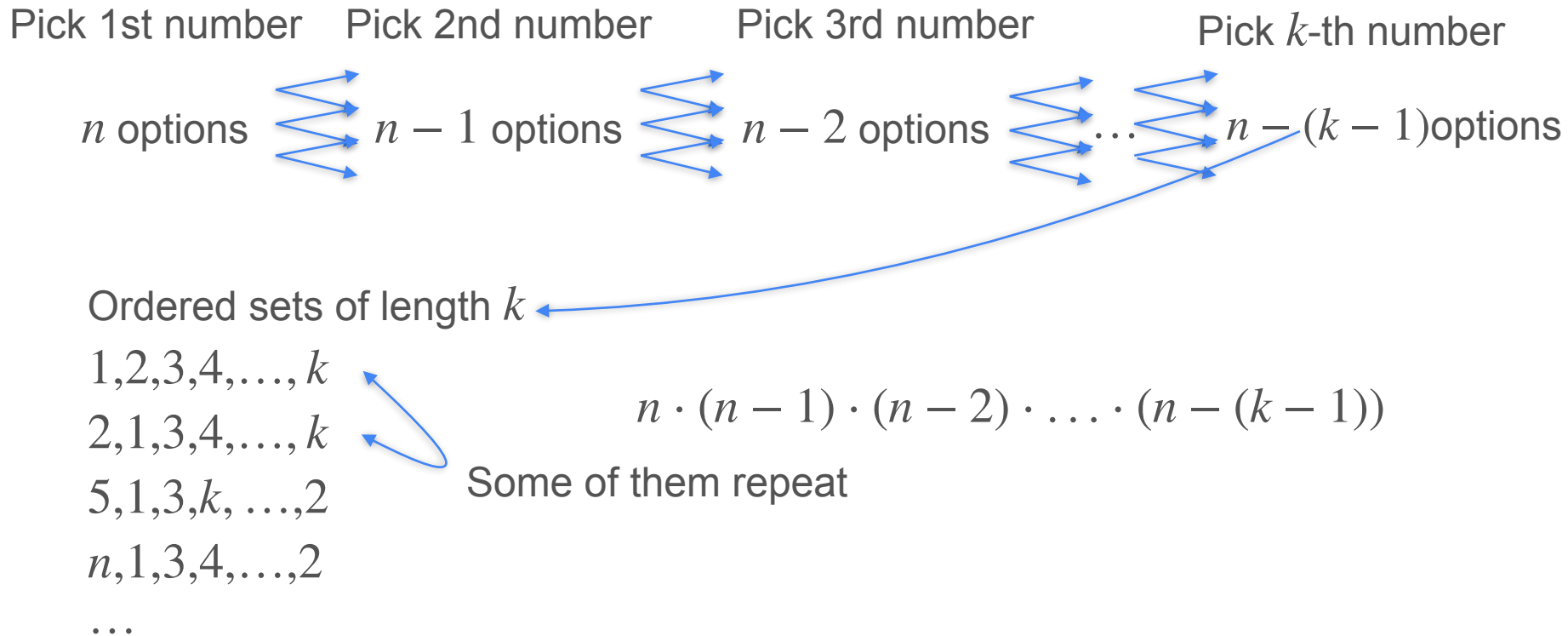


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Probability Distributions

**(Optional)
Binomial Coefficient**

Binomial Coefficient



Binomial Coefficient

	Pick 1st	Pick 2nd	Pick 3rd	Pick 4th
	4 options	3 options	2 options	1 option
1,2,3,4				
1,2,4,3				
1,3,2,4				
1,3,4,2				
...				
4,3,2,1				

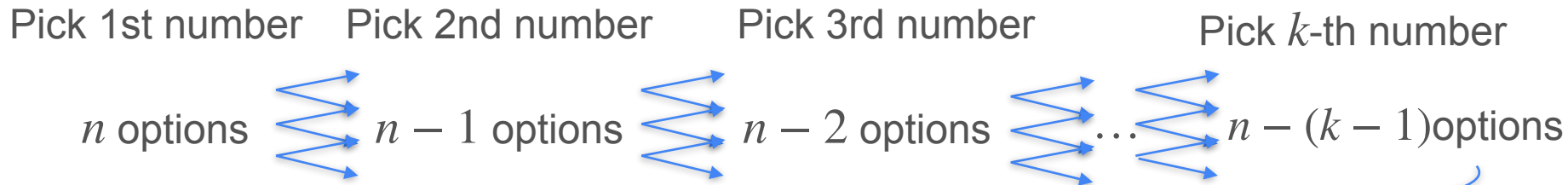
$$4 \cdot 3 \cdot 2 \cdot 1 = 4!$$

For five numbers:

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5!$$

General solution: $k!$

Binomial Coefficient



Ordered sets of length k

→ Unordered sets of length k

1, 2, 3, 4, ..., k

2, 1, 3, 4, ..., k

5, 1, 3, k , ..., 2

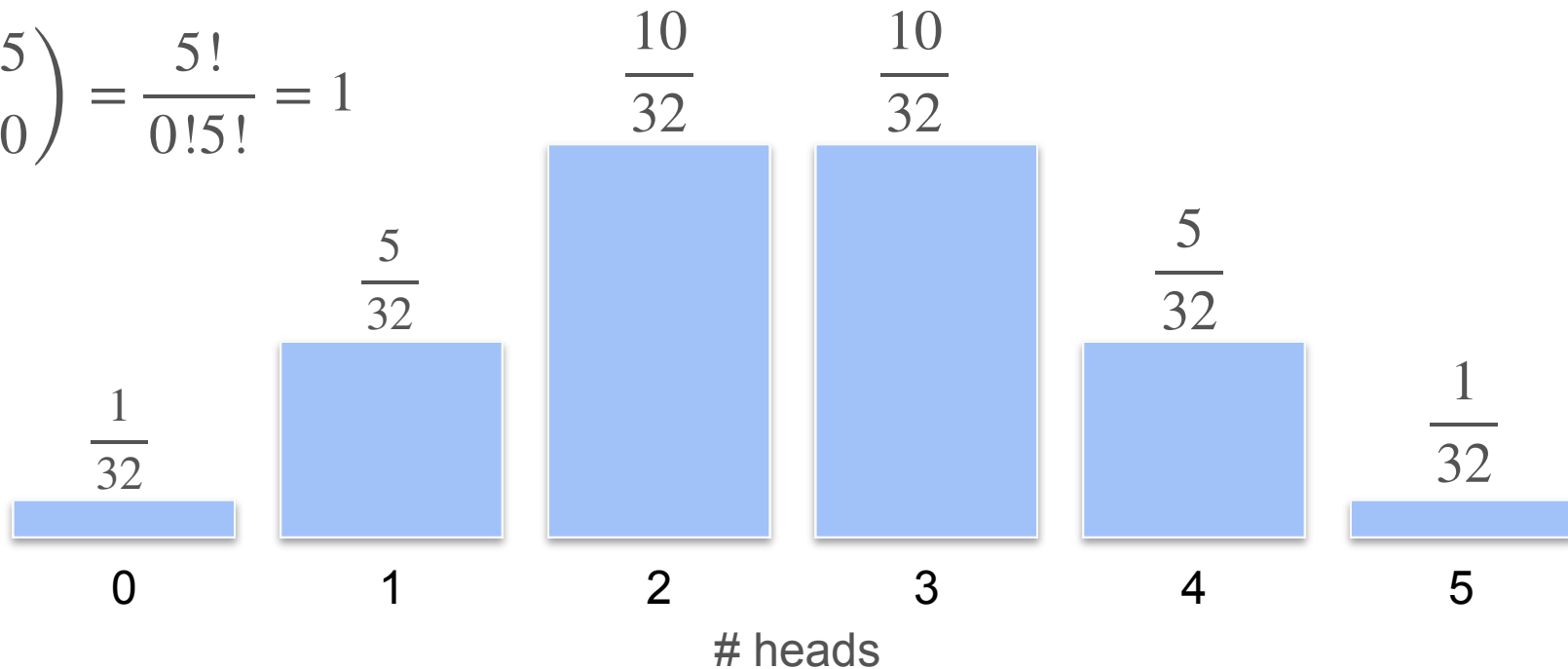
n , 1, 3, 4, ..., 2

...

$$\frac{n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - (k - 1))}{k!} = \frac{n!}{(n - k)! k!} = \binom{n}{k}$$

Binomial Distribution: Fair Coins

$$\binom{5}{0} = \frac{5!}{0!5!} = 1$$



Binomial Distribution: Fair Coins



50 %

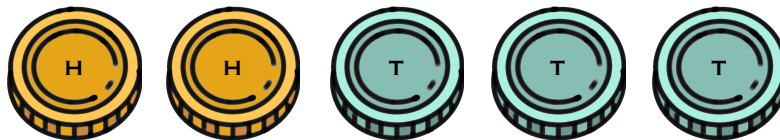


50 %



$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5$

$$= \frac{1}{32}$$



$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5$

$$= \frac{1}{32}$$

Binomial Distribution: Biased Coins



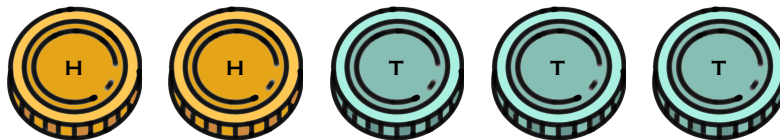
30 %



70 %



$$0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 = 0.00243$$



$$0.3 \cdot 0.3 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.01323$$

Binomial Distribution: Biased Coins

	$0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 = 0.3^5 \cdot 0.7^0$
	$0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 \cdot 0.7 = 0.3^4 \cdot 0.7^1$
	$0.3 \cdot 0.3 \cdot 0.3 \cdot 0.7 \cdot 0.7 = 0.3^3 \cdot 0.7^2$
	$0.3 \cdot 0.3 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.3^2 \cdot 0.7^3$
	$0.3 \cdot 0.7 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.3^1 \cdot 0.7^4$
	$0.7 \cdot 0.7 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.3^0 \cdot 0.7^5$

$$= 0.3^k \cdot 0.7^{n-k}$$

Binomial Distribution: Biased Coins



$$0.3^5 \cdot 0.7^0$$



$$0.3^4 \cdot 0.7^1$$



$$0.3^3 \cdot 0.7^2$$



$$0.3^2 \cdot 0.7^3$$



$$0.3^1 \cdot 0.7^4$$



$$0.3^0 \cdot 0.7^5$$

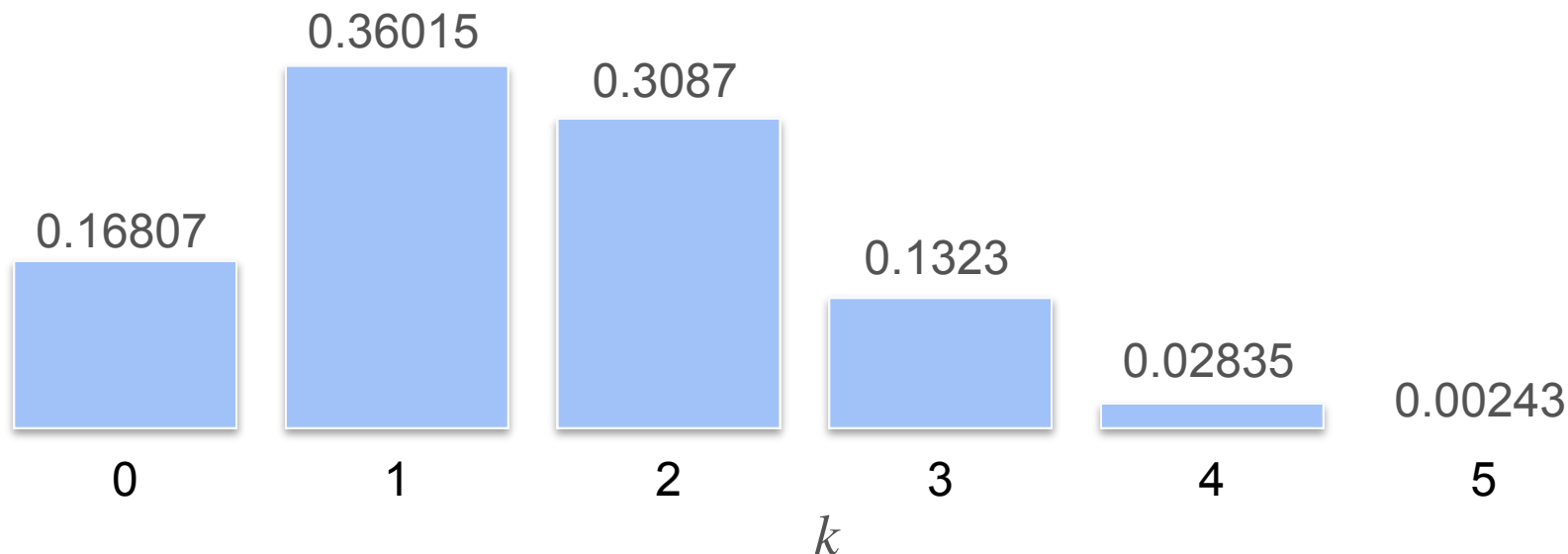
$$= 0.3^k \cdot 0.7^{n-k} \rightarrow \binom{n}{k} 0.3^k \cdot 0.7^{n-k}$$

Account for all
possible orders of
heads and tails

Binomial Distribution: Biased Coins

$$= \binom{n}{k} 0.3^k \cdot 0.7^{n-k}$$

$$\begin{aligned} n &= 5 \\ p &= 0.3 \end{aligned}$$





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Probability Distributions

Bernoulli Distribution

Bernoulli Distribution

X = Number of heads

Flip a coin



$$P(X = 1) = 0.5$$

Success



$$P(X = 0) = 0.5$$

Failure

X = Number of 1's

Throw a dice



$$P(X = 1) = \frac{1}{6}$$

Success

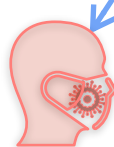


$$P(X = 0) = \frac{5}{6}$$

Failure

X = Number of sick patients

Flip a coin



$$P(X = 1) = p$$

Success



$$P(X = 0) = 1 - p$$

Failure

$X \sim \text{Bernoulli}(p)$ ← p is the parameter of the Bernoulli distribution

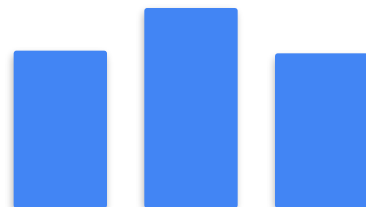
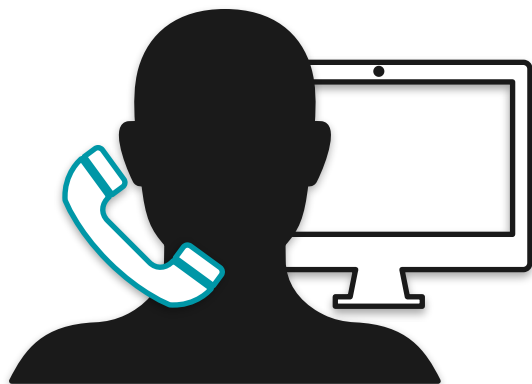


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Probability Distributions

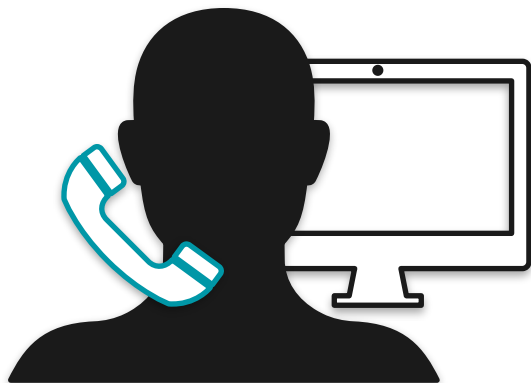
Probability Distributions (Continuous)

From Discrete to Continuous Distributions



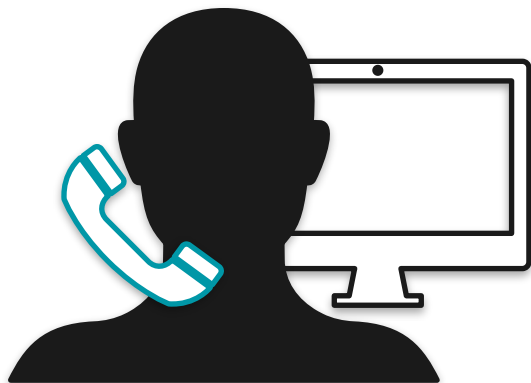
Waiting time: 1 2 3 (min)

From Discrete to Continuous Distributions



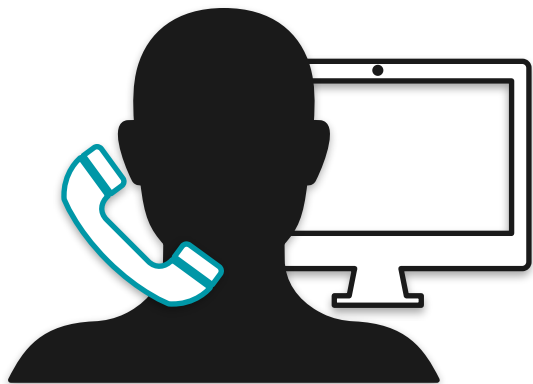
Waiting time: 1 1.01 2 3 (min)

From Discrete to Continuous Distributions

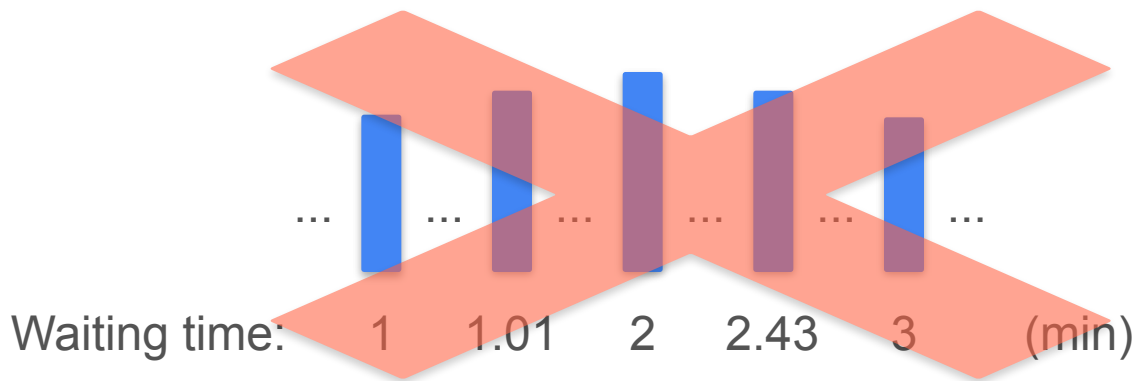


Waiting time: 1 1.01 2 2.43 3 (min)

From Discrete to Continuous Distributions



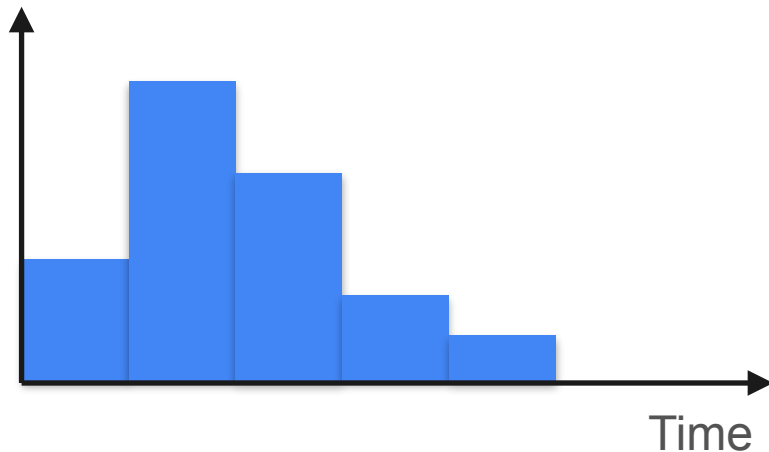
This is a continuous distribution!



What is the probability that you will wait EXACTLY one minute for the call?

Answer: ZERO

From Discrete to Continuous Distributions



$P(\text{between 0 and 1 mins})$

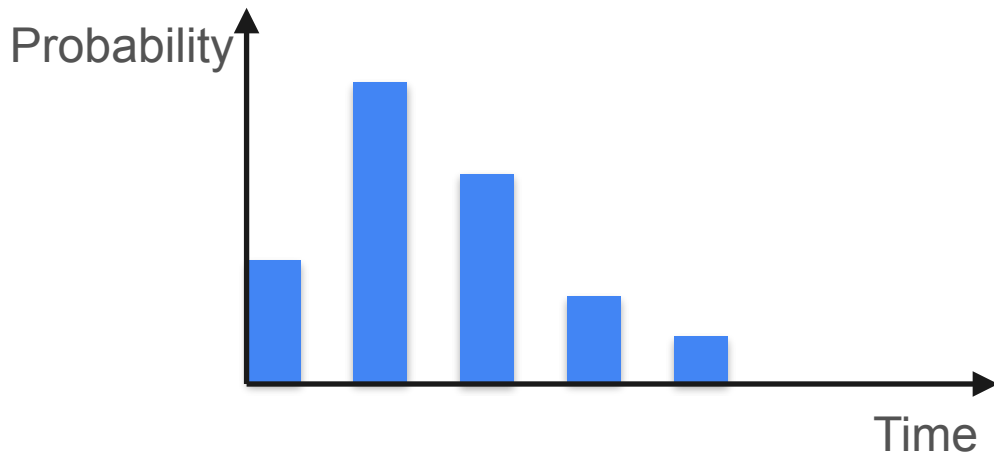
$P(\text{between 1 and 2 mins})$

$P(\text{between 2 and 3 mins})$

$P(\text{between 3 and 4 mins})$

$P(\text{between 4 and 5 mins})$

From Discrete to Continuous Distributions



$P(\text{between 0 and 1 mins})$

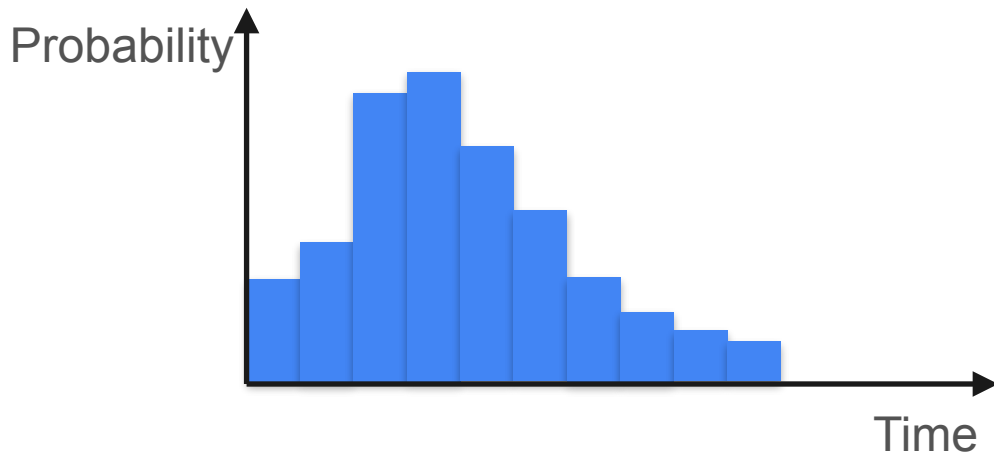
$P(\text{between 1 and 2 mins})$

$P(\text{between 2 and 3 mins})$

$P(\text{between 3 and 4 mins})$

$P(\text{between 4 and 5 mins})$

From Discrete to Continuous Distributions



$P(\text{between 0 and 0.5 mins})$

$P(\text{between 0.5 and 1 mins})$

$P(\text{between 1 and 1.5 mins})$

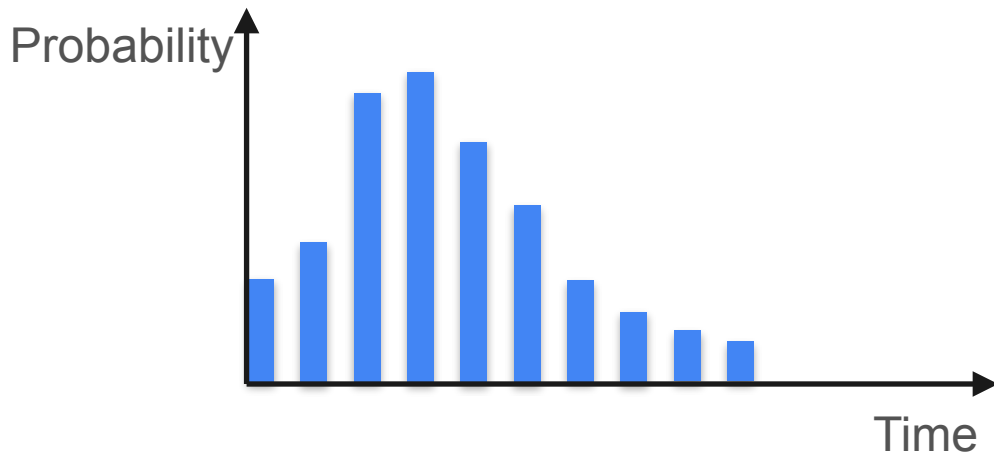
$\vdots$

$P(\text{between 3.5 and 4 mins})$

$P(\text{between 4 and 4.5 mins})$

$P(\text{between 4.5 and 5 mins})$

From Discrete to Continuous Distributions



$P(\text{between 0 and 0.5 mins})$

$P(\text{between 0.5 and 1 mins})$

$P(\text{between 1 and 1.5 mins})$

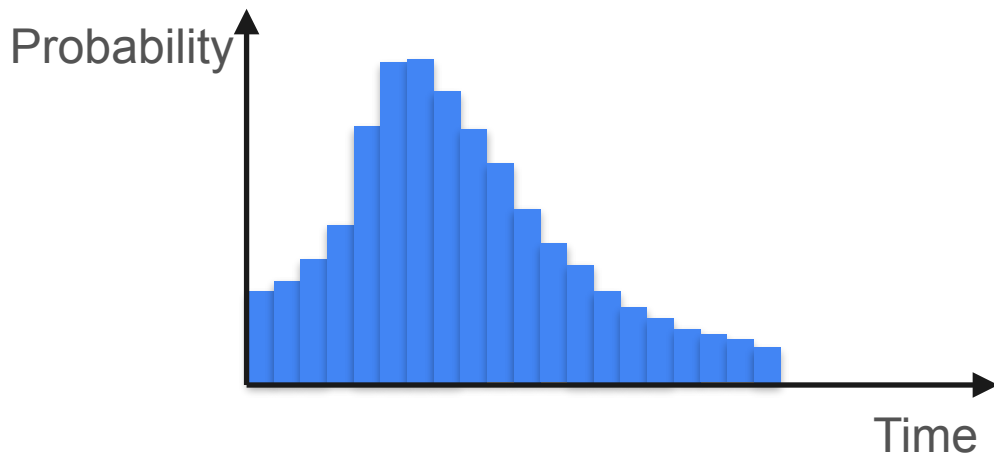
$\vdots$

$P(\text{between 3.5 and 4 mins})$

$P(\text{between 4 and 4.5 mins})$

$P(\text{between 4.5 and 5 mins})$

From Discrete to Continuous Distributions



$P(\text{between } 0 \text{ and } 0.25 \text{ mins})$

$P(\text{between } 0.25 \text{ and } 0.5 \text{ mins})$

$P(\text{between } 0.5 \text{ and } 0.75 \text{ mins})$

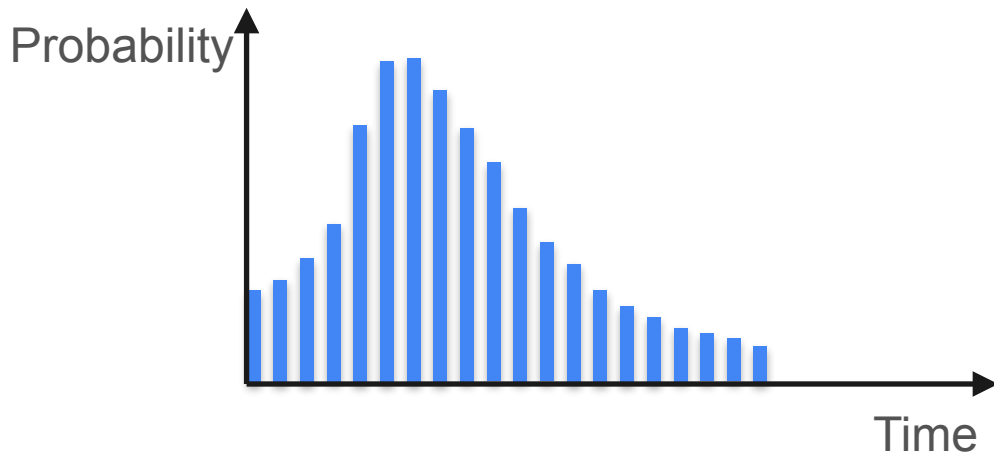
$\vdots$

$P(\text{between } 4.25 \text{ and } 4.5 \text{ mins})$

$P(\text{between } 4.5 \text{ and } 4.75 \text{ mins})$

$P(\text{between } 4.75 \text{ and } 5 \text{ mins})$

From Discrete to Continuous Distributions



$P(\text{between } 0 \text{ and } 0.25 \text{ mins})$

$P(\text{between } 0.25 \text{ and } 0.5 \text{ mins})$

$P(\text{between } 0.5 \text{ and } 0.75 \text{ mins})$

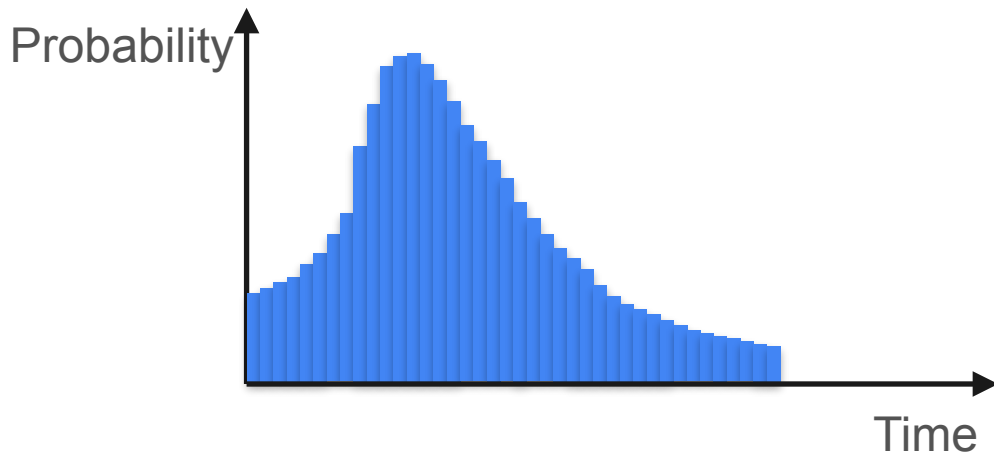
$\vdots$

$P(\text{between } 4.25 \text{ and } 4.5 \text{ mins})$

$P(\text{between } 4.5 \text{ and } 4.75 \text{ mins})$

$P(\text{between } 4.75 \text{ and } 5 \text{ mins})$

From Discrete to Continuous Distributions

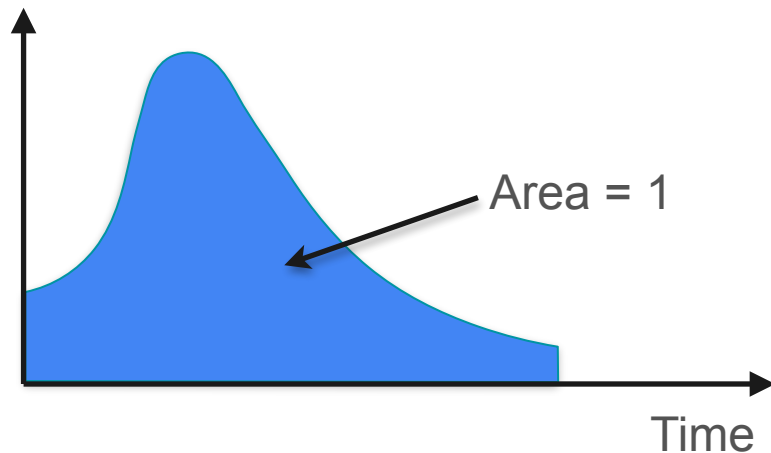


$P(\text{between } 0 \text{ and } 0.125 \text{ mins})$

$\vdots$

$P(\text{between } 4.875 \text{ and } 5 \text{ mins})$

From Discrete to Continuous Distributions



- Discrete:
 - Sum of heights equals 1
- Continuous:
 - Area under the curve equals 1

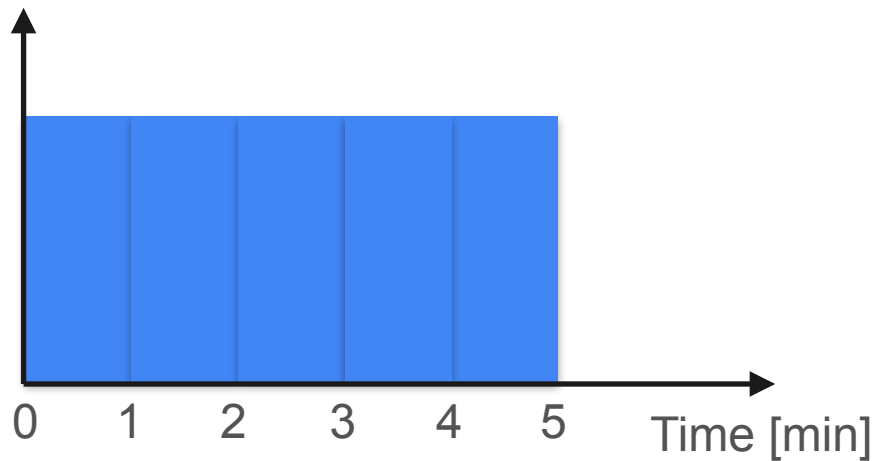


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Probability Distributions

Probability density function

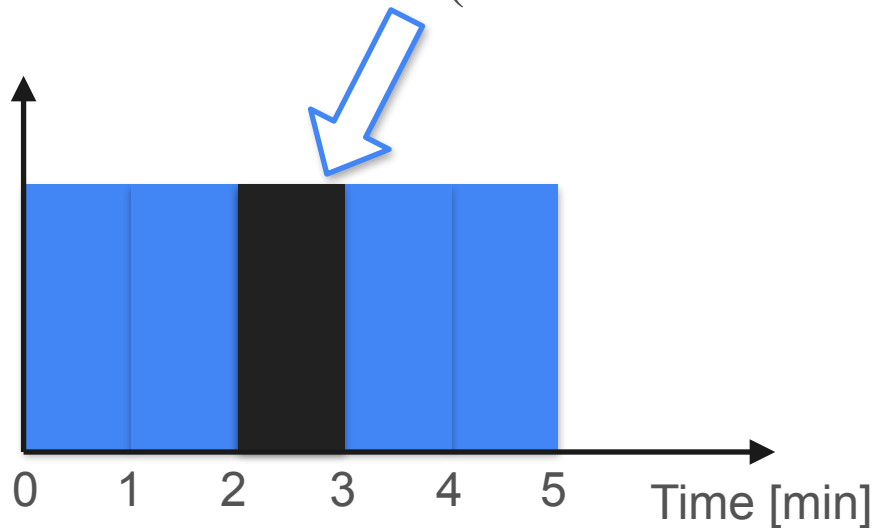
Probability Density Function



Probability Density Function



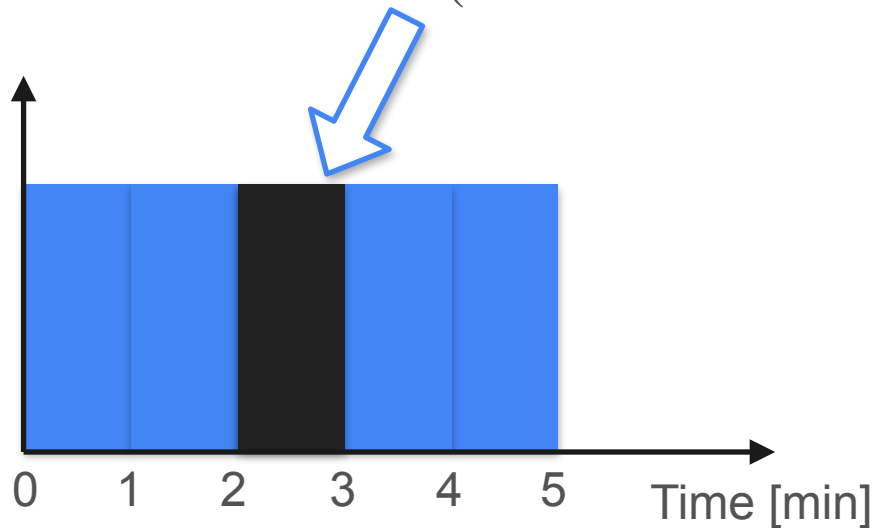
$P(\text{between 2 and 3 min}) = ?$



Probability Density Function



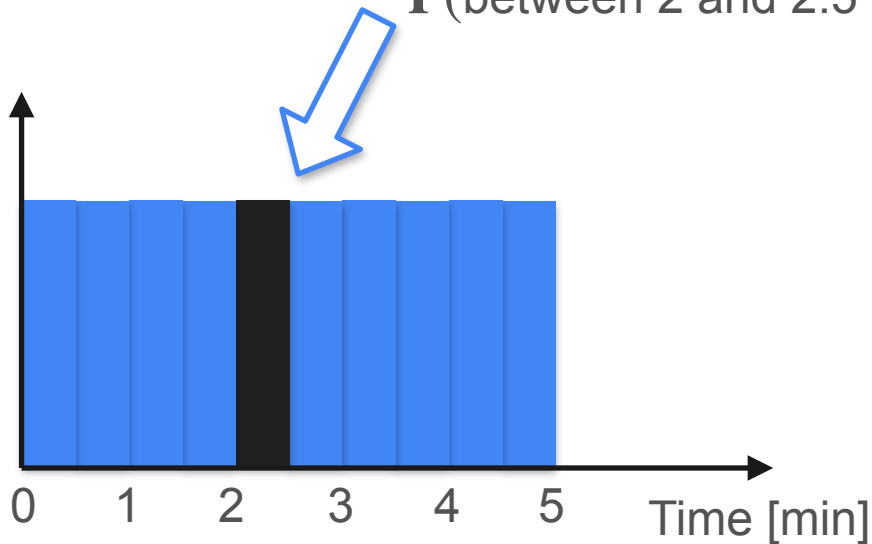
$$P(\text{between 2 and 3 min}) = 1/5$$



Probability Density Function



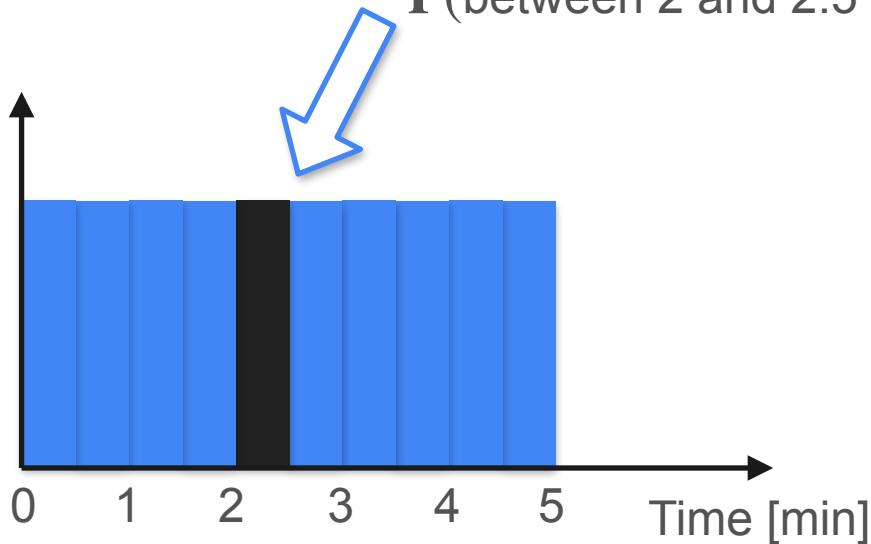
$P(\text{between 2 and 2.5 min}) = ?$



Probability Density Function



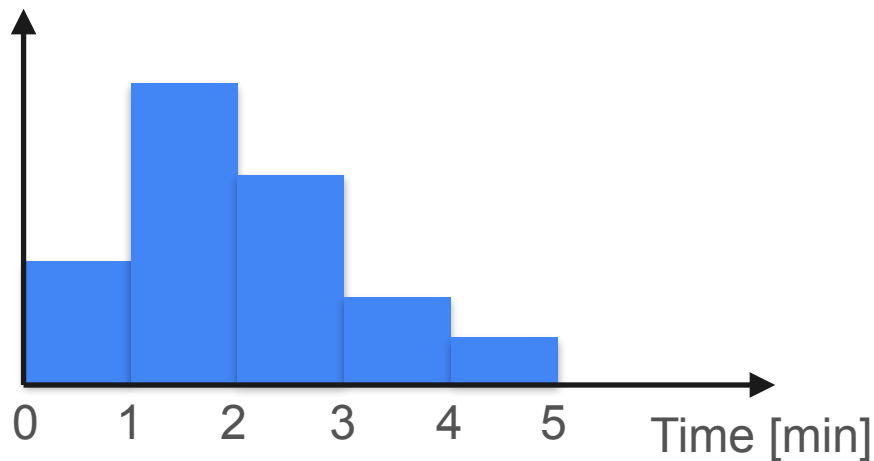
$$P(\text{between 2 and 2.5 min}) = 0.1$$



Probability Density Function



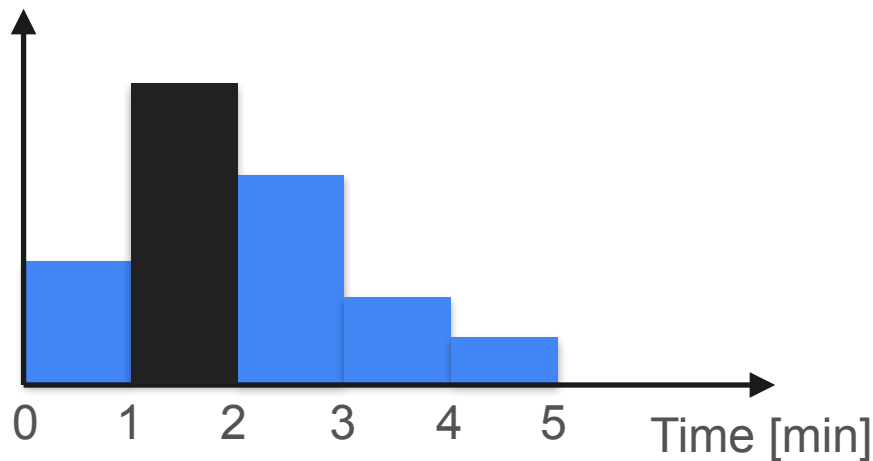
$P(\text{between 1 and 2 min})$



Probability Density Function



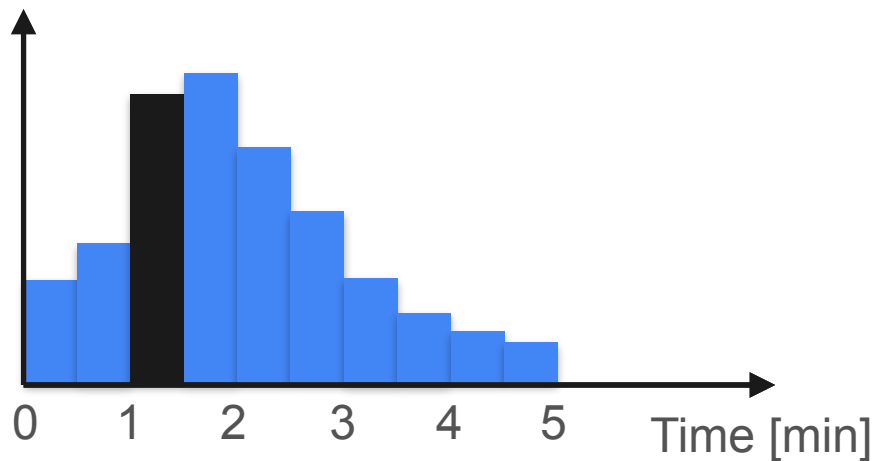
$P(\text{between 1 and 2 min})$



Probability Density Function



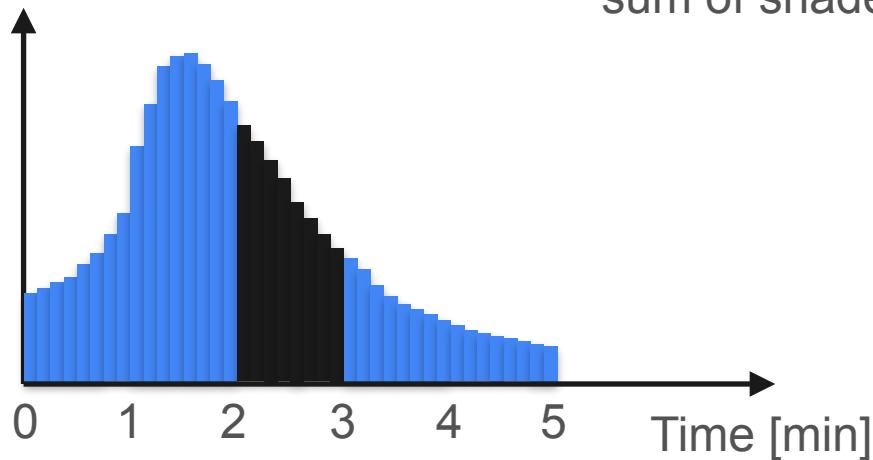
$P(\text{between 1 and 1:30})$



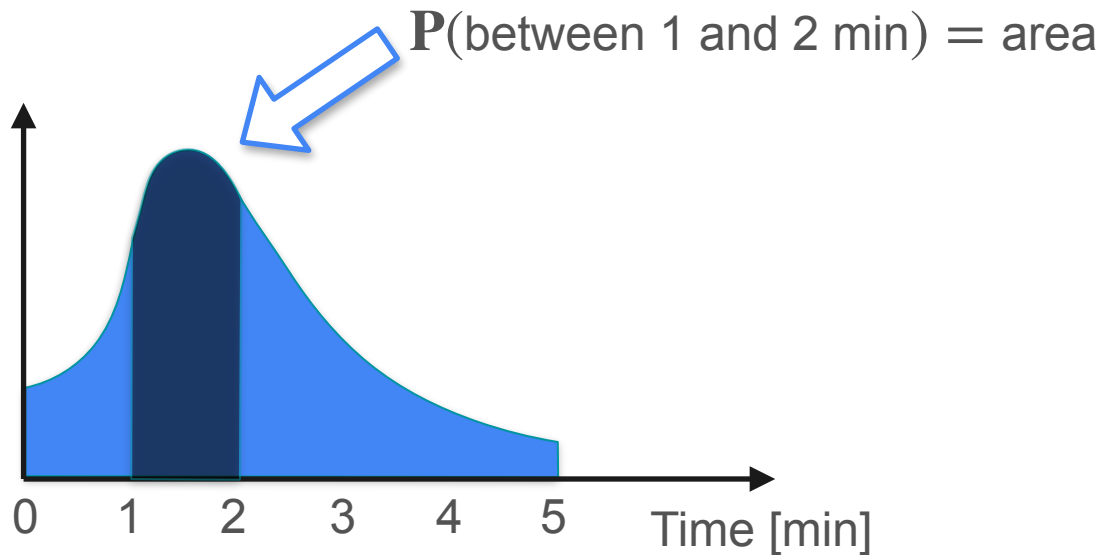
Probability Density Function



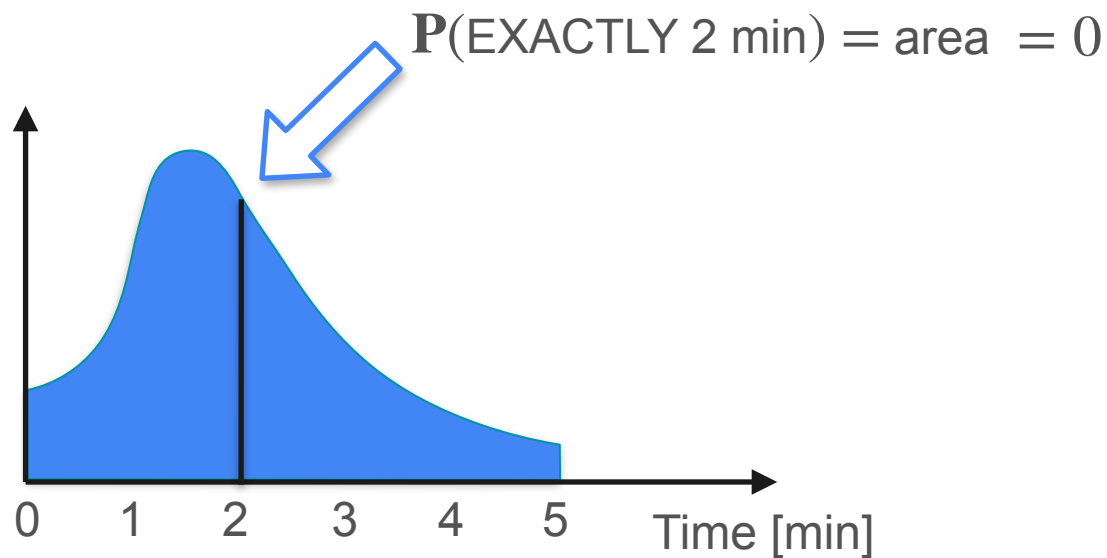
$P(\text{between 2 and 3 min}) =$
sum of shaded areas



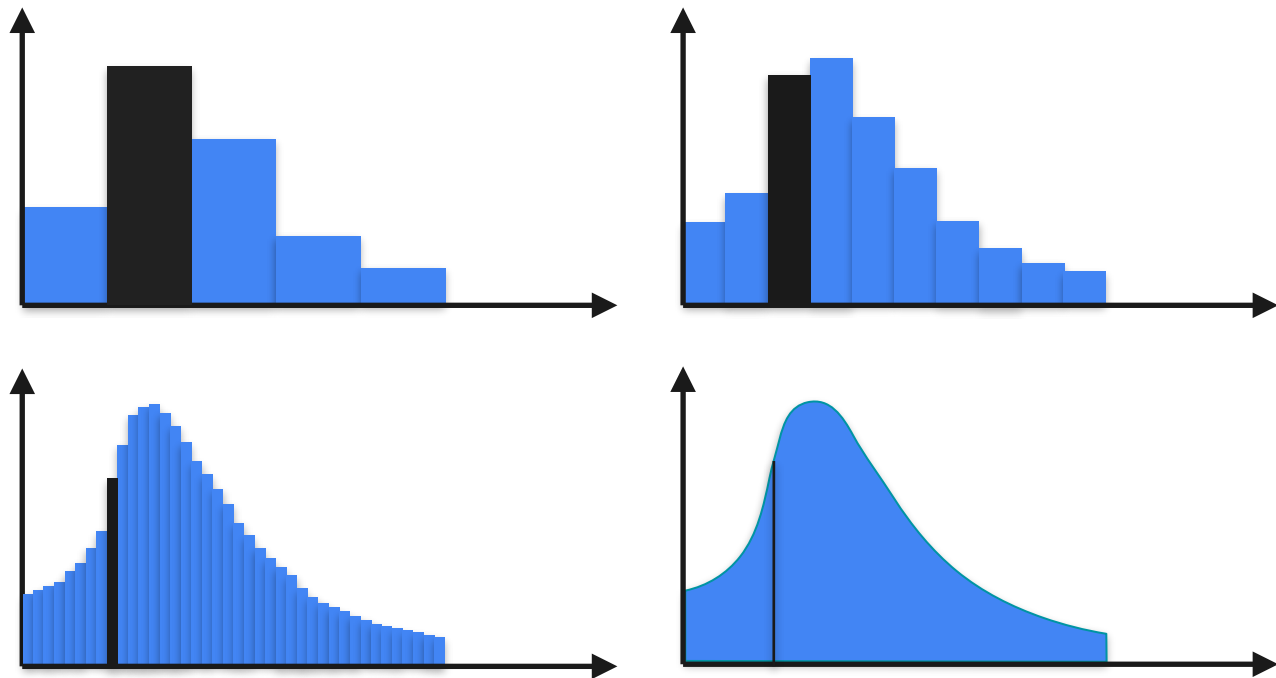
Probability Density Function



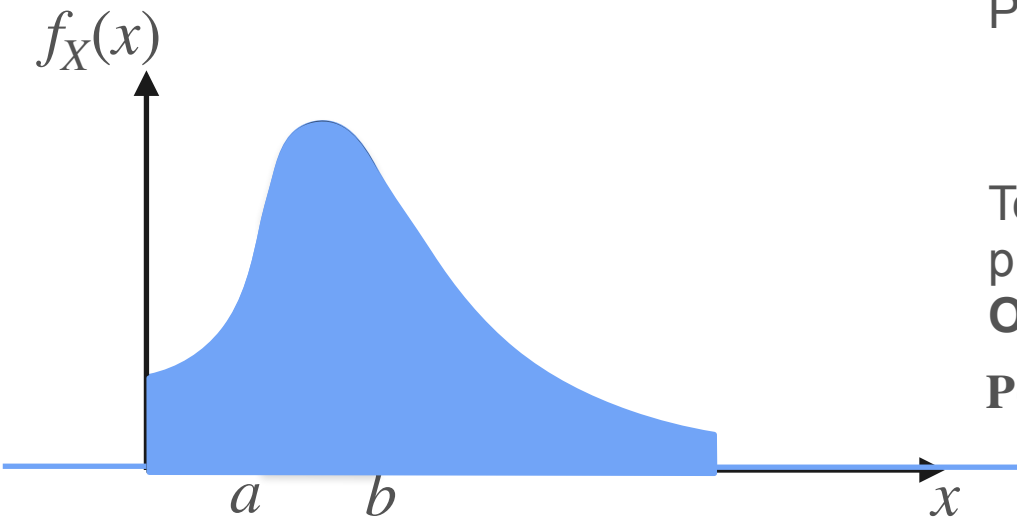
Probability Density Function



Probability Density Function



Probability Density Function: Formal Definition



Probability Density Function (PDF)

$$f_X(x)$$

Tells you the rate you accumulate probability around each point.

Only defined for continuous variables!

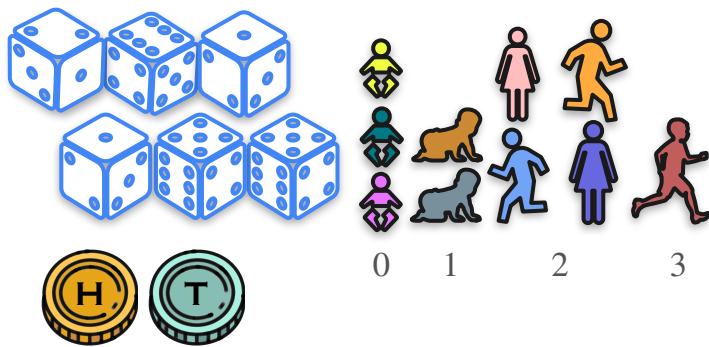
$P(a < X < b) = \text{area under } f_X(x)$

$f_X(x)$ needs to satisfy:

- It is defined for all numbers
- $f_X(x) \geq 0$
- Area under $f_X(x) = 1$

Discrete and Continuous Random Variables

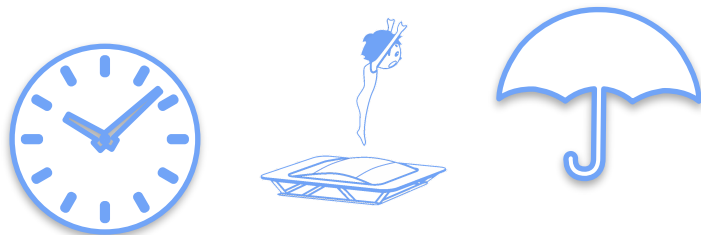
Discrete random variables



Can take only a **finite** or at most countable number of values

$$\text{PMF: } p_X(x) = \mathbf{P}(X = x)$$

Continuous random variables



Takes values on an interval (infinite possibilities!)

$$\begin{aligned} \text{PDF: } f_X(x) \\ \mathbf{P}(X = x) = 0 \quad \forall x \end{aligned}$$

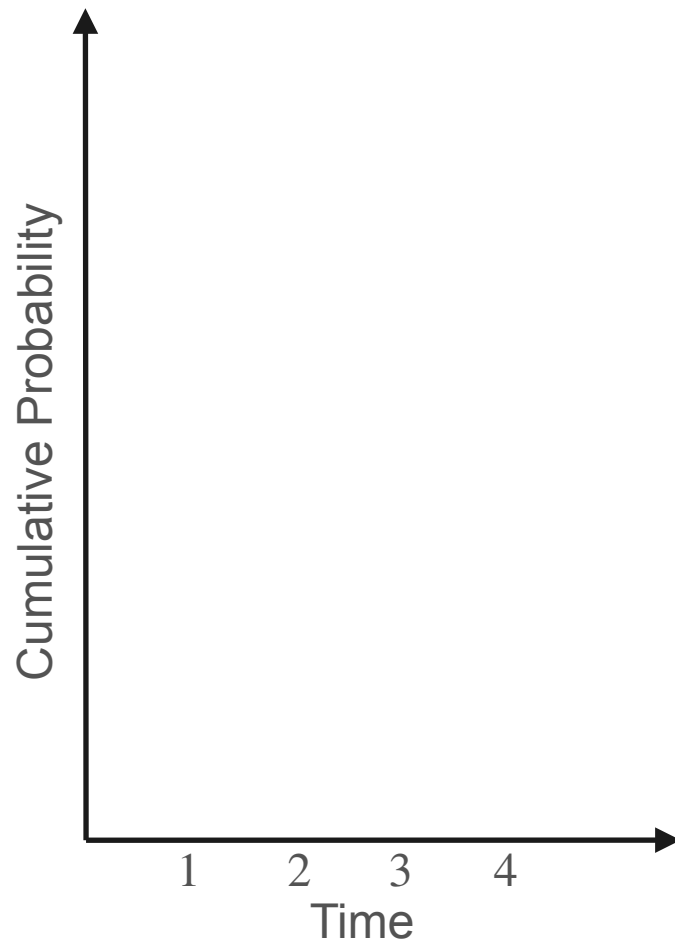
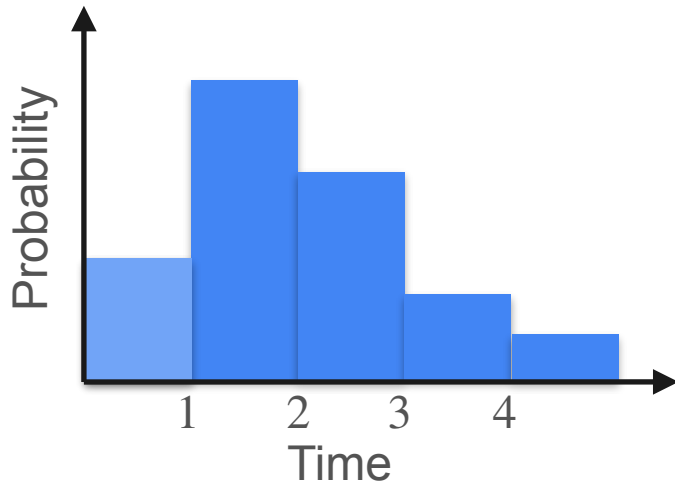


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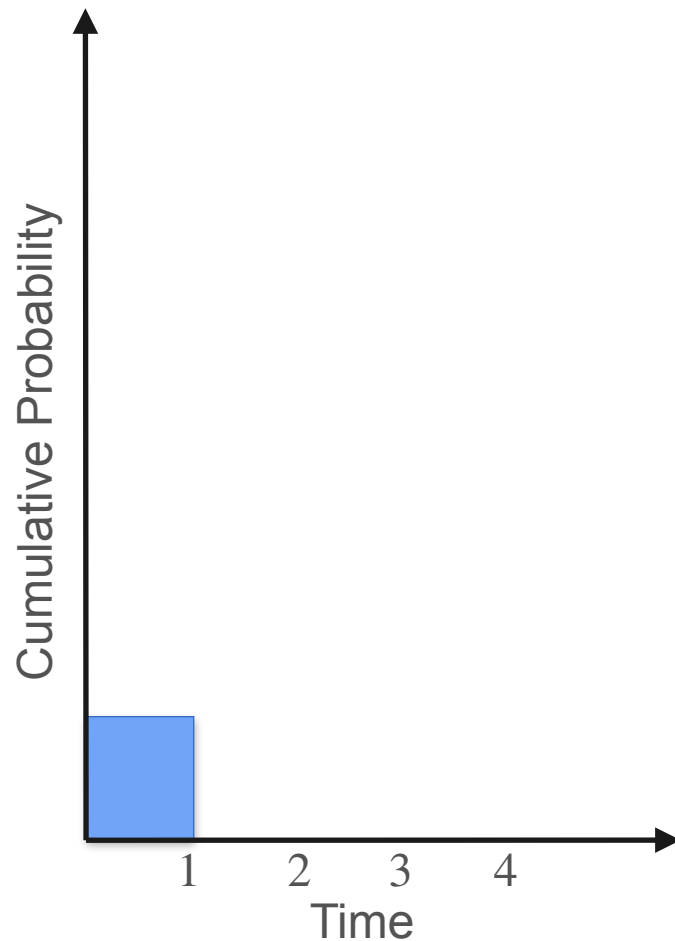
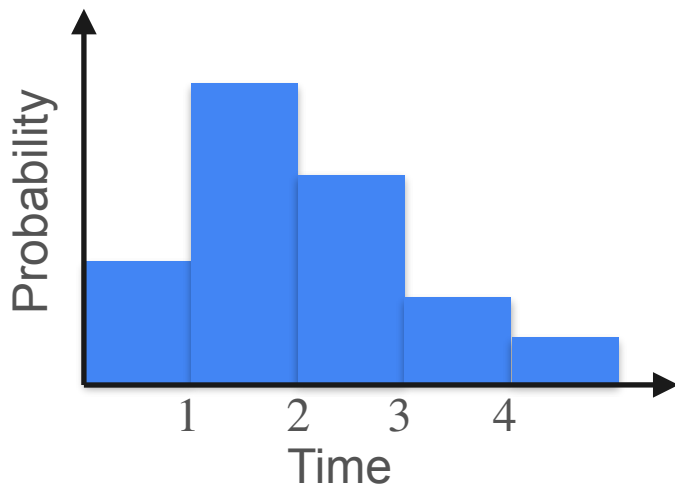
Probability Distributions

Cumulative Distribution Function

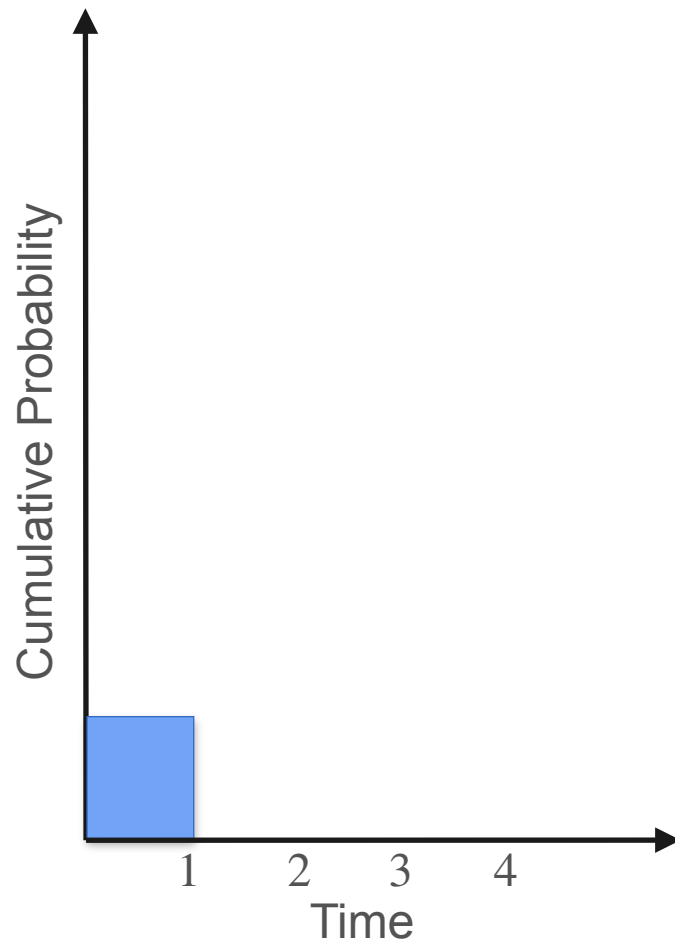
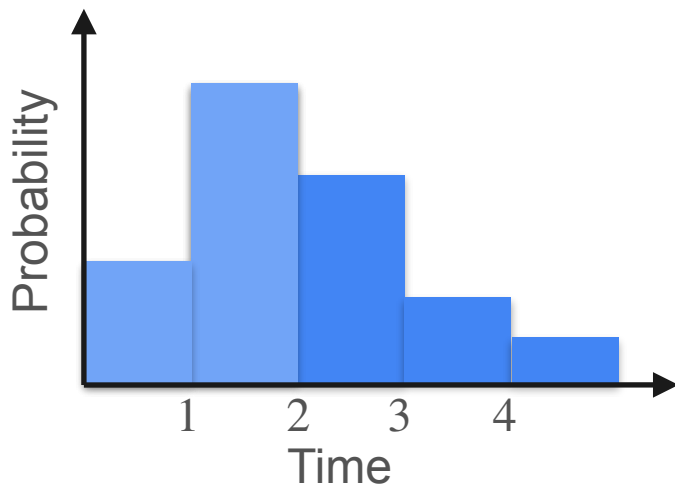
Cumulative Distribution



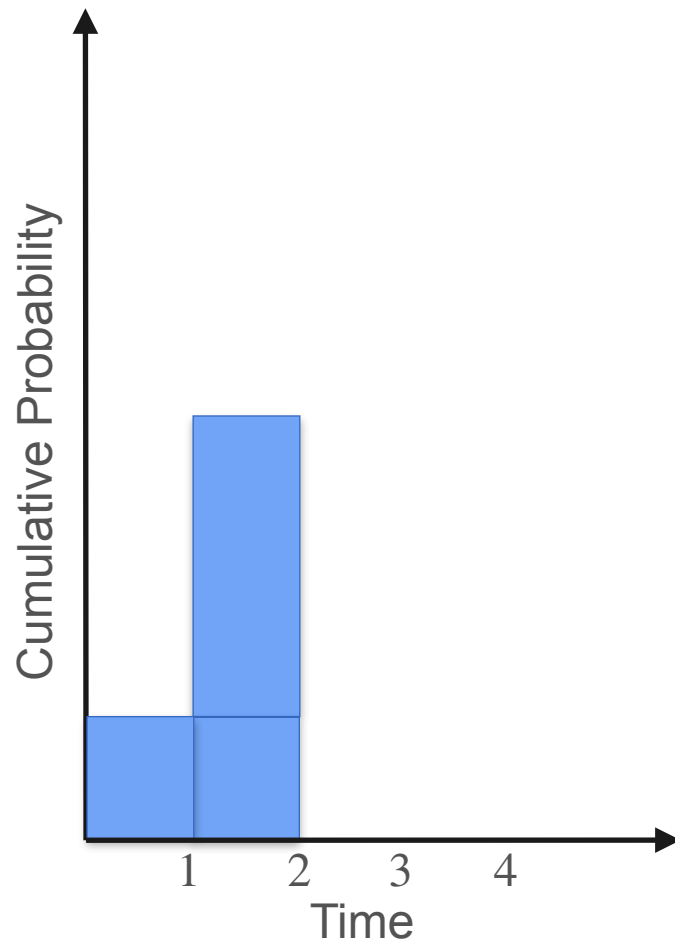
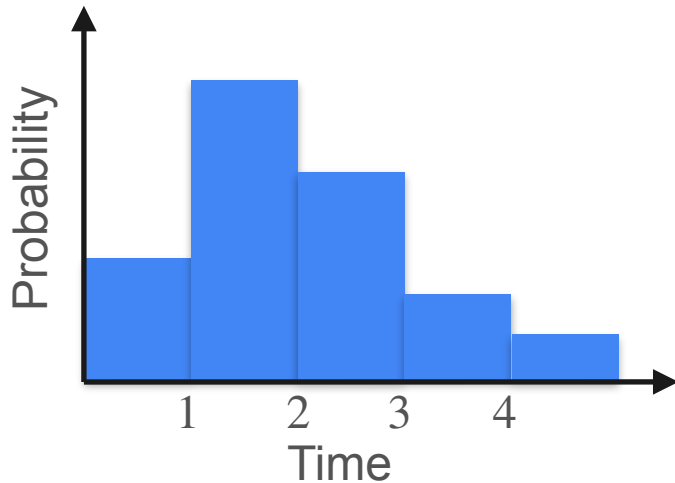
Cumulative Distribution



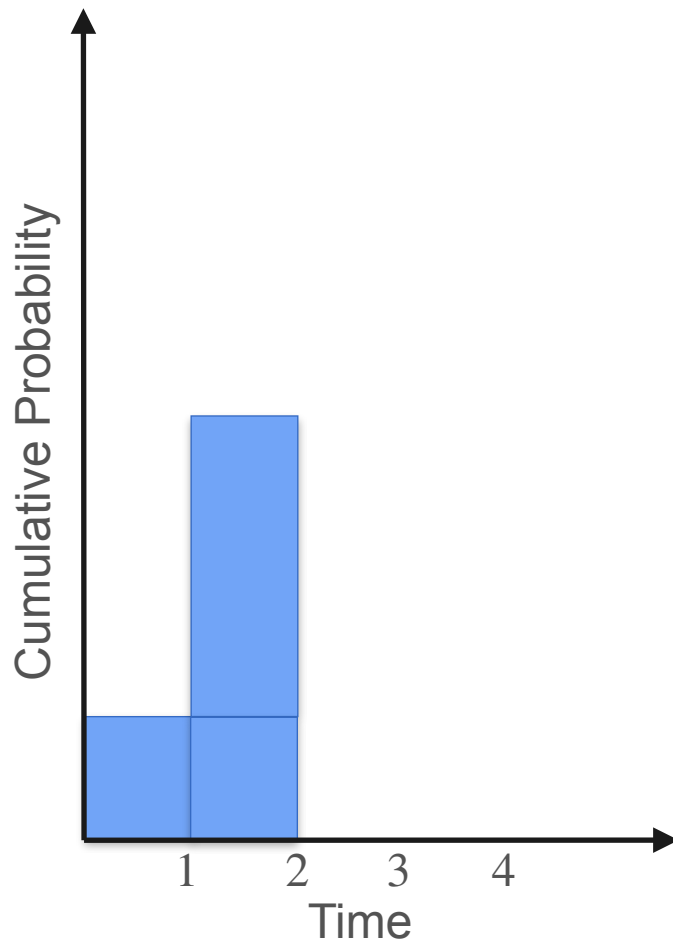
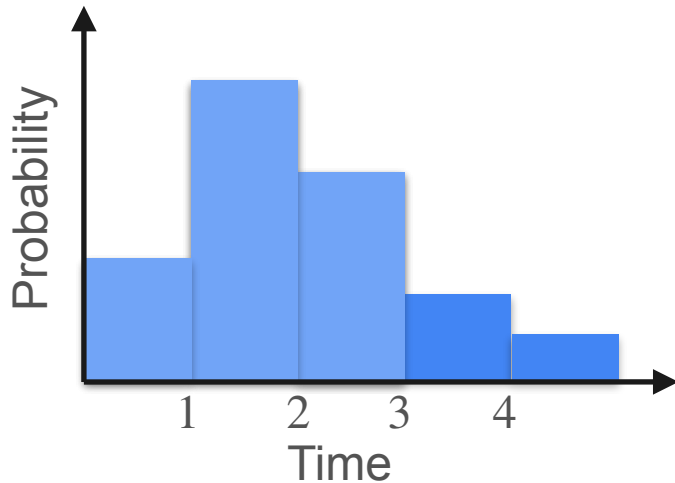
Cumulative Distribution



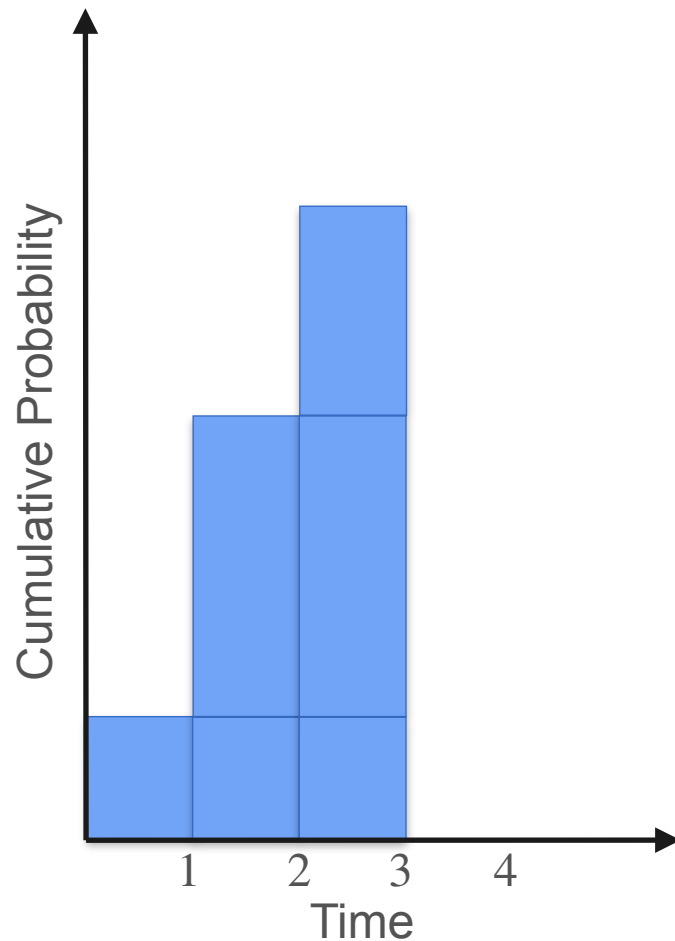
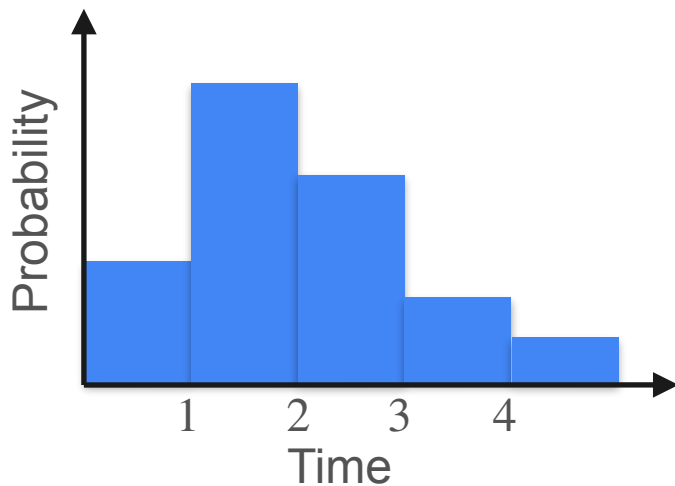
Cumulative Distribution



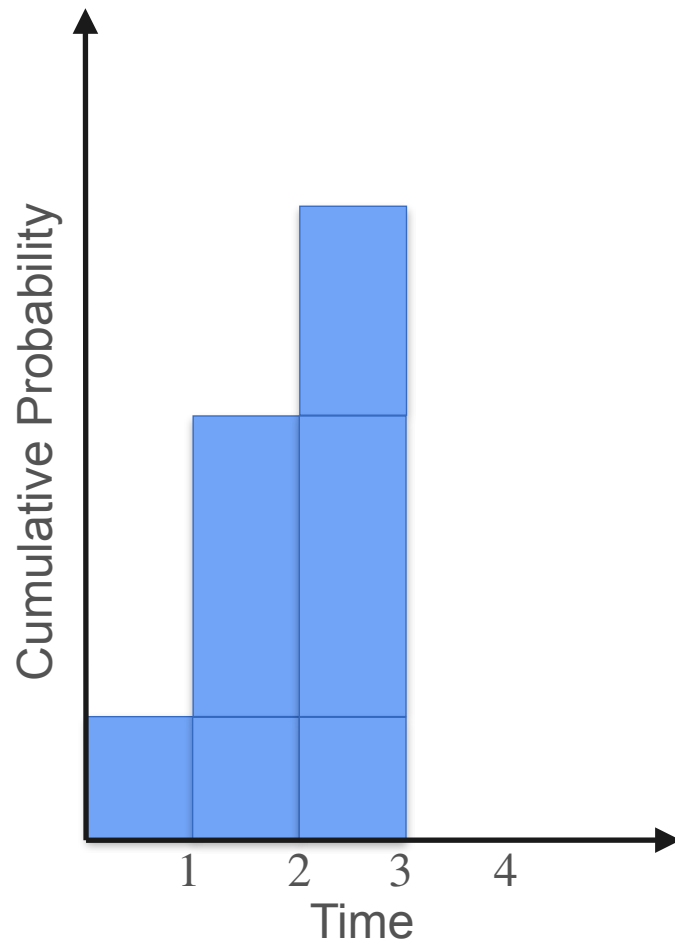
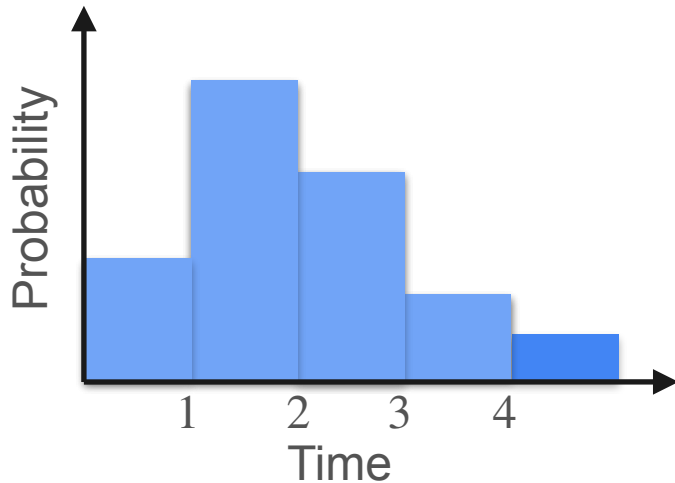
Cumulative Distribution



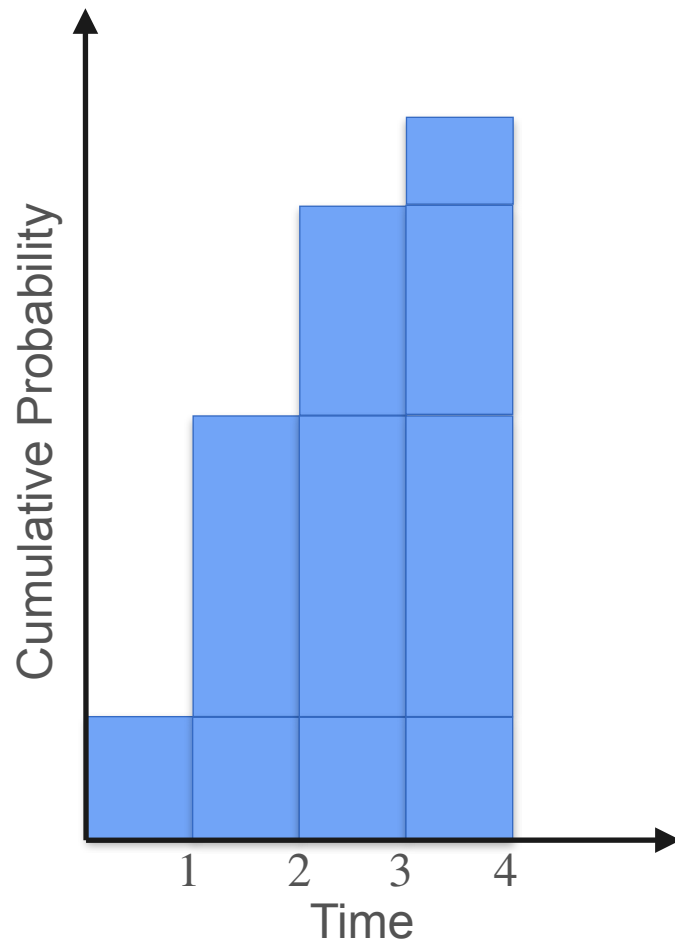
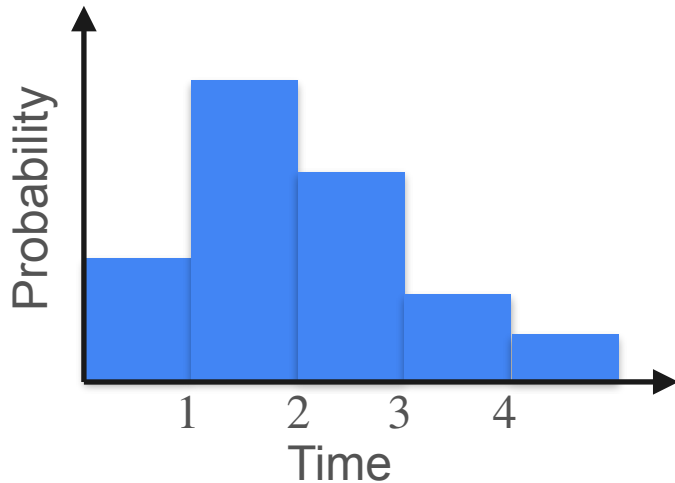
Cumulative Distribution



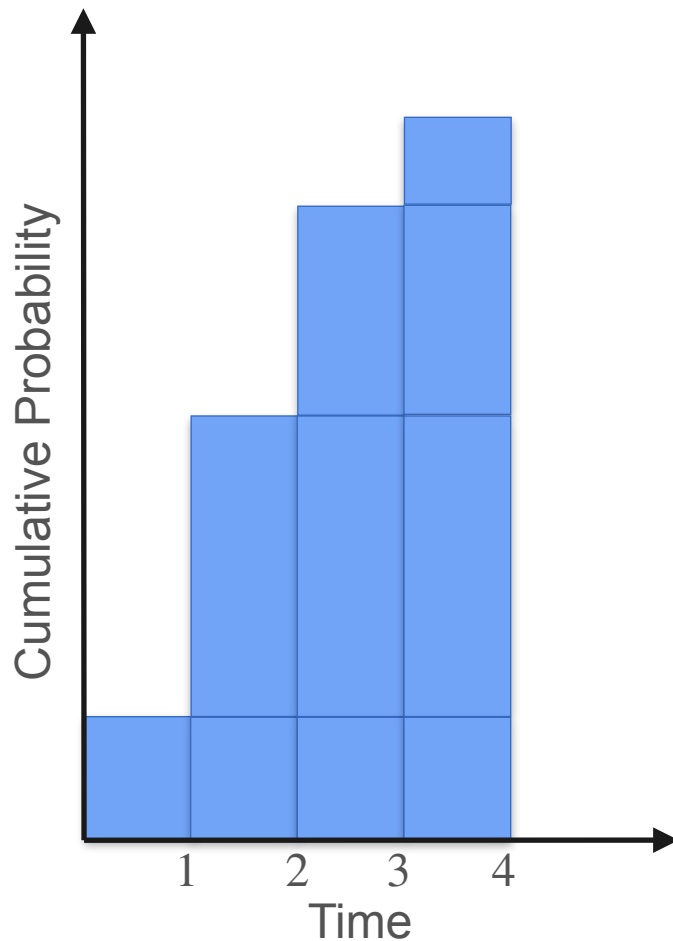
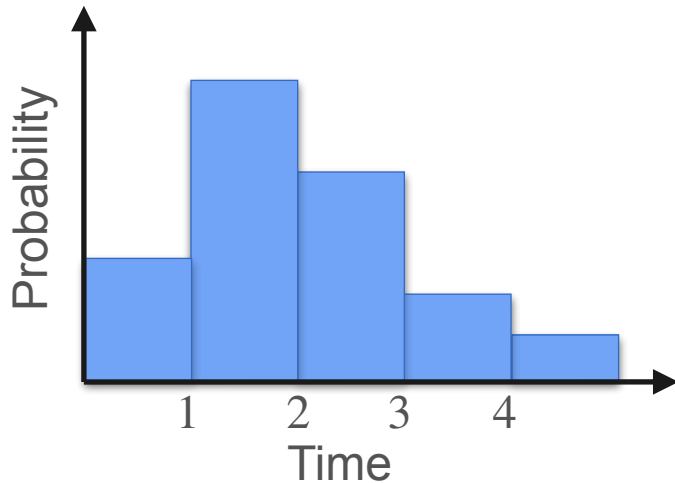
Cumulative Distribution



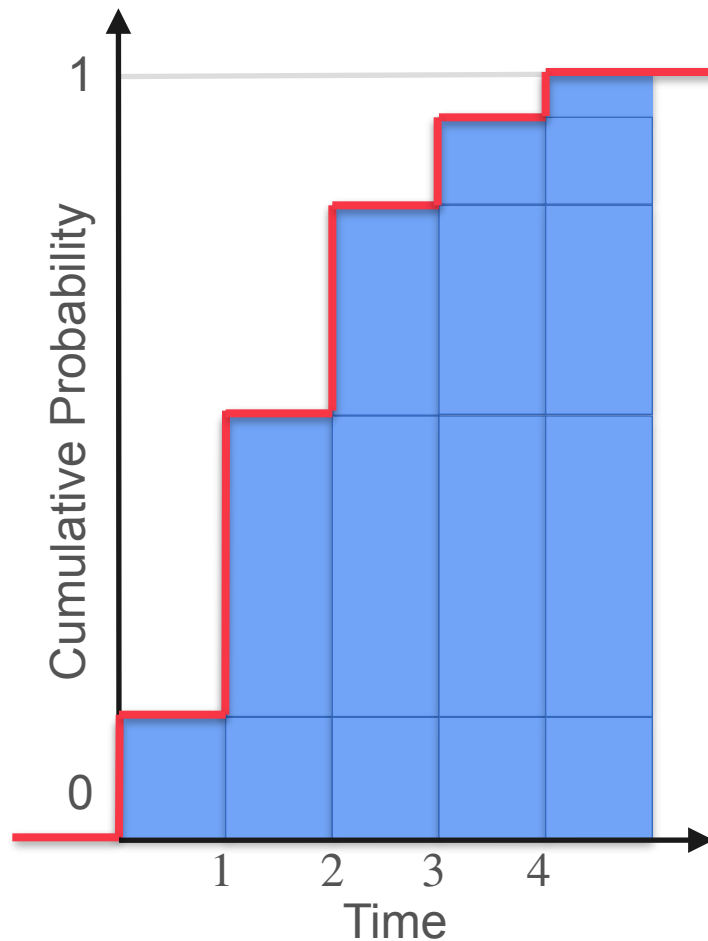
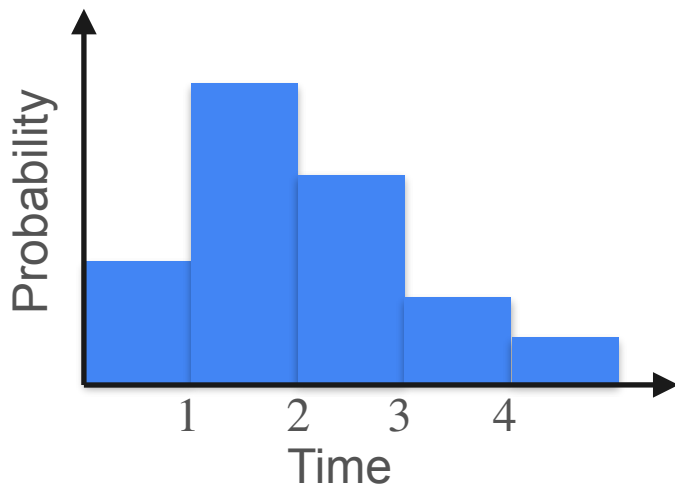
Cumulative Distribution



Cumulative Distribution

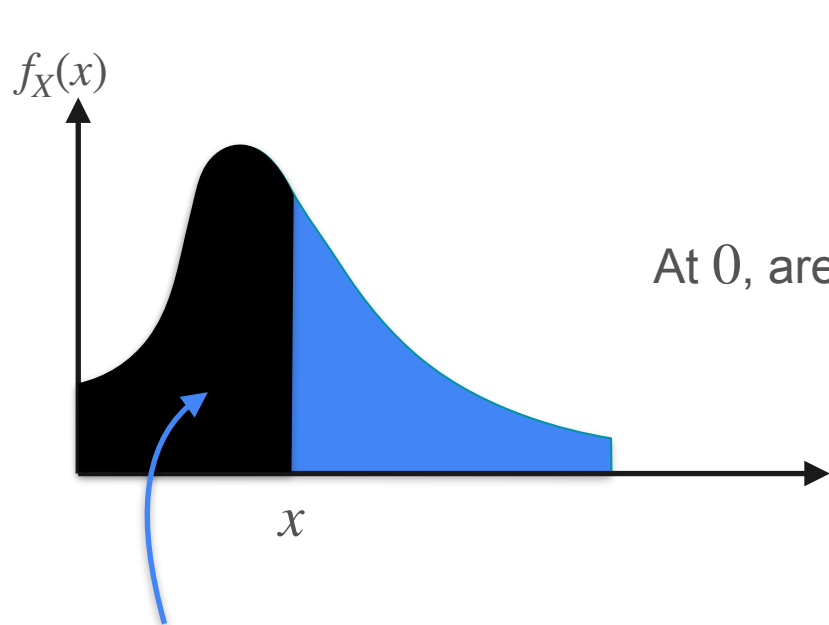


Cumulative Distribution

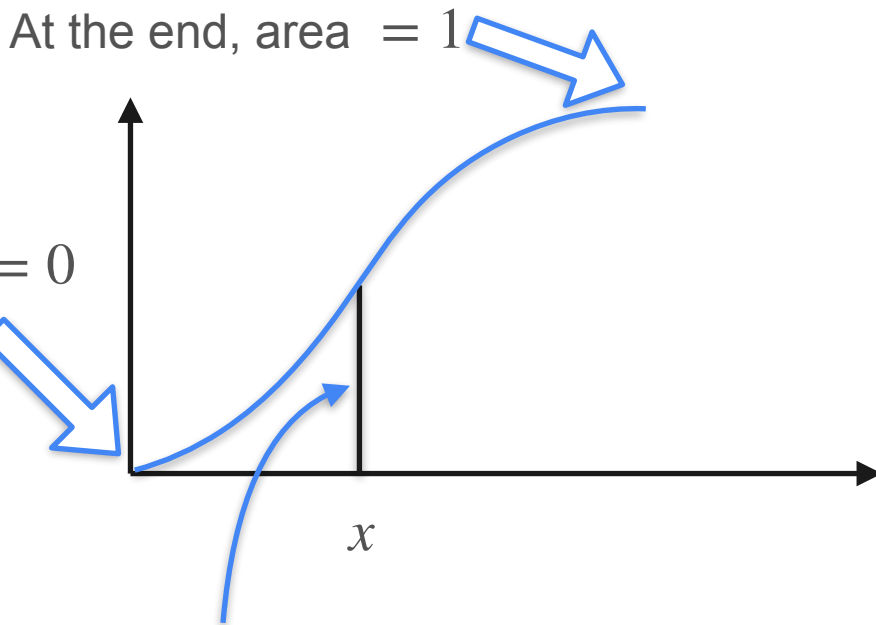


Cumulative Distribution

CDF: Cumulative distribution function



At 0, area = 0



$P(\text{less than or equal to 2 minutes}) = 0.5$

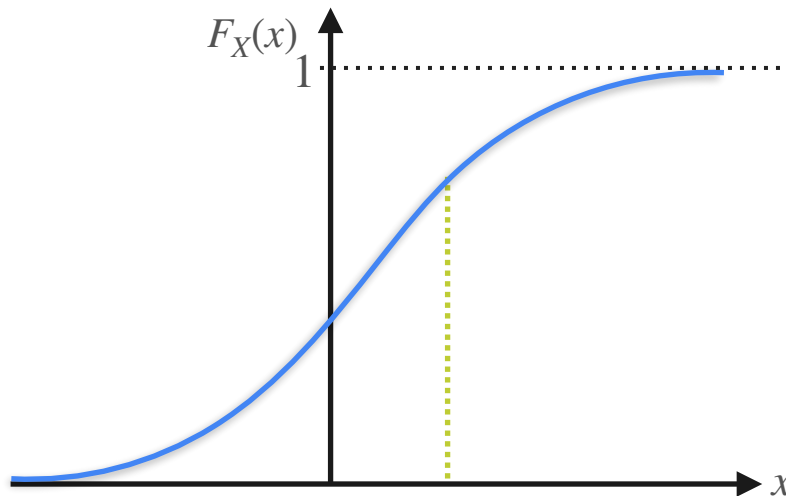
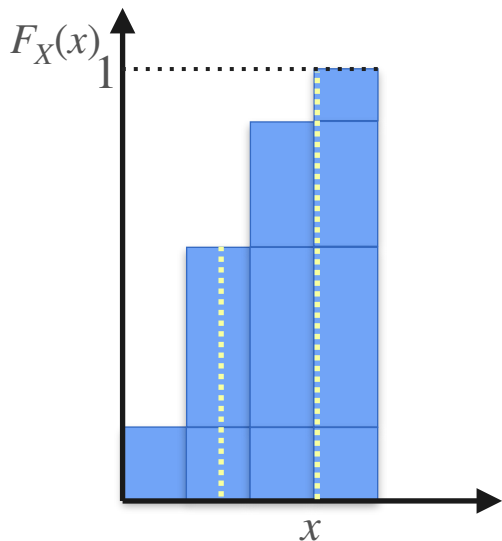
$P(\text{less than or equal to 2 minutes}) = 0.5$

Cumulative Distribution Function: Formal Definition

The CDF shows how much probability the variable has accumulated until a certain value

That means that

$$F_X(x) = \mathbf{P}(X \leq x) \leftarrow \text{It is defined for every real number}$$

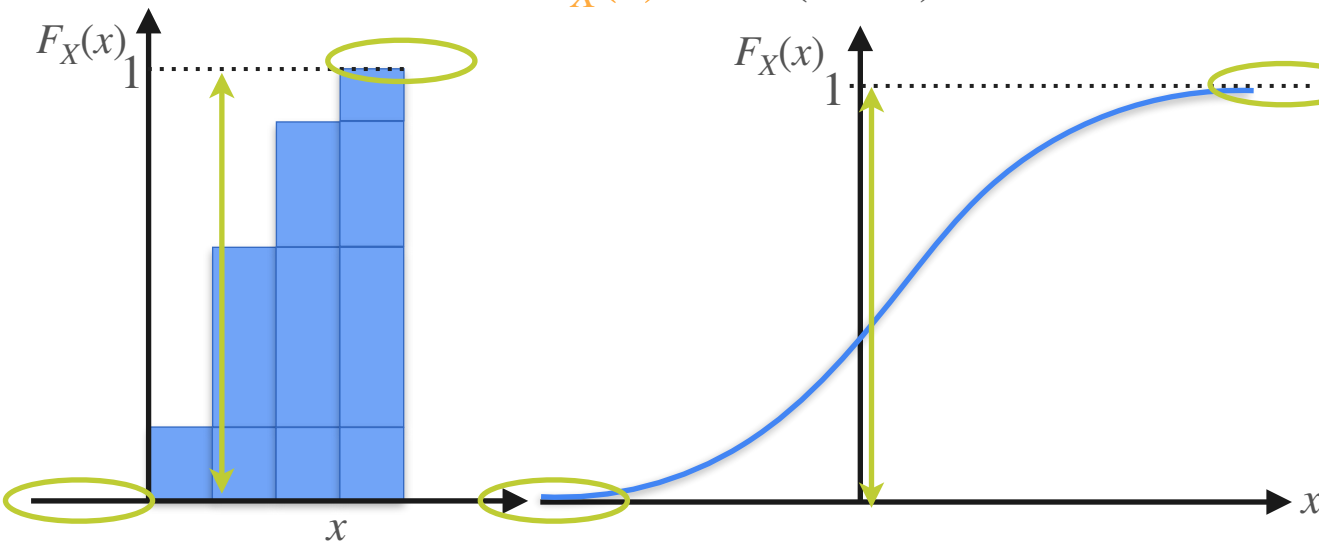


Cumulative Distribution Function: Formal Definition

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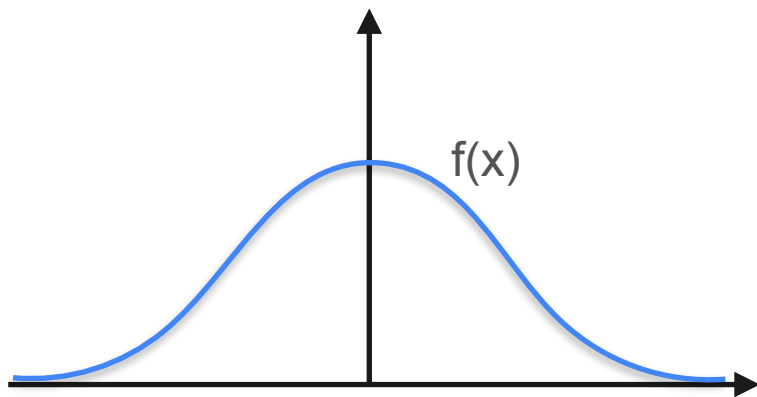


Properties

- $0 \leq F_X(x) \leq 1$
- Left “endpoint” is 0
- Right “endpoint” is 1
- **Never decreases**

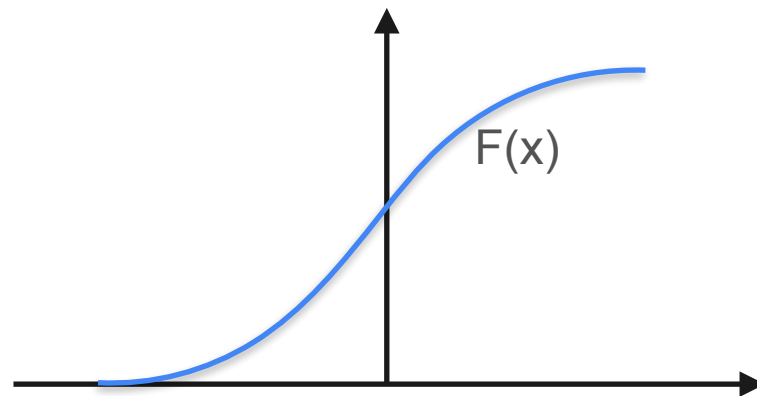
PDF and CDF Summary

PDF



- area = 1
- Always positive

CDF



- left “endpoint” is 0
- right “endpoint” is 1
- (endpoints can be at infinity)
- Always positive and increasing

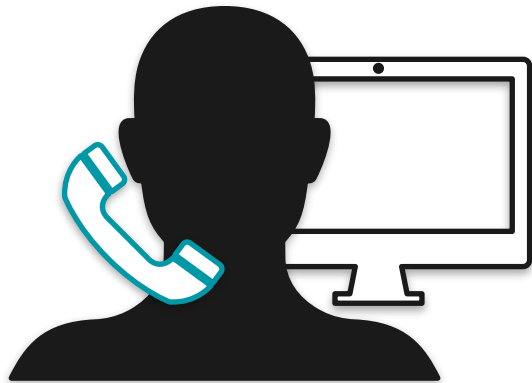


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Probability Distributions

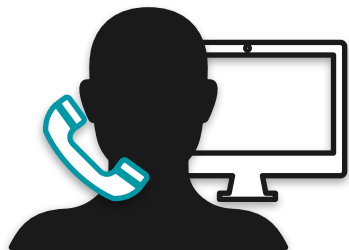
Uniform Distribution

Uniform Distribution: Motivation

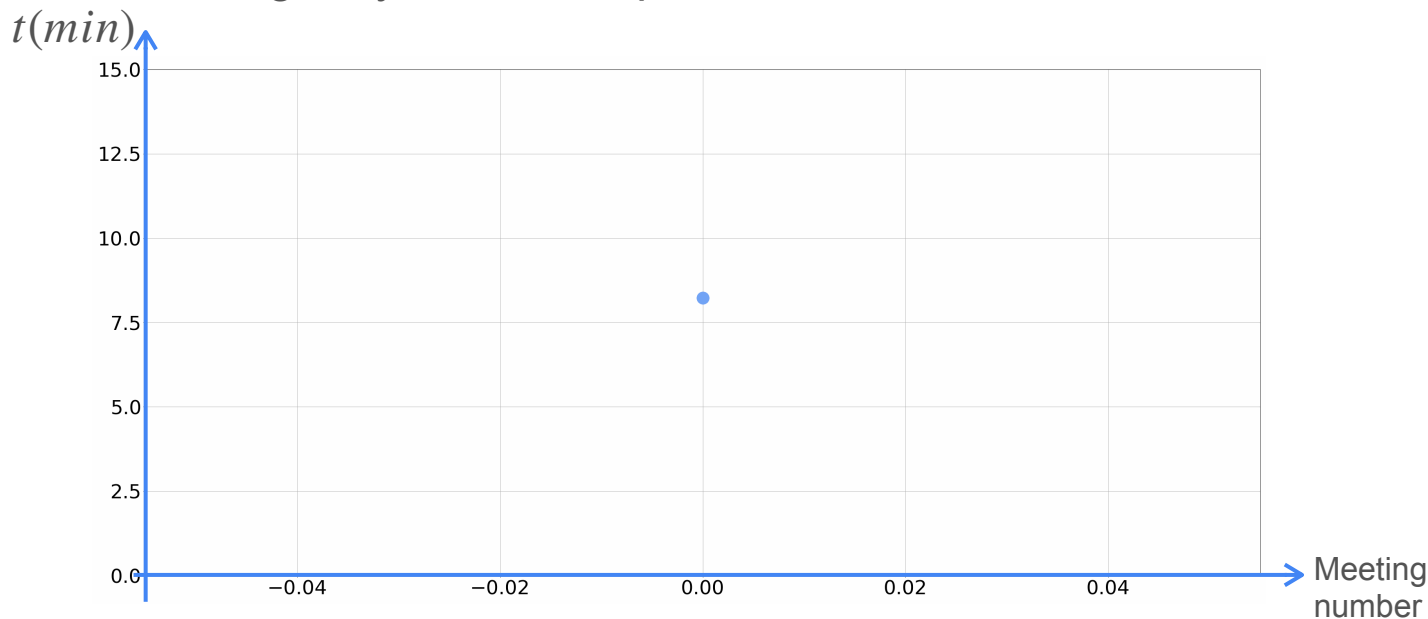


You're calling a tech support line.
They can answer any time between zero and 15 minutes and if they don't answer in this time, the line is disconnected.

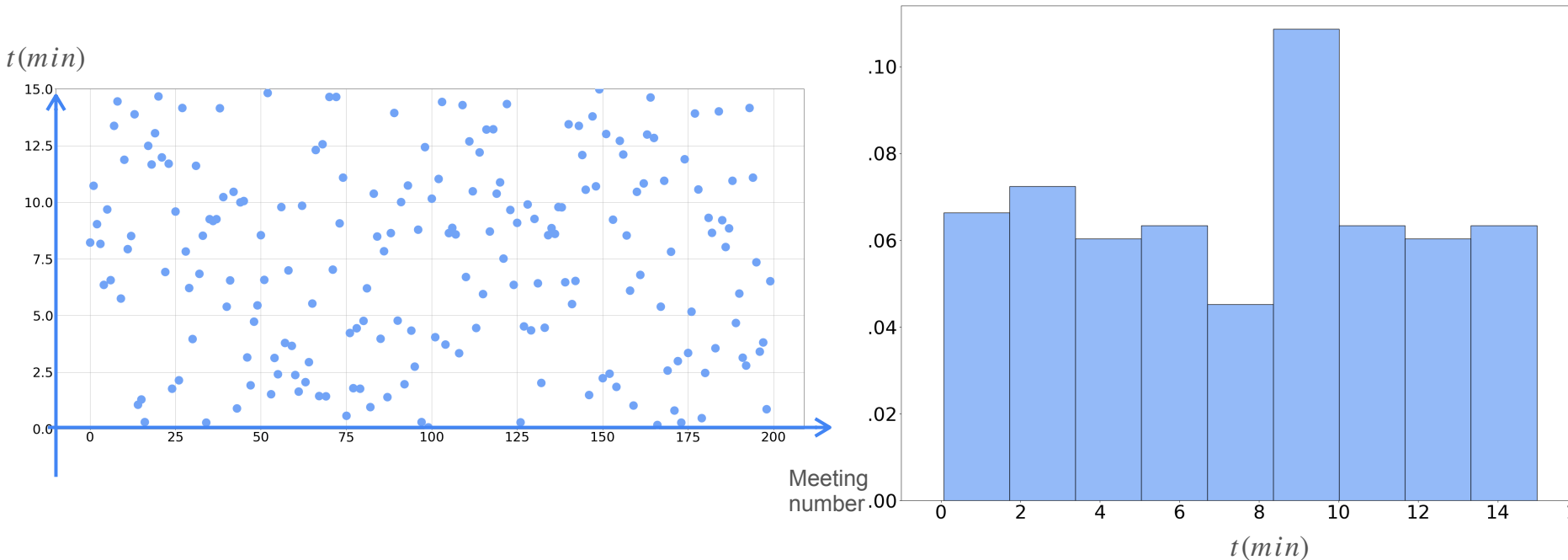
Uniform Distribution: Motivation



Last 200 times you called them, you took down notes of how long they took to respond



Uniform Distribution: Motivation



Uniform Distribution: Motivation

T: time (in minutes) you have to wait

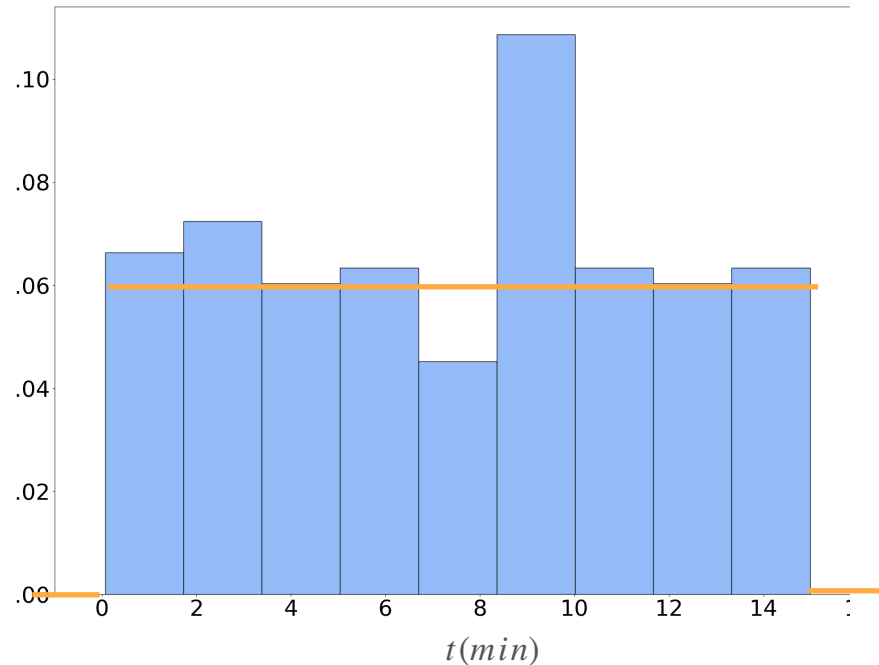
Any value between 0 and 15 minutes must have the same frequency of occurrence.



The pdf must be constant for all values in the interval (0,15)



Which constant? $\rightarrow 15 \times h = 1 \rightarrow h = \frac{1}{15} = 0.06$



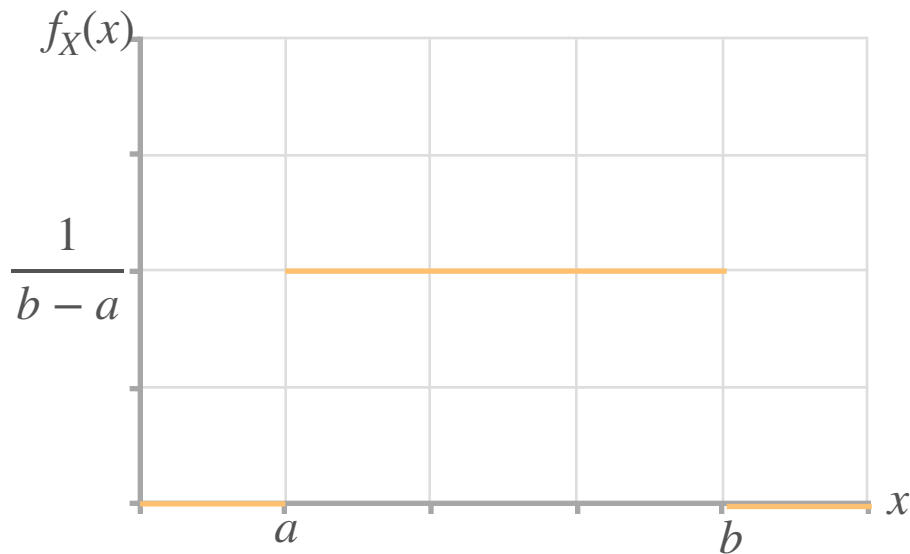
Uniform Distribution: Model

A continuous random variable can be modeled with a **uniform** distribution if all possible values lie in an interval and have the **same frequency** of occurrence

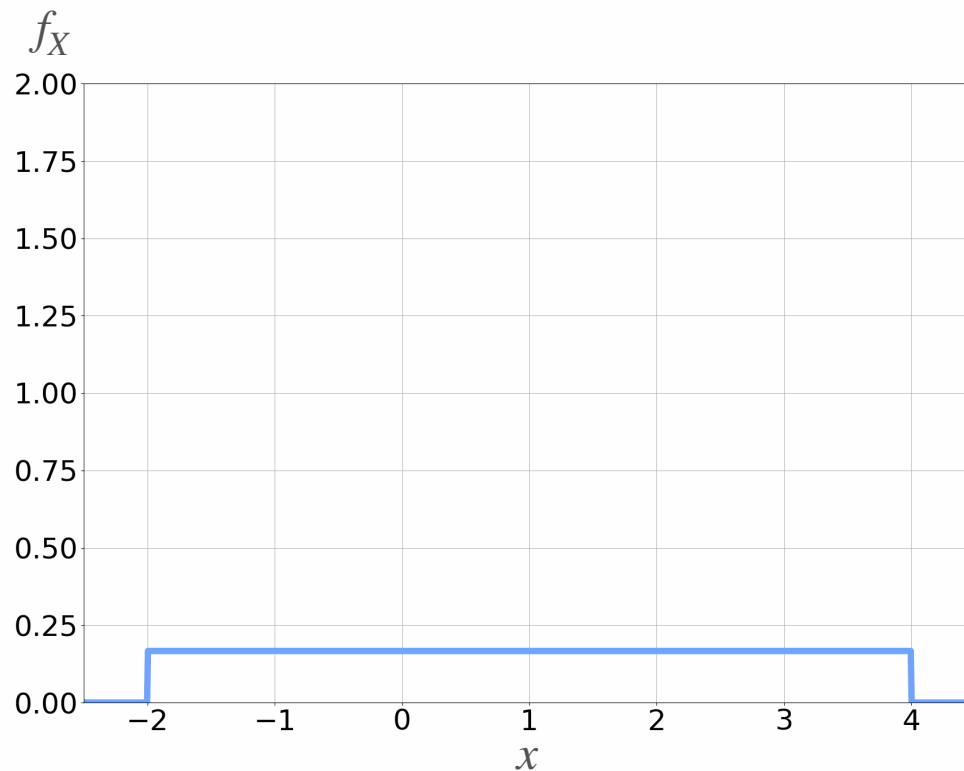
Parameters:

- a : beginning of the interval
- b : end of the interval

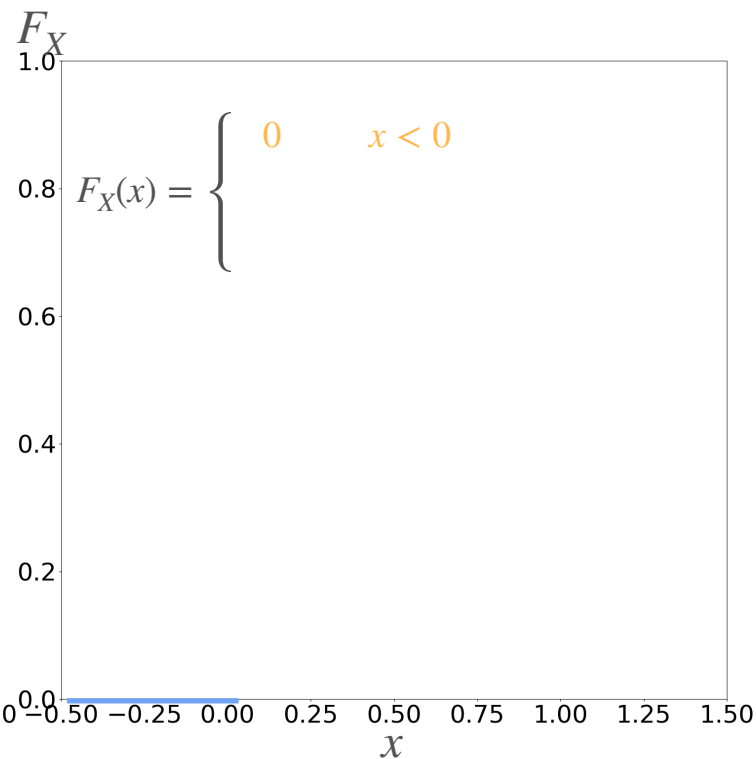
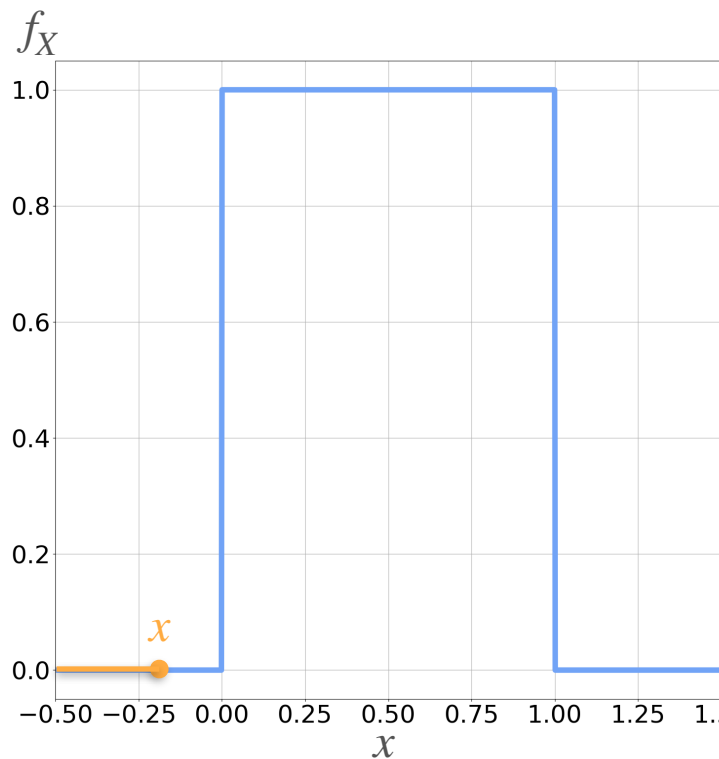
$$f_X(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & x \notin (a, b) \end{cases}$$



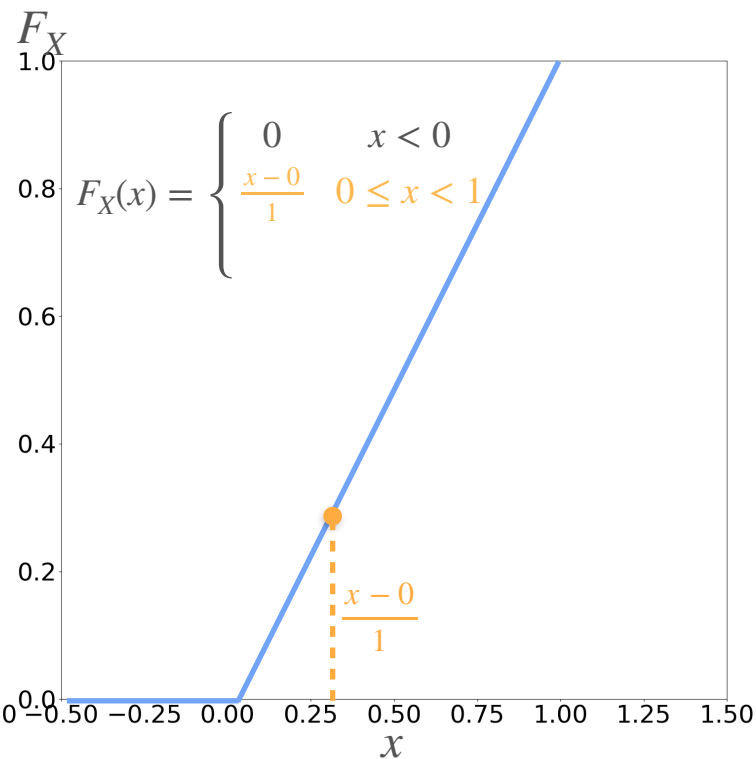
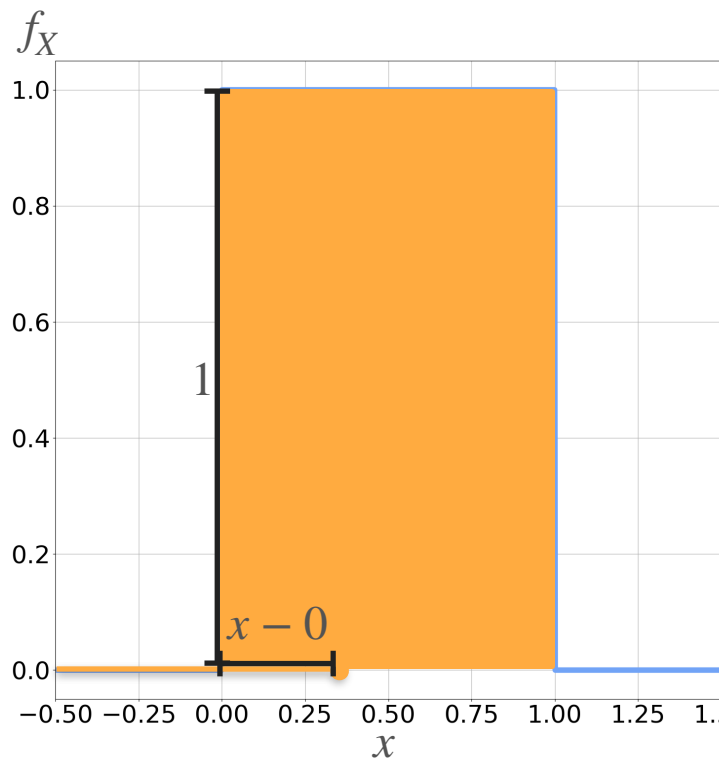
Uniform Distribution: PDF



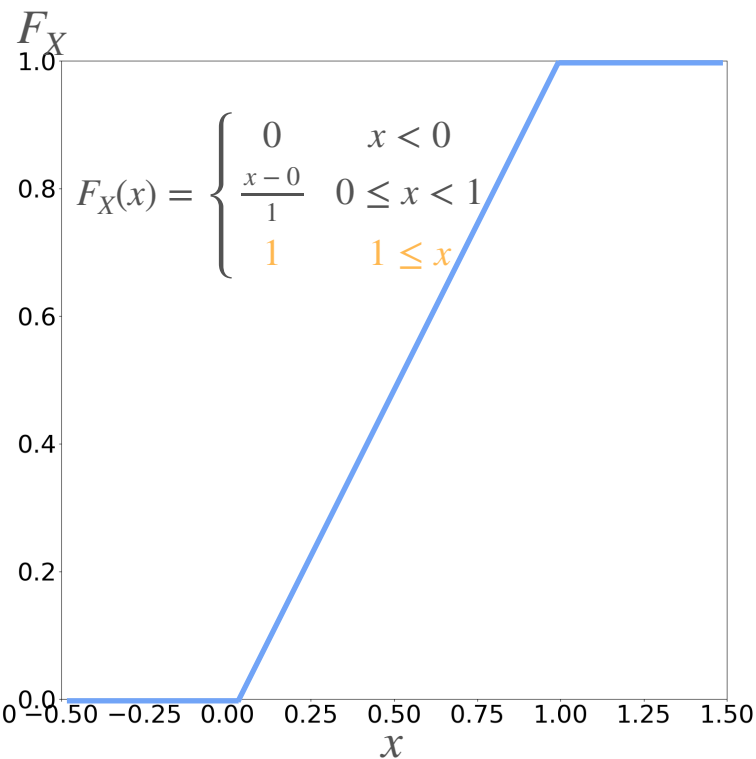
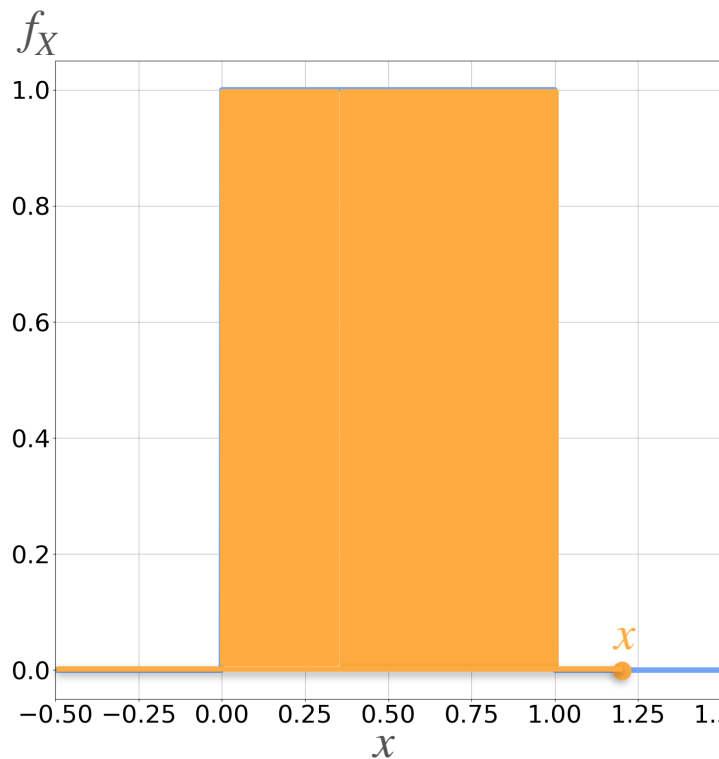
Uniform Distribution: CDF



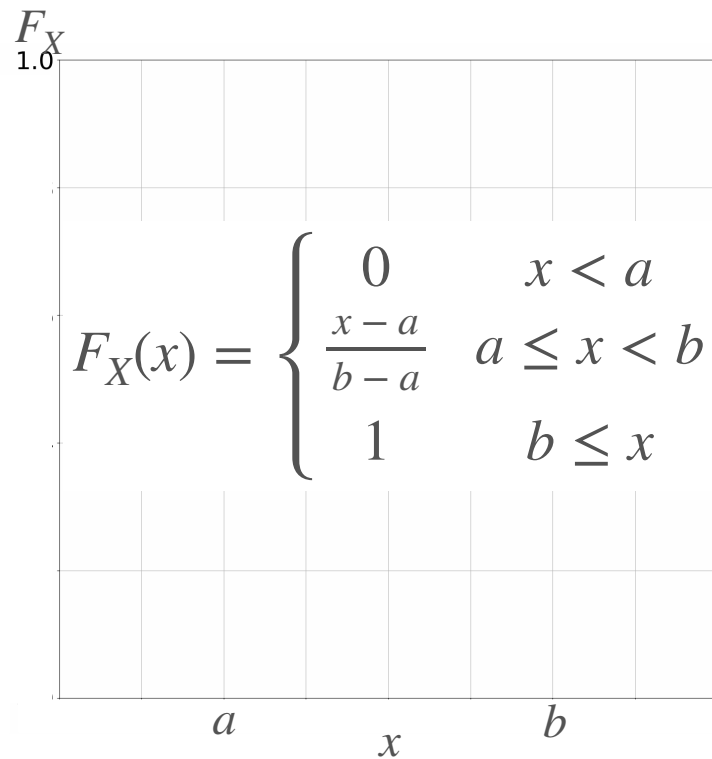
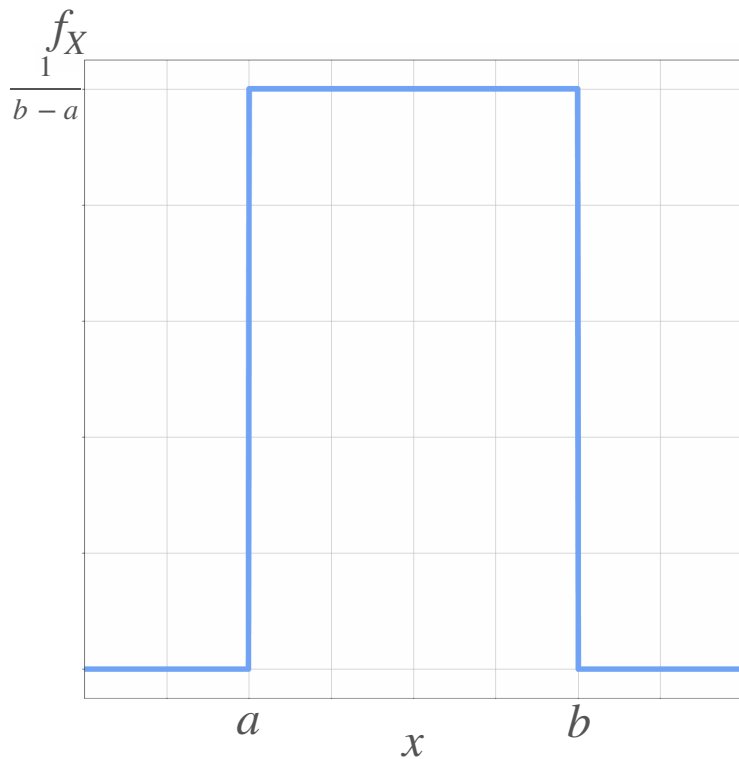
Uniform Distribution: CDF



Uniform Distribution: CDF



Uniform Distribution: CDF



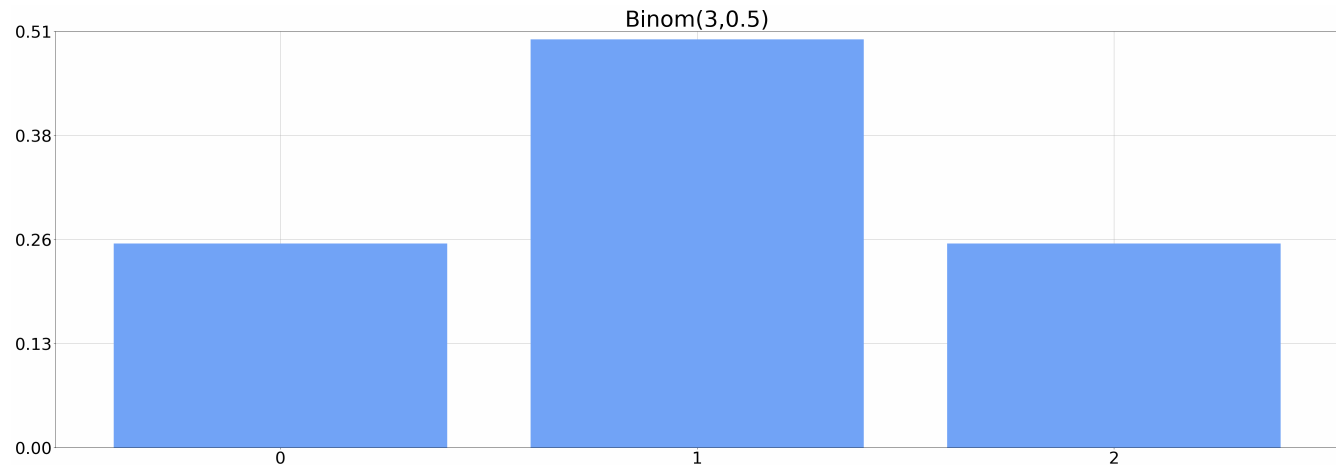


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Probability Distributions

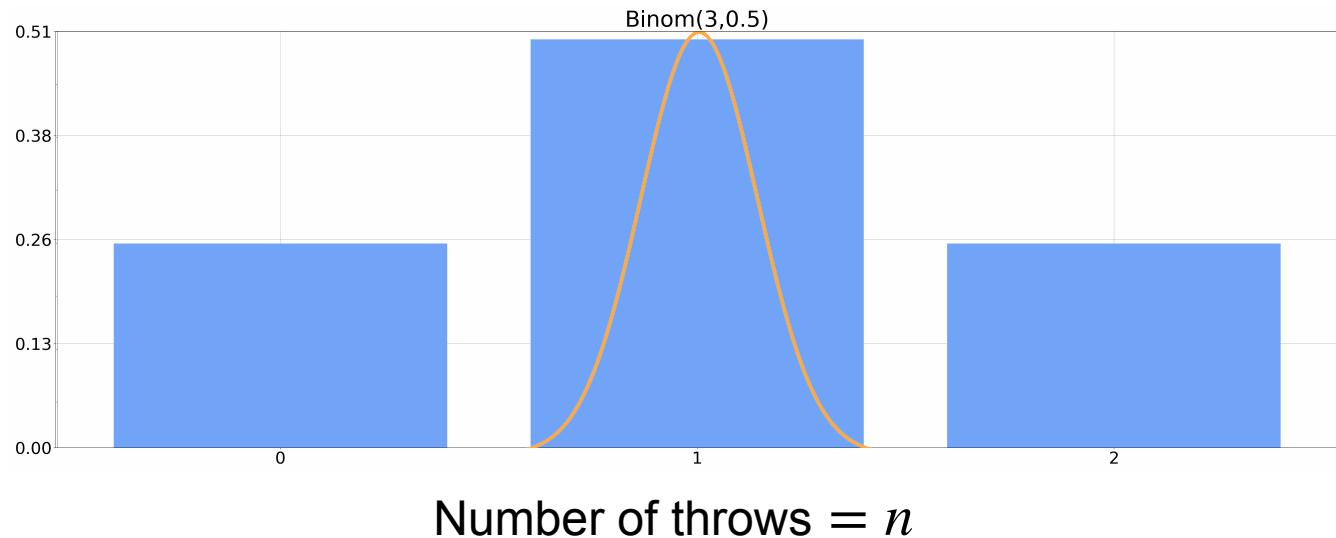
Normal distribution

Binomial Distribution With Very Large n

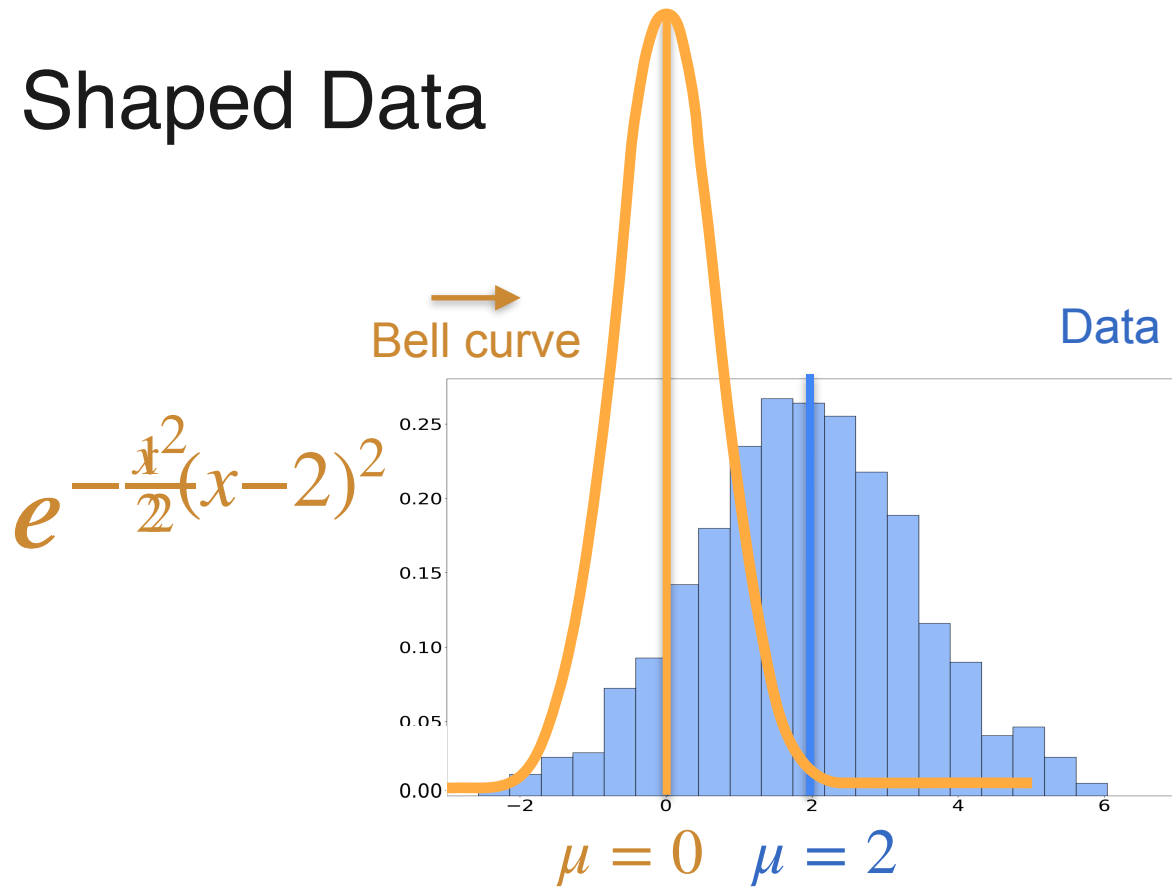


Number of throws = 2

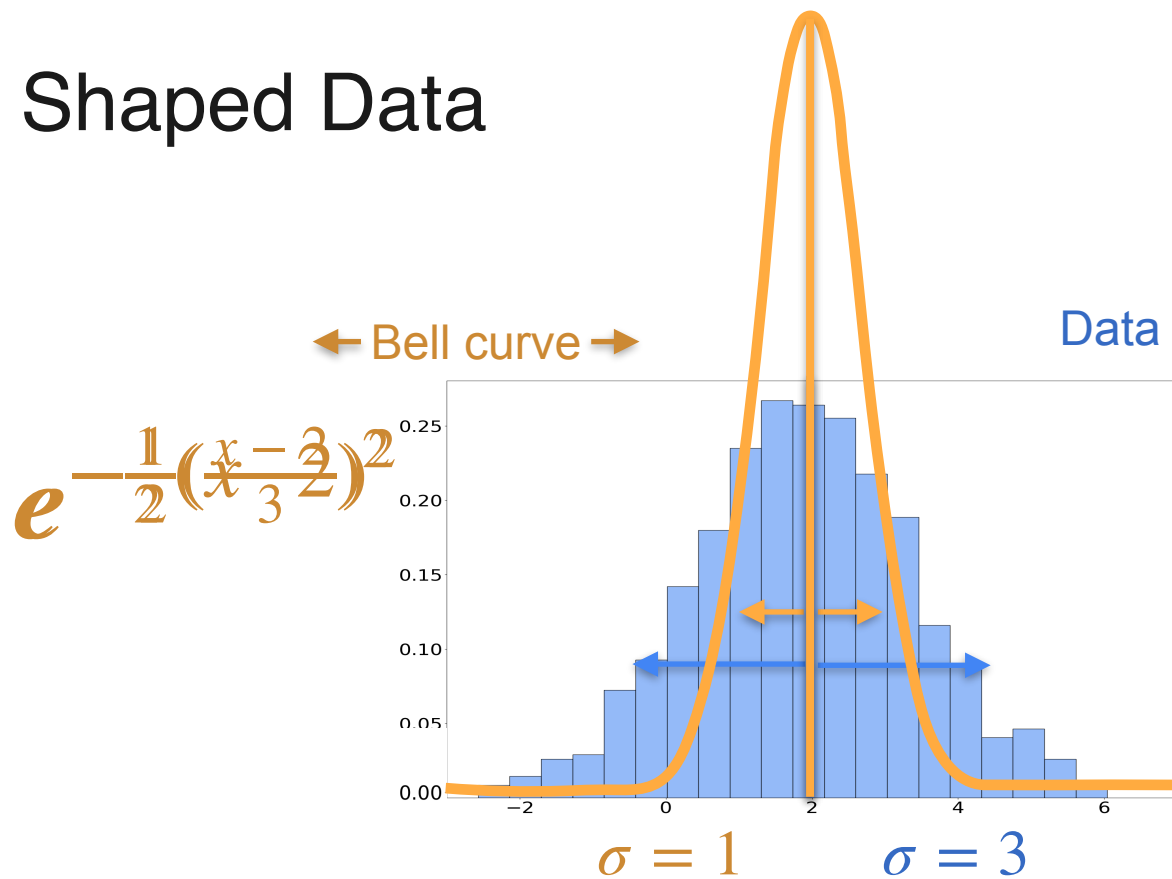
Binomial Distribution With Very Large n



Bell Shaped Data



Bell Shaped Data

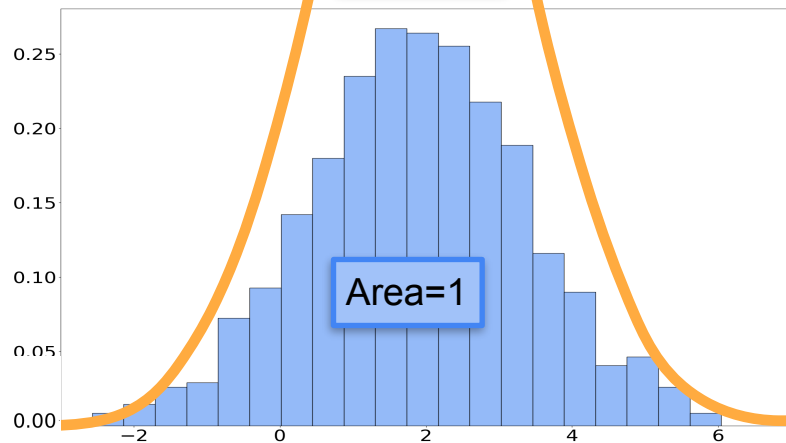


Bell Shaped Data

$$\frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2}$$

Bell curve

Data



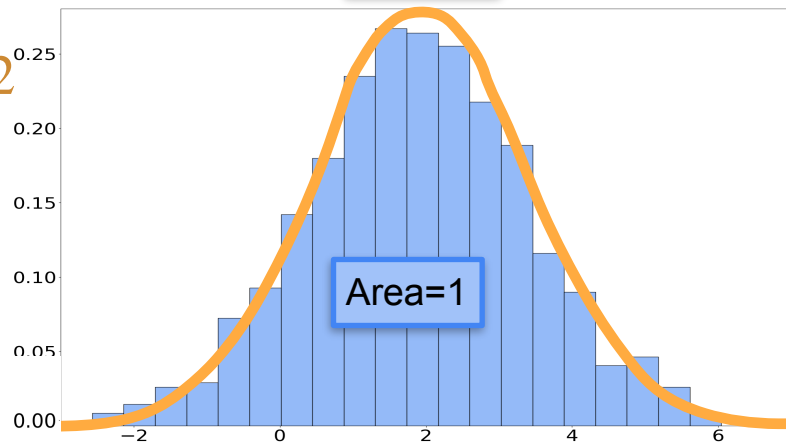
Bell Shaped Data

$$\frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2}$$

Bell curve

Area=1

Data



Bell Shaped Data

Mean = μ

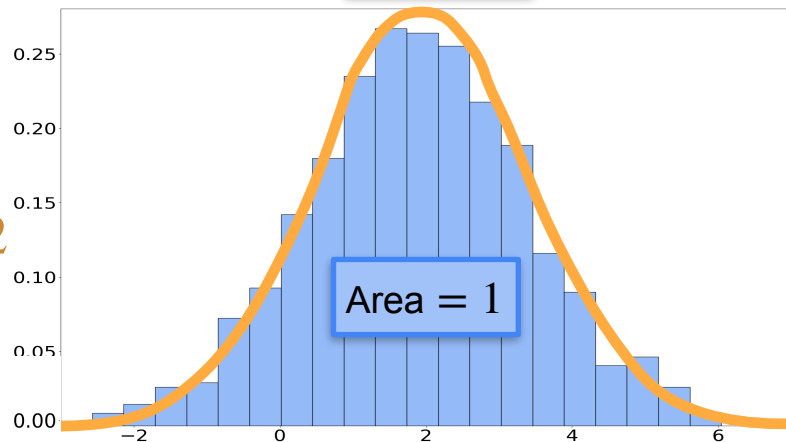
Standard deviation = σ

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Bell curve

Area = 1

Data



Bell Shaped Data

Mean = μ

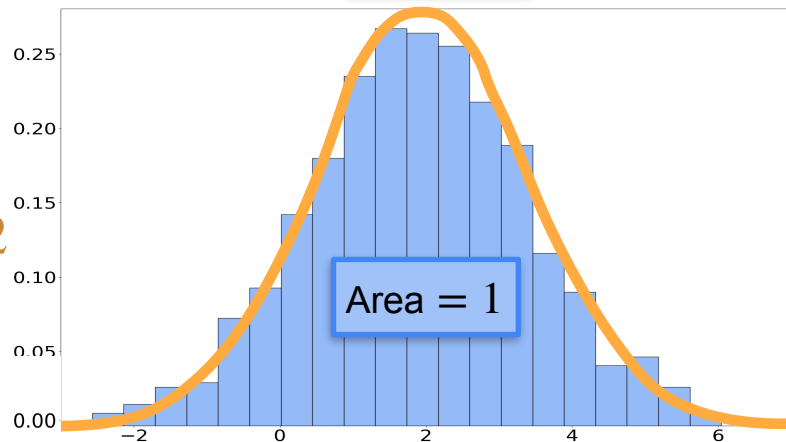
Standard deviation = σ

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Bell curve

Area = 1

Data



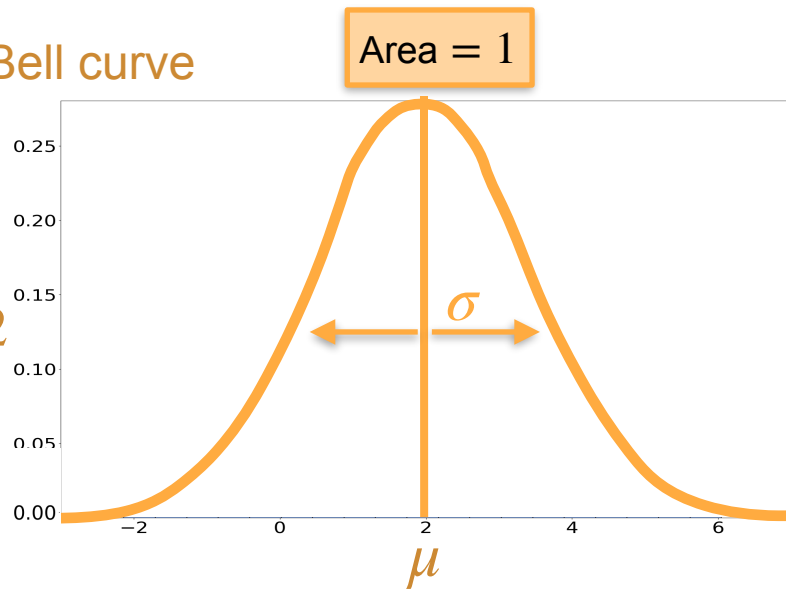
Normal Distribution

Mean = μ

Standard deviation = σ

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Bell curve



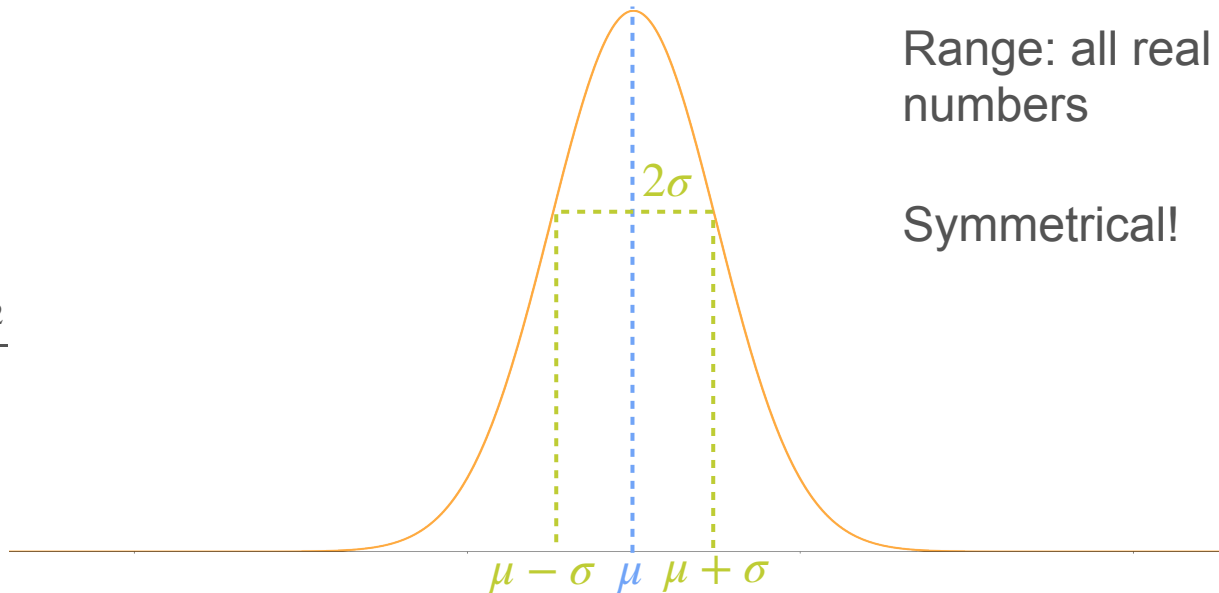
Normal Distribution

Parameters:

- μ : center of the bell
- σ : spread of the bell

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Scaling constant

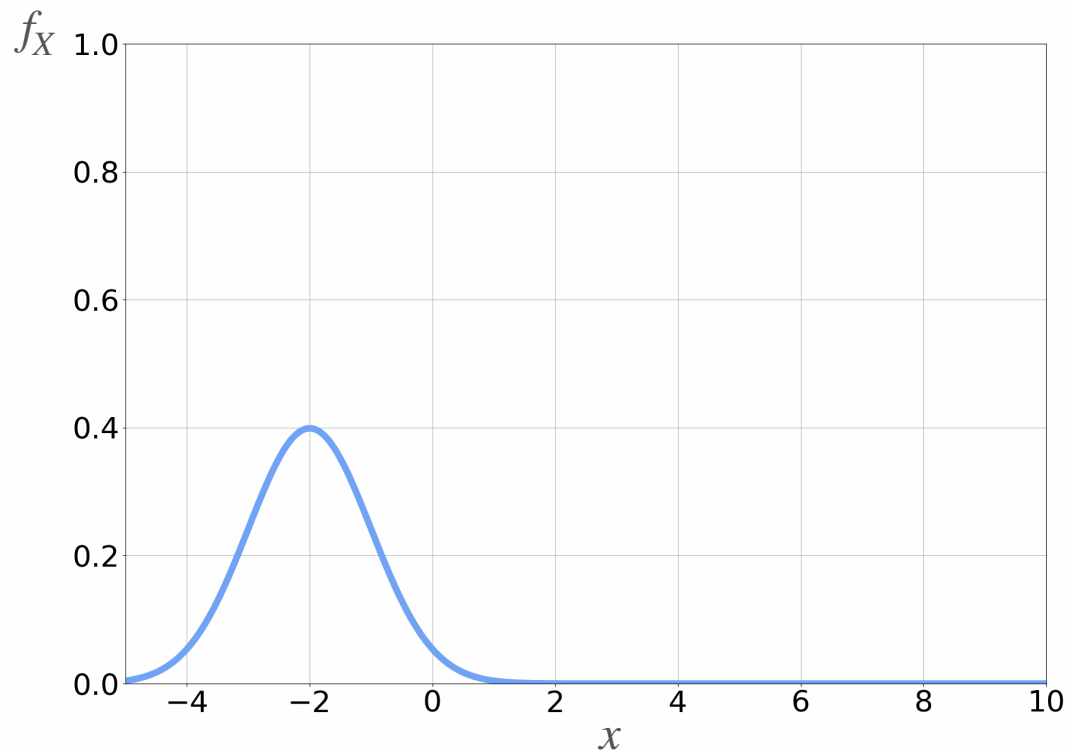


Normal Distribution

Parameters:

- μ : center of the bell
- σ : spread of the bell

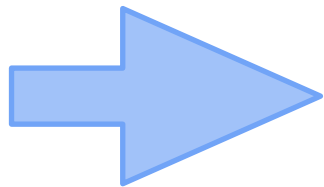
$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$



Normal Distribution - Notation

Parameters:

- μ : center of the bell
- σ : spread of the bell



$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

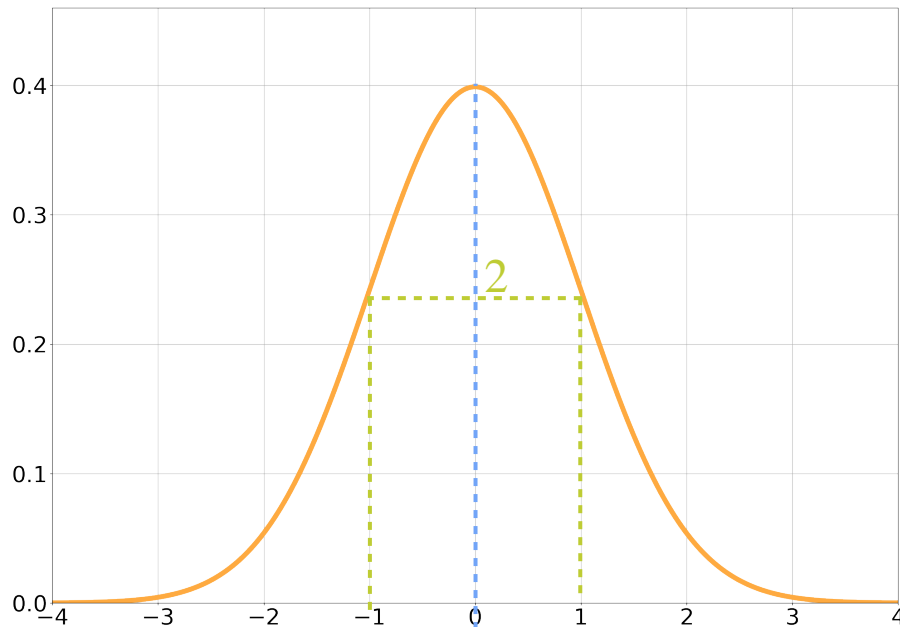
$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Standard Normal Distribution

Parameters:

- μ : 0
 - σ : 1
- $$X \sim \mathcal{N}(0, 1^2)$$

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi} \cdot 1} e^{-\frac{1}{2} \frac{(x-0)^2}{1^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2} \end{aligned}$$



Standardization

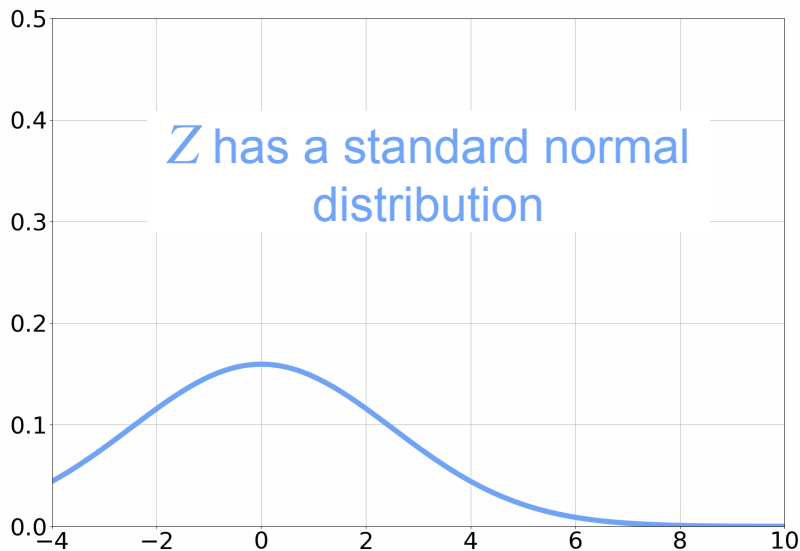
There's a really easy way to convert any normal distribution to the standard one!

X distributes normally with

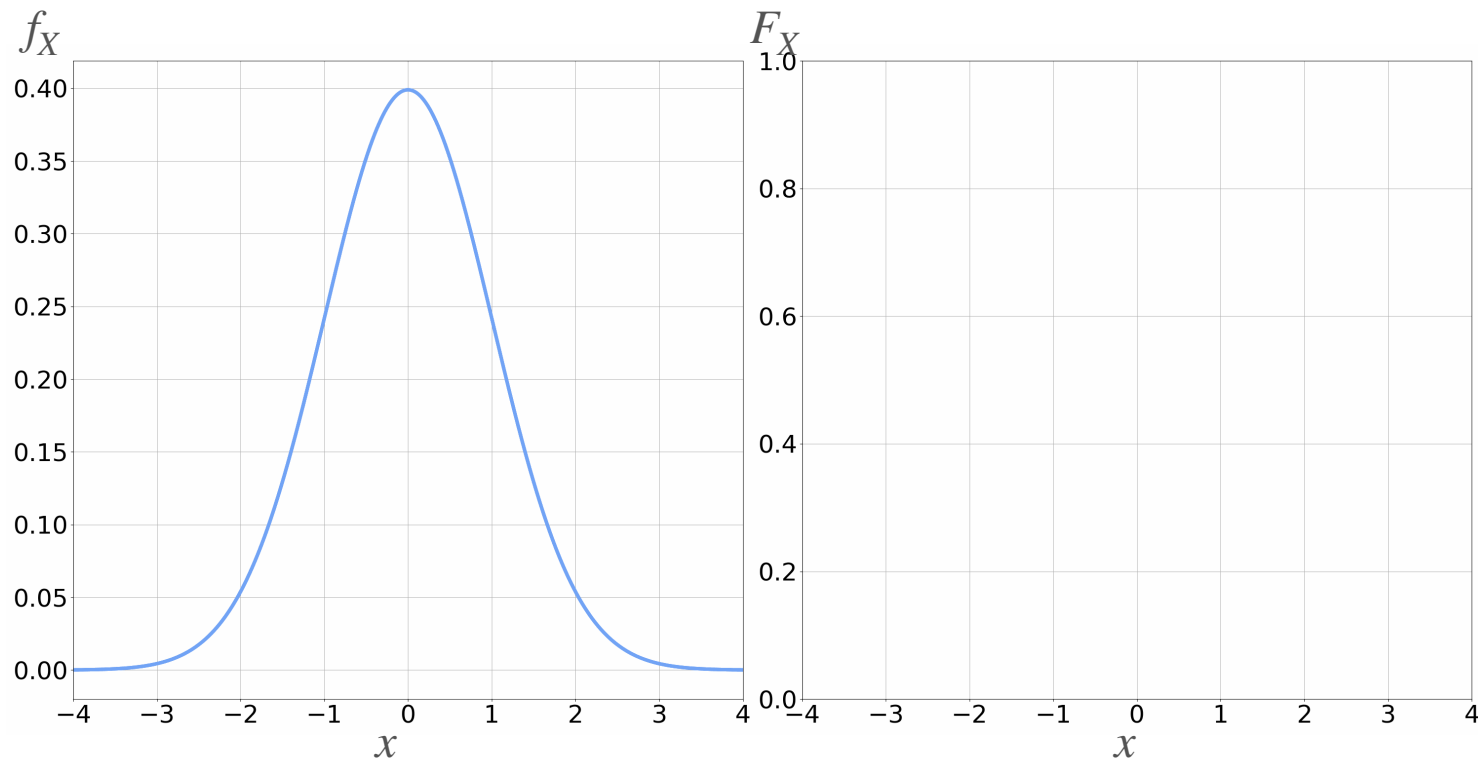
$$\mu = 2, \sigma = 2.5$$

$$Z = \frac{X - \mu}{\sigma}$$

Standardization is crucial to compare variables of different magnitudes!

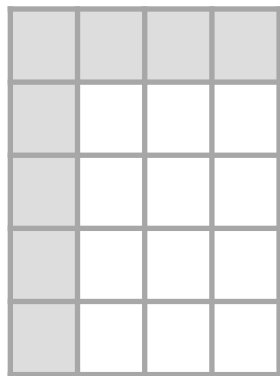


What Does the CDF Look Like?



Computing Probabilities From the PDF

This math can't be done by hand

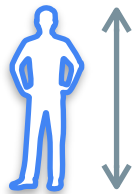


In the old days, people used tables of data



Now, you can use the help of some software to do the approximate area under the curve for you!

Normal Distribution: Applications



Height



Weight



IQ



Noise in a
communication channel

In general, characteristics that are the sum of many independent processes

Many models in ML are designed under the assumption that the variables follow a normal distribution



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Probability Distributions

(Optional)
Chi-squared distribution

Chi-Square Distribution: Motivation



Communication channel

Noise

Interference from other devices

Obstructions like walls, trees, etc.

Atmospheric conditions: rain, humidity, etc

Electrical interference, i.e. from power lines

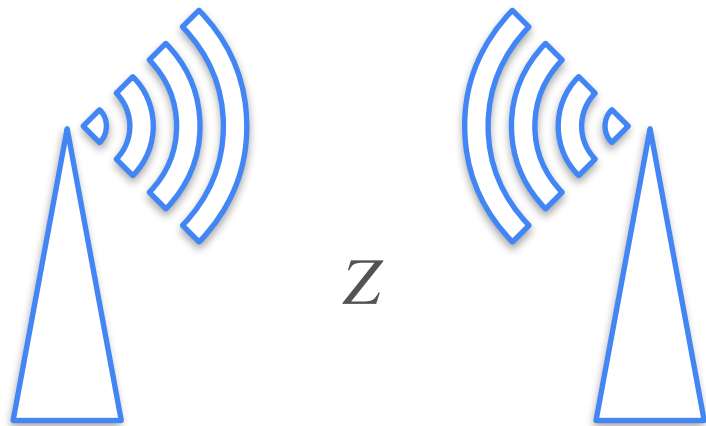
Others



Message sent: 10010

Message received: $10010 + Z$

Chi-Square Distribution: Motivation



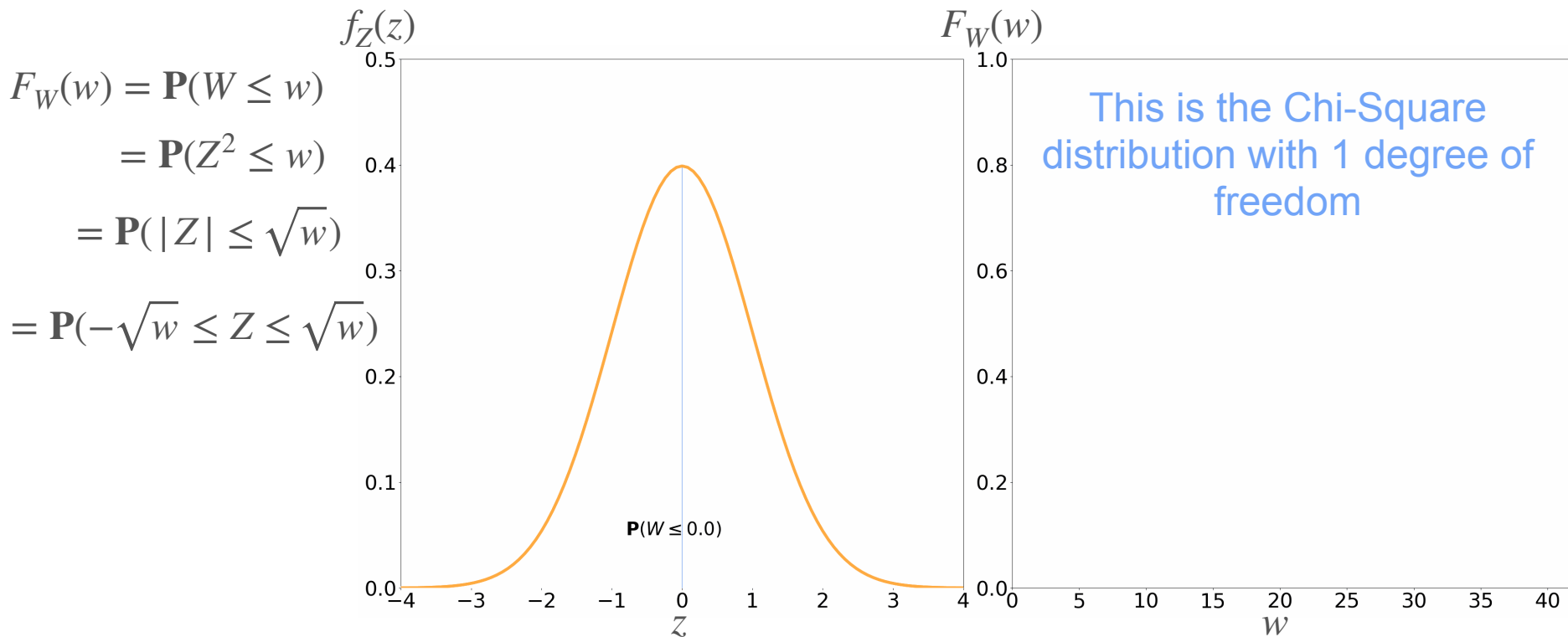
The communication channel has noise with a standard normal distribution

What is the **power** of the noise in the channel?

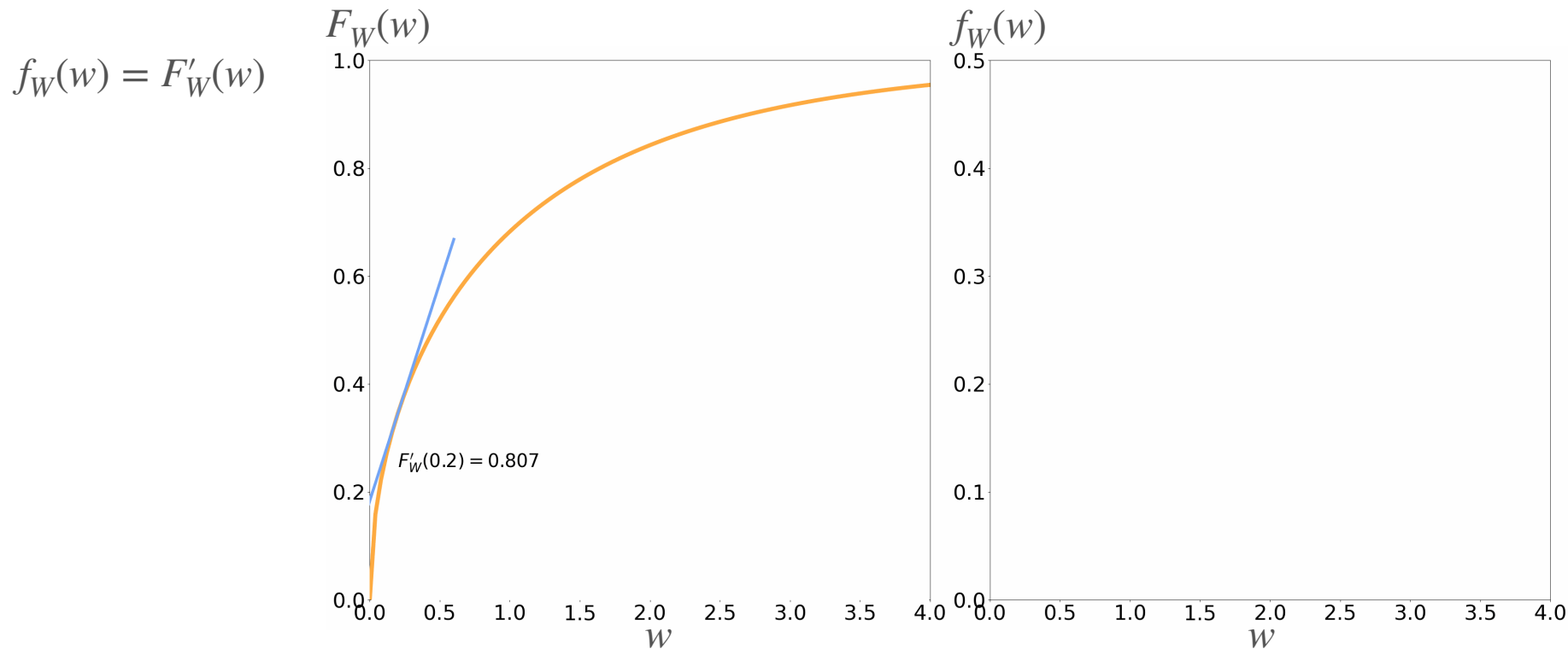
$$W = Z^2$$

What is the distribution of W ?

Chi Square Distribution



Chi Square Distribution



Chi-Square Distribution

Accumulated power over 2 transmissions?

$$W_2 = Z_1^2 + Z_2^2$$

Chi-Square with 2 df

Accumulated power over 5 transmissions?

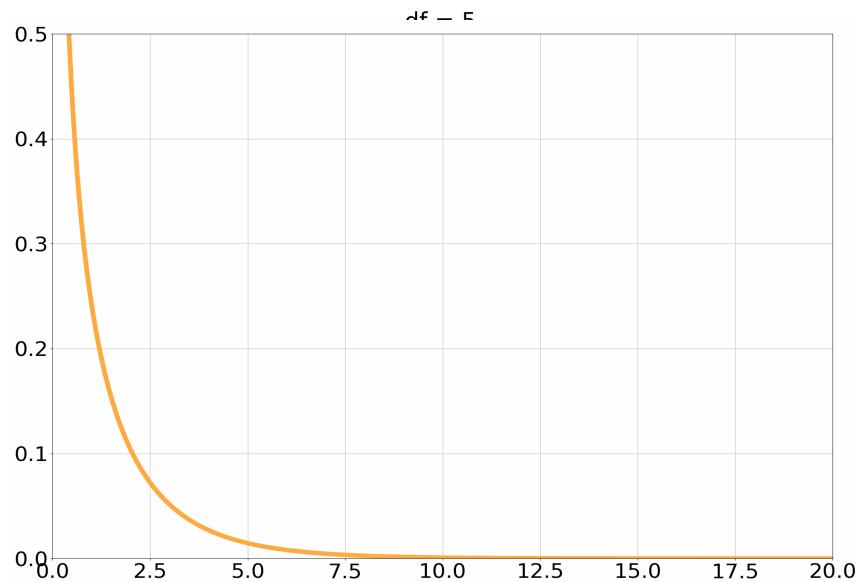
$$W_5 = Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 + Z_5^2$$

Chi-Square
with 5 df

Accumulated power over k transmissions?

$$W_k = \sum_{i=1}^k Z_i^2$$

Chi-Square with k df



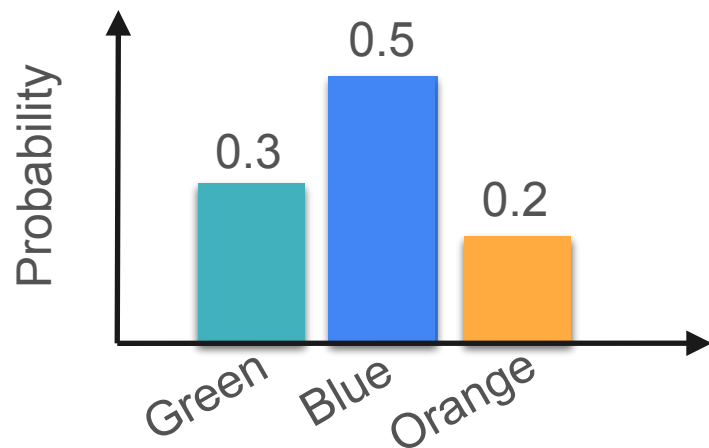


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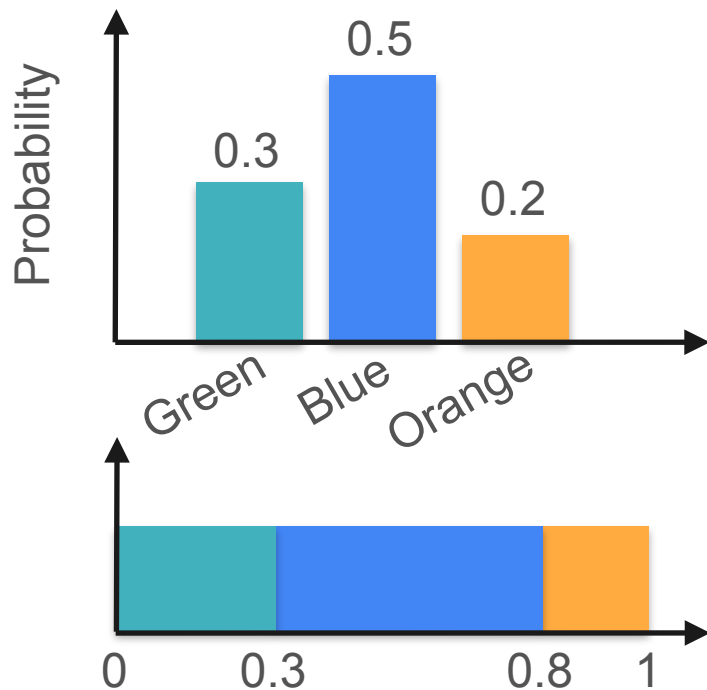
Probability Distributions

Sampling from a Distribution

Sampling From a Distribution

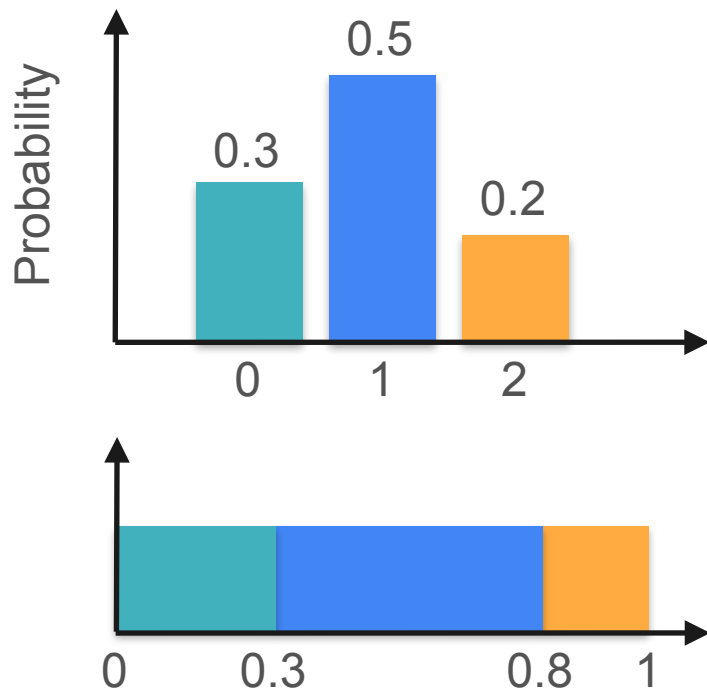


Sampling From a Distribution



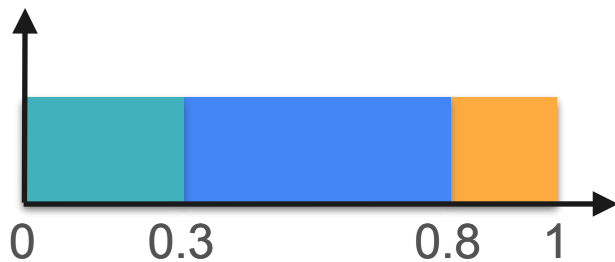
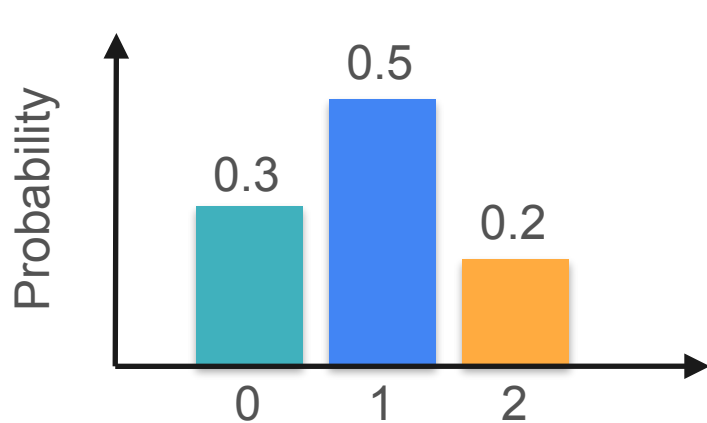
- **Step 1:** generate a random number between 0 and 1
- **Step 2:** find out which interval the number belongs to
 - $[0, 0.3)$
 - $[0.3, 0.8)$
 - $[0.8, 1]$
- **Step 3:** Assign an outcome based on the interval

Sampling From a Distribution

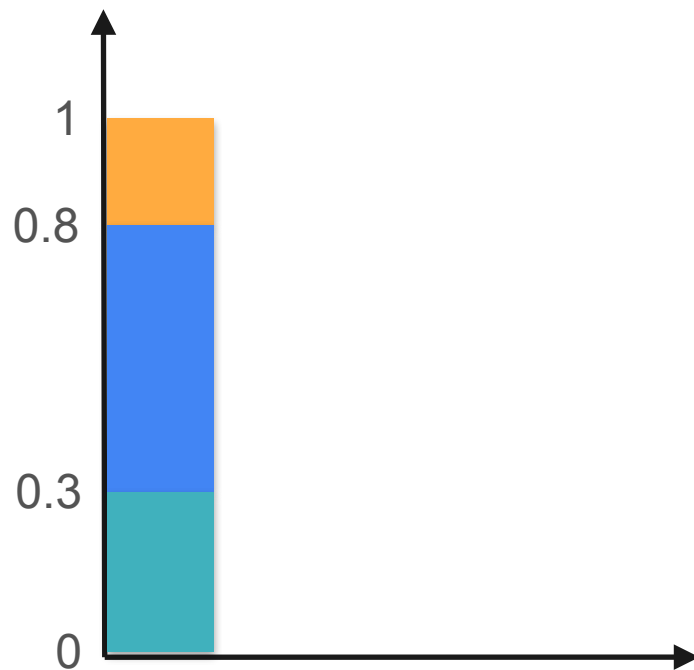
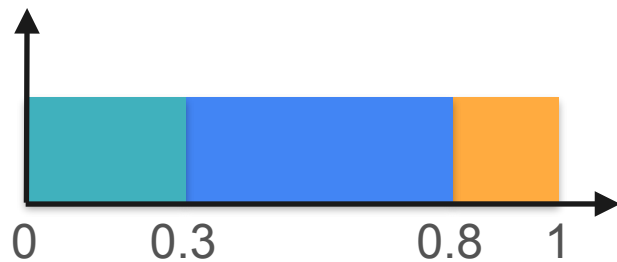
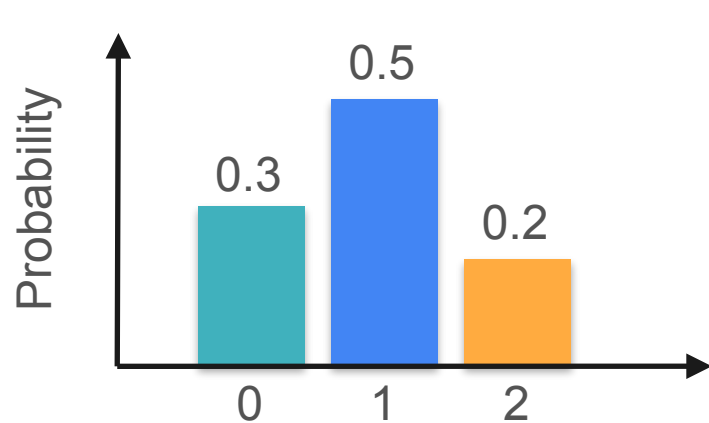


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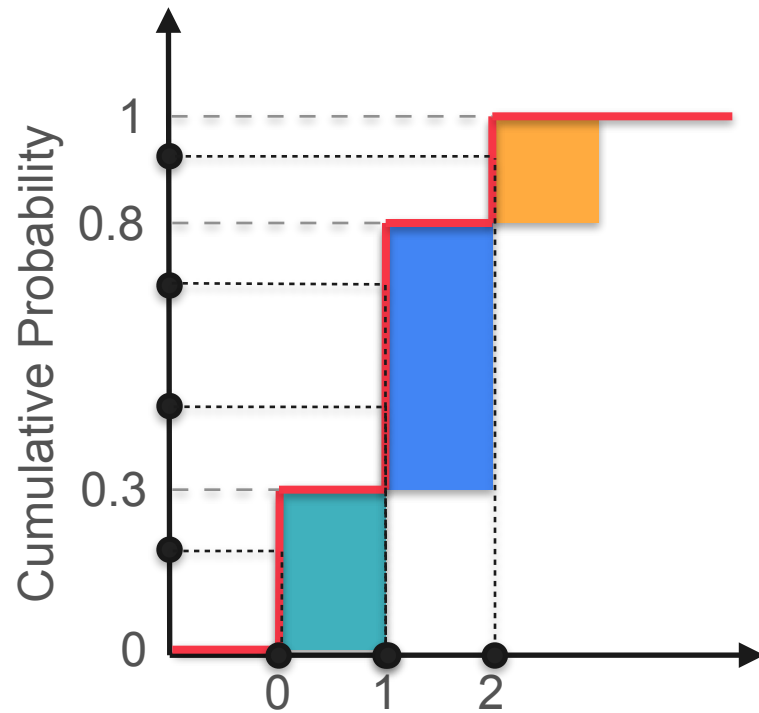
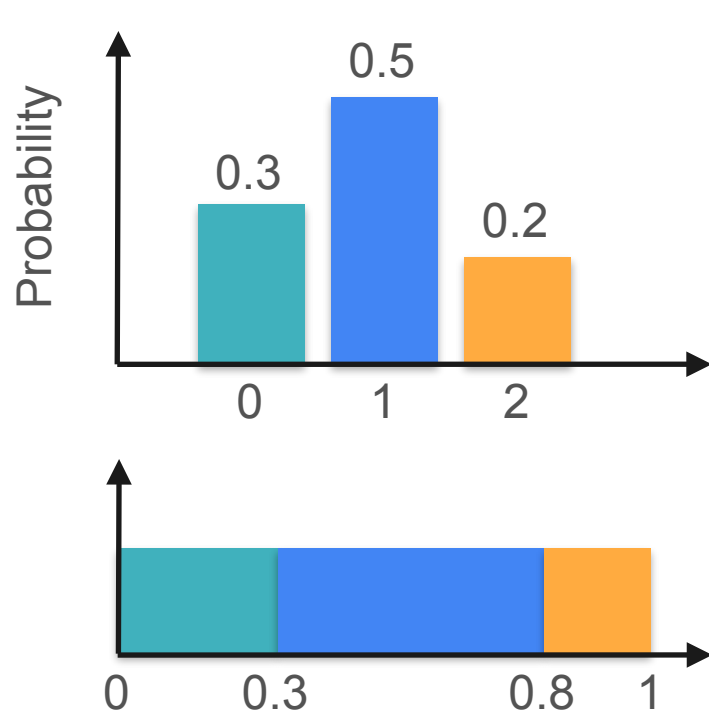
Sampling From a Distribution



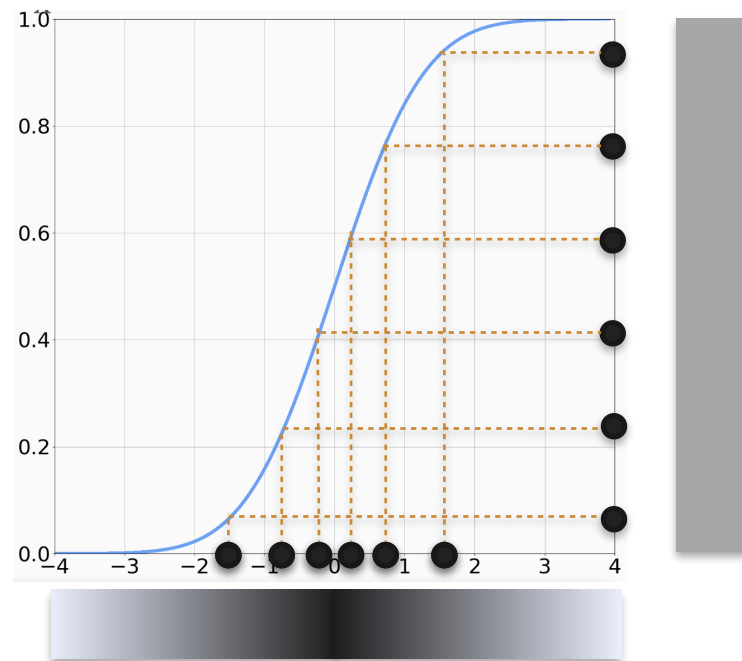
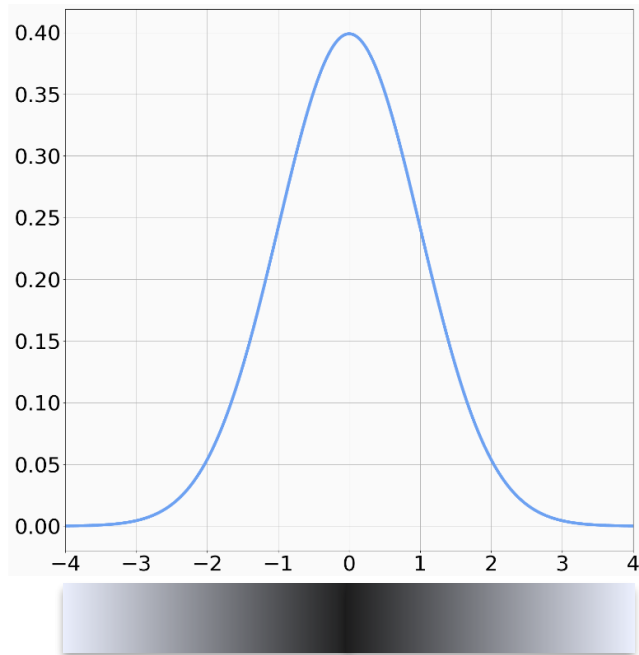
Sampling From a Distribution



Sampling From a Distribution



Sampling From a Normal Distribution





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Probability Distributions

Conclusion

Week 1 - Conclusions

Talking head